# Development of Thermoplastic Biomaterials for Thermoformable Clear Aligners

by

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*Mieleni minun tekevi,*
*aivoni ajattelevi*
*lähteäni laulamahan,*
*saa'ani sanelemahan.*

- Elias Lönnrot, *Kalevala*, Runo I

*"I am driven by my longing,*
*and my mind impels me*
*to begin my singing,*
*to start my reciting."*
— English translation by Keith Bosley

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# OTHER RESEARCH MERITS

The following manuscripts have been prepared based on work carried out during the course of this doctoral thesis and are intended for submission to peer-reviewed journals following the thesis defense. Material from these manuscripts is reproduced, in whole or in part, within the corresponding chapters of this thesis as indicated below; however, as these manuscripts had not been submitted for publication at the time of thesis submission, they are not included in the *Published and submitted content* chapter.

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# LIST OF ACRONYMS

| | |
|---|---|
| **3PB** | 3-Point Bending test |
| **ATR-FTIR** | Attenuated Total Reflectance Fourier-Transform Infrared spectroscopy |
| **bFGF** | basic human Fibroblast growth factor |
| **CAB** | Cellulose Acetate Butyrate |
| **CAT** | Clear Aligner Treatment |
| **CBCT** | Cone Beam Computerised Tomography |
| **CBDO** | 2,2,4,4-Tetramethyl-1,3-cyclobutanediol |
| **CGRP** | Calcitonin Gene-Related Peptide |
| **CHDM** | 1,4-Cyclohexanedimethanol |
| **coPES** | Co-Polyester |
| **DAPI** | 4',6-diamidino-2-phenylindole |
| **DTC** | Direct-to-customer |
| **DLP** | Digital Light Processing |
| **DMTA** | Dynamic Mechanothermal Analysis |
| **DOR** | Degree of Ordering of the hard TPU segment |
| **DPS** | Degree of Phase Separation |
| **DSC** | Differential Scanning Calorimetry |
| **DZ** | Fractional Double Zener model |
| **EG** | Ethylene Glycol |
| **EU** | European Union |
| **EVA** | Ethylene-Vinyl Acetate |
| **FBS** | Fetal Bovine Serum |
| **FDI** | World Dental Federation |
| **FEM** | Finite element method |

**FFD** Fused Filament Deposition

**GP** Guiu-Pratt logarithmic model

**HDI** Hexamethylene Diisocyanate

**hGFs** Human Gingival Fibroblasts

**hTPU** High-hardness Thermoplastic Polyurethane

**IL-1$\beta$** Interleukin-1$\beta$

**IL-6** Interleukin-6

**KWW** Kohlrausch-Williams-Watts model

**LCA** Life cycle assessment

**LDH** lactate dehydrogenase enzyme

**MDI** Methylene Diphenyl Diisocyanate

**ML** Multilayer

**mTPU** Middle-hardness Thermoplastic Polyurethane

**NCD** Non-Communicable Disease

**PBS** Phosphate-Buffered Saline

**PBT** Polybutylene terephthalate

**PC** Polycarbonate

**PCL** Polycaprolactone

**PCTG** Polycyclohexylenedimethylene Terephthalate Glycol-modified

**PDL** Periodontal ligament

**PE** Polyethylene

**PEF** Polyethylene furanoate

**PEIT** Poly(ethylene-co-isosorbide) terephthalate

**PEN** Polyethylene Naphthalate

**PET** Polyethylene Terephthalate

**PET-PEN** PET-PEN Copolymer

**PETG** Polyethylene Terephthalate Glycol-modified

**PMMA** Polymethyl Methacrylate

**PP** Polypropylene

**PVC** Polyvinyl Chloride

**RM** Reduced elastic Modulus

**ROS** Reactive Oxygen Species

**SLA** Stereolithography

**SP** Substance P

**sTPU** Soft-hardness Thermoplastic Polyurethane

**TDI** Toluene Diisocyanate

**Tg** Glass transition Temperature

**TGA** Thermogravimetric Analysis

**THF** Tetrahydrofuran

**TNF-$\alpha$** Tumour Necrosis Factor-$\alpha$

**TPA** Terephthalic Acid

**TPU** Thermoplastic Polyurethane

**USA** United States of America

**UV** Ultra-violet light

**VA** Vinyl Acetate

**vGP** van Grup-Palmen graph

**VST** Vicat Softening Temperature

**WHO** World Health Organization

# 1. STATE OF THE ART

Malocclusion and dental misalignment, whether genetic or multifactorial in origin, precipitate a range of disorders that vary in severity and systemic impact. Beyond compromising periodontal function, these conditions may contribute to gastrointestinal issues and psychosocial dysfunctions rooted in esthetics and self-perception, potentially leading to social isolation, as illustrated in the Figures 1.1 and 1.2.[1], [2], [3], [4], [5]

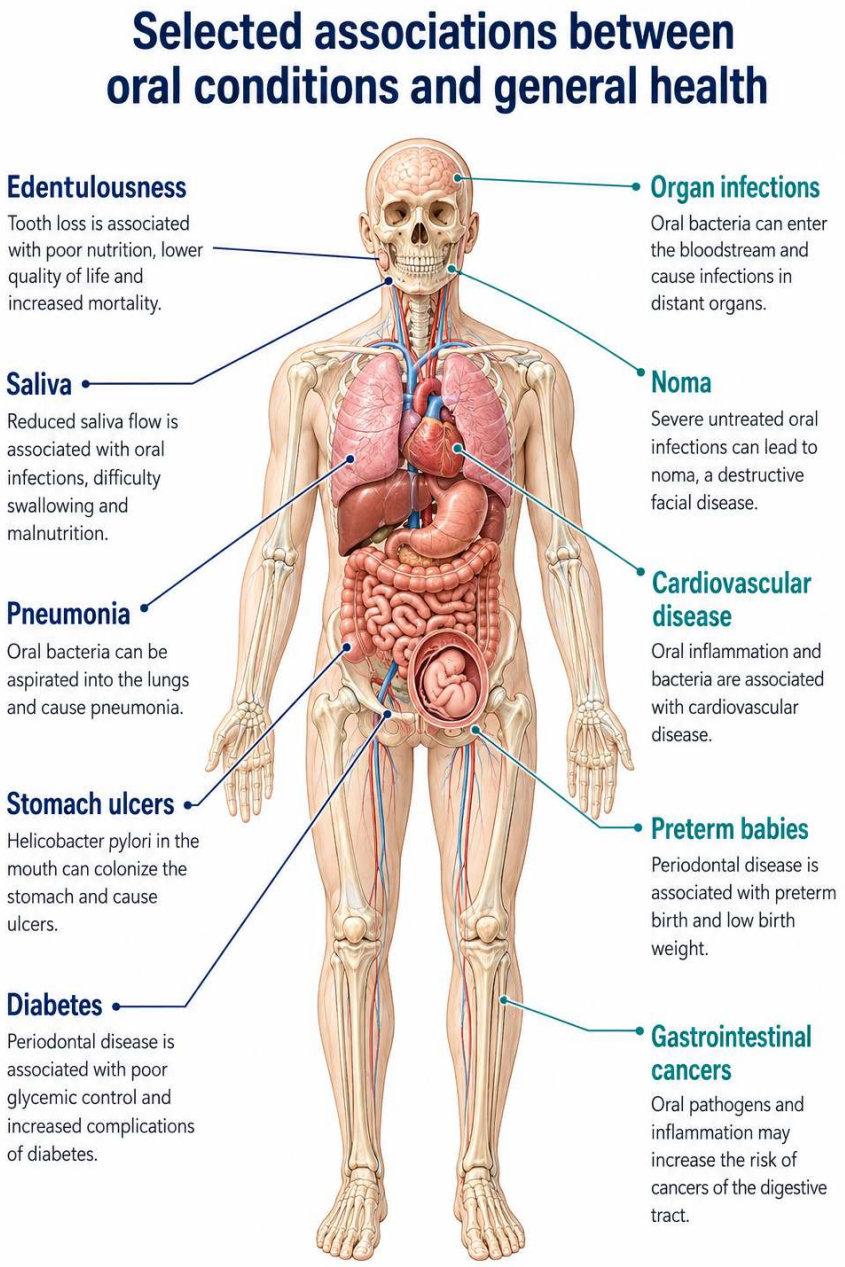



Figure 1.1: Diseases with an oral relationship. Adapted from [1]

Malocclusions represent one of the most prevalent oral dysfunctions globally, affecting approximately 56% of the population, with heightened severity observed in children and adolescents aged 4 to 17 [6], [7]. The incidence is currently on the rise and is more widespread than in previous centuries, a trend attributed to shifts in lifestyle, the industrialization of food, and modern dietary habits [1], [2], [8], [9]. Consequently, the World Health Organization (WHO) has recognized oral-related pathologies as a significant epidemic [1], [2], [9]. These diseases are a substantial burden on global healthcare systems. Evidence demonstrates that hospital emergency admissions due to oral complications have doubled in the USA over the last decade. Furthermore, oral-related illnesses have escalated to become the fourth most expensive diseases to treat worldwide; as of 2012, they ranked as the third most costly in the EU (Figure 1.3), surpassed only by coronary diseases and diabetes. Notably, both of these conditions possess an oral health component [10], [11], [12], resulting in a synergic effect between comorbidities, such as the cycle depicted in Figure 1.2 [1], [2].

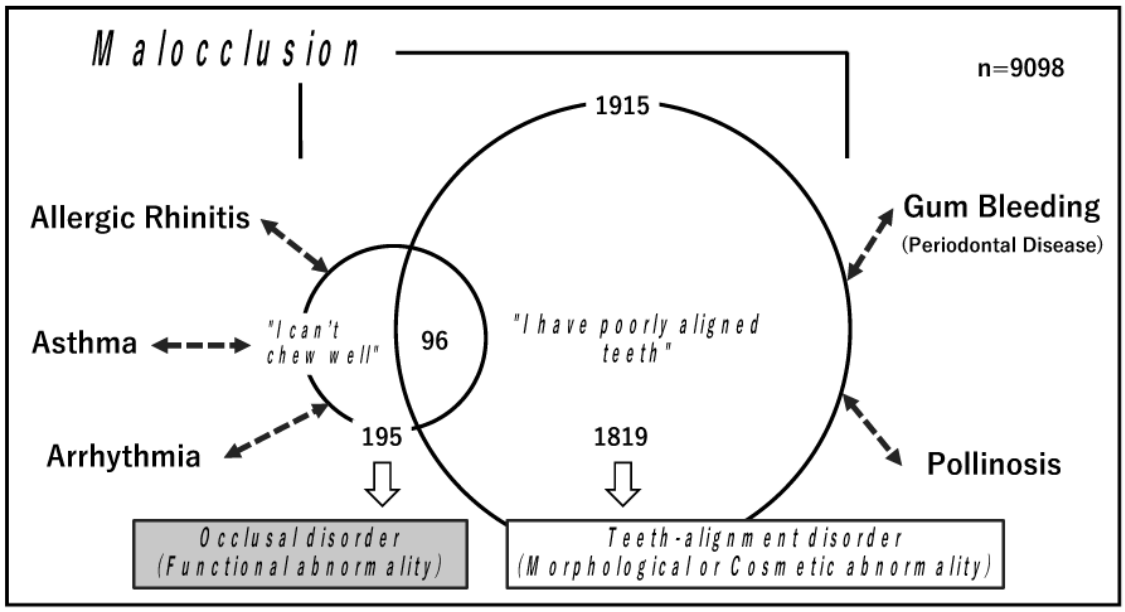
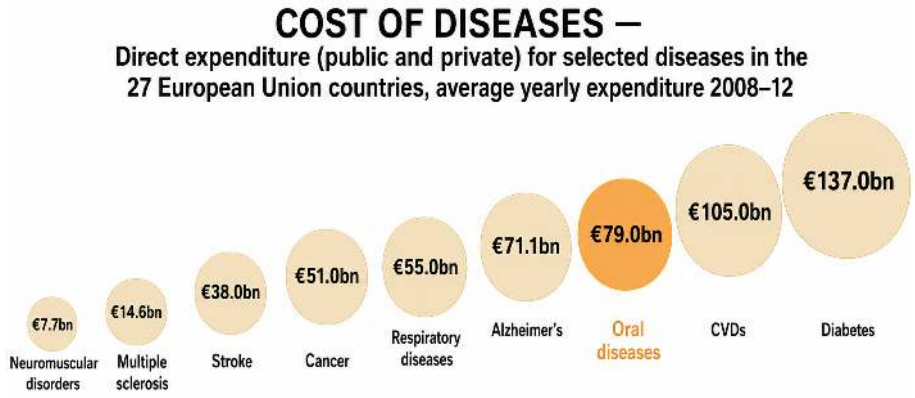



Figure 1.2: Synergic relationship between malocclusions and different illnesses[3]

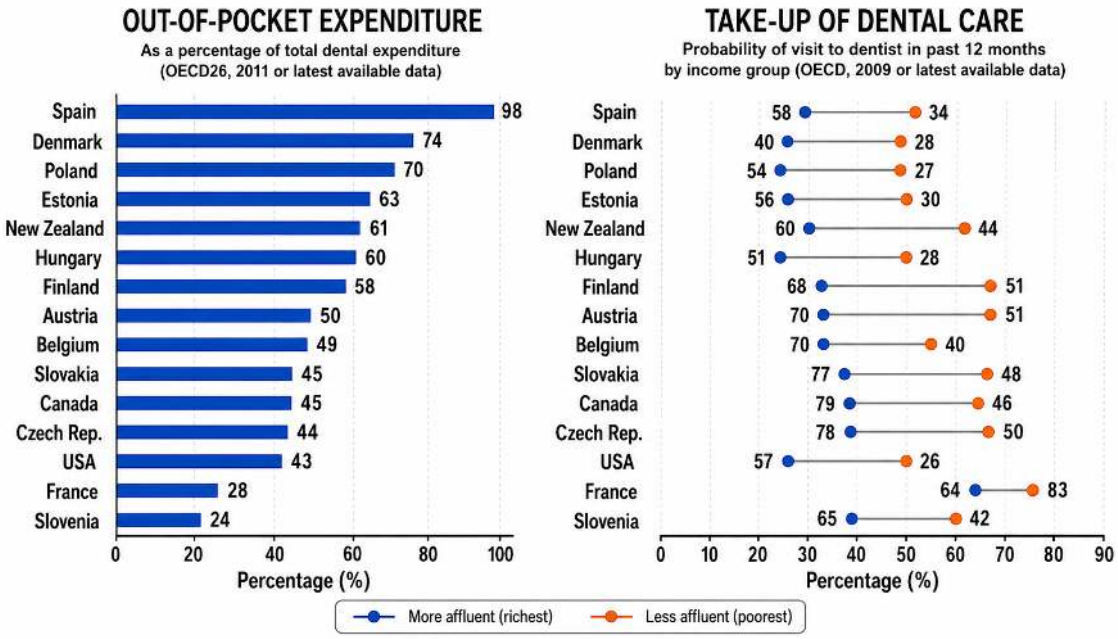
The WHO and the World Dental Federation (FDI) estimated that between 1990 and 2019, the population affected by oral diseases increased by 1 billion, reaching 3.5 billion worldwide, roughly half of the world population. Since 1990, oral-related diseases have been the most prevalent Non-Communicable Disease (NCD) that affect humanity, and thus healthcare systems should take charge of oral health interventions.[1], [2].

Oral health conditions also impose a substantial economic burden and reveal marked disparities between developed and developing nations, as well as among high-, middle-, and low-income populations. In the USA, direct spending on oral healthcare is about 50 US $ per capita annually, while productivity losses linked to oral problems amount to 42 US $ per capita per year. In contrast, in regions such as South Asia and Africa,

where per capita expenditure is only 0.5 US $ per year, the associated losses can be even greater. Globally, most oral health costs are paid directly out-of-pocket by patients. Within the EU, which has the highest overall coverage of oral healthcare costs worldwide, governments fund on average 45% of treatment expenses. In certain countries, such as Slovenia and France, public coverage reaches up to 80%, whereas in others, including Spain and Denmark, patients bear nearly the entire cost of treatment (Figure 1.3).[1]




(a) Total annual expenditure




(b) Out-of-pocket expenditure

Figure 1.3: Expenditure in oral health within the EU and the Organisation for Economic Co-operation and Development (OECD) (2011). Adapted from [1]

Oral diseases are unevenly distributed within populations, following a marked socioeconomic gradient in which the greatest burden and severity are concentrated among socially and economically vulnerable groups throughout the life of the disease. Clear Aligner Treatment (CAT) offers a potentially game-changing approach to reducing these inequities in oral healthcare, due to its non-invasive application, proven clinical effective-

ness, and economic efficiency. When delivered under strict professional guidance, this approach can move beyond correction of malocclusions to encompass a wider range of oral health conditions. [1], [2]

### 1.1. Biomechanics of tooth motion

The correction of malocclusions relies on the application of continuous, light forces to the tooth surfaces to guide teeth into their intended therapeutic positions [13], [14], [15], [16], [17]. This force-driven process initiates a biological response involving simultaneous bone resorption and formation. Once the force is applied, distinct mechanical conditions arise within the Periodontal ligament (PDL) (see Figure 1.4 for a schematic representation of the PDL), specifically a tension zone and a compression zone (Figure 1.4). In the tension zone, osteoblasts are activated, resulting in new bone formation, whereas in the compression zone, osteoclastic activity is stimulated, leading to bone resorption [8], [13], [18], [19]. In the framework of CAT, it is crucial to differentiate between the magnitudes of the forces that are applied. Traditionally, orthodontic force levels are grouped into two main categories: orthopedic and orthodontic forces.[8]

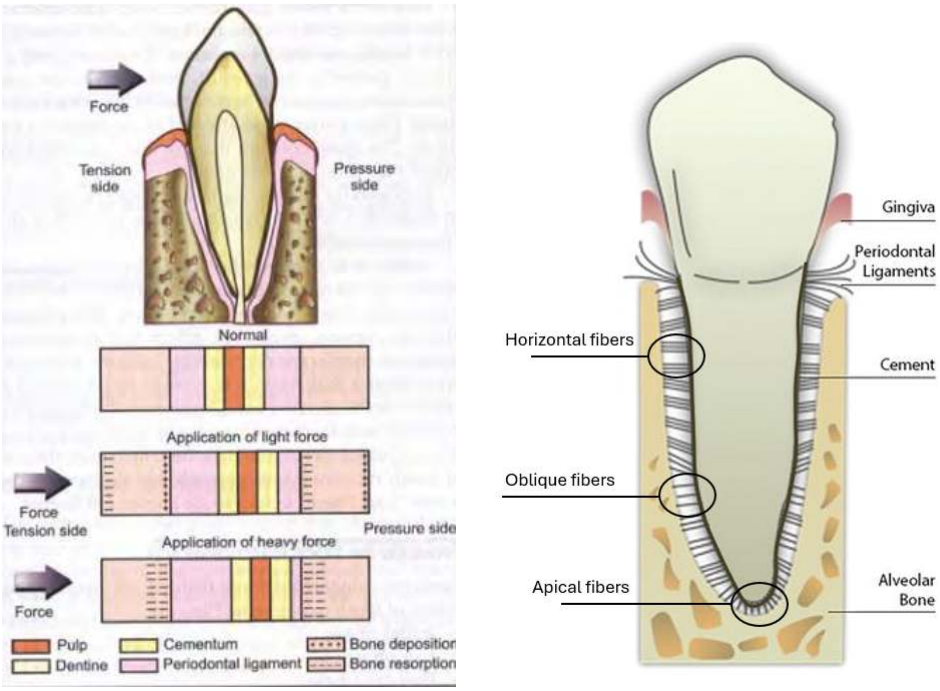
Orthopedic forces are relatively high (2–5 N) and are capable of inducing skeletal changes in the craniofacial complex. In orthodontics, such forces are typically delivered by appliances like headgear, employed particularly in pediatric patients to modulate and redirect bone growth. Orthodontic forces, in contrast, act primarily on the teeth to produce tooth movement and are generally about an order of magnitude lower than orthopedic forces (0.5–1 N).[20], [21]. For CAT, recommended force magnitudes currently lie between 0.2 N and 1.2 N, with an emphasis on avoiding stresses exceeding approximately 3 kPa.

These force ranges are those commonly used in conventional orthodontic therapy, and a clear understanding of them is essential to prevent soft tissue injury or unwanted patterns of bone remodeling during treatment. Appropriate force levels elicit the correct biological signaling pathways that govern tooth movement. Although three principal theories of tooth movement have been proposed, the actual process likely reflects a combination of all three.

The precise optimal force values are highly case-dependent, as they largely depend on the condition of the PDL and the degree of bone mineralization. These, in turn, can be influenced by genetic background, sex, and an array of phenotypic and environmental factors, including diet, systemic diseases, prior trauma, pain sensitivity and perception, and substance use or addiction, among others.

### 1.1.1. Pressure-Tension theory

This theory proposes that, on the tension side, blood flow is increased, which raises $O_2$ levels and lowers $CO_2$ levels in the PDL as a result of altered circulation. The concept relies on the Venturi effect: blood vessels on the tension side dilate and blood velocity decreases, thereby promoting more efficient exchange of substances between the PDL and the bloodstream. In this region, the PDL fibers are also elongated, leading to greater cellular proliferative capacity and enhanced fibrillar regeneration, ultimately improving tissue connectivity and guiding it toward a more specialized tissue type [8], [13], [18], [19], [22], [23], [24] (for a graphical explanation see figure 1.4).




(a) Tensile and compression areas[13] (b) Periodontal ligament (PDL) [19]

Figure 1.4: Pressure-tension theory and Periodontal ligament

On the compression side, opposite effects are observed. The process begins with the formation of an aneurysm or a constriction of the blood vessels, leading to increased blood flow velocity and reduced exchange of substances. Consequently, cell proliferation decreases, and tissue becomes disorganized, with a loss of specific tissue fibers due to the loosening of both fibers and cells. This compromised area is subsequently cleared by macrophages, osteoclasts, and other neighboring cells that remove the necrotic tissue. In addition to the mechanical and vascular changes, orthodontic force also elicits a neural response via the sensory nerve fibers innervating the periodontal ligament, which release two major neuropeptides: Substance P (SP) and Calcitonin Gene-Related Peptide (CGRP). SP induces vasodilation and enhances vascular permeability, intensifying the inflammatory response and influencing osteoblast and osteoclast function; CGRP serves as a strong vasodilator and supports bone formation in the tension area, providing

a complementary role in regulating orthodontic tooth movement [19]. Figure 1.5 summarizes these and other biological responses elicited by the appropriate application of force to a tooth.

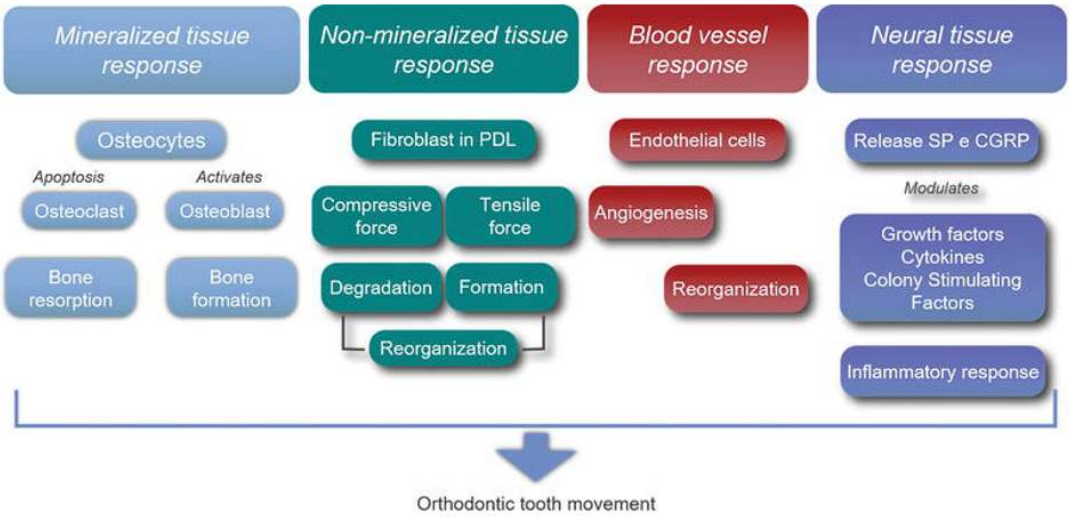



Figure 1.5: Biological responses triggered by correct force application over a tooth [19].

Already in 1932, it was calculated that the stresses applied must remain below the capillary blood flow pressure (2–2.5 MPa)[25] in order to prevent generalized necrosis on the compression side caused by ischemia of the soft tissues. This limit is expressed as a pressure (force per unit area) rather than as a net force because the biologically relevant mechanical stimulus is the stress distributed over the PDL surface, not the total load acting on the crown. The PDL presents an average surface area of roughly $150-200\,mm^2$ per tooth, although this varies widely with tooth category, root shape, and root development stage [8]. Consequently, typical orthodontic forces used in practice (0.2–1.2 N) translate into PDL stresses on the order of 1–8 kPa when spread across the entire ligament area, remaining well below the capillary pressure limit. In reality, however, the aligner does not contact the entire PDL surface homogeneously. Instead, forces are delivered through discrete contact regions between the aligner and the crown and are then redistributed through the root and surrounding alveolar bone. As a result, local stress peaks within the PDL can be considerably higher than the computed average, which makes the pressure threshold a more relevant clinical design parameter than the sole consideration of the overall force magnitude [8], [26]. This conceptual framework also accounts for the characteristic time course of orthodontic tooth movement, typically represented as a displacement curve with 3 to 4 distinct phases, as illustrated in Figure 1.6.

1. **Initial phase**: Shortly after the force is applied, the tooth undergoes rapid displacement within the PDL space. This phase extends from about 24 hours up to 2 days.

2. **Lag period**: Tooth movement then slows or stops for approximately 20 to 30 days

due to hyalinisation of the PDL in the compression zone. No additional movement occurs until all necrotic tissue has been removed.

3. **Acceleration phase**: Tooth movement resumes and increases in a non-linear manner. This phase represents a transition towards a linear pattern of movement.

4. **Linear phase**: A steady rate of movement is reached. In this stage, bone formation proceeds at a rate that does not depend on the magnitude of the applied force, and as long as a minimal force level is maintained, tooth movement can theoretically continue indefinitely.

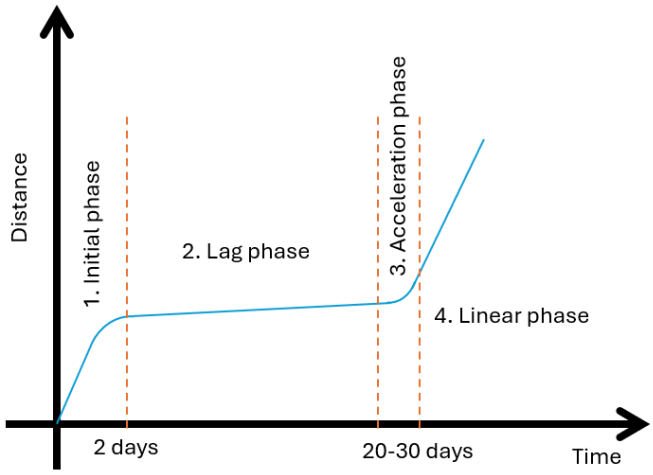
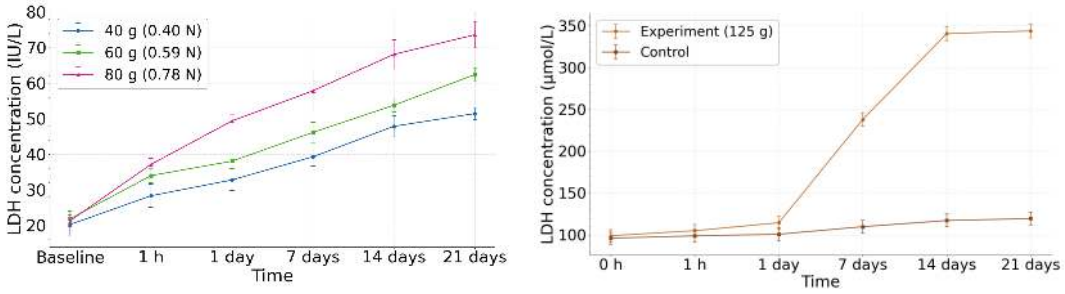



Figure 1.6: Orthodontic displacement curve

Almost all displacement occurs during the last two phases of the curve, followed in magnitude by the first phase (Figure 1.6). This pattern is actually consistent with findings from CAT, which indicate that the bulk of tooth movement is achieved within the first 48–72 h of treatment, after which the aligner primarily serves to stabilize and refine the final position. These phenomena have been quantified in patients fitted with metallic appliances delivering relatively light forces of 125 g (1.25 N), by assessing salivary LDH concentrations. LDH is an intracellular enzyme released into the extracellular space when the cell membrane is compromised; its level in saliva therefore reflects the extent of mechanical injury sustained by the periodontal tissues around the loaded tooth. For each patient, a contralateral tooth not subjected to orthodontic load served as an internal control, so that baseline LDH arising from physiological cell turnover could be subtracted from the experimental readings. This approach isolated the specific contribution of the applied orthodontic force to periodontal tissue damage and removed variability in baseline LDH concentrations between individuals. In that study, the results are presented in Figure 1.7. The LDH concentration increased from 99 $\mu mol/L$ at baseline to 340 $\mu mol/L$ after 14 days, remained at this level until the study endpoint at 21 days, while the control tooth stayed stable at around 120 $\mu mol/L$ throughout [27]. In a subsequent comparable study,

forces between 40 g (0.40 N) and 80 g (0.78 N) were applied, as depicted in Figure 1.7. A similar monotonic rise in LDH was observed at all force magnitudes, but no plateau was reached within the 21-day observation window [28]. The rate of increase in LDH appears to be largely unaffected by the force magnitude within the tested range, implying that the rate of tissue damage—and, by extension, the speed of tooth movement once initiated—may be relatively force-independent in this interval, although this interpretation still requires verification in specifically designed studies. Collectively, these findings indicate that lower forces result in reduced cumulative tissue damage while remaining sufficient to induce tooth movement, and that higher forces lead to an earlier onset of the position-stabilization phase, thereby complicating precise positional control.




(a) LDH concentration under three force levels (40, 60 and 80 g). Data from [28]

(b) LDH concentration in experiment and control groups (force: 125 g). Data from [27]

Figure 1.7: Salivary LDH enzyme concentration as a marker of periodontal tissue damage during orthodontic force application using metallic fixed appliances

To date, no investigation has quantified salivary LDH levels in patients undergoing treatment with clear aligners, so a direct comparison between the tissue damage caused by thermoplastic appliances and that associated with metallic fixed appliances remains unresolved. Nonetheless, the force magnitudes reported in the aforementioned studies offer a valuable biological reference framework for aligner material design: forces below 80 g (0.78 N) seem adequate to achieve tooth movement while limiting cumulative periodontal tissue damage. It should be emphasised, however, that directly transferring findings from fixed metallic appliances to clear aligners is problematic. The two systems differ fundamentally in their spatial force distribution along the dental arch, the intermittent character of aligner wear, the viscoelastic force relaxation typical of thermoplastic materials, and the proportion of periodontal ligament engaged at each stage of treatment [26], [29].

It must be taken into account that a single movement in CAT requires between 14 and 20 days, which is only sufficient to avoid entering the acceleration phase. As a result, the tooth position remains unstable and highly responsive to applied forces. Additional applications of force will move the tooth further in the intended direction. However, the opposite can also occur: if insufficient bone regeneration takes place in the tension zone, the tooth may relapse toward its original position. This phenomenon underlies the need

for a “retention” phase in treatment. During this phase, a clear aligner is used that delivers only the minimal force required to prevent further movement, and it has to be worn for an extended duration.[18]

The pressure-tension theory relies exclusively on histological findings and, although it corresponds reasonably well with experimental observations, it predicts force levels that are far greater than required. Evidence shows that hyalinization occurs only when minimal forces are applied; however, during bodily tooth movement, only limited hyalinization is detected. Therefore, hyalinization is considered a sign of tissue breakdown that should be avoided by reducing the magnitude of the applied forces[18], [30]. Moreover, the classical concept of distinct compression and tension zones within the PDL has been challenged, as it has been demonstrated that the PDL behaves as a continuous, hydrostatic medium in which any applied load is distributed uniformly throughout all of its regions.[18], [31]

### 1.1.2. Bone reformation theory

In 1897, it was proposed that forces on a tooth are transmitted through the root to the bone, bending it and creating tension and compression that induce local bone resorption and formation; these effects increase when deformation is prolonged.[32], [33]

This theory explains the slow movement of crowded teeth, which requires substantial bone bending, the faster tooth movement after extraction, and the easier remodeling in children, whose less calcified bone is more flexible.[18] It also aligns with Wolff’s law, which predicts bone responses to mechanical stimuli.[34]

However, the bone reformation theory conflicts with a key principle of orthopedics, which holds that compression in bone stimulates growth, whereas tension promotes resorption. This discrepancy stems from opposite definitions of compression and tension in orthopedics and orthodontics: in orthopedics, a “compressed” PDL refers to increasing bone concavity beneath it, whereas in orthodontics it refers to the actual mechanical state of the PDL.[18]

### 1.1.3. Dental piezoelectricity theory

This theory suggests that when a force is exerted on a tissue, an electric potential field is generated along that tissue, resulting in specific alterations in the cell membranes of the resident cells. For example, negative potentials promote osteoblastic activity, whereas positive potentials enhance osteoclastic activity. This phenomenon has been experimentally confirmed in studies using light forces and externally applied electric fields during tooth remodeling, where increased cellular activity and a higher cell count in the PDL, along with accelerated tooth movement, have been observed. Nevertheless, this mechanism alone does not fully account for the overall process of tooth movement. [18], [23],

[24]

### 1.2. Types of movements in orthodontics

How the force is applied is also crucial, as it indicates the main deformation state of the devices, the way they transmit force to the tissue, and the benefits of CAT compared with conventional treatments. In orthodontics, there are seven principal types of tooth movement, illustrated in Figure 1.8.[18], [35]:

1. **Controlled tipping**: The crown moves in one direction while the root undergoes minimal displacement in the opposite direction, achieved by a force acting perpendicular to the crown surface combined with a moment perpendicular to the tooth axis.

2. **Uncontrolled tipping**: The most basic type of tooth movement, involving a force applied directly to the crown without a counteracting moment, causing the root to move in the opposite direction of the crown with no rotational control.

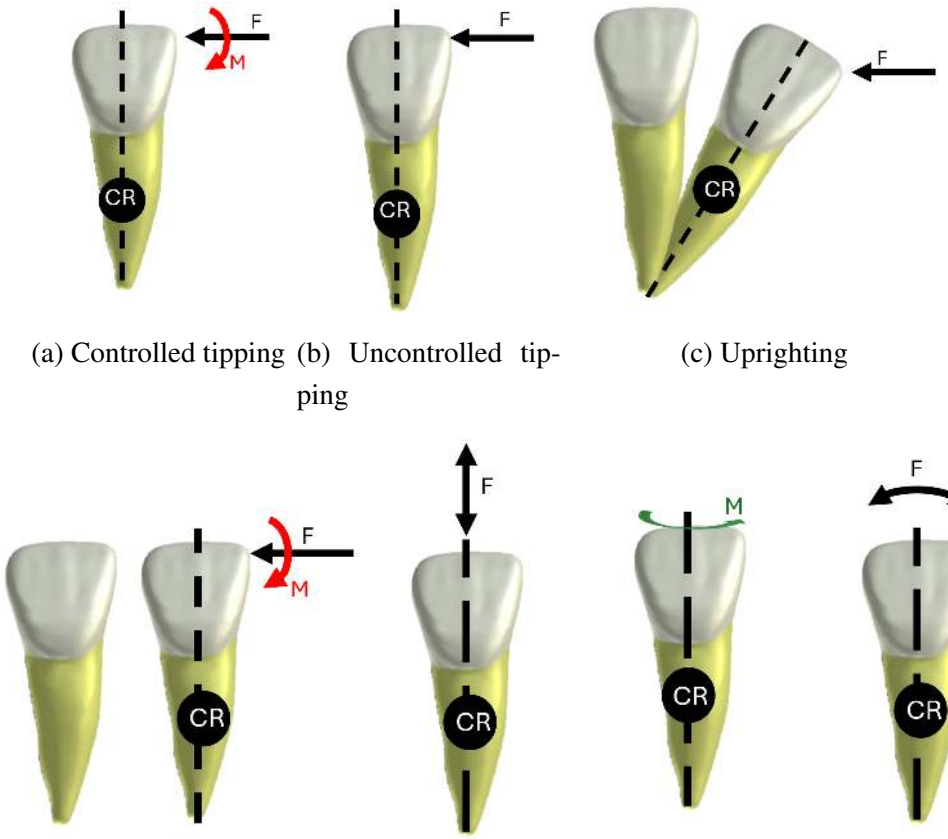
3. **Uprighting**: Corrects teeth that have been tipped mesiodistally by applying a moment that returns the root to its correct axial position.

4. **Bodily movement (translation)**: By combining a force with a bending moment perpendicular to the tooth axis, both crown and root move the same distance and in the same direction.

5. **Extrusion and intrusion**: These movements displace the tooth along its vertical axis, either extruding it out of or intruding it into the alveolar socket.

6. **Rotation**: Aligns the tooth about its long axis by applying a moment in a plane perpendicular to the tooth axis.

7. **Torqueing**: Applies a bending moment perpendicular to the tooth axis to regulate the axial inclination of the root relative to the crown, independently of crown position.




(a) Controlled tipping (b) Uncontrolled tipping (c) Uprighting

(d) Translation (e) Extrusion/intrusion (f) Rotation (g) Torqueing

Figure 1.8: Orthodontic movements and force systems

As illustrated in Figure 1.8, most tooth movements involve a force applied perpendicular to the crown tip combined with a bending moment. By contrast, extrusion and intrusion are clearly the least efficient movements, since the aligner must exert forces in a plane parallel to the crown, rather than perpendicular to it, and it lacks an adequate contact surface to do so. Consequently, these movements require attachments, typically in the form of small elements bonded directly to the tooth surface with epoxy, which create the necessary surface for the aligner to deliver the force.

At present, it is often claimed that most orthodontic movements are still insufficiently predictable with CAT for it to be considered fully reliable, and that certain malocclusions cannot be managed with this approach, even though all movement types are theoretically achievable. Although the technique has existed for more than 20 years and is widely used worldwide, it is still undergoing development; many critical assessments are based on earlier experiences, when numerous movements were either impossible or poorly predictable. Recent evidence indicates that the most predictable displacements are small-range tooth movements and rotations, with overall efficiencies between 80 and 96 %, whereas vertical movements such as extrusions and intrusions performed without attachments remain the least efficient, with success rates below 30 %[36], [37], [38]. It has

been reported that 80 % of treatments with CAT need midcourse corrections, 17 % are converted to fixed metallic appliances or alternative therapies, and 3 % are discontinued altogether [16]. The main reasons are:

- Suboptimal selection of the plastic material.
- Treatment planning carried out by an inexperienced technician or under no medical supervision.
- Insufficient clinical follow-up.
- Poor patient adherence during treatment.

The last factor typically involves patients not wearing the aligners for the prescribed number of hours, which may, in turn, be related to overly rigid plastics that provoke pain or discomfort [36], [37], [39]. Patient compliance is currently one of the principal challenges in clear-aligner therapy, and there is still no reliable way for patients to verify that they have actually worn their aligners for the recommended duration. The future of clear aligners is likely to entail further digitalization and a move away from rigidly standardized protocols and appliance designs.

This point is actually the reason why clear aligners are not suitable for extended use in children, as they require the full cooperation of the patient. Dr. Gilberto Salas, an eminence in CAT, lists the following 5 points as drawbacks of the current CAT[40]:

1. **Lack of robust biomechanical assessment by the professional**: Many clinicians still depend on oversimplified or outdated biomechanical concepts, often limiting their evaluation to a 2D analysis of forces, whereas CAT demands a comprehensive assessment of all forces and their interactions with the different tissues.
2. **Suboptimal use of plastics and attachments**: This mainly involves aligners that are poorly designed or incorrectly positioned, as well as plastic materials whose elastic properties do not adequately match the requirements of the treatment.
3. **Mistakes in simultaneous movements**: Certain combinations of tooth movements can cause the aligner to lose tracking or generate an occlusal imbalance. This aspect must be carefully considered by the clinician during treatment planning.
4. **Insufficient customization to the patient's actual needs**: The forces needed for tooth movement vary widely and can be influenced by systemic conditions, age, and other individual factors. Therefore, it is more appropriate to tailor the viscoelastic properties of the plastic to each patient. With current knowledge, we can better anticipate biological requirements by considering both the patient's medical history and their bone density. In principle, this would allow the clinician to choose a more suitable force range for each case by adjusting the aligner's properties.

5. **Exclusive reliance on standardized brand protocols**: At present, clinicians are restricted by fixed protocols that prevent the individualized adjustment of parameters for each treatment or treatment stage. This problem stems from mass-production approaches that have largely disregarded the specific needs of the end patient.

A further consideration, raised by Salas [40], concerns the integration of orthodontic expertise throughout the full treatment planning process. Effective clear aligner therapy requires the direct involvement of a qualified orthodontist at every stage, as the biomechanical complexity of tooth movement cannot be adequately addressed through standardised protocols or delegated to technicians working from incomplete clinical information. Systems in which virtual treatment planning is carried out by non-clinical personnel operating on the basis of limited patient data and prescriptive instructions are inherently ill-suited to accommodate the range of biomechanical challenges that may arise during treatment, and represent a structural limitation on the quality and safety of the therapeutic outcome [40].

Since the COVID-19 pandemic, a Direct-to-customer (DTC) model of orthodontic practice has become increasingly widespread, characterized by fully remote care without in-person professional supervision. In this model, the patient photographs their own dentition with a smartphone and submits the images to a technician who plans the tooth movements; the aligners are then shipped directly to the patient at comparatively low cost, with no clinical examination performed at any stage of the process [41], [42]. As illustrated in Figure 1.9, the DTC model grew rapidly following the COVID-19 pandemic before collapsing with the bankruptcy of its principal operator, SmileDirectClub, in December 2023 — a trajectory that stands in marked contrast to the continued growth of clinician-supervised clear aligner therapy over the same period.

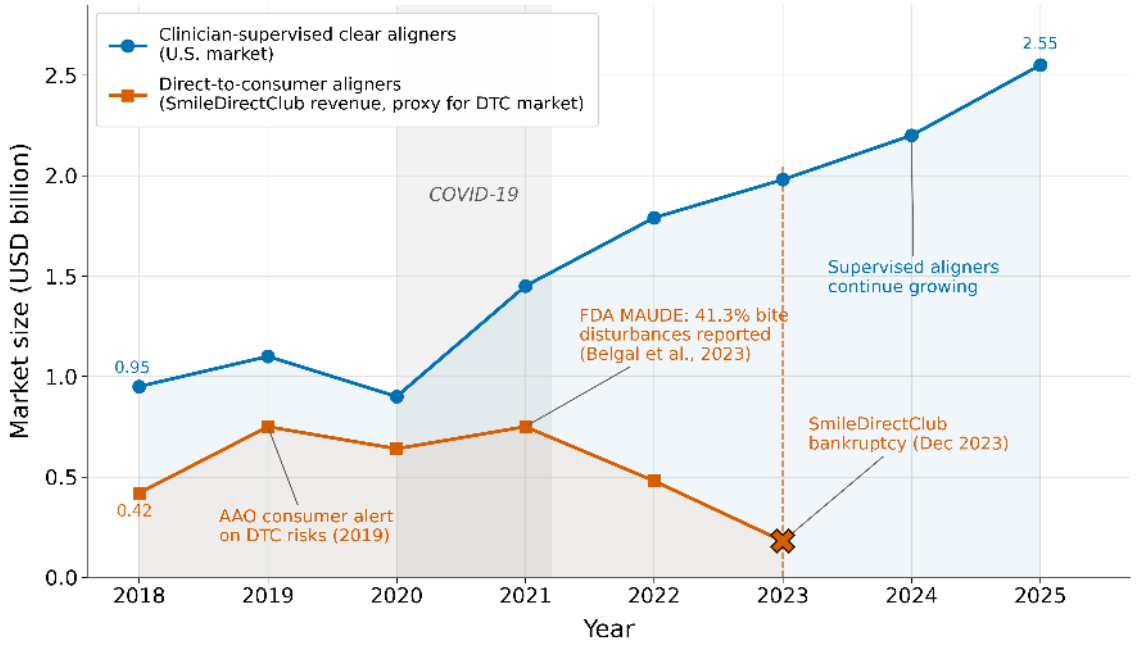



Figure 1.9: Revenue of the DTC clear aligner model (proxied by SmileDirectClub annual revenue) versus the clinician-supervised clear aligner market in the USA from 2018 to 2025. Data from [41], [43], [44], [45], [46]

This approach has led to, and further exacerbated, numerous oral health problems for patients, particularly in the USA [41], [42], [43]. An evaluation of the FDA MAUDE adverse event database revealed that 41.3 % of reported complications related to DTC aligners involved occlusal or bite disturbances, 29.8 % involved orofacial pain, and 26.6 % involved some form of periodontal damage, with a substantial proportion of these outcomes being irreversible [41], [43]. Similar adverse events have also been reported for professionally supervised aligner therapy, underscoring that these risks are greatly magnified in the absence of clinical oversight [43]. These complications arise not only from the absence of a qualified professional directing treatment planning, but also from reliance on non-professionals to obtain the initial dental records and, most critically, from the complete lack of clinical supervision throughout treatment [42], [44]. It is therefore argued that highly trained orthodontists must be involved at every stage of the treatment planning process and that patients must have direct access to them at all times.

### 1.3. Phases of an orthodontic treatment and properties of the materials

There are mainly two ways to plan orthodontic treatment in CAT, and the choice between them depends on the professional and the specific case. The first option starts by eliminating all crowding, and maintaining that alignment involves either arch expansion or contraction. The second method is the opposite, starting with aligners that perform expansions or contractions and finishing with crowding. Finally, in all cases, there is a retention phase that typically lasts for the patient's lifetime.[47], [48]

What is clear is that there exists a phase that comprises very accurate movements and small high deformations in tension, bending, and torsion at different points of the device; and a second and third phase that comprise lower deformations, mainly in bending. This is the reason why top brands in the sector have decided to use two types of thermoplastic instead of one.[47]

Historically, only one plastic was used, so in the high deformation phase, excessive force was applied, resulting in pain, discomfort, and inaccurate treatments, as well as undesired hyalinization that hindered subsequent movements. Then, softer plastics were introduced with the primary goal of reducing the elastic modulus of the material and allowing for more precise control of the force over time, resulting in greater consistency in these materials.

Another advantage of using a less elastic and more viscous material for teeth crowding is that the material can adapt better to the teeth's topography, making the actual force closer to what the orthodontist has planned. Many CAT brands include devices called gummies or chewers (see Figure 1.13) to fitting their appliances to the mouth. These are small, rubbery, hollow cylinders that the patient must bite or chew when wearing an appliance for the first time to help drive the appliance into its position. It is very common for orthodontists to hear patients claim that the device is too stiff and difficult to place or that it even jumps from their mouth when in place. Thus, it seems clear that the viscous properties are of greater importance than was expected.

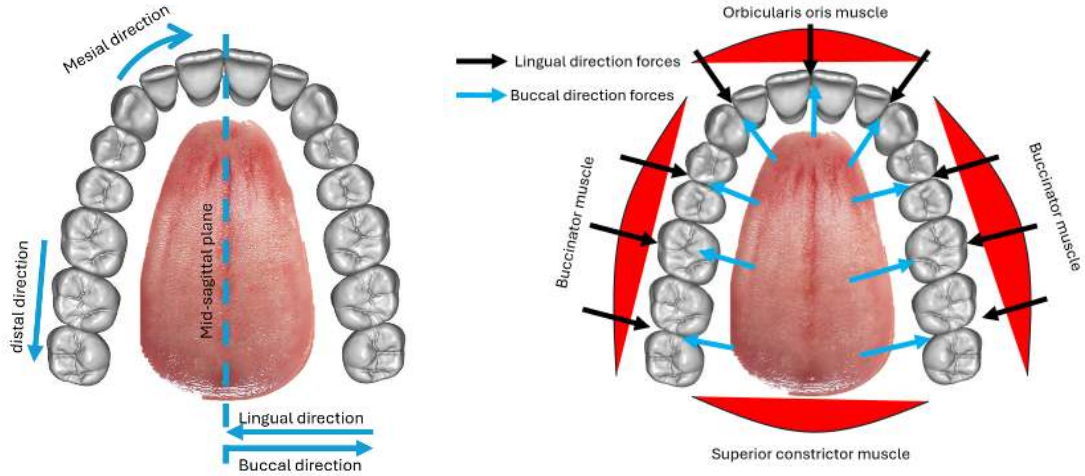
We must also consider that CAT is based on the stress relaxation properties of the plastic used. Stress relaxation refers to the process by which a material dissipates force under a constant strain over time. Elastic materials, such as metals, can maintain a constant stress level over time under such circumstances, but the viscous components in plastics allow the stresses to drop indefinitely. The rate of this drop depends on the viscous and elastic properties of the material; in general, the more elastic, the higher the initial force, and the better the material can maintain it.

### 1.3.1. Retention phase and dental arch equilibrium forces

The last phase of retention is due to the equilibrium of forces in the mouth. The shape of the dental arch is due to an equilibrium of forces between the loads that the muscles in the lips and cheeks exert in the lingual direction and the forces that the tongue exerts in the opposite direction, as depicted in Figure 1.10 [8], [49]. This equilibrium is dynamic, as it can be affected by various factors, including breathing, eating, drinking, speech, abnormal habits such as thumb sucking in children, and even human actions like playing a wind instrument [49]. Thus, sometimes an exogenous force is needed to equilibrate the forces.

A simple example is the opposite effects of thumb-sucking and playing a wind instrument. When kids suck their thumbs, the forces from the cheeks, which are caused by

a muscle called the buccinator, surpass those of the tongue and result in maxillary constriction [8], [50]. Playing a wind instrument has the opposite effect and causes maxillary contraction[51], [52].




(a) Anatomic sketch of the mouth (b) Equilibrium of forces in the mouth

Figure 1.10: Anatomy and forces in the mouth

Thus, orthodontics involves remodeling an equilibrated dental arch into another in which the musculoskeletal forces are no longer in equilibrium and, therefore, an external force is always required to maintain this equilibrium

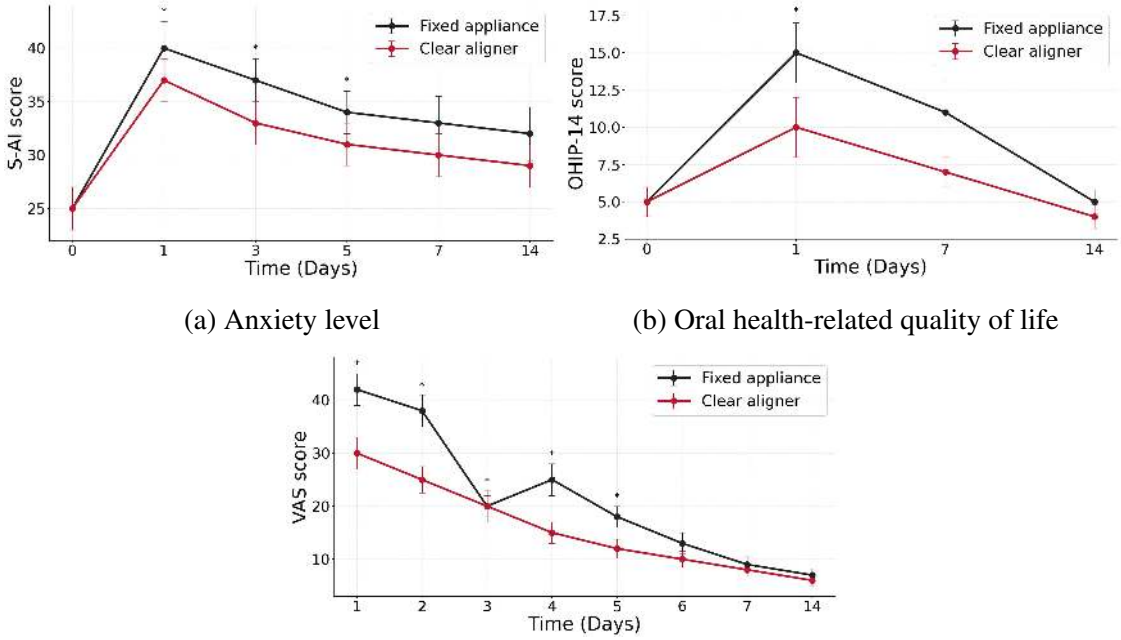
. Another fact to consider is that even after a dental movement is achieved, the bone remains unstable. The newly created bone is a young and not fully mineralised structure, so under occlusal forces it may bend or finch backwards. The advances in materials have led to new approaches in addressing oral health and malocclusions, which will be discussed in the next section.

## 1.4. Historical review on malocclusion treatments

Malocclusion treatment dates from pre-roman times when the idea was to strengthen the teeth using gold or silver appliances fixed directly to the mouth[53], [54]. Since then, metal braces have dominated malocclusion treatment, with only the material changing from soft metal to hard austenitic steel wires introduced in the 19th century and Nickel-Titanium memory shape wires introduced in the 20th century [54]. The first plastic, or more exactly, rubber gum-massager device, was developed by Orrin Remensnyder in 1926 [55] for performing minor dental movements and was popularized by Dr.Kesling in the 1940s [56]. The first transparent plastic mouthguards were manufactured in the 1960s for professional athletes and were later introduced into the general market by Robert J. Ponitz [57], thanks to advancements in mass production techniques.

The main enhancement of CAT occurred in 1997, thanks to two Stanford University

students who laid the groundwork for modern CAT [58], [59], based on 3D images and additive manufacturing . The treatment begins by scanning the patient's oral cavity using an intraoral scanner or by digitizing clay impressions of the mouth. Later, all the dental movements are planned by a professional using specialized software, which takes into account the maximum forces the tooth can support. Then, the whole treatment is sliced into a series of subsequent steps, which are converted into 3D images that are further reproduced by a resin 3D printer. Next, the dental impressions from the 3D printer are used as molds over which a clear thermoplastic is thermoformed. Finally, the thermoformed shape is cut either by hand using a trimmer or by a laser cutting machine, cleaned, sterilized, and handed to the patient [26], [39].




(a) Anxiety level

(b) Oral health-related quality of life

(c) Pain level

Figure 1.11: Comfortability of fixed metal appliances against clear aligners. Adapted from [60]

The main advantage of CAT for the patient is the "invisible" nature of the treatment, as it is based on clear plastic that should be almost invisible, and if well planned, not even noticed by the user, being at the same time less painful and producing less anxiety than metallic appliances, as seen in Figure 1.11. Thus, it implies a less invasive treatment, but also a more hygienic one because the device can be removed for cleaning and, in some cases, even re-sterilized, as depicted also in Figure 1.11. It is also said to be faster and more convenient option than fixed metallic appliances, since it requires fewer follow-up sessions. Conventional treatment based on metal appliances requires weekly or biweekly visits to tighten or change metal arcs that exert force due to glued or cemented structures in the crowns [60]. Another fact to bear in mind is that nickel sensitization, although uncommon, is more frequent and severe than plastic allergy, which is reported to produce only Xerostomia (dry mouth), which can be alleviated by drinking water more frequently

[61], [62], [63].

Although clear aligners are generally reported to be more comfortable for most patients than metal fixed braces, they are still far from perfect. A 29 % of the patients report low or even no pain sensed during treatment using clear aligners, 55 % report mild pain, 15 % moderate pain, and only 1 % high or unbearable pain [64] as we can observe in Figure 1.12.

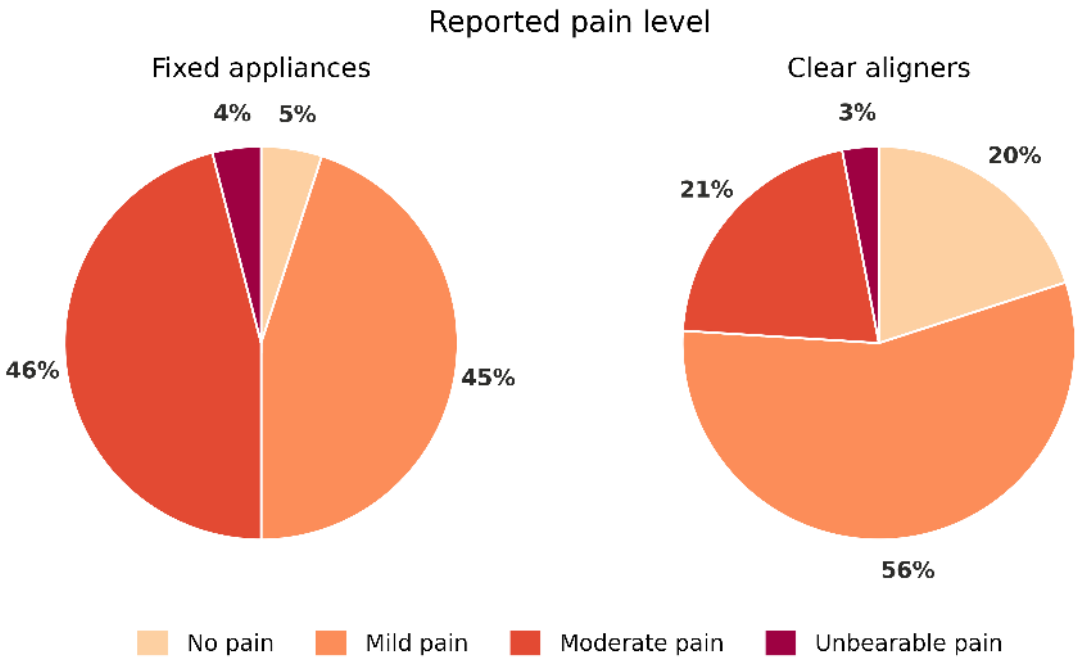



Figure 1.12: Pain level comparison between clear aligners and metal appliances. Data from [64]

### 1.5. Plastic devices used in CAT

Although this thesis focuses specifically on clear aligners, a wide variety of plastics are used in orthodontics and more specifically in CAT for different purposes. Here we provide a summary of the most important plastics for CAT, along with a more graphical list in the figure 1.13 :

- **Clear aligners**: They are active devices that move teeth to their corresponding position based on a programmed misfit with the dentition
- **Dental Retainers/Essix**: Passive devices that keep the position of the tooth once established. They can also be used to perform simple collective movements like arch expansions or retractions.
- **Templates and resins for attachment placement**: Templates are temporal devices that the professional uses to place attachments. They must be highly transparent to

avoid any interaction with the Ultra-violet light (UV) used for curing the adhesive that bonds the attachment to the tooth. Templates are usually made of the same plastics as aligners or retainers but in a thinner version, and are disposed of after use. The UV resin or dental glue used to bond the attachments is usually a polyurethane-based UV curing resin.

- **Elastic bands**: Elastic bands are used to perform tooth extractions. They are made of elastomer or rubber.
- **Chewers**: They are small hollow plastic cylinders that the patient must bite or chew to guide the aligner or retainer to its correct position during the first days of treatment.
- **Mouthguards**: Plastics with high dissipation used in sports or to protect teeth from bruxism. They are usually much thicker than aligners or retainers and are made of a soft plastic.
- **3D printed models**: 3D printed models of each dental movement planned in treatment. The majority of the models are made of thermosetting resins by UV curing.

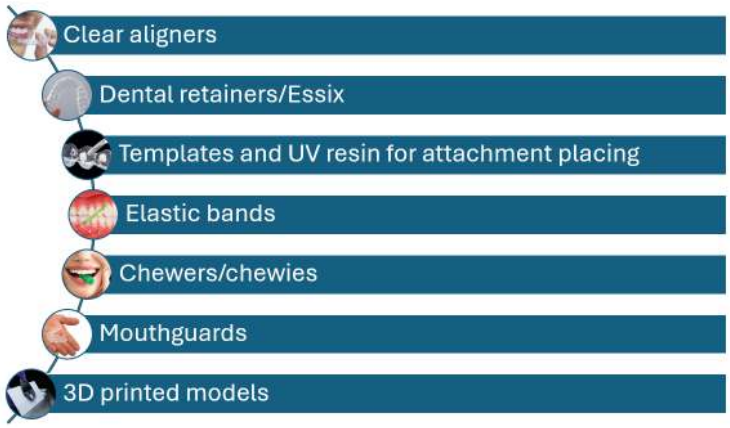



Figure 1.13: Plastics used in Clear Aligner Treatment (CAT)

## 1.6. Thermoplastics used for clear aligners

CAT relies on the mechanical, physical, biological, thermal, chemical, and organoleptic properties of the clear thermoplastic used. The polymer should be able to withstand the action of chemicals and biological agents that are mechanically aided, generating a hostile environment. It must also be optically transparent, non-toxic, and comfortable to wear. Comfort relates to the potential for harming oral tissues and thus to the forces applied to them. As we have mentioned, there is a distribution of optimal forces, and the manner in which the aligner delivers the force over the teeth is complex, so it is not easy to set the perfect mechanical and thermal features for an ideal CAT plastic before it is in

treatment testing, and the perfect plastic for a specific case can be the least optimal or cause unforeseen effects in another case.

The main mechanical factor currently used for selecting a material for CAT is Young's modulus. It has been demonstrated by computer simulations that 530 MPa is sufficient to perform minor and major movements [65], [66]. However, most brands in the sector use stiffer plastics that may reach up to four times that value.

But the force not only depends on the elastic modulus but also on the aligner thickness. It is said that the thinner the aligner, the more comfortable it will be, not only in terms of force but also in terms of the patient's feeling. The idea is hence clear: increase the elastic modulus to decrease the aligner thickness. But actually, too-thin aligners are easily punctured when used or tend to break in a fragile manner. The standard thicknesses for aligner sheets before thermoforming are 0.6 and 0.76 mm, while for retainers and mouthguards, they are 1 and 1.2 mm, respectively. We must include a secondary type of device called templates.

The second mechanical parameter when selecting a material for aligners is called stress relaxation or "force decay". Every time the aligner is introduced in the mouth, it is charged with an elastic force due to its misfit with the dentition. The initial elastic force decays over time due to the plastic's viscous properties. Significant stress relaxation is undesirable, as the aligner would only exert an active force in the first hours of treatment, rendering it a useless device for the remainder of the time.

A third fact that is not commonly considered in orthodontics is the softening point of the aligner. The oral temperature ranges from 36 to 37 ºC with peaks ranging from 50 ºC to 4 ºC due to hot and cold drinks and meals. Although these temperatures are far from the softening point of common thermoplastics, they are enough to cause it after 20 hours of use. In the same sense, secondary relaxations should be regarded as desirable, as they can provide sufficient deformability to allow for a comfortable first fitting of the aligner.

The most used plastics in CAT and their properties will be depicted in the next section.

### 1.6.1. Polyethylene Terephthalate (PET), Polyethylene Terephthalate Glycol-modified (PETG) and other Co-Polyester (coPES)

PET is the seventh most available and used thermoplastic in the world nowadays [67], [68] (see Figure 1.14) thanks to its excellent transparency, good resistance to chemicals and gasses, good mechanical resistance, and low price. It is the most common and widely used Co-Polyester (coPES)[26], [69], [70], [71]

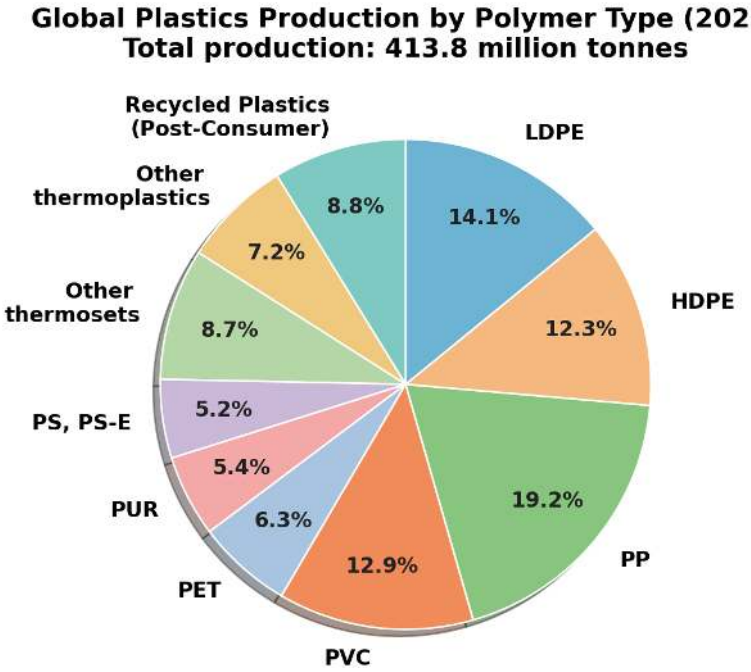



Figure 1.14: Annual plastic production by type in 2023. Data from [68]

In Clear Aligner Treatment, its glycolated derivative (PETG) is more commonly used than bare PET, because the crystallinity of PET can hinder its transparency and increase the cost of manufacturing. Generally, commercial PET is the result of the polymerization reaction of Terephthalic Acid (TPA) and Ethylene Glycol (EG). To produce fully amorphous PETG, the EG is partially substituted with a high molecular mass glycol called 1,4-Cyclohexanedimethanol (CHDM). This molecule, besides being bulkier, has aliphatic rings that hinder the crystallization of PETG. If the substitution is below 30% molar, crystallization is not avoided but is slowed down [72], [73]. Between 30 and 50%, the final product can be called PETG, and above 50% molar, the properties change and lead to another amorphous thermoplastic called Polycyclohexylenedimethylene Terephthalate Glycol-modified (PCTG)[73]. Recently, another improvement to this plastic has appeared on the market. The building blocks are the same as PCTG, but the EG is replaced with another bulky aliphatic cyclic diol, usually 2,2,4,4-Tetramethyl-1,3-cyclobutanediol (CBDO), which adds even more stiffness to the polymer and increases the glass transition temperature above 100 ºC. This plastic is known as Tritan-type copolyester, a brand name of the product. A graphical explanation of the PET derivatives can be seen in Figure 1.15.

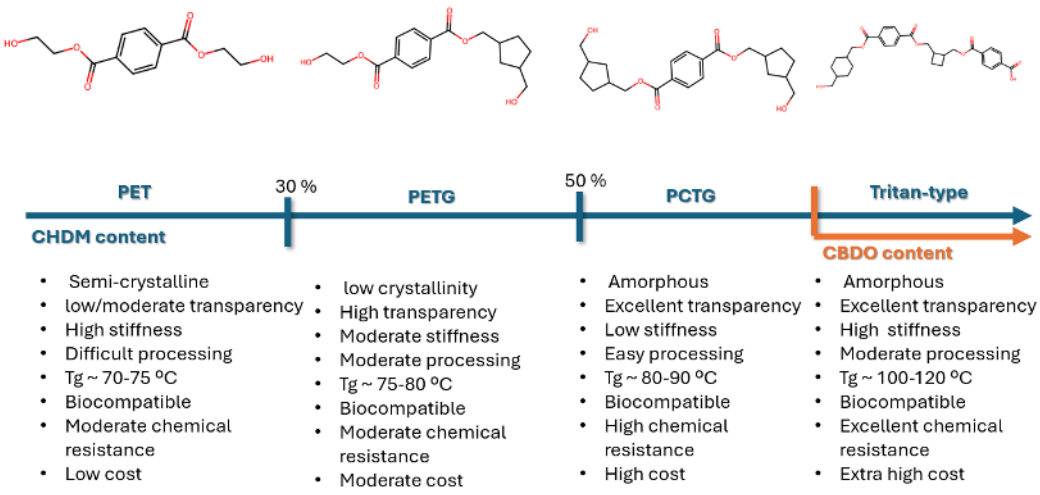



Figure 1.15: PET, PETG, PCTG and Tritan copolyester chemical structure and properties

Even though it seems to fulfill the perfect properties for CAT, PETG usage is declining because it has too-high an elastic modulus and poor stress relaxation. This means that the aligner made out of PETG tends to generate pain and discomfort and ceases to apply force after 2 days [74], [75], [76]. The reason for that decay could be the energy curve needed for orthodontic treatment, which increases in the first days of treatment until reaching a peak of around two days, as can be seen in Figure 1.16. After that, the energy suffers an exponential decay until reaching a remaining low value[14]. The energy peak at two days may be high enough to cause a complete softening of the material.

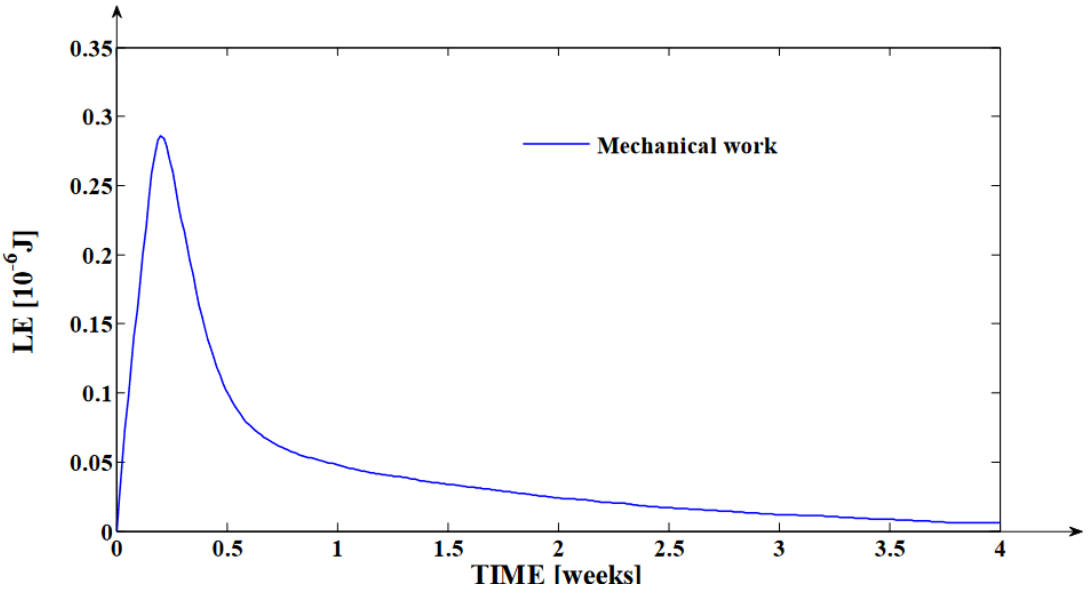



Figure 1.16: Evolution of mechanical work during orthodontic treatment [14]

Also, despite the low price of PET and PETG, their economic variability is high [77],

[78] (see Figure 1.17), and the companies try to avoid using them. Recently, it has been discovered that PET leachates can act as endocrine disruptors and that certain production routes may be contaminated by metalloids such as antimony, so its use in healthcare is closely regarded [79], [80]. Although PCTG and Tritan copolyester are said to be safer and have more optimal properties for clear aligners than PETG, their use is currently not extensive, primarily due to their higher price and lower availability compared to PETG [81], [82]. Another reason for the lack of information on PCTG in dentistry is that PCTG is commonly misidentified and sold as PETG or "modified PETG". Tritan copolyester is more commercially common than PCTG, but it is also used for other purposes, such as reusable water bottles or as a replacement for Polycarbonate (PC). However, information regarding its usage in dentistry has not been found, mainly due to its cost and novelty.

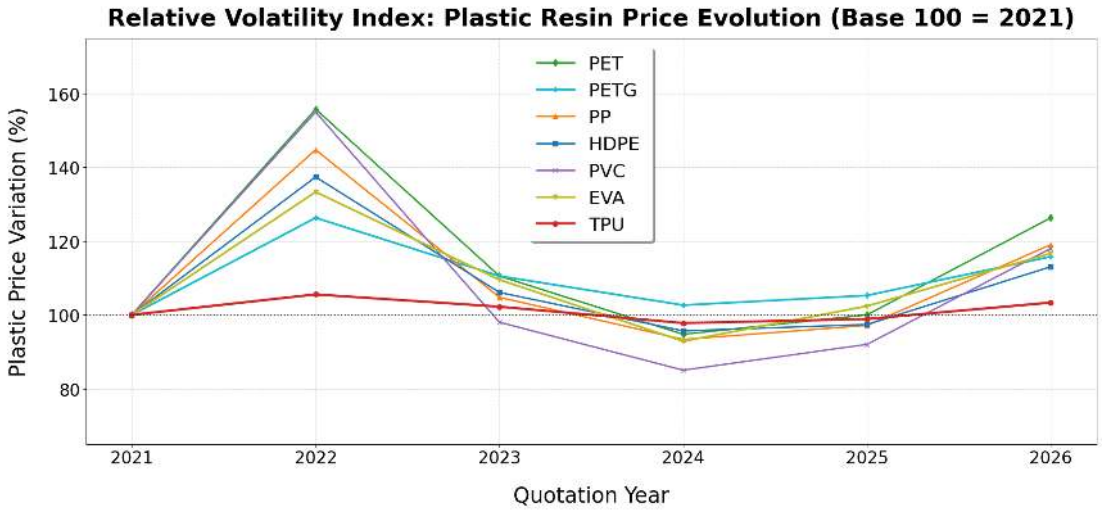
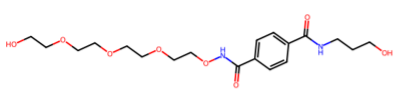



Figure 1.17: Price variation of the main commercial plastics between 2021 and 2026. Data from [77]

### 1.6.2. Thermoplastic Polyurethane (TPU)

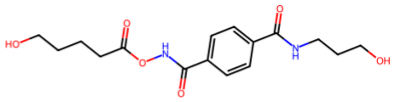
TPU is a broad family of plastics composed of soft and hard segments. The main polymer chain is bonded through the reaction of a diisocyanate with a short-chained diol, another diisocyanate, and a long-chained diol. By varying the ratio of components, their chemical structure, or processing parameters, the properties can be tailored. This broad range of possibilities is what makes it the eighth most common polymer commercially[67], [68] (see Figure 1.14).

Currently, two main families of TPU exist commercially: one with an ester-type long diol chain and another with an ether-type, both depicted in figure 1.18. Both have similar properties, but the ester has enhanced tensile and flexural properties, improved wear and solvent resistance, and better properties at high temperatures. The ether TPU, in contrast, displays high strength, resistance to hydrolysis, high resilience, and good properties at low temperatures.[83], [84], [85]



**Polyether TPU**

- Highly flexible
- Excellent hydrolisis and chemical resistance
- Good abrasion resistance
- Moderate thermal stability
- Low elastic modulus and hardness
- Good processability



**Polyester TPU**

- moderate flexibility
- Moderate hydrolysis and chemical resistance
- Excellent abrasion resistance
- High thermal stability
- Moderate elastic modulus and hardness
- bad processability

Figure 1.18: polyether and polyester TPU chemical structures and properties

Regarding its use in dentistry, TPU, in general, is less chemically resistant than any coPES, less biocompatible, and more cases are reported of TPU allergy or adverse reactions than with any other plastic [86]. It is also more susceptible to coloring agents such as coffee, tea, red wine, nicotine, or commercial sodas, but its mechanical properties are more suitable for treatment, its organoleptic feel is better, and the probability of sensitization or allergy is lower.[26], [71], [87], [88] although it is still reported in dentistry [61], [62] . The major advantage of the TPU lies in stress relaxation, which being slightly better than that of coPES in general, and a lower elastic modulus[71], so the forces are slight and continued, which are the main suspected features of its major comfort for most patients[15], [17], [74], [86].

To further distinguish the different types of TPU based on their mechanical properties, the following classification may clarify:

- **Soft-hardness Thermoplastic Polyurethane (sTPU)**: Shore D hardness below 40
- **Middle-hardness Thermoplastic Polyurethane (mTPU)**: Shore D hardness between 40 and 75
- **High-hardness Thermoplastic Polyurethane (hTPU)**: Shore D hardness over 75

Although several Shore scales exist, TPU materials are generally labelled using only the A and D scales: Shore A for the softer grades and Shore D for the harder ones. Conveniently, interconversion tables between the two scales are available.

### 1.6.3. Ethylene-Vinyl Acetate (EVA)

Ethylene-Vinyl Acetate (EVA) is a copolymer of ethylene and Vinyl Acetate (VA), in which the VA content typically ranges from 10 to 50 % by weight, determining the

mechanical character of the material: low VA contents produce a relatively stiff, semi-crystalline plastic, whereas higher VA contents yield a soft, rubbery elastomer with properties approaching those of thermoplastic polyurethane. In dentistry, the most widely used grades are the elastomeric formulations with VA contents of approximately 10-30 %, which provide the flexibility and resilience required for intraoral appliances [71].

One of the most distinctive processing characteristics of EVA is its ability to be self-adapted without conventional thermoforming: mouthguards fabricated from this material are softened by immersion in hot water at approximately 65°C for a few minutes, after which the patient bites the appliance directly to produce a custom fit. Although this approach is currently restricted to passive retention devices, the same thermoplastic behaviour could in principle be exploited in active appliances to facilitate initial insertion and improve the seating accuracy of the device at the beginning of each treatment stage [89].

EVA represents a compelling alternative to TPU for elastomeric applications in orthodontics: it can match the elastic compliance of TPU across a comparable range of hardnesses, is generally less demanding to process and more widely available as a commodity material, exhibits adequate optical transparency, and ranks among the most biocompatible thermoplastics in clinical use [71]. Its principal limitations for active aligner applications are twofold: first, its relatively low elastic modulus restricts the magnitude of orthodontic forces that can be generated; and second, EVA films frequently present significant optical haze, arising from the semi-crystalline nature of the polyethylene phase and the scattering of light at crystalline domain boundaries, which may compromise the aesthetic requirements of clear aligner therapy [71], [89]. Despite the mechanical parallels with TPU and its processing and biocompatibility advantages, references to the use of EVA specifically in CAT remain scarce and largely confined to early literature, suggesting that this material class has been largely overlooked in the development of modern aligner systems, a gap that the present work seeks to begin addressing

### 1.6.4. Other minor thermoplastics used

The earliest vacuum-formed orthodontic appliances, developed during the 1960s and 1970s, incorporated a broader range of thermoplastics than those currently in use, including Polyvinyl Chloride (PVC), Polyethylene (PE), Polypropylene (PP), Polycarbonate (PC), Polymethyl Methacrylate (PMMA), and Cellulose Acetate Butyrate (CAB). The first vacuum-formed aligner, introduced by Ponitz in 1971 [57], was fabricated from a material called Biocryl [90], a composite of CAB, polyurethane, polyvinyl acetate-polyethylene, polycarbonate-cycolac, and latex, which was subsequently superseded by the Co-Polyester that dominate the market today. A chronological summary of clear aligner development, including the materials employed and the key technological milestones at each stage, is presented in Figure 1.19.

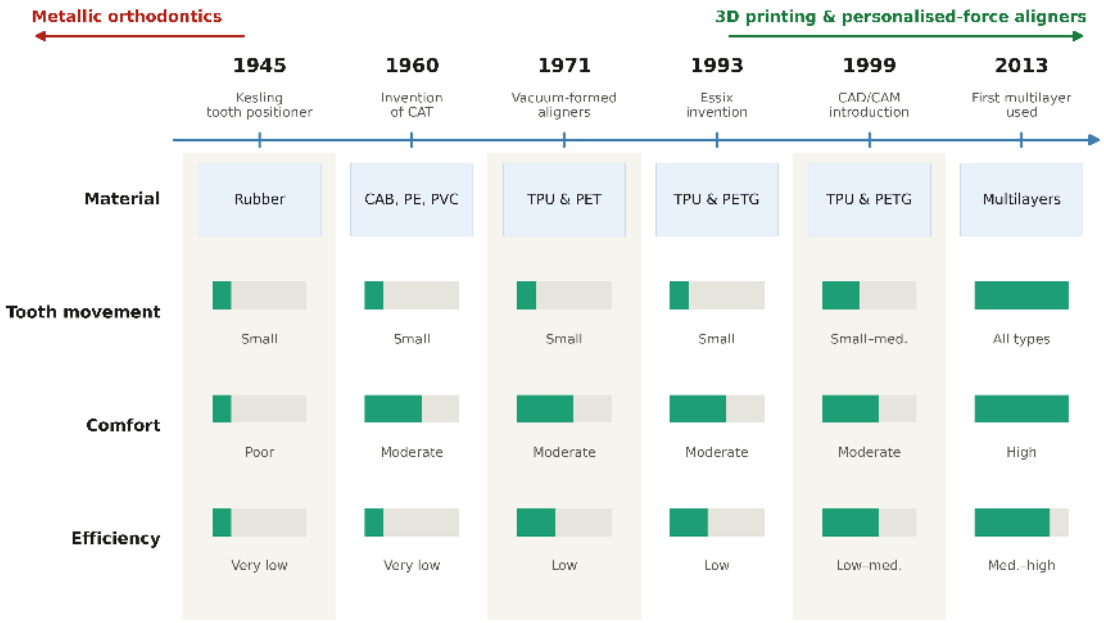



Figure 1.19: Clear aligner history and materials

The vast majority of these early thermoplastics have since been discontinued for active aligner fabrication, owing to a combination of excessive stiffness, inadequate elastic recovery, or insufficient transparency under clinical use conditions. Their application is now largely confined to passive devices such as retainers, mouthguards, sports guards, and occlusal splints.

Polycarbonate represents a particularly instructive case: despite its use in premium clear aligners during earlier decades, it has been largely abandoned due to its high cost, excessive rigidity, susceptibility to rapid hydrolysis, and poor long-term dimensional stability in the intraoral environment [86], [91]. A similarly marginal status applies to PVC: published data on PVC-based clear aligners are scarce [91], [92], and no systematic cytotoxicity assessment has been reported for this material in an orthodontic context — a significant omission given the well-documented toxicological concerns associated with PVC and its plasticizer additives [93], [94]. The available evidence suggests that PVC-based devices produce thicker appliances with mechanical performance broadly comparable to PETG, but at the cost of substantially reduced patient comfort during wear [92].

Polyethylene and Polypropylene were also explored in early aligner systems but have been definitively displaced: their elastic moduli are insufficient to generate the orthodontic forces required for active tooth movement, and both materials exhibit progressive loss of optical transparency under intraoral conditions, rendering them unsuitable for the esthetic demands of clear aligner therapy [71], [86]

### 1.6.5. Multilayer (ML) and blends

Recently, some clear aligner brands have decided to take a side path by combining various types of thermoplastics in a multilayer product. The final material is said to gather all the benefits of its components, but the structure of the multilayer and its components should be wisely selected.

The main and most successful approach is to use a soft inner layer sandwiched between two stiff outer layers. This approach ensures full protection of the inner layer from chemicals and staining agents and, as the manufacturers claim, provides mechanical isolation of both sides of the aligner, which makes the treatment more effective. Also, the soft layer would ensure that the mechanical force is continuous over time.

Another factor to consider when choosing multilayers is that by selecting the different layers carefully[95], a shape memory effect may be achieved, which is said to be an interesting feature for clear aligners that may facilitate more effective and faster treatments[96], [97], [98], [99], [100]. The idea is based on combining a plastic with a well-defined glass transition or melting point (switching phase) that defines the temporal shape of the device, and another plastic that memorizes the permanent shape of the device (permanent phase). The switching phase can be any amorphous or crystalline polymer, and the permanent phase is usually an elastomer[95], [101]. There are two clear ways to introduce shape memory polymers in dentistry. The first would be to make two orthodontic steps with one device. The patient would receive the device in its temporal shape as movement 1. Then, after the prescribed time, the patient should immerse the clear aligner in a hot water bath that shifts the aligner's shape into movement 2. This would decrease the number of aligners used but would require extra effort on the patient's part in performing the phase shifting when required [97], [100]. Although the advantages of shape memory polymers have yet to be proven, commercial solutions already exist to adapt them to the patient using a device that should be filled with water and can control the phase shifting temperature of the clear aligner [102].

The second approach to introduce shape memory polymers in clear aligners would be to fix the current dentition shape of the patient as a temporary shape. Then, the aligner would phase shift in response to the oral environment. The only advantage of this approach is that the initial fitting of the aligner would be completely painless the first time the patient uses the device, which seems like a scarce advantage to adapt such a complex material for use [39].

Regarding biocompatibility, the studies have found that these combinations show low to moderate cytotoxicity in almost all cases, with almost no instances of biological reaction or sensitization during treatment [103], [104], [105]. Interestingly, the number of inflammation and allergic reactions has increased in patients using Invisalign clear aligners [43], [61], [106], [107], [108] , which is the most common brand of aligners worldwide. Researchers have noted that this increase may be attributed to the presence of isocyanate on the surface of Invisalign aligners, resulting from the use of TPU as the outer layer [61],

[107]. In comparison, the second and third most commonly used aligner brands (Zendura FLX and Spark) utilize polyester as an outer layer, with rare cases of adverse reactions.

Previous studies in multilayer simulation have demonstrated the superiority of multilayers over monolayers, and that the soft-to-hard ratio of the layers should be at least 50 % or higher to obtain a real advantage in treatment [109]. These studies lack a practical counterpart, as they do not examine the real-world production and manufacturing of this type of material, its economic viability, or its use in treatment.

Another approach, less explored, is the blending of the different thermoplastics. It is less researched because different polymers, in general, are non-miscible, so they will only produce a material with high heterogeneities, lower mechanical properties, brittle behavior, and high opacity or yellowing after blend extrusion [110], [111]. These characteristics are generally undesirable for clear aligners. As a matter of fact, the blending of TPU and PETG is not suitable for mass production[112], because both are said to be incompatible, whereas PC and Polybutylene terephthalate (PBT) are compatible with TPU [113], [114]. A brief study suggests that the ideal features can be achieved by blending TPU, PETG, and PC; however, it is based on limited experimental data, and the aligners were produced through injection molding rather than extrusion and thermoforming. [69]

### 1.6.6. Clear aligner manufacturing process

Currently, almost all devices on the market are manufactured by thermoforming transparent plastic sheets over 3D-printed models[115], [116]. Clear aligner manufacturing comprises a series of steps that begin once the patient visits the dental clinic and undergoes a mouth scan. Once that happens, the orthodontists or dentists usually send the scanners to a third-party company that processes and plans the treatment. Sometimes, if the case is complex, Cone Beam Computerised Tomography (CBCT) is also needed, along with dental scanners, which provide better accuracy. However, CBCT is not used in all cases due to concerns about radiation exposure and the complexity of the processing[117], [118], [119].

After that, a trained professional plans the treatment aided by dedicated software. That software slices the treatment into small and easy steps, and generates a three-dimensional image of the teeth's position at the end of each step. Hence, the step images are typically 3D printed using Stereolithography (SLA) or Digital Light Processing (DLP). The main advantages of these techniques over other 3D printing methods, such as Fused Filament Deposition (FFD), are their high accuracy and printing velocity. However, the main disadvantages are that the models cannot be recycled or reused, and the material is more expensive and difficult to handle.

After printing, the models are polymerised by Ultra-violet light and cleaned with suitable solvents to eliminate free monomers that had not cured during printing and polymerisation. Once cleaned, the models are used as molds for thermoforming the devices using

vacuum-assisted desktop thermoformers, such as the one shown in Figure 1.20. Thermoforming consists of heating a thermoplastic sheet above its glass transition. Usually, the plastic heats 70 ºC to 90 ºC above its Glass transition Temperature (Tg), depending on its rheological properties. While the plastic is heating, the model is sprayed with a substance that aids the demolding after thermoforming. The most commonly used substance is simply a mixture of hand soap and water, although more professional solutions are available.[115]

Once heated and melted, the thermoplastic is clamped over the model, and vacuum is applied to form the thermoplastic to the mold. Clear aligner sheets are commonly protected by a sacrificial, ultra-thin plastic sheet glued to one or both sides of the plastic. This sacrificial layer not only protects the material during shipping but also prevents any dirt or free monomer present in the model from contaminating the clear aligner device. Hence, after thermoforming, the thermoplastic must be demolded and peeled from that layer before cutting the device.

There have been recent claims from professionals and patients regarding why companies cannot or are unwilling to provide the molds used for thermoforming instead of discarding them. Actually, the molds are too revealing of the planning process and the biomechanical basis used to perform the planning. Sending the printed molds or the STL back to the orthodontist or the patient gives them space to question the entire process. In some cases, molds are not available because they are evidence of patent infringement or the type of technology used. At that point, clear aligners are a rather closed and not-transparent sector, in which lawsuits between competing companies are currently common. Professionals in the sector claim that the current protective environment and the fear of plaintiffs [120], [121], [122] and commercial war state among companies is stalling the sector.




Figure 1.20: clear aligner manufacturing process and end-of-life. Adapted from [39]

After thermoforming, demolding, and peeling, the device is cut out from the thermoplastic sheet either manually using a hand-held or an automatic milling machine or a $CO_2$ laser cutter. Finally, the devices are trimmed if needed, cleaned, sterilised, and sent again to the dentists or to the patient directly. A graphic summary of the production and end-of-life process for clear aligners is shown in Figure 1.20.

Each treatment typically consists of 20 to 60 aligners that must be worn for 10 to 30 days, approximately 20 to 22 hours per day, and only removed for eating or cleaning. After usage, each aligner is discarded as general waste; therefore, this technique generates a significant amount of waste that is currently not recycled. Not only from used aligners but also from the molds that are only used ones, scrap from cutting the thermoforming shapes, and inaccuracies and errors in the manufacturing chain [62], [123], [124], [125]. Some brands in the sector have started to collect the aligners that patients should bring back voluntarily when visiting the doctor to recycle the waste into other products[126], [127]. In principle, the materials in an aligner as a medical device are non-recyclable due to health security reasons unless a costly sterilisation procedure is included[128]. Due to the loss of properties, the materials are not able to be used as aligner materials again, but can find other valuable uses, for example, as textile fibres or as 3D printing filament [129], [130], although other companies recently have developed biodegradable alternatives [131], [132]. Many experts in CAT are claiming to enforce obligatory recycling aligners, even changing the medical device's law or creating a new embodiment in the law specifically for these types of devices, including how to properly sterilize and recycle them.[127], [131], [133], [134]

A high percentage of this waste could be eliminated if, instead of thermoformed aligners, fully 3D-printed aligners were used; however, the properties of 3D printing resins still fall short of reaching that point [115], [116], [135], [136]. There is a product already approved in Europe for producing 3D-printed aligners directly; However, professionals still prefer to use thermoformed aligners due to their properties, [137] and other resins that can already produce reliable mouth guards and retainers[138], [139]. 3D printed aligners not only would reduce waste but would also decrease the errors and inaccuracies that arise from thermoforming and laser cutting, implying a faster manufacturing process.

## 1.7. Mathematical modelling of viscoelastic materials in orthodontics

Accurate modelling of tooth movement requires constitutive descriptions of both the PDL and the thermoplastic aligner, as both materials exhibit pronounced time-dependent mechanical behaviour. Under physiological conditions, all forces acting on the tooth are transmitted and attenuated by the PDL, whose viscoelastic response governs the temporal profile of orthodontic loading. The same time-dependent character applies to the aligner thermoplastic, whose stress relaxation under a fixed strain determines how rapidly the delivered force decays during the wearing period. Any constitutive model intended for use in orthodontic simulation must therefore satisfy a set of fundamental physical require-

ments: causality, monotonic energy dissipation, asymptotic equilibrium at a non-negative residual modulus, thermal dependence consistent with the time–temperature superposition principle, complete monotonicity of the relaxation function, and the ability to interconvert between time- and frequency-domain representations [140], [141], [142]. A detailed discussion of these requirements and their mathematical foundations is provided in Appendix A.

Classical viscoelastic models (such as the generalised Maxwell model and its Prony series representation, the Kohlrausch-Williams-Watts model (KWW) stretched exponential, and the Guiu-Pratt logarithmic model (GP)) have been widely employed in the polymer literature, and in less extense in the orthdodontic literature [74], [143], [144], [145]. The generalised Maxwell model is the most commonly implemented in commercial finite-element codes, as it is computationally straightforward and can be fitted to experimental data with acceptable accuracy [142], [146]; however, it requires a large number of parameters and does not yield physically interpretable quantities such as the continuous relaxation spectrum or the molecular weight distribution [147]. The KWW function provides a compact empirical description of stress relaxation but lacks a closed-form relaxation spectrum and cannot be extended to arbitrary loading histories via the Boltzmann superposition principle [146], [148]. The GP model, derived for thermally activated dislocation glide in metals, predicts a stress that decreases without bound as $t \to \infty$, violating the requirement of a well-defined equilibrium modulus, and is formally nonlinear, precluding time-frequency interconversion [142], [149]. Neither the KWW nor the GP model can therefore serve as a general constitutive equation suitable for finite-element implementation under complex, time-varying loading histories.

Fractional viscoelastic models overcome these limitations by introducing the springpot (Scott-Blair element) [150], [151], a rheological element whose constitutive equation interpolates continuously between a purely elastic spring ($\alpha = 0$) and a purely viscous dashpot ($\alpha = 1$) as shown by the Equation 1.1

$$\sigma(t) = \eta^* \, D^{\alpha}[\varepsilon(t)], \quad 0 < \alpha < 1 \tag{1.1}$$

Combinations of springs, dashpots, and springpots yield compact fractional models — most notably the fractional Zener model and the Fractional Double Zener model (DZ) — that capture the broad, continuous relaxation spectra of polymers and soft tissues with few physically interpretable parameters [140], [152]. A key advantage of these models is that they yield the continuous relaxation time spectrum $H(\tau)$ analytically, enabling direct comparison with molecular weight distributions and providing a rigorous physical interpretation of each parameter [140], [146]. The fractional order $\alpha$ encodes the degree of molecular cooperativity and the breadth of the relaxation distribution, and carries direct physical meaning as a measure of the non-Markovian character of the material response [153], [154]. Despite their demonstrated applicability to a wide range of thermoplastics and biological soft tissues, fractional models have rarely been applied to PDL mechanics or aligner thermoplastics in the orthodontic literature [145], representing a significant methodological gap that the present work begins to address.

Knowledge of the relaxation time spectrum is of direct clinical relevance in aligner therapy: a material with excessively short relaxation times will dissipate its stored elastic energy before the prescribed wearing interval has elapsed, while one with unduly long relaxation times may sustain elevated forces beyond the point at which bone remodelling is complete, increasing the risk of root resorption and overcorrection [29], [155]. The optimal aligner material should therefore retain a mechanically significant relaxation modulus throughout the full change interval — a requirement that motivates the master curve analysis and DZ model characterisation performed in Chapters 4 and 5 of this thesis.

# 2. OBJECTIVES OF THE THESIS

The following objectives were defined as the main objectives of the thesis:

1. To characterize and understand the physical, chemical, optical, mechanical, and biological properties of commercially available thermoplastic materials in the clear aligner industry, establishing a comprehensive reference framework against which novel materials can be evaluated.

2. To obtain raw thermoplastic materials from commercially reliable and stable sources that fulfill the mechanical, thermal, optical, and biological requirements identified for clear aligner fabrication.

3. To develop a semi-industrial method to process raw thermoplastic materials into multilayer clear aligner sheets. The method must be scalable to manufacture the quantities required by Secret Aligner and flexible enough to accommodate changes in material composition or target properties.

4. To characterize the manufactured multilayer materials and evaluate whether their mechanical, viscoelastic, and biological properties are comparable to or superior to those of commercial clear aligner sheets, and if possible, to manufacture real clear aligners from selected configurations.

5. To develop a rational quantitative design framework that allows the final properties of a multilayer device to be predicted from the properties of its constituent raw materials, and that is flexible enough to accommodate new material combinations in the future.

6. To evaluate the biocompatibility and intraoral stability of the homemade multilayer materials, including their cytotoxic potential, oxidative and inflammatory response, and physicochemical stability under simulated intraoral conditions.

7. To study the economic feasibility of in-house multilayer thermoplastic sheet production at a scale appropriate for a mid-sized European manufacturer such as Secret Aligner, with sufficient flexibility to be scaled upward or adapted to different multilayer configurations as the company's production requirements evolve.

Besides these objectives, other secondary objectives were also defined:

1. To study the environmental impact of clear aligner therapy, including a quantitative life cycle assessment of the full manufacturing and clinical workflow and a comparative analysis with conventional fixed appliance treatment.

2. To characterise the range of forces and deformations experienced by both the thermoplastic aligner and the dentition during clear aligner treatment, through instrumented in vitro measurements and open-source finite element simulation, and to validate the simulation framework against experimental data.

3. To identify niches for improvement of current aligner thermoplastics and emerging trends in the sector that could help Secret Aligner improve its competitiveness in a market currently dominated by large multinational suppliers, and to propose scientifically grounded strategies for the rational design of next-generation aligner materials.

# 3. MATERIALS AND METHODS

### 3.0.1. Commercial clear aligner materials

Commercial materials for clear aligners were obtained by Secret Aligner S.L. from 14 different sources worldwide. Up to 30 different clear aligner materials were tested or received for characterisation, from which 12 different multilayer configurations were identified, comprising PETG, PCTG, and TPU grades in various combinations. A complete inventory of all materials received is provided in Table B.1 (Appendix B). The dataset reflects current market trends in clear aligner thermoplastics, which are dominated by multilayer architectures produced across multiple regions, with PETG being progressively replaced by PCTG and TPU grades — particularly for retainer applications. Secret Aligner was also able to supply finished aligners from two leading commercial brands (Invisalign and Spark); these companies do not disclose the composition of their materials and do not supply non-thermoformed sheets, but their multilayer architectures were characterised by cross-sectional nanoindentation as described in Chapter 4.

### 3.0.2. Raw material sourcing

After conducting the initial screening of materials and considering the requirements for multilayer manufacturing, an attempt was made to reproduce the physical properties of Zendura FLX, thereby altering them in a controlled manner. Two types of thermoplastics were obtained and divided into hard thermoplastics used as outer layers and soft thermoplastic elastomers used as inner layers. The general requirements imposed for all the materials obtained were high transparency, susceptibility to thermoforming, a medical or food-grade certificate, and commercial availability.

The requirements imposed on the hard layers were a high Glass transition Temperature of at least 100 ºC, with a low or absent crystalline structure and a cold crystallization process. Also, the elastic modulus range was set between 900 and 2000 MPa. Taking into account those requirements and excluding materials still under development, the following materials were attempted to be obtained from commercial sources:

- **PCTG/Tritan copolyester**: Three different grades of Tritan copolyester were obtained directly from Eastman, although similar materials were attempted to be obtained from SK chemicals without success. At least 5 Kg of each grade was obtained. No other PETG derivative type was found commercially that fulfilled the necessary requirements.
- **Polyethylene Naphthalate (PEN) and PET-PEN Copolymer (PET-PEN)**: Pure PEN and PET-PEN copolymer were obtained directly from Indorama Ventures. 5

Kg of both polymers were obtained. Other PEN derivatives were attempted to be obtained from Tekra without success.

- **Poly(ethylene-co-isosorbide) terephthalate (PEIT)**: It was impossible to obtain any source of PEIT due to its low commercial availability. It was impossible to contact Roquette Materials, the only supplier of such material, to obtain PEIT samples.
- **Polyethylene furanoate (PEF)**: It was impossible to obtain any source of PEF in time for the tests from Avantium Technologies, the only supplier of such material, to analyze the samples PEF.
- **High-hardness Thermoplastic Polyurethane (hTPU)**: 1 Kg of hTPU was obtained from the company Filament2Print.

For comparison reasons, we decided to include PETG in the hard layers, although it does not fulfill some of the listed requirements. The complete list of hard thermoplastics obtained is presented in Table 3.1.

| material | Composition | Hardness | VST (ºC) | Source | Origin |
|---|---|---|---|---|---|
| Tritan MP100 | PCTG | 75 | 115 | Eastman | Netherlands |
| Tritan MP150 | PCTG | 75 | 110 | Eastman | Netherlands |
| Tritan MP200 | PCTG | 75 | 140 | Eastman | Netherlands |
| PETG | PETG | 75 | 75 | Eolas Prints | Spain |
| PEN | PEN | 85 | 125 | Indorama Ventures | USA |
| PENco | PEN-PET | 85 | 130 | Indorama Ventures | USA |
| TPU Hardness + | TPU ether | 83 | 164 | Filament2Print | Spain |

Table 3.1: List of all the hard thermoplastics obtained. Hardness is measured in shore D scale. VST stands for Vicat softening temperature

Regarding the soft layers, mostly thermoplastic elastomers were sourced, and no additional requirements were imposed beyond the general ones. Hence, the materials sourced were basically different types of soft and medium-hardness TPU and EVA, as listed in Table 3.2.

| material | Composition | Hardness | VST (ºC) | Source | Origin |
|---|---|---|---|---|---|
| EVA 1828 | EVA | 29 | 44 | Hanwha | South Korea |
| Laripur 9060 | TPU ether | 33 | 110 | COIM | Italy |
| TPU 93A | TPU ester | 41 | 165 | Eolas Prints | Spain |
| Estane-F95A | TPU ester | 46 | - | Filament2Print | Spain |
| Laripur 5260 | TPU ether | 52 | 120 | COIM | Italy |
| Estane-F98A | TPU ester | 60 | - | Filament2Print | Spain |
| Laripur 71D60 | TPU ether | 70 | 162 | COIM | Italy |

Table 3.2: List of all the soft thermoplastic elastomers obtained. Hardness is measured in shore D scale. VST stands for Vicat softening temperature

### 3.0.3. Differential Scanning Calorimetry (DSC)

DSC traces were obtained using a melting-cooling-melting standard procedure on TA Q200 equipment, from room temperature or from -90 ºC when needed, to 250 ºC with a heating/cooling rate of 10 ºC/min. Aluminum non-hermetic pans and between 5 and 10 mg of hand-cut samples were used.

### 3.0.4. Thermogravimetric Analysis (TGA)

TGA curves were obtained using a TA Q50 instrument. The experiments consisted of heating a plastic piece of less than 10 mg, placed over a platinum pan in a controlled atmosphere to 900 ºC with a heating rate of 10 ºC/min. Two different types of experiments were used: one in an oxidative atmosphere (air) and another in an inert atmosphere (nitrogen). The parameters obtained were :

- T5 - Temperature at which the weight has decreased for a 5%
- T10 - Temperature at which the weight has decreased for a 10%
- T50 - Temperature at which the weight has decreased for a 50%
- Tmax - Temperature of the maximum of the weight loss derivative with respect to the temperature
- Tonset - Onset temperature of the weight loss defined as a drop of 2 dB in the weight loss

After analyzing the curves, the activation energy of the process was obtained based on the Coats-Redfern theory.[156], [157].

In the case of TPU samples, an in-house Python script was developed to perform Gaussian deconvolution of the derivative TGA curve and estimate the soft and hard seg-

ment mass fractions from the relative peak areas (available at https://github.com/Ozyman1998/Thesis/tree/main/TPU_TGA_ANALYZER).

#### 3.0.5. Dynamic Mechanothermal Analysis (DMTA)

The DMTA curves were obtained using a DMTA TA Q800 device in tensile configuration, from room temperature to 160 ºC at a heating rate of 3 ºC/min. The frequency was set to 1 Hz and the amplitude to 20 $\mu$m. The samples were dog-bone type samples cut by laser with a length of 10 mm, a width of 1.5 mm and a nominal thickness in the straight testing region.

In cases where cryogenic DMTA measurements were required, cooling to -120 ºC was achieved using liquid nitrogen, and the experiment was performed with the same exact parameters as mentioned before.

### 3.1. Elastic modulus testing

Prior to performing stress relaxation measurements, tensile tests and a 3-Point Bending test (3PB) were conducted on some of the materials. The experiments were carried out using the DMTA equipment with the same type of samples as in the DMTA and rectangular samples in 3PB. The bending samples were cut by laser to be 30 mm in length, 12 mm in width, and of a nominal thickness. The tests consisted of setting a force ramp of 3 N/min up to 18 N (the maximum force the equipment can perform). The tests were conducted at 37 ºC, so an equilibration and homogenization of the temperature in the chamber was performed before each experiment. From these tests, the elastic modulus and the strain ranges were obtained to perform the stress relaxation experiments.

Regarding the tensile configuration, the samples were dog-bone type with the same dimensions as already mentioned. The force ramp was set to 5 N/min, and a thermal equilibration of the chamber was performed before starting the test.

#### 3.1.1. Stress relaxation

Stress relaxation measurements were conducted using the DMTA equipment in tensile and 3PB. The stress relaxation involved equilibrating the samples in the DMTA chamber to 37 ºC and waiting for 5 minutes to homogenize the temperature. The sample was then stretched to a predefined strain, and the strain was held constant for a set time (usually 3 hours) while the force was monitored. After the stress relaxation time, the sample was allowed to deform freely under a small load (1 mN), and the strain was monitored for 5 minutes of recovery at 37 ºC.

### 3.2. Uniaxial tensile test

A uniaxial tensile test was performed using Instron equipment with a load cell of 100 N in tensile configuration. The same samples as in DMTA in tensile configuration were used. The test consisted of setting a deformation velocity of 1 mm/min at room temperature until rupture or a deformation of 200%.

### 3.3. Attenuated Total Reflectance Fourier-Transform Infrared spectroscopy (ATR-FTIR)

spectra were obtained using a Nicolet iS50. Before each measurement, the crystal was cleaned with ethanol and a background spectrum was collected. Each measurement consisted of the average of 64 scans. Chemical identification was performed using the spectral library available in the instrument software, complemented with the open-source online tool OpenSpecy. Python scripts were also developed to characterise TPU materials chemically based on diagnostic band criteria reported in the literature; these scripts identify the soft segment type (polyester or polyether), the diisocyanate (MDI, TDI, or HDI), and compute the degree of phase separation (DPS) and degree of ordering (DOR) from carbonyl band deconvolution. The scripts are openly available at https://github.com/Ozyman1998/Thesis/tree/main/ATR-FTIR.

All spectra were normalised using reference bands established in the literature. For copolyester materials, the terephthalate out-of-plane deformation band (725 $cm^{-1}$) was used as the normalisation reference; for TPU materials, the aromatic C–H deformation band (1413 $cm^{-1}$) was used instead. For cell culture medium spectra, the ATR-FTIR measurement was performed by placing a droplet of liquid directly onto the crystal and acquiring the spectrum immediately thereafter; in this case, the O–H stretching band (3250 $cm^{-1}$) was used as the normalisation reference.

### 3.4. UV-VIS spectroscopy

UV-VIS spectroscopy was performed to assess the optical transparency of the plastic sheets using a Shimadzu UV-2600i spectrometer, with barium sulfate as the baseline reference and air as the blank. Only planar specimens of defined dimensions are compatible with this system; accordingly, 5 $\times$ 5 $cm$ square samples were cut manually prior to measurement. Python scripts were developed to extract quantitative transparency metrics from the raw spectra, including average transmittance in the UVB ( 280-315 nm ) and UVA (315 - 400 nm) ranges, visible transmittance (400 - 700 nm) , cut-off wavelengths at $T = 10$ %, $T = 50$ % and $T = 90$ % and an approximate yellowing index. The scripts are openly available at https://github.com/Ozyman1998/Thesis/tree/main/UV-VIS.

### 3.5. Vickers hardness

The Vickers hardness was measured over plain disks stacked until reaching 6.5 mm in thickness using 0.245 N indentation force for 15 seconds. The hardness value was computed directly from the machine software. At least 3 different indentations in different parts of the plastics were performed.

### 3.6. Nanoindentation

Prior to performing the nanoindentation measurements, the selected materials were cut into small squared samples and placed into a plastic pin. Hence, the plastic pin was completely covered by epoxy resin (EpoFix Resin & Hardener, Stuerers). The resin was prepared by mixing 7 parts of epoxy with 1 part of hardener by weight and stirred gently by hand, avoiding bubbling. The resin with the materials was then left to cure for 24 hours at room temperature and unmolded afterwards, resulting in epoxy coins. The coins were ground using a parallel plate guide until reaching the surface of the materials on both sides. Later, the epoxy coins were polished on both surfaces using polyamide plates until obtaining a perfectly smooth surface. A graphical example of the pucks can be seen in Figure 3.1.



Figure 3.1: Sample Pucks after measuring

Once the samples were prepared, they were glued to a metallic sampler holder and introduced into the nanoindentation equipment. The equipment used was a Nanotests Vantage (Micro Materials LTD) and the indenter was a Berkovich type indenter. Since the Berkovich indenter has the same area-to-depth ratio as the Vickers' indenter, the hardness values obtained by nanoindentation are comparable to those obtained by Vickers hardness

testing.

The experiment consisted of indenting the materials on their cross section using a force of 25 mN with a loading and unloading velocity of 25 mN/s and a holding time between loading and unloading of 60 seconds. The distance between consecutive indentations was set to 50 $\mu$m. In some cases, such as thin films, the distance was reduced to 40 $\mu$m to account for errors. Decreasing that distance further resulted in an underestimation of the properties of the materials due to accumulated errors. Hence, it was decided to avoid decreasing the indentation separation further.

To fit the indentation area, the Berkovich equation with one additional term was used (equation 3.1). [158], [159]

$$A(h_c) = 24.5h_c + 900h_c \tag{3.1}$$

being $h_c$ the penetration depth recorded by the equipment.

The Reduced elastic Modulus (RM) was obtained by fitting the unloading curve. The plastic work was computed as the area between the loading and unloading curves, and the elastic work as the area under the unloading curve,[158], [159], [160] as can be seen in Figure 3.2.

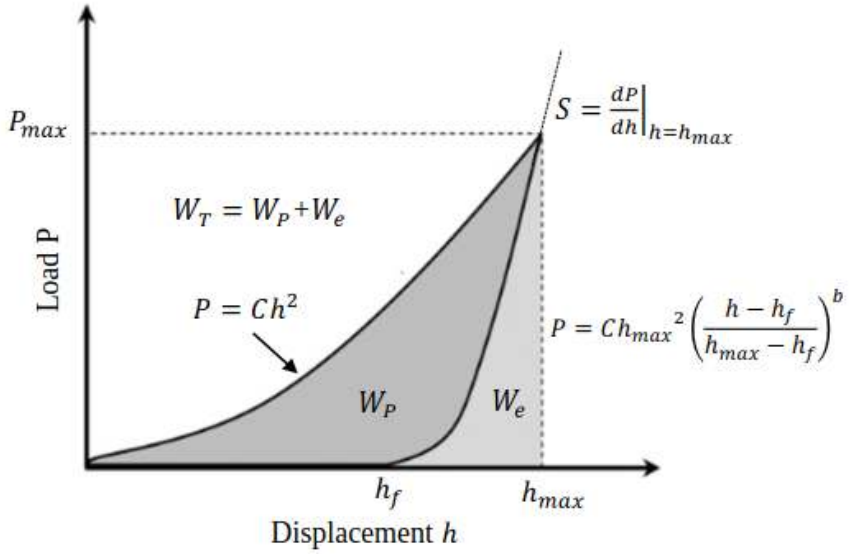



Figure 3.2: Typical nanoindentation curve with all the associated parameters [161]

In Figure 3.2, the S parameters is the contact rigidity, a parameters related with the reduced modulus and the area of the contact (A) by the equation 3.2. [161], [162]

$$RM = \frac{\sqrt{\pi}}{2\zeta}\frac{S}{\sqrt{A}} \tag{3.2}$$

where $\zeta$ is a geometrical correction factor that depends on the indenter geometry. However, for Berkovich indenter $\zeta$ is close to 1.

Finally the elastic modulus of the material was computed using a general relationship (Equation 3.3), assuming certain degree of error in the estimation.

$$E \simeq \frac{RM}{3} \tag{3.3}$$

All the calculations were performed by the equipment software in an automated manner.

In monolayers, at least 10 different indentation measurements separated by 50 $\mu$m were done. In multilayers, the profiling of the cross section was done first transversally to the cross section, setting 50 $\mu$m of separation between indentations. This process was repeated at least 3 times. Later, each individual layer was tested in its midplane, performing 10 indentations separated by 50 $\mu$m at least.

Due to a lack of accuracy and time, it was decided not to perform any TPU mapping in order to obtain the hard and soft segment distribution.

### 3.7. Water sorption

At least three samples of each material were exposed to partial immersion in distilled water for 30 days at 37°C. Specimens of 30 mm in length and 10 mm in width were cut by laser from as-received flat sheets; for thermoformed materials, specimens of equivalent dimensions were cut by hand from the lower lateral face of the wedge-shaped thermoforming mold (Figure 4.20), which corresponds to the region of greatest and most uniform material deformation. In the case of thermoformed mateials the experiment was restricted to 15 days after the observation that after 10 days the curves reach a plateau. After cutting, all specimens were weighed using a precision balance. Specimens were placed individually in glass containers filled with distilled water to a level covering half of the specimen length, thereby achieving partial immersion, and stored in a ventilated oven at 37°C. At each designated time point, specimens were removed from the containers, blotted dry with absorbent paper to remove surface water, and immediately weighed. Water absorption was calculated according to Equation 3.4.

$$\text{Absorption}(\%) = \frac{W(t) - W(0)}{W(0)} \times 100 \tag{3.4}$$

where $W(t)$ is the sample weight at time t and $W(0)$ is the initial dry weight. The absorption curves were fitted using the fractional diffusion model of Ritger and Peppas [156], [157], whose physical basis and mathematical derivation are provided in Appendix A (Equation A.25).

Following immersion, ATR-FTIR spectra and DSC traces were acquired on all immersed specimens to assess potential chemical degradation and microstructural changes relative to non-immersed controls, following the protocols described in Sections 3.3 and 3.0.3 respectively. DMTA characterisation of immersed specimens was performed only for materials in which DSC revealed detectable differences relative to the non-immersed state, following the protocol described in Section 3.0.5.

## 3.8. Extrusion

Before extrusion, all the materials were dried using a vacuum oven at 85 ºC and 70 mBar of vacuum pressure. Hard layer materials (Tritan MP100, MP150 and MP200) were dried for 7 hours, while soft layers (EVA,TPU85A, TPU52D and TPU70D) were dried for 3 hours.

### 3.8.1. Molding tests

After drying, molding tests were performed on some of the materials to see the flowability at extrusion temperature. For this, a hot plate press was used along with a square mold of 150 x 150 mm and 0.2 mm thickness. 6.5 g of each material was melted to produce plain sheets.

The molding pressure was set to 5 MPa and the molding time to 4 minutes. The temperatures were 270 ºC for all the hard layers, 190 ºC for TPU85A and TPU52D, and 215 ºC for TPU 70D. The pressure was held at 5 MPa during cooling as well.

Molding tests were performed in the Centre Català del Plàstic, in Barcelona.

### 3.8.2. Extrusion of hard layers

The hard layers (MP100, MP150, and MP200) were extruded in the Centre Català del Plàstic in Barcelona using a dual screw extruder. The extrusion conditions were designed following the Tritan copolyester extrusion manual provided directly by Eastman.

The machine used was a Kneter 25x36D Collin Lab & Pilot Solutions, with a screw diameter (D) of 25 mm and a screw length of 900 mm (L/D = 36). Six heating elements were used in the screw barrel, plus one in the nozzle (see heating scheme table C.1). A flathead nozzle of 100 mm in width and an aperture of 0 - 1 mm was employed. For hard layers, the nozzle aperture was set to 0.5 mm.

The materials were spooled after extrusion using a hot roll-calender (CR72T Collin lab & Pilot solutions) with a roller diameter of 72 mm and a roller width of 185 mm. The distance between rollers in the calender was set to 0.3 mm for hard layers extrusion.

### 3.8.3. Extrusion of soft layers

Two types of soft layer extrusions were done. The first one was done at the Centre Català del Plàstic in Barcelona using a single screw extruder only for TPU extrusion. The extrusion conditions were designed under the advice and recommendations made by COIM. The machine was an E20T Collin Lab & Pilot Solutions extruder with a screw diameter (D) of 20 mm and a screw length (L) of 500 mm (L/D = 20). Three heating elements were used in the screw barrel, plus a heating element at the nozzle (see extrusion profile table

C.1). The same type of flathead nozzle and spooling system as in hard layers was used. The nozzle apertures used were 0.2 mm, 0.25 mm, 0.3 mm, and 0.5 mm. The distances between rollers used were 0.1 mm, 0.15 mm, 0.2 mm, and 0.3 mm.

The second type of extrusions was performed at the Instituto de Ciencia y Tecnología de Polimeros (ICTP) in Madrid, using an Xplore MC40 with a CFL 65 Cast film line. In this case, TPU and EVA were extruded. The Xplore MC40 was equipped with a 0.2 mm flat die, and the screws were set corotating to avoid high torque generation. EVA was extruded at 180 ºC, TPU85A at 195 ºC, TPU52D at 205 ºC, and TPU70D was attempted to be extruded at 220 ºC. The screw speed was set to 40-60 rpm, and the spool velocity was adapted to obtain the required thickness.

## 3.9. Biocompatibility and intraoral stability testing

### 3.9.1. Materials

The materials evaluated in the biocompatibility and intraoral stability study comprised two Tritan copolyester monolayers (MP100 and MP200), two polyether TPU soft layer monolayers (TPU 85A and TPU 52D), four homemade multilayer configurations (100/85A, 100/52D, 200/85A and 200/52D), and the commercial multilayer reference material Maflex Premium (referred to as 'Commercial' in all figures in Chapter 6). The specific combinations of materials and assays performed are summarised in Table 6.1. TPU monolayer materials were excluded from artificial saliva absorption testing owing to their low stiffness, which prevented retention of specimen geometry during immersion. Indirect cytotoxicity testing was restricted to monolayer materials and the commercial reference.

### 3.9.2. Sample preparation

Three sample geometries were prepared depending on the assay. For indirect cytotoxicity testing, square specimens ($2 \times 2\ cm^2$, thickness 0.7 mm) were cut by laser and sterilised using ethylene oxide gas prior to use. For direct cytotoxicity and cell-material interaction assays, disc-shaped specimens (8 mm diameter and 0.7 mm thickness) were likewise prepared by laser cutting and sterilised by ethylene oxide. For artificial saliva absorption experiments, rectangular specimens (30 mm in length, 8 mm in width and 0.7 mm in thickness) were prepared by laser cutting; sterilisation was not required for this assay. All samples were cleaned with absolute ethanol prior to cutting to remove surface contaminants. Extracts for indirect cytotoxicity testing were prepared at a surface area-to-volume ratio of $0.6\ cm^2/mL$ by immersing the sterilised square specimens in complete cell culture medium and incubating at 37°C for 1, 3 and 7 days.

### 3.9.3. Artificial saliva absorption and thermomechanical stability

The artificial saliva absorption protocol followed the same sample preparation, immersion geometry, weighing procedure, and post-immersion characterisation scheme described in the section 3.7, with the following differences: the immersion medium was artificial saliva (Artificial Saliva Modified Fusayama/Meyer, Esslab, UK) instead of distilled water; the immersion period was 17 days instead of 30 days, as preliminary water absorption experiments confirmed that saturation was reached within 10 days for all materials. The experiment was restricted to copolyester monolayers (MP100 and MP200), homemade multilayer configurations, and the commercial reference material — TPU monolayer materials were excluded as their low stiffness prevented retention of specimen geometry during immersion. Post-immersion DSC and DMTA characterisation was performed on all materials regardless of the magnitude of observed DSC changes, given the clinical relevance of even modest thermomechanical alterations under simulated intraoral conditions. The absorption curves were fitted using the fractional diffusion model of Ritger and Peppas [156], [157], whose physical basis and mathematical derivation are provided in Appendix A (Equation A.25).

### 3.9.4. Cell culture

Human Gingival Fibroblasts (hGFs) (Ref. 300703, Cytion, Germany) were cultured in complete medium consisting of a 1:1 mixture of DMEM and Ham's F12 (Biowest, France), supplemented with 10 % Fetal Bovine Serum (FBS) (Pan Biotech, Germany), 10 ng/mL basic human Fibroblast growth factor (bFGF) (ThermoFisher, USA), 10 $\mu g/mL$ insulin (ThermoFisher, USA), and 1 % penicillin/streptomycin (Biowest, France). Cultures were maintained at 37°C in a humidified atmosphere with 5 % $CO_2$ and approximately 95% relative humidity [163]. All experiments were performed using cells between passages 9 and 15.

### 3.9.5. Indirect cytotoxicity assay

Indirect cytotoxicity testing was performed in accordance with ISO 10993-5 [164]. Sterilised samples were immersed in complete cell culture medium at a surface area-to-volume ratio of 6 $cm^2/ml$ and incubated at 37°C for 1, 3 and 7 days to generate time-dependent conditioned extracts. At each time point, the conditioned medium was collected and used immediately for cytotoxicity testing. hGFs were seeded into 96-well plates at a density of $10,000$ $cells/cm^2$ and allowed to attach for 24 h, after which the culture medium was replaced with the corresponding conditioned extract. Cells were incubated with the extracts for 24 h at 37 °C in a humidified 5 % $CO_2$ atmosphere. Cells cultured in complete medium without extract served as the viability control (100 % viability). Mitochondrial metabolic activity was assessed using a resazurin-based assay (Resazurin Cell Viability Assay, Canvax, Spain; 10 $\mu L$ resazurin per 100$\mu L$ culture medium

per well) after 3 h incubation at 37 °C. Absorbance was measured at 570 nm using a microplate reader (Tecan Infinite 200 PRO). Results were normalised to the positive control and expressed as percentage viability relative to the untreated control.

### 3.9.6. Direct cytotoxicity and cell-material interaction

Disc-shaped samples were cleaned, sterilised and placed individually in 48-well plates. hGFs were seeded directly onto the material surface at $10,000\ cells/cm^2$ in complete culture medium and incubated at 37°C with $CO_2$. At days 1, 3 and 7, culture medium was collected for subsequent LDH and ELISA analyses. Cell proliferation was assessed at each time point using the resazurin assay described above. At day 7, LDH activity was quantified using a commercial kit (CyQUANT LDH Cytotoxicity Assay, Thermo Fisher Scientific, USA) following the manufacturer's instructions: 50 $\mu L$ of sample was mixed with 50 $\mu L$ reaction mixture, incubated 30 min at room temperature in the dark, stopped, and absorbance measured at 490 and 680 nm. A positive cytotoxicity control was included at each time point by treating cells on standard tissue culture plates with 1 % Triton X-100 for 45 min. For morphological evaluation, cells cultured for 1 and 7 days were fixed with 4 % paraformaldehyde in PBS for 30 min, permeabilised with 0.2 % Triton X-100 in PBS for 45 min, and stained with rhodamine-phalloidin (Alexa Fluor 660, 1:400; Thermo Fisher Scientific) for the actin cytoskeleton and 4',6-diamidino-2-phenylindole (DAPI) (1:2000; Thermo Fisher Scientific) for nuclei. Fluorescence imaging was performed using an Olympus FV4000 confocal microscope at × 10 magnification

### 3.9.7. Reactive Oxygen Species (ROS) assay

Intracellular ROS production was evaluated using the H2DCFDA Cellular ROS Assay Kit (HB7375, Helo Bio). hGFs were seeded in black 96-well plates at $35,000\ cells/cm^2$ and cultured for 48 h, after which the medium was replaced with material extracts and cells were incubated for a further 24 h. Positive ROS controls consisted of cells treated with 6.7 $\mu M$ pyocyanin for 2 h; negative controls were maintained in complete medium without extracts. For detection, cells were washed and incubated with 20 $\mu M$ H2DCFDA in assay buffer for 45 min at 37°C in darkness, then washed twice before measurement. Fluorescence intensity was measured at excitation/emission wavenumbers of 485/535 nm using a microplate reader. ROS levels were corrected for the negative control and normalised to the positive control. Qualitative assessment was performed by confocal microscopy at × 10 magnification and optical microscopy at × 4 magnification.

### 3.9.8. Cytokine quantification by ELISA

Concentrations of Interleukin-6 (IL-6), Tumour Necrosis Factor-$\alpha$ (TNF-$\alpha$) and Interleukin-1$\beta$ (IL-1$\beta$) were quantified in culture supernatants collected after 7 days of direct culture

using commercial ELISA kits (Human IL-6 HUFI00180, Human TNF-$\alpha$ HUFI00262, Human IL-1$\beta$ HUFI03370; all from Assay Genie, Ireland), following the manufacturer's instructions. Absorbance was measured at 450 nm and cytokine concentrations were calculated from standard curves prepared according to the kit specifications. Results are expressed in pg/mL. The lower and upper limits of quantification (LLOQ and ULOQ) were determined from the standard curves and used to classify results as detectable, below the LLOQ, or above the ULOQ.

#### 3.9.9. Statistical analysis

All experiments were performed in triplicate using three independent material samples and repeated in at least two independent assays. Data are presented as mean ± standard deviation (SD). Outliers were identified and removed using the iterative Grubbs test ($\alpha = 0.1$), applied independently to each sample group; this conservative threshold was selected to minimise false outlier removal in small sample sizes ($n = 3-6$), in accordance with ISO 10993 guidelines [164]. Statistical analyses were performed in Python (version 3.12). One-way ANOVA followed by Tukey's HSD post hoc test was used to assess differences among material groups; Student's t-test was used for pairwise comparisons between each material and the corresponding spontaneous control. A significance level of $p < 0.05$ was applied throughout. Groups sharing the same letter in figures were not significantly different (Tukey HSD); asterisks indicate significant differences relative to the spontaneous control (Student's t-test)

# 4. SCREENING OF COMMERCIAL MATERIALS USED IN CAT

### 4.0.1. Introduction

Secret Aligner sources its commercial materials from various suppliers accumulated over the years, along with chemistry and property data provided by the respective distributors.

Throughout this period, the company's primary material was Zendura FLX, a multilayer product that, according to its manufacturer, features a sandwich structure comprising an outer copolyester layer and a soft inner polyurethane elastomer core. The rationale behind this screening was twofold: to characterize the properties of thermoplastics commonly used in clear aligner fabrication and to establish a straightforward, reproducible method for comparatively evaluating the fundamental attributes of such materials using Zendura FLX as the reference standard.

At the beginning of this project, Zendura FLX was the only multilayer material available on the market, and Secret Aligner was required to procure it directly from Bay Materials at a considerable cost. Beyond the purely economic burden, several additional risk factors warranted attention: the inherent volatility of the euro–dollar exchange rate, the potential impact of tariffs imposed by the European Commission on imports from the USA, and ongoing concerns related to lead times and supply chain reliability. Further compounding these challenges were Bay Materials' limited commercial flexibility and accessibility, as well as Secret Aligner's apprehension over the possibility of abrupt price increases or an unforeseen halt in the production or sales of this specific material due to patent-related considerations[120], [121], [122].

Accordingly, one of Secret Aligner's primary objectives was to identify alternative suppliers offering more competitive pricing. This goal was achieved relatively early on; nevertheless, the characterization of materials from commercial sources remained a routine task throughout the project. Over the course of this work, more than 30 distinct clear aligner materials were evaluated, including 13 multilayer materials and a range of monolayer variants based on hTPU, PETG, or PCTG (see Table B.1).

Multilayer clear aligner materials are currently manufactured in multiple regions around the world; however, only a few companies produce them within Europe, generally at a higher cost than their international counterparts. The following list compiles all countries of origin of companies supplying multilayer clear aligner materials that were identified, contacted, or from which material samples were received:

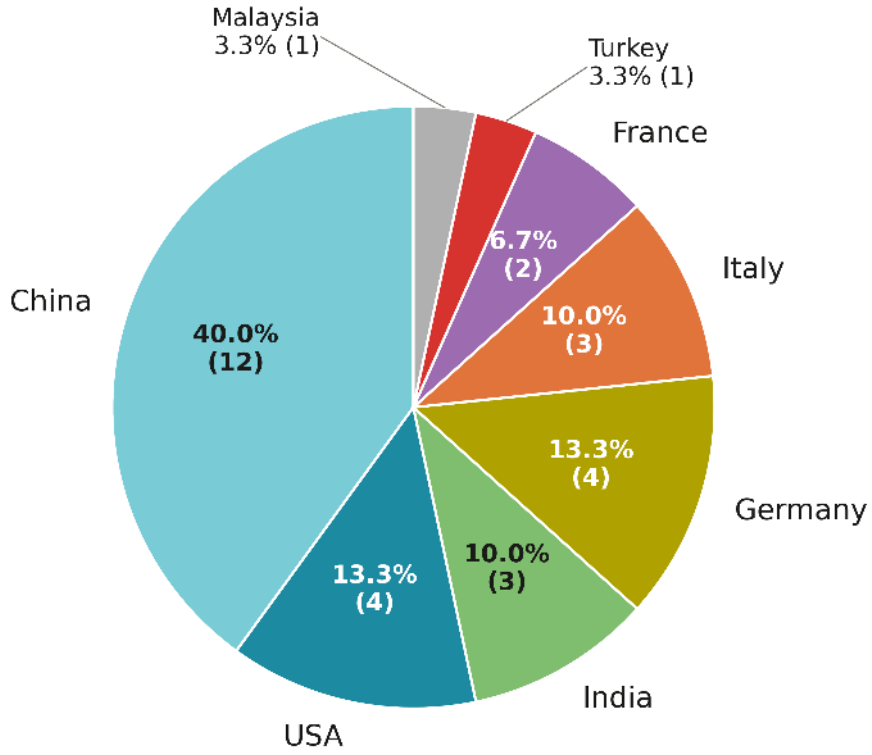



Figure 4.1: Country of origin of multilayers. The numbers below the percentages represent the total number of multilayer products.

As illustrated in Figure 4.1, China currently dominates the multilayer clear aligner sheet market, accounting for 40 % of identified suppliers, followed by the USA (13.3 %) and Germany (13.3 %). The remaining suppliers are distributed across India, Italy, France, Malaysia, and Turkey, reflecting a broad but uneven global manufacturing landscape. A notable pattern emerges from this distribution: most major industrialised nations host at least one supplier of multilayer thermoplastic sheets for thermoformed clear aligners, suggesting that the technology has reached sufficient maturity to support local manufacturing ecosystems in developed markets. In southern Europe, suppliers are present in Italy and France; however, European providers tend to operate primarily within their domestic markets, limiting their commercial reach and making cross-border sourcing within Europe less straightforward than reliance on established Asian distributors.

Furthermore, European suppliers are generally not price-competitive relative to Chinese and Indian manufacturers, which further constrains their appeal for companies operating across multiple markets. This situation is of direct relevance to Secret Aligner, which currently operates in Spain, Portugal, and France, and for which access to competitively priced multilayer materials within the European market (particularly in the Iberian Peninsula, where no domestic supplier has been identified) would reduce dependence on geographically distant supply chains. Concerns surrounding potential tariffs, supply chain

disruptions, and currency exchange volatility associated with dominant Chinese suppliers remain pertinent and are increasingly significant in the context of the current geopolitical and economic climate.

Caution must be exercised with respect to the intellectual property landscape for multilayer aligner materials. Bay Materials LLC currently holds active patents in the United States that prevent other companies from manufacturing multilayer products for clear aligners with a Dual Shell structure, defined as two outer rigid layers and an inner compliant core, for dental use[165], [166], [167]. The equivalent European patent application (EP4076955[168]) remains pending, and the examining division of the European Patent Office has already issued a communication raising objections to the original claims due to lack of novelty and extended industrial processability of such materials, which the applicant has partially addressed through amendments; however, the application has not yet been granted and its final scope remains uncertain. Bay Materials has actively enforced its US patents through litigation against domestic competitors [169], [170], while pursuing Chinese manufacturers has proven significantly more difficult and costly, which largely explains why the majority of multilayer aligner sheet producers are based in China.

It should be noted, however, that the patent protection is not unlimited in scope. The claims specifically cover constructions in which the outer layers have an elastic modulus between 1000 and 2500 MPa and the inner elastomeric core has an elastic modulus between 50 and 500 MPa, regardless of the fabrication method employed. From a broader commercial perspective, the enforceability of these patents in Europe is currently limited for several compounding reasons. First, the European patent application, remains pending and has no legal effect until granted. Second, even upon grant, a European patent does not confer uniform pan-European protection: it must be validated and maintained separately in each designated member state, requiring Bay Materials to initiate independent national litigation in each jurisdiction where infringement is alleged — a process that is both costly and strategically complex. Third, Bay Materials' litigation strategy to date has focused exclusively on the US market, where their patent position is strongest and their principal commercial competitors operate; no enforcement actions in Europe have been identified.

Taken together, these factors suggest that the European market currently presents a more favorable legal environment for the development and commercialization of novel multilayer aligner materials than the United States, provided that the specific material combinations employed fall outside the scope of the claims as finally granted.

### 4.0.2. Materials and methods

To perform a first material screening, the plastics in the Table 4.1 were obtained by Secret Aligner.

| commercial name | Brand | Thickness (mm) | Chemistry | recommended use |
|---|---|---|---|---|
| Duran | Scheu-Dental | 0.76 | PETG | aligner/retainer |
| Biolon | Dreve | 0.76 | PETG | aligner/retainer |
| Leone | Leone | 1 | PETG | aligner/retainer |
| Attachment | Tristar | 0.31 | PETG | Template |
| Prismlab C | Prismlab | 0.76 | PCTG | Aligner/retainer |
| Zendura AT | Bay materials | 0.31 | TPU | Template |
| Smilekeeper | Tristar | 0.76 | TPU | Aligner/retainer |
| PU flex | Taglus | 0.76 | TPU | Aligner/retainer |
| Essix C+ | Dentsply Sirona | 1 | PP/PE | Guards/retainers |
| Zendura FLX | Bay materials | 0.76 | ML | Aligners |
| Filtzwiesser | Filtzwiesser | 0.76 | ML | Aligners |
| Maflex Premium | Maflex | 0.76 | ML | Aligners |
| Maflex Premium Plus | Maflex | 0.76 | ML | Aligners |

Table 4.1: List of materials supplied by Secret aligner

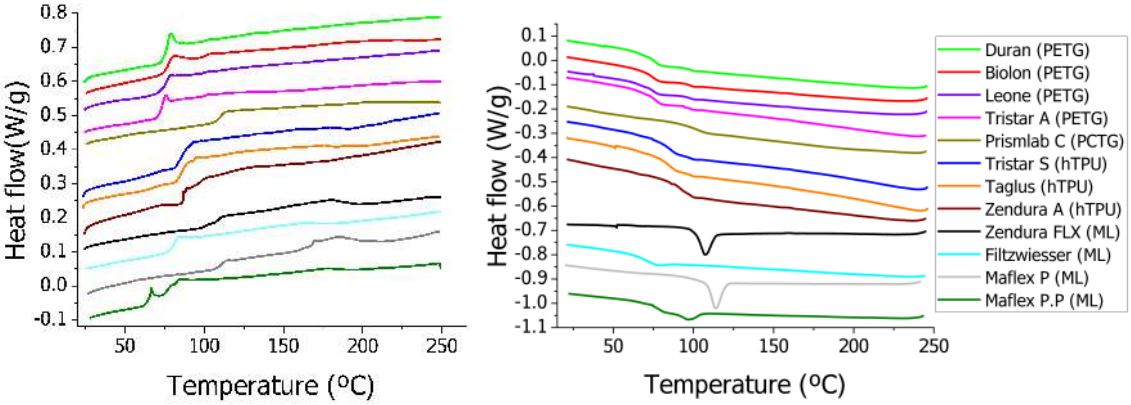
This initial screening was restricted to non-thermoformed materials for three reasons. First, testing thermoformed samples would introduce additional variables associated with both the thermoforming process itself and the 3D printing of the corresponding molds. It is true that flat, uniform specimens can be obtained by thermoforming over prismatic or cubic molds with perfectly planar faces (See figure 4.20)[171], [172]; however, uncertainties related to machine settings, thermoforming temperature, and mold fabrication methodology would still persist. Second, extracting flat, adequately sized specimens from finished aligners is not feasible, given the curved geometry and absence of planar surfaces inherent to these devices. Third, one of the overarching aims of this screening is to establish a reference database and testing framework for evaluating new materials in the future — materials that, in most cases, are supplied in non-thermoformed form.

### 4.1. Results on as-received materials

As shown in Figure 4.2, the vast majority of the materials tested are highly amorphous. For the sake of clarity, the DSC traces of Essix C have been omitted from the figure; however, they are characteristic of polypropylene, exhibiting a melting point of 150 ºC on heating and a crystallization peak of 117 ºC on cooling. Unlike the other materials, Essix C is a highly crystalline polymer. Its degree of crystallinity was calculated from the crystallization enthalpy measured upon cooling, yielding $\Delta H_c = 70 J/g$. Using the fusion reference enthalpy for 100% crystalline isotactic polypropylene of 207 J/g [173], [174], a degree of crystallinity of approximately 34% was obtained.

Several lines of evidence support the assignment of Essix C as isotactic polypropylene (iPP). First, the melting and crystallisation peaks at 150 ºC and 117 ºC, respectively, which

are characteristic of the $\alpha$-crystalline phase of iPP [173], [174], and are inconsistent with syndiotactic polypropylene (sPP), which — depending on the degree of syndiotacticity — typically exhibits melting points in the range of 130–160 ºC [175], [176]. Furthermore, sPP is rarely encountered in commercial plastic applications, making its use in a thermoformable dental aligner sheet unlikely. Atactic polypropylene (aPP) was also excluded, as it is essentially amorphous[177].




(a) Heating traces

(b) cooling traces

Figure 4.2: DSC traces in as-received materials

The Tg of the materials also provide useful information regarding their chemical nature. The Tg values recorded for PETG fall within the range of 75-80 ºC, consistent with values reported in the literature for this copolyester[178], [179]. For the hTPU materials, the relatively high Tg values observed in Figure 4.2 are consistent with a polyester-based TPU with a significant hard segment content, as polyester soft segments — owing to their higher polarity and stronger intermolecular interactions compared to polyether counterparts — generally yield higher Tg values[180], [181]. Finally, Prismlab C+ exhibits a notably higher Tg of 111 ºC, indicative of a PET derivative with a more rigid backbone compared to standard PETG.

In the multilayer materials Zendura FLX, Maflex Premium, and Filtzwiesser, each layer within the structure exhibits a clearly distinct thermal behaviour. During heating, the first expected thermal event would be the Tg of the soft segments of the TPU component. However, this transition was not observed, even though DSC experiments were conducted down to - 90 ºC. This absence can be attributed to the relatively low TPU content within the multilayer structure, combined with the inherently broad and low-intensity nature of this transition, which renders it difficult to detect by DSC[83], [182].

The Tg of the hard segments should in principle appear at higher temperatures; however, its detection by DSC is challenging for three concurrent reasons [83], [182]. First, the hard segment Tg in TPU is intrinsically difficult to observe by DSC and may not produce a discernible signal even under optimal conditions. Second, the relative concentration of the TPU hard segment within the overall multilayer structure is low, further

reducing the magnitude of the transition. Third, this transition is often obscured by overlapping thermal events, such as melting endotherms of adjacent layers or, in the case of multilayer materials, by the Tg of the hard outer layer itself. It should also be noted that, when microphase mixing between soft and hard segments is sufficiently extensive — that is, when the two phases are miscible rather than segregated — a single broad Tg may emerge in place of two distinct transitions, as the material behaves as a homogeneous phase with a single intermediate glass transition temperature [183]. Conversely, well-segregated TPU materials present two distinct Tg values corresponding to the soft and hard segment phases respectively, making the individual transitions in principle more detectable but in practice still difficult to resolve when the hard segment concentration is low [182], [183].

The Tg of the hard outer layer constitutes the third thermal event observed upon heating a multilayer material. In Zendura FLX and Maflex Premium, this transition is analogous to that of Prismlab C whereas in Filtzwiesser it more closely resembles the behaviour of a standard PETG. The fourth and final thermal event on heating corresponds to the melting of the hard segment crystals within the TPU, which occurs in the range of 180-200 ºC. This process is strongly dependent on the chemical nature of the TPU used in each multilayer, as well as on the processing conditions employed during manufacture[83], [182].

Upon cooling from 250 ºC, the only thermal event discernible in the cooling traces of multilayers shown in Figure 4.2 is the crystallisation of the TPU hard segments. As with the melting process, this transition is governed by the chemical nature of the TPU, the relative concentration of hard and soft segments, and the thermal history of the material [83], [182].

Maflex Premium Plus has been excluded from the comparative analysis described above. According to information provided by the manufacturer, this multilayer shares the same architecture as Invisalign, consisting of soft TPU outer layers encasing a rigid inner core [184], as analyzed below (Figure 4.32). In Figure 4.2, only the first heating scan and the subsequent cooling are shown for this material. During the first heating, a thermal event is observed at approximately 60 ºC that is absent in the second heating scan. This irreversible event is tentatively attributed to the melting of Polycaprolactone (PCL) crystals, which may constitute the soft segments of a polyester-based TPU [83], [182]. A second thermal event, a Tg at approximately 80 ºC, is present in both the first and second heating scans, suggesting it is an intrinsic material transition rather than an artefact of thermal history. While this transition most likely originates from the hard layer, its precise chemical assignment cannot be unambiguously determined from DSC data alone; the most probable candidates are either PETG or a hard-segment TPU domain. Finally, the crystallisation exotherm observed upon cooling is tentatively attributed to the hard segments of the TPU component, in analogy with the behaviour described above for Maflex Premium.

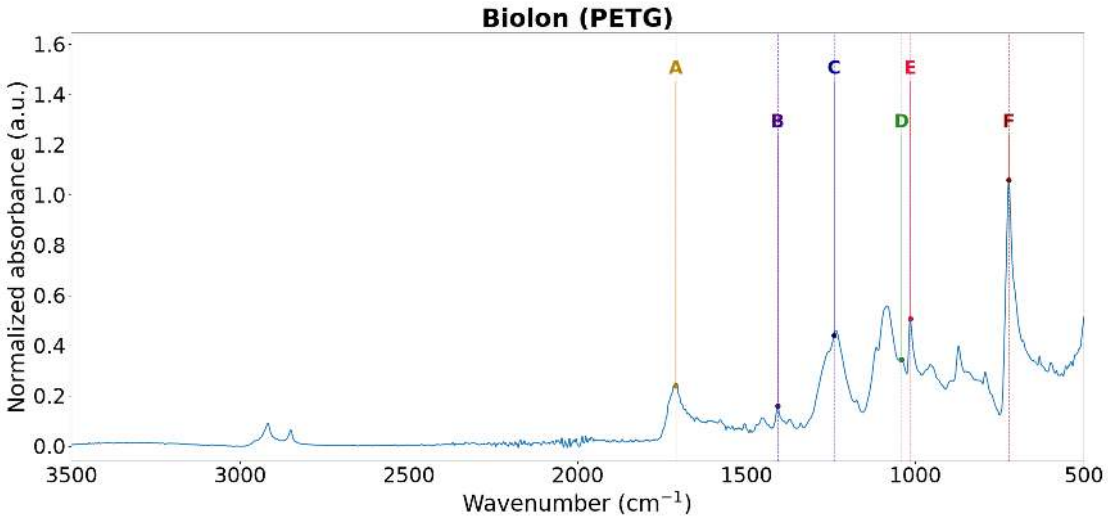
It may also be noted that the crystallisation processes in Maflex Premium Plus are more broadly distributed across the temperature range than in the other multilayer materials, resulting in a less defined exotherm. This suggests a more heterogeneous crystallisation behaviour, likely reflecting the superposition of multiple processes of different nature occurring over a wider temperature window, consistent with the more complex layered architecture of this material.

A characteristic enthalpy relaxation peak is observed in the DSC heating traces of all PETG materials in the vicinity of the glass transition, clearly indicative of physical aging [185], [186]. Physical aging is particularly relevant in the context of clear aligners, as is known to increase stiffness and brittleness while reducing toughness and, in some cases, promoting densification or even crystallisation of amorphous polymers[186], [187]. Physical ageing arises when an amorphous polymer is cooled too rapidly following thermal processing or shaping, trapping the macromolecular chains in a thermodynamically non-equilibrium glassy state. Given sufficient thermal energy and time, the chains undergo slow segmental relaxation towards a lower-energy configuration — a process manifested as an endothermic peak superimposed on the Tg visible in the DSC first heating traces [185], [186].

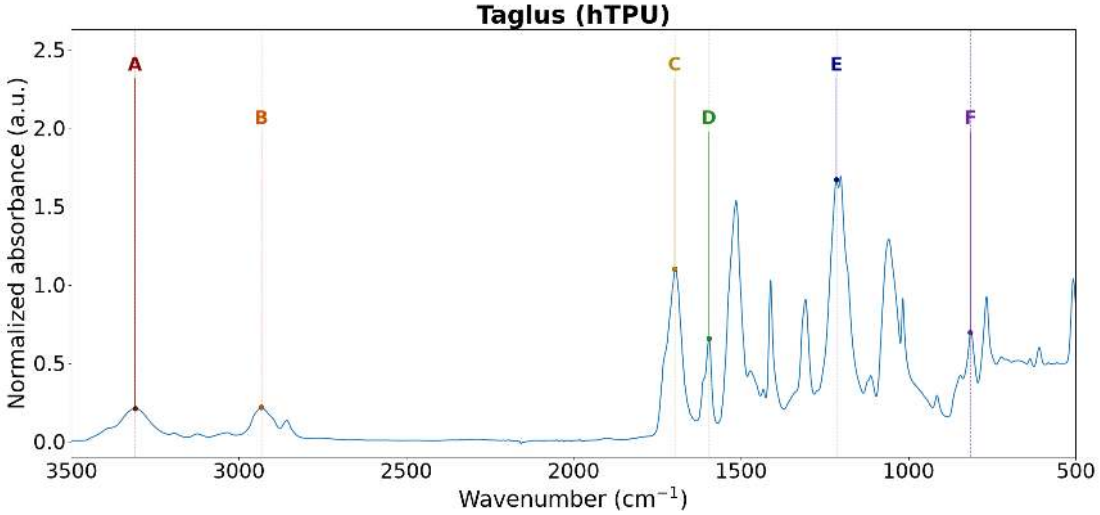
Notably, no ageing peak is observed in Prismlab C (PCTG). This may be attributed to its chemical composition: in this material, a proportion of the Ethylene Glycol (EG) units characteristic of standard PETG is replaced by a bulkier diol. which introduces greater steric hindrance and conformational rigidity into the polymer backbone [72], [188]. As a result, the chain mobility is hindered, and the free volume in the plastic is less readily compacted, raising the energetic barrier to physical aging, being this material more resistant to enthalpic relaxation than conventional PETG.

The chemical nature of all materials was subsequently confirmed by ATR-FTIR spectroscopy. The spectra of the PETG (Biolon, Duran, Leone, Tristar A) and PCTG (Prismlab C) materials are consistent with a standard copolyester backbone, as shown in Figure 4.3. The characteristic bands of the terephthalate segment were identified at 1709 $cm^{-1}$ (C=O ester stretching)[72], 1240 $cm^{-1}$ (asymmetric C–O–C stretching)[189] and 725 $cm^{-1}$ (aromatic C–H out-of-plane deformation, confirming 1,4-ring substitution)[190],being this last one the band used for normalising the spectra[190]. The presence of 1,4-Cyclohexanedimethanol (CHDM) as a comonomer was further confirmed by the bands at 1014 $cm^{-1}$ and 1040 $cm^{-1}$ attributed to the ring deformation modes of the cis and trans conformers of CHDM, respectively [72], [190]. The ratio of the trans to cis band absorbances ranged from 0.62 to 0.67 across all PETG samples, indicating a consistent predominance of the cis conformer regardless of material origin. This result is consistent with the known synthesis of CHDM via catalytic hydrogenation[72], which inherently favours cis enrichment, and suggests that cis/trans composition is not a differentiating variable among the commercial PETG grades studied. The higher cis content is associated with greater chain irregularity, reduced crystallisation tendency, and the optical transparency characteristic of these materials[191], [192], all of which are relevant properties for their application in clear or-

thodontic aligners. All hTPU materials were identified as polyester-based TPU, consistent with the DSC results discussed above.




(a) PETG Spectra. Band assignments: **A** = 1709 cm$^{-1}$, C=O ester stretching; **B** = 1407 cm$^{-1}$, aromatic ring stretching; **C** = 1240 cm$^{-1}$, C-O-C asymmetric stretching; **D** = 1040 cm$^{-1}$, CHDM trans conformer; **E** = 1014 cm$^{-1}$, CHDM cis conformer; **F** = 722 cm$^{-1}$, C-H aromatic out-of-plane bending




(b) hTPU Spectra. Band assignments: **A** = 3310 cm$^{-1}$, N-H stretch (urethane); **B** = 2932 cm$^{-1}$, $CH_2$ stretching (soft segment); **C** = 1698 cm$^{-1}$, C=O H-bonded (ordered hard segment); **D** = 1596 cm$^{-1}$, Amide II / C=C MDI; **E** = 1215 cm$^{-1}$, C-N stretch (urethane); **F** = 815 cm$^{-1}$, MDI para-substituted aromatic ring.

Figure 4.3: ATR-FTIR in as-received materials

Regarding the external layers in the multilayers, in the case of Zendura FLX, Maflex Premium, and Filtzwiesser, all of them were identified as PETG or PCTG based, whereas Maflex Premium Plus was identified as TPU.

From the ATR-FTIR spectra of the TPU samples (Figure 4.3), additional structural in-

formation can be extracted beyond simple chemical identification, namely the Degree of Phase Separation (DPS) and the Degree of Ordering of the hard TPU segment (DOR).[193], [194], [195], [196], [197]

Phase separation in TPU arises from the thermodynamic incompatibility between the hard and soft segments, which tend to microphase-segregate into distinct domains.[194], [198]. The covalent bonds linking both segments prevent the macrophase separation typical of polymer blends, instead yielding a nanoscale-domain morphology in which glassy or semicrystalline hard-segment domains act simultaneously as physical crosslinks and reinforcing fillers within the soft-segment matrix[199], [200]. The DPS quantifies the extent of this segregation by relating the fraction of hydrogen-bonded carbonyl groups to the total carbonyl content (see Equation 4.1) [195], [196]. A high DPS indicates that hard segments are effectively aggregated into cohesive domains, reinforcing the soft matrix and conferring the characteristic elastomeric behavior of TPU [197].

$$DPS\,(\%) = \frac{\text{Hydrogen-Bonded Carbonyls}}{\text{All carbonyls}} = \frac{A_{1698} + A_{1715}}{A_{1698} + A_{1715} + k * A_{1730}} * 100 \tag{4.1}$$

where $A_{1698}$, $A_{1715}$, and $A_{1730}$ are the integrated areas of the corresponding Gaussian components obtained from the deconvolution of the carbonyl region (1650-1760 $cm^{-1}$), and $k \simeq 1.2$ is a molar absorption coefficient ratio accounting for the difference in extinction between free and hydrogen-bonded carbonyl groups. [197]

The hydrogen-bonded carbonyl population can be further divided into two sub-types[194], [196]. The band at 1698 $cm^{-1}$ corresponds to ordered hydrogen-bonded carbonyls, associated with crystalline hard-segment domains, while the band at 1715 $cm^{-1}$ corresponds to disordered hydrogen-bonded carbonyls located at the interphase between soft and hard domains or within non-crystalline hard-segment regions[197], [201]. Finally, the band at 1730 $cm^{-1}$ is associated with free carbonyls, which are those not engaged in hydrogen bonding under quiescent conditions but susceptible to bond formation upon mechanical deformation[195], [202] . The DOR isolates the contribution of the crystalline fraction to the total carbonyl content:

$$DOR(\%) = \frac{\text{Hydrogen-Bonded ordered Carbonyls}}{\text{All carbonyls}} = \frac{A_{1698}}{A_{1698} + A_{1715} + k * A_{1730}} * 100 \tag{4.2}$$

A high DOR implies a well-developed crystalline hard-segment microstructure, which further enhances the mechanical reinforcement of the soft matrix. Consequently, DOR and DPS are closely correlated: a high DOR is both a consequence and a driver of a high DPS.

Regarding the chemical nature of the soft segment, ester and ether-type TPUs can be distinguished by their characteristic absorption patterns. Ester-type soft segments exhibit strong absorptions at 1730 $cm^{-1}$ (C=O stretching of the ester carbonyl), and 1170 $cm^{-1}$ (C−O−C stretching), together with mid intensity bands in the 1200-1240 $cm^{-1}$ region.

Ether-type soft segments, in contrast, lack the 1730 $cm^{-1}$ absorption; their spectra are dominated by a strong band at 1110 $cm^{-1}$ (C–O–C stretching of ether linkages), accompanied by medium-intensity absorptions in the 1060-1080 $cm^{-1}$ region. [197], [203], [204], [205]

The chemistry of the hard segment can likewise be identified spectroscopically. The present analysis was restricted to the three diisocyanates that dominate commercial TPU production: Methylene Diphenyl Diisocyanate (MDI), Toluene Diisocyanate (TDI), and Hexamethylene Diisocyanate (HDI). While the catalogue of isocyanates used in TPU synthesis is broad, these three account for virtually the entire market: MDI-based TPUs represent approximately 60 % of total volume, TDI-based TPUs around 25 %, and aliphatic hard segments the remaining 15 %, of which HDI constitutes about 95 %.[206], [207]

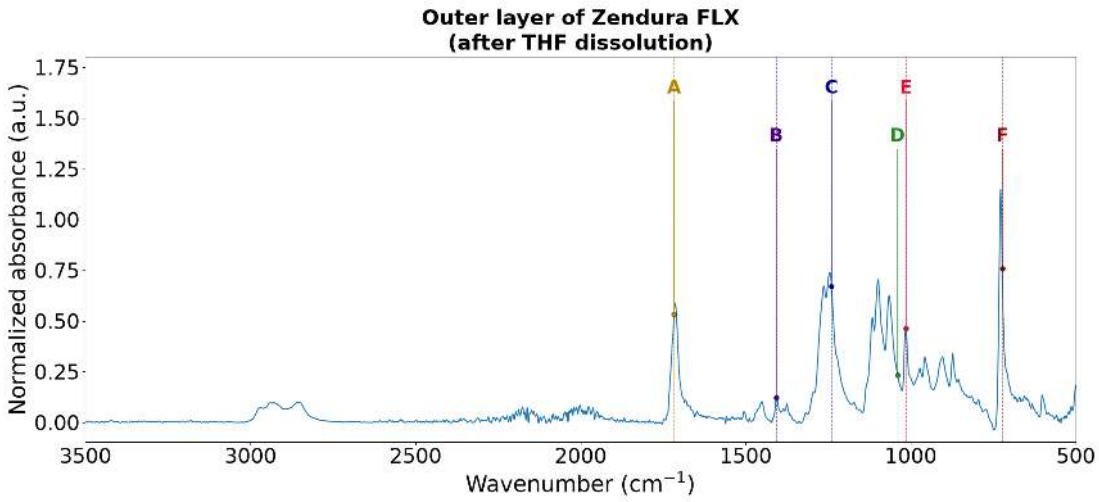
MDI is unambiguously identified by strong absorptions at 815 $cm^{-1}$ out-of-plane C–H deformation of the para-disubstituted aromatic ring), 1510 $cm^{-1}$ , and 1596 $cm^{-1}$ (aromatic C=C stretching). TDI produces a weaker or absent band at 815 $cm^{-1}$ and a shift of the dominant aromatic stretching to 1614 $cm^{-1}$ ; owing to the spectral similarity between MDI and TDI, unambiguous discrimination can sometimes require complementary techniques. HDI, being aliphatic, is characterised by the complete absence of aromatic signals and by prominent absorptions at 720-730 $cm^{-1}$ ($CH_2$ rocking of the hexamethylene chain), 1465 $cm^{-1}$ ($CH_2$ bending) and 2850 and 2920 $cm^{-1}$ (symmetric and asymmetric $CH_2$ stretching) being also the aromatic C–H region (3030-3100 $cm^{-1}$) absent in this case. [205], [208]

The ATR-FTIR analysis of the carbonyl region yielded DPS values ranging from 88 to 99 % and DOR values from 80 to 89 % across all hTPU samples. These values are consistent with, and in several cases exceed, those reported in the literature for high-performance polyester-MDI TPUs, where DPS typically falls in the 75-95 % range and DOR in the 60-80 % range[195], [197], [202]. The high DPS indicates that the hard-segment domains are well-segregated from the soft matrix, forming a cohesive physical network that acts as both a reinforcing filler and a reversible crosslink under deformation — a structural feature that underpins the elastic recovery and force delivery capacity essential in orthodontic aligners [196]. The elevated DOR further implies that a large fraction of these hard-segment domains has adopted a crystalline or highly ordered packing, which increases the stiffness of the hard phase and thereby enhances its reinforcing efficiency within the soft matrix.[209]

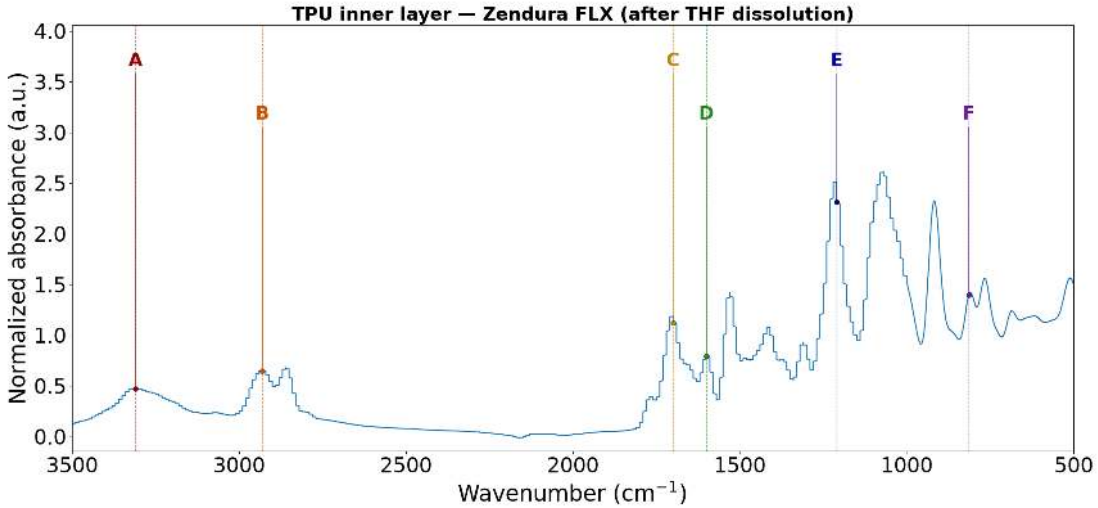
Regarding chemical composition, all monolayer TPU samples were identified as polyester-based (classification confidence 89-95 %) and MDI-based (85-88 %). The polyester soft segment is characterized by stronger dipole-dipole interactions than its polyether counterpart, which promotes a greater extent of microphase separation and typically yields higher DPS values — consistent with the results obtained here.[203], [205] From a clinical perspective, the choice of soft segment type has direct implications for intraoral durability: hydrolytic degradation of the ester linkages under prolonged exposure to saliva and its enzymatic components is a recognized concern for polyester-based aligner materials [26]

and the high degree of phase separation observed here — which reduces the accessibility of the ester bonds to water and salivary esterases by confining the soft segment within well-protected domains — may mitigate, but not eliminate, this susceptibility [26], [179]. The MDI-based hard segment, characterized by the symmetric para-substituted aromatic ring, is known to promote stronger and more regular hydrogen bonding compared to TDI, contributing to the high DOR values observed and to greater long-term dimensional stability of the aligner under cyclic thermal and mechanical loading in the oral environment[196], [208]. The DPS, DOR and chemical identification of TPU materials were determined using purpose-built Python scripts, which are publicly online available in a thesis repository

ATR-FTIR analysis of the outer surface of the Maflex Premium Plus multilayer sheet yielded that the outer layer in this multilayer is composed of a polyester-based TPU with high phase separation (DPS = 90 %) and moderate ordering (DOR = 43 %). The hard phase in this material is primarily composed of MDI.




(a) Outer Copolyester layer. Band assignments: **A** = 1709 $cm^{-1}$, C=O ester stretching; **B** = 1407 $cm^{-1}$, aromatic ring stretching; **C** = 1240 $cm^{-1}$, C-O-C asymmetric stretching; **D** = 1040 $cm^{-1}$, CHDM trans conformer; **E** = 1014 $cm^{-1}$, CHDM cis conformer; **F** = 722 $cm^{-1}$, C-H aromatic out-of-plane bending.




(b) Inner TPU layer. Band assignments: **A** = 3310 $cm^{-1}$, N-H stretching (urethane); **B** = 2932 $cm^{-1}$, $CH_2$ stretching (soft segment); **C** = 1698 $cm^{-1}$, C=O H-bonded (ordered hard segment); **D** = 1596 $cm^{-1}$, Amide II / C=C MDI stretching; **E** = 1215 $cm^{-1}$, C-N stretching (urethane); **F** = 815 $cm^{-1}$, MDI para-substituted aromatic ring.

Figure 4.4: ATR-FTIR spectra in the Zendura FLX layers after dissolution in THF

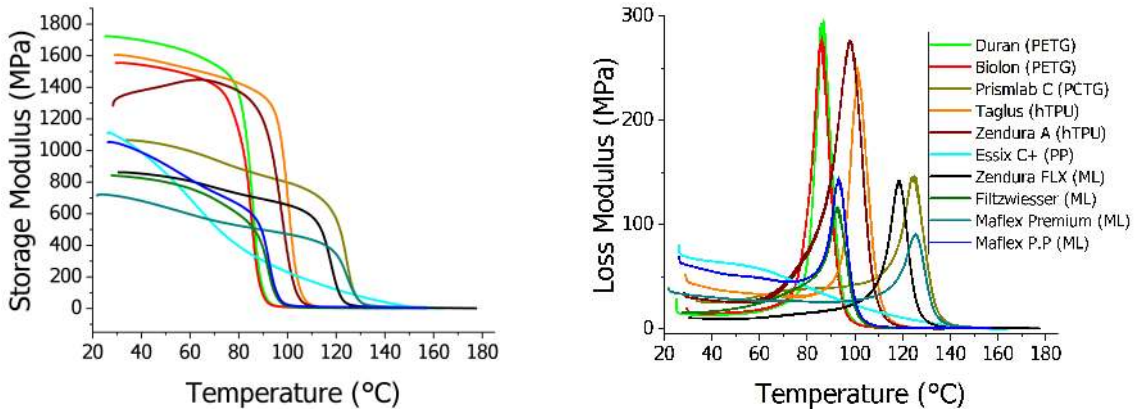
To characterize the TPU core layer of the Zendura FLX multilayer material, selective dissolution was attempted. A small piece of the sheet was immersed in Tetrahydrofuran (THF) for seven days at room temperature. THF is a well-established polar aprotic solvent known to dissolve thermoplastic polyurethanes while leaving polyesters intact under ambient conditions [210], [211]. After this period, the copolyester outer layers had delaminated and were floating freely in the solvent; they were collected and set aside. The remaining THF solution was then transferred to a Petri dish and left to evaporate under a

fume hood for 24 hours, after which a thin yellowish film had formed on the dish surface.

However, ATR-FTIR analysis of the recovered film revealed intense residual THF absorption bands, as may be seen in Figure 4.4, most notably the ring breathing mode at 913 $cm^{-1}$ and the C−O stretching band at 1072 $cm^{-1}$, indicating significant solvent retention due to the high intensity of those peaks. Due to this fact, the identification of the TPU in the middle layer of Zendura FLX as polyether is unreliable. In general, since the dissolution method clearly affects the plastic and the final film was too thin, it is not possible to render any classification of the soft segment by this method. However, the dissolution does not affect the hard segment [192]which is clearly identified as MDI-based.

The copolyester layers recovered by physical delamination were also characterized by ATR-FTIR. Their spectra were unaffected by the dissolution process, as seen in Figure 4.4, showing no evidence of solvent retention or chemical modification, and were visually indistinguishable from those of the PETG and PCTG reference materials analyzed in this study (Biolon, Duran, Leone, and Tristar S for PETG; Prismlab C for PCTG) across all diagnostic spectral regions.

These results demonstrate that THF immersion at room temperature constitutes a viable method for the selective chemical delamination of copolyester/TPU multilayer aligner sheets, exploiting the differential solubility of TPU and terephthalate-based copolyesters in this solvent[210], [211].This approach enables independent spectroscopic characterization of each layer without interlaminar cross-contamination and may prove useful both for the analytical assessment of multilayer aligner materials before and after clinical use, and as a preparative step in potential end-of-life recycling strategies for composite thermoplastic orthodontic devices. However, the present study is limited to chemical characterization by ATR-FTIR; further work is needed to assess whether prolonged THF immersion affects the thermal and mechanical properties of the recovered copolyester layers, including any changes in Tg, elastic modulus, or stress relaxation behavior that could be relevant to their potential reuse or recycling.




(a) Storage modulus

(b) Loss modulus

Figure 4.5: DMTA curves in as-received materials

More differences are observable in the DMTA curves in Figure 4.5. It may be observed that in conventional materials, such as PETG and hTPU, in the oral-temperature range (20 ºC to 60 ºC), the storage modulus is relatively high compared to multilayers or Prismlab C (PCTG) . This parameter governs the initial force sensation that the devices produce once they are inserted. Regarding the loss modulus at oral temperatures, it indicates the deformability of the device. Devices must deform slightly to fit perfectly to the dentition and hence apply the forces in the exact points where they were planned by the orthodontist.

It has been reported that PETG-based aligners are more difficult to insert than hTPU-based ones, a difference attributable to the higher loss modulus of hTPU materials at oral temperature, which confers greater flexibility during seating [29], [212]. At oral temperature (37 ºC), PETG materials exhibited the highest storage modulus values, ranging from 1251 to 1705 MPa, with hTPU materials displaying comparable values (1472-1595 MPa), followed by Prismlab C (1064 MPa), Essix C+ (1023 MPa), and the multilayer materials (691-1009 MPa). Upon heating to 60 ºC - a temperature transiently reached in the oral cavity following ingestion of hot beverages[213] - all materials showed a decrease in storage modulus, consistent with the viscoelastic softening that accompanies increasing proximity to the glass transition[171]. The magnitude of this decrease varied markedly between the material groups. PETG materials showed the smallest and most uniform decreases (-42 to -85 MPa, with a storage modulus slope between 37 and 60 ° C ranging from -1.8 to -3.7 MPa/ºC), reflecting the large thermal margin between the service temperature and their glass transition (Tg ~ 80 °C). Among hTPU materials, the decrease ranged from -71 to -125 MPa with slopes between -3.4 to -5.5 MPa/ºC. Prismlab C and most multilayers behaved similarly to PETG (-52 to -95 MPa, with slopes ranging from -3 to -4 MPa/ºC). However, Maflex Premium Plus multilayer showed a steeper decrease in storage modulus (-192 MPa, with a slope of -8.3 MPa/ºC). The highest thermal sensitivity of all materials tested was recorded for Essix C+ (-330 MPa, with a slope of -14.3 MPa/ºC).

The anomalously steep thermal response of Maflex Premium Plus can be rationalized in terms of the intrinsic viscoelastic properties of its constituent layers rather than its architecture. Under uniaxial tension, the macroscopic storage modulus of a multilayer laminate follows an iso-strain rule of mixtures [214], [215], [216]. Because the storage modulus of the outer soft layers in Maflex Premium Plus is negligible relation to that of the hard core, the external layers contribute minimally to the macroscopic stiffness, and the thermal response of the laminate is effectively governed by the central hard layer. Consequently, the elevated thermal sensitivity of Maflex Premium Plus (-8.3 MPa/ºC) cannot be attributed to its atypical multilayer structure (soft/hard/soft) but must reflect the intrinsic thermal dependence of the central layer itself. The mechanical properties of thermoplastic polyurethanes are known to be strongly dependent on the hard-to-soft segment ratio and on the degree of microphase separation, both of which modulate the effective glass transition temperature and the width of the viscoelastic transition region[217], [218]. It is

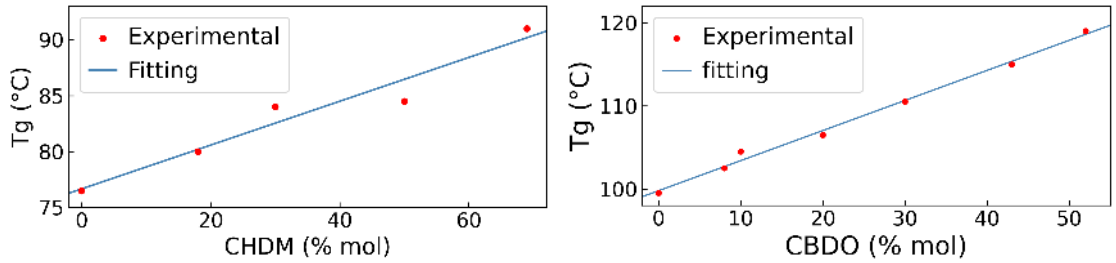
therefore proposed that the central layer in Maflex Premium Plus is a hTPU.

Talking about the Tg of the polymers obtained from DMTA measurements, as shown in figure 4.5, the PETG materials all have a Tg near 85 ºC, which is relatively close to oral temperature. It has been established that the mechanical properties of a thermoplastic remain essentially stable in service, provided that the operating temperature does not exceed one third of the glass transition temperature of the material[142], [219]. For a service temperature of 37 ºC, this criterion implies a minimum Tg of 111 ºC. Among the materials tested, PETG, hTPU, Filtzwiesser, and Maflex Premium Plus all present Tg values below this threshold, indicating that their mechanical response may be subject to a non-negligible viscoelastic contribution under intraoral conditions — a finding consistent with the decreases in storage modulus observed between 20 and 60 ºC discussed above.

The molar content of CHDM and CBDO in the PETG materials can be estimated from Tg, which is taken as the peak temperature of the loss factor in the DMTA curves. Based on published data [73], [220] relating Tg to copolymer composition, an empirical relationship given by equation 4.3 has been derived:

$$Tg(C) = 76.6 + 0.25x_{CHDM} + 0.36x_{CBDO} \tag{4.3}$$

where $x_{CHDM}$ and $x_{CBDO}$ denote the molar fractions of CHDM and CBDO respectively, and the intercept 76.6°C corresponds to the Tg of pure amorphous PET [171], [221]. Both comonomers act as positive modifiers of Tg (Figure 4.6), with CBDO exerting a greater influence per mole fraction owing to the higher conformational rigidity of its bicyclic structure relative to the cyclohexane ring of CHDM [73], [220].




(a) Variation of the Tg in PETG with respect the CHDM content. Data from [220]

(b) Variation of the Tg in PETG with respect to the CBDO content. 0 point corresponds to a composition of 81 % CHDM and 19 % EG. Data from [73]

Figure 4.6: PET derivatives chemical composition

The Tg values of 85.1-88.6 °C measured for all tested PETG materials (see Table 4.2) are consistent with a CHDM content of approximately 30 mol %, the most common commercial formulation, or an equivalent combination of both comonomers [72], [222]. It should be noted that an equivalent Tg could in principle be achieved with approximately 26 mol % CBDO alone; however, full optical transparency requires a minimum

CBDO substitution of 40 mol % to suppress crystallisation, which would simultaneously raise Tg to approximately 91°C and substantially increase material cost [223]. The outer copolyester layers of Zendura FLX and Maflex Premium, which present substantially higher Tg values of 118 °C and 124 °C respectively (Table 4.2), cannot be explained by CHDM substitution alone and require the simultaneous presence of both CHDM and CBDO in the formulation — consistent with their identification as PCTG or Tritan-type copolyesters rather than standard PETG [72], [73]

| Material | Composition | Tg (ºC) |
|---|---|---|
| Duran | PETG | 88.6 |
| Biolon | PETG | 85.6 |
| Leone | PETG | 85.1 |
| Tristar A | PETG | 85.4 |
| Prismlab C | PCTG | 121.8 |
| Tristar S | hTPU | 96.1 |
| Taglus | hTPU | 100.2 |
| Zendura A | hTPU | 98.1 |
| Essix C+ | PP | - |
| Zendura FLX | ML | 118.3 |
| Filtzwiesser | ML | 85.9 |
| Maflex Premium | ML | 124.5 |
| Maflex Premium Plus | ML | 93.2 |

Table 4.2: Glass transition (Tg) obtained from DMTA measurements taken as the temperature at the maximum of the loss modulus curve

A complementary representation of the DMTA results is provided by the Cole-Cole plot, in which the loss modulus is plotted as a function of the storage modulus, as shown in Figure 4.7. In this representation, materials located in the lower right region of the graph exhibit a predominantly elastic response (high storage modulus and low loss modulus) whereas materials in the upper left region display a more viscous, energy-dissipative character. The distance from the origin is proportional to the overall stiffness of the material, while the angle subtended with the abscissa reflects the loss factor ($tan(\delta) = E''/E'$), and thus the relative contribution of viscous dissipation to the mechanical response [142]. This plot therefore allows the simultaneous visualization of both the stiffness and the viscoelastic character of the materials, and facilitates the grouping of materials with similar mechanical behavior independently of temperature, an advantage over the conventional E' against temperature representation, in which thermal effects and intrinsic material differences are superimposed. [171]

In a conventional amorphous thermoplastic, the Cole-Cole plot yields an arc whose geometry encodes the viscoelastic relaxation spectrum of the material [142]. A perfect semicircular arc corresponds to a single relaxation time (Debye model, see Appendix

A), whereas broadened or asymmetric arcs indicate a distribution of relaxation times, described by the Cole-Cole and Havriliak-Negami models respectively [142], [219]. In principle, the shape parameters of these models can be extracted from the arc geometry and used to reconstruct the full stress relaxation curve via inverse Laplace transformation, providing a complete characterisation of the time-dependent mechanical response without additional experiments. However, this approach requires frequency-domain data or multifrequency temperature sweeps to apply the time–temperature superposition principle; a single-frequency temperature scan, as performed here, does not provide sufficient information to identify the relaxation time distribution unambiguously [142]. Accordingly, the arc geometry is characterised here through directly measurable empirical parameters.

The diameter of the semicircle — equivalent to the difference between the glassy and rubbery storage moduli $\Delta E' = E'_{glassy} - E'_{rubbery}$, is the geometrical parameter of greatest clinical relevance. It spans the range of storage modulus values encountered over the service life of the aligner, from the relatively high value characteristic of a freshly inserted device to the lower value of an aligner that has undergone the stress relaxation associated with clinical use. This range directly governs the evolution of the orthodontic force delivered to the tooth over the wear period: materials with a larger diameter exhibit a more pronounced force decay between insertion and removal, whereas materials with a narrow arc maintain a more consistent stiffness — and therefore more predictable force delivery — throughout the prescribed wear cycle. [29], [213].

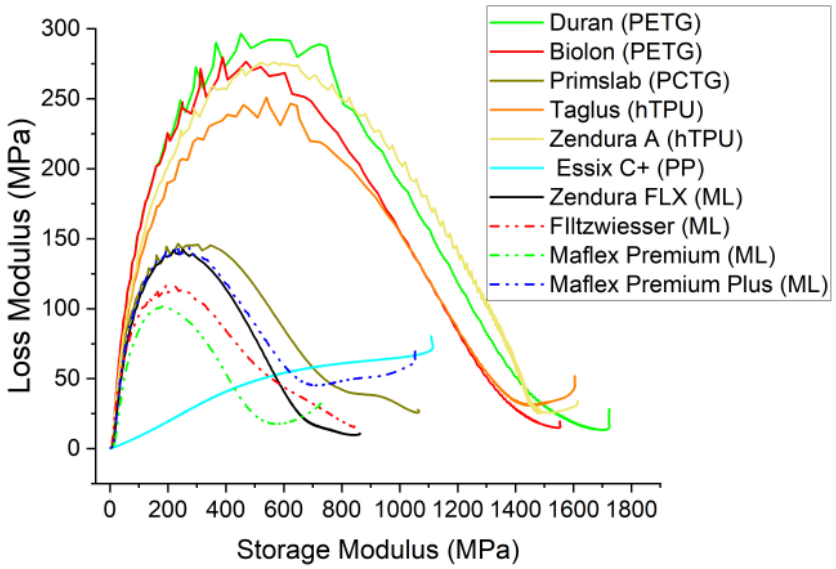



Figure 4.7: Cole-Cole Plot in as-received material

As may be seen in Figure 4.7, all materials exhibit Cole-Cole curves of similar shape, consistently skewed towards the low storage modulus region, with the loss modulus peak displaced towards the rubbery plateau. This skew reflects a broad and asymmetric distribution of relaxation times weighted towards long relaxation times: the majority of poly-

mer chain segments undergo cooperative relaxation rapidly upon crossing the glass transition — giving rise to the steep ascending branch of the arc — whereas a significant fraction of segments, constrained by steric or topological restrictions along the backbone, exhibit substantially longer relaxation times, producing the extended tail towards the glassy plateau. This distribution of relaxation times has direct consequences for the temporal profile of force decay during clinical use: the short-relaxation-time fraction governs the rapid initial stress decay observed in the first hours after insertion [29], while the long-relaxation-time fraction — though individually slower — sustains a progressive, low-rate force loss throughout the remainder of the wear cycle, as these segments continue to relax on timescales commensurate with the 14-day prescription period.

The breadth and intensity of this relaxation process is captured by the area enclosed by the Cole-Cole arc (Table 4.3), equivalent to the integral $\int E''dE'$ over the full glass transition, which reflects both the magnitude of the storage modulus drop $\Delta E' = E'_g - E'_r$ and the maximum dissipation $E''_{max}$. This parameter discriminates systematically between chemical groups. PETG materials exhibit the largest areas, consistent with their fully amorphous structure, negligible rubbery modulus ($E'_r = 0.05 - 1.5$ MPa) and large relaxation strength ($\Delta E' = 1552$-$1730$ MPa). The hTPU materials show intermediate values, with $\Delta E' = 1460$-$1600$ MPa and slightly elevated rubbery moduli ($E'_r = 3.8$-$4.6$ MPa), reflecting the partial redistribution of dissipation across the hard and soft segment relaxations characteristic of thermoplastic polyurethane microstructure. The multilayers present the smallest areas of all groups studied, a direct consequence of their laminated architecture: the compliant inner layer contributes a non-negligible rubbery modulus that elevates the lower bound of the transition and substantially compresses $\Delta E'$ to 707 - 842 MPa in Zendura FLX, Filtzwiesser, and Maflex Premium with and $E'_r$ between 1.1 and 7.8 MPa. Regarding to Maflex Premium Plus, shows larger $\Delta E' = 1050$ MPa and $E'_r = 3.5$ MPa, consistent with the higher intrinsic relaxation strength of the hTPU interior layer. It should be noted that the arc area is not an independent parameter — being largely determined by $\Delta E'$ and $E''_{max}$, and is therefore treated here as a qualitative descriptor of the overall transition rather than as a primary comparative metric.

These differences carry direct clinical implications for orthodontic force delivery. The viscoelastic nature of thermoplastic aligners means that the force exerted upon insertion decreases progressively over time through stress relaxation, even before any tooth movement occurs [29]. Materials with a larger $\Delta E'$ exhibit a broader dynamic range of mechanical response, which translates into a greater absolute magnitude of force decay over the wear cycle. Conversely, the systematically smaller $\Delta E'$ values of the multilayers are consistent with the lower stress relaxation rates and more stable long-term force delivery reported for multilayer materials in the literature [29], [213], suggesting that their compressed relaxation strength contributes to a more predictable orthodontic force profile throughout the prescribed wear period.

| Material | Chemistry | E' at 37 ºC (MPa) | E" at 37 ºC (MPa) | Area ($GPa^2$) |
|---|---|---|---|---|
| Duran | PETG | 1706 | 13.5 | 0.29 |
| Biolon | PETG | 1548 | 15.0 | 0.25 |
| Leone | PETG | 1580 | 24.5 | 0.26 |
| Tristar A | PETG | 1182 | 10.3 | 0.24 |
| Prismlab C | PCTG | 1065 | 26 | 0.08 |
| Tristar S | hTPU | 1493 | 40.4 | 0.18 |
| Taglus | hTPU | 1627 | 50.2 | 0.21 |
| Zendura A | hTPU | 1582 | 42.3 | 0.21 |
| Essix C+ | PP | 1023 | 65.7 | - |
| Zendura FLX | ML | 857 | 10.1 | 0.06 |
| Filtzwiesser | ML | 830 | 16.3 | 0.05 |
| Maflex Premium | ML | 684 | 25.2 | 0.03 |
| Maflex Premium Plus | ML | 1009 | 59.9 | 0.06 |

Table 4.3: DMTA Parameters in as-received materials

Another useful representation is the van Grup-Palmen graph (vGP), in which the complex modulus ($|E^*|$) is plotted against the phase lag ($\delta = arctan(E''/E')$), as shown in Figure 4.8. This representation is closely related to the Cole-Cole plot but offers additional insight into the molecular weight distribution, polydispersity, and relaxation processes of polymers of the same chemical family [224], [225]. The presence of additives, low-concentration secondary polymers, or contaminants manifests itself as characteristic shoulders or discontinuities in the vGP curve [226], while the absence of such features is indicative of highly linear thermoplastic architectures with narrow molecular weight distributions. However, it should be noted that meaningful comparisons are restricted to materials of identical chemical architecture, since the shape of the vGP curve depends intrinsically on the chemistry of the backbone [225].

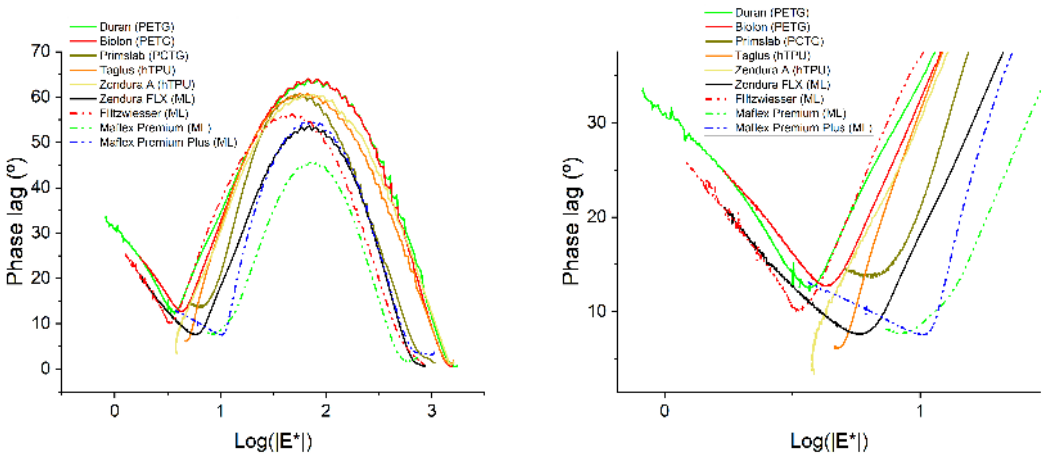
In both the Cole-Cole and the vGP representations, the temperature increases from right to left along the horizontal axis. Both plots were here constructed from fixed-frequency temperature sweep experiments; an equivalent representation can in principle be obtained from a frequency sweep at fixed temperature, provided that the material is thermorheologically simple — that is, that temperature and frequency affect the relaxation processes equivalently via the time–temperature superposition principle [142]. This condition may not hold for all thermoplastics: semi-crystalline polymers or polymer blends may exhibit thermorheologically complex behaviour, in which the superposition principle is only approximately valid [142], [227].

Each vGP curve in Figure 4.8, displays a primary maximum in phase lag associated with the glass transition, followed in most cases by a minimum corresponding to the entanglement plateau, where the elastic contribution of physical chain entanglements dominates the viscoelastic response [224], [225]. A phase lag of 0° corresponds to a perfectly

elastic solid and 90° to a purely viscous fluid, with intermediate values characteristic of viscoelastic behaviour.

The height of the phase lag maximum is directly related to the breadth of the relaxation time distribution [142], [225]: higher values indicate that chain segments relax within a narrower temperature interval, corresponding to a more homogeneous molecular environment, whereas lower values reflect a broader, more heterogeneous distribution. PETG (Biolon, Duran, Leone and Tristar A) and PCTG (Prismlab C) materials exhibit the highest maxima at approximately 65°, consistent with their fully amorphous, chemically homogeneous backbone. The hTPU (Zendura A, Taglus and Tristar S) materials show intermediate values of 58-60°, reflecting the broader distribution introduced by the coexistence of hard and soft segments. The multilayers (Zendura FLX, Maflex Premium Plus, Maflex Premium and Filtzwiesser) present the lowest maxima at approximately 52-48º, a consequence of their heterogeneous laminated architecture comprising chemically distinct layers with different relaxation temperatures. From a clinical perspective, a broader relaxation time distribution implies a more gradual and temporally distributed force decay during the wear cycle, consistent with the more stable long-term force delivery reported for multilayer materials in the literature [29].

The position of the entanglement minimum along the $log(|E^*|)$ axis reflects the entanglement plateau modulus $E_n^0$, inversely related to the molecular weight between entanglements, $M_e$,though comparisons are only meaningful within the same chemical family [225]. Within the PETG group, all materials show the minimum at $log(|E^*|) \simeq 0.5$, suggesting similar entanglement densities. Prismlab C (PCTG) shows the minimum displaced to slightly larger values of $log(|E^*|)$, attributable to the greater conformational rigidity of its cyclic diol comonomers, relative to EG in PETG, which restricts backbone segmental mobility and increases the effective entanglement plateau modulus [219]. The entanglement minimum is absent in the hTPU materials within the accessible modulus range; instead, their vGP curves decrease monotonically towards lower $log(|E^*|)$ values, indicating that the entanglement plateau is displaced below the experimental window. The phase lag of the hTPU curves in this region falls below that of the other materials, suggesting that their entanglement minimum, once reached, will correspond to a lower phase lag - implying a more pronounced elastic character at the plateau, consistent with the robust physical crosslink network provided by the hard segment domains [142].




(a) Full Van Grup-Palmen graph

(b) entanglements minimum

Figure 4.8: Van Grup-Palmen graphs

The breadth of the vGP curve also carries implications for thermoformability. A broader curve — reflecting a wider distribution of relaxation times — indicates that the material softens more gradually with increasing temperature, which translates into a wider and more forgiving thermoforming window. In this respect, the multilayer laminates, which exhibit the broadest vGP curves among all groups studied, are expected to be more tolerant of temperature variations during thermoforming than the PETG materials, whose narrower curves indicate a more abrupt transition between the glassy and rubbery states. Conversely, the sharper transition of the PETG group, while making thermoforming more temperature-sensitive, also ensures more reproducible mechanical properties at service temperature, since the material operates well within its glassy plateau at 37°C.

A primary design goal for next-generation aligner materials is to deliver lower, more sustained orthodontic forces over the full wear cycle, reducing patient discomfort while maintaining sufficient rigidity for controlled tooth movement. From a viscoelastic standpoint, this requires shifting the distribution of relaxation times towards longer values — so that stress relaxation occurs gradually rather than abruptly in the first hours after insertion — while simultaneously maintaining a sufficiently broad distribution to ensure that the device retains enough stiffness for gentle, comfortable insertion. However, shifting relaxation times towards longer values necessarily implies either increasing the average molecular weight or employing a chemically stiffer backbone, both of which tend to increase the elastic modulus and hence the forces exerted by the device. This represents a fundamental materials design trade-off: reducing the force decay rate and increasing molecular weight or backbone rigidity are competing objectives that cannot be optimized independently. Understanding the elastic modulus range accessible within each material family is therefore essential for contextualizing the viscoelastic behavior described above.

Uniaxial tensile tests were performed on all materials to this end, and the results are presented in Table 4.4.

| Material | Chemistry | E (MPa) | Group | $\varepsilon_y$ (%) | Group |
|---|---|---|---|---|---|
| Duran | PETG | 1214.6 ± 29.9 | ab | 5.5 ± 0.2 | e |
| Biolon | PETG | 1064.3 ± 39.8 | c | 5.7 ± 0.3 | e |
| Leone | PETG | 941.8 ± 24.1 | d | 5.4 ± 0.4 | e |
| Tristar A | PETG | 1146.2 ± 65.2 | b | 5.0 ± 0.3 | e |
| Prismlab C | PCTG | 936.2 ± 47.0 | d | 7.5 ± 0.6 | cd |
| Tristar S | hTPU | 1213.7 ± 30.5 | ab | 8.1 ± 0.3 | bc |
| Taglus | hTPU | 1226.1 ± 40.6 | ab | 8.1 ± 0.3 | bc |
| Zendura AT | hTPU | 1255.2 ± 25.5 | a | 6.3 ± 0.4 | de |
| Zendura FLX | ML | 659.4 ± 34.9 | f | 9.3 ± 0.2 | b |
| Maflex Premium | ML | 613.4 ± 16.7 | fg | 8.6 ± 0.1 | bc |
| Maflex Premium Plus | ML | 838.3 ± 74.5 | e | 5.3 ± 0.2 | e |
| Filtzwiesser | ML | 661.3 ± 50.4 | f | 5.2 ± 0.2 | e |
| Essix C+ | PP | 543.5 ± 46.7 | g | 14.7 ± 2.6 | a |

Table 4.4: Uniaxial tensile test results. Values are reported as mean ± standard deviation ($n$ = 7-8). Within each parameter, materials sharing a letter are not significantly different (Tukey's HSD, $\alpha$ = 0.05).

The two most important parameters obtained from uniaxial tensile testing are the elastic modulus and the yield strain. It is highly unlikely that a device will yield upon insertion, as the stresses involved are relatively low and the applied strains are typically restricted to between 0.1 and 2 % [143], [228], [229]. However, prolonged and repeated use may cause the material to yield and undergo permanent deformation, compromising the accuracy of tooth movement and the overall treatment outcome [230]. The elastic modulus governs the initial force sensation transmitted to the tooth and surrounding periodontal ligament, and should therefore be relatively low to ensure patient comfort [231]. The yield strain, on the other hand, defines the extent of the elastic range — that is, the maximum deformation the material can sustain without permanent deformation. As shown in Table 4.4, the elastic moduli of the materials studied range widely, from 543.45 MPa in Essix C+ (PP) to 1255.03 MPa in Tristar S (hTPU), reflecting the broad spectrum of polymer chemistries employed in current clear aligner systems. As also observed in Figure 4.4 and Table 4.4, Zendura FLX, Maflex Premium and Essix C+ successfully reduce the elastic modulus while maintaining a comparatively high yield strain, suggesting that these materials may offer a favorable balance between flexibility and elastic recovery.

As shown in Table 4.4, PETG and hTPU materials exhibit comparable elastic moduli in some cases but markedly different yield strains. Among the PETG materials, Duran, Biolon, Leone and Tristar A display elastic moduli between 941 and 1155 MPa and yield strains ranging from 5.01 to 5.70 %, indicating a relatively stiff but moderately elastic behavior. PETG can generally be stretched up to approximately 5 % before undergoing plastic deformation, which limits its elastic working range during clinical use [88]. In contrast, hTPU materials (Taglus, Zendura AT and Tristar S) exhibit higher elastic moduli

— between 1213 and 1255 MPa — alongside considerably higher yield strains of 6.25 to 8.11 %, suggesting a stiffer yet more elastically resilient response. The PCTG material (Prismlab C) presents an intermediate behavior, with an elastic modulus of 952 MPa and a yield strain of 7.5 %, combining moderate stiffness with an extended elastic range. It is worth noting that Essix C+ (PP) stands out with the lowest elastic modulus (543 MPa) and the highest yield strain (14.65 %), indicating a markedly more flexible and deformable character compared to the rest of the materials studied.

ML materials present particularly interesting mechanical profiles. Zendura FLX and Maflex Premium display elastic moduli of 659 and 618 MPa, respectively, with yield strains of 9.31 and 8.56 % — values that place them within the range of hTPU and PCTG in terms of elastic deformability, despite their considerably lower stiffness. In contrast, Maflex Premium Plus and Filtzwiesser exhibit elastic moduli of 838 and 664 MPa and lower yield strains of 5.27 and 5.18 %, respectively, approaching the behaviour of PETG. In the case of Filtzwiesser, the manufacturer confirmed that the outer copolyester layer is PETG, which likely explains the reduced yield strain. Regarding Maflex Premium Plus, the company stated that the material shares the same multilayer structure as that used by Invisalign and analysed in section 4.7. Furthermore, chemical and thermal analysis indicates that this multilayer consists of outer soft TPU layers surrounding a hard inner core, likely composed of hTPU. Although the hTPU materials examined here exhibit yield strains between 6.25 and 8.11 %, lower values have been reported in the literature for this polymer family [232], [233], [234]. It can therefore be concluded that the outer layer composition appears to govern the yield strain response in multilayer systems. Materials with PETG outer layers, such as Filtzwiesser, exhibit yield strains comparable to monolithic PETG (5.18 %), while those with PCTG outer layers, such as Zendura FLX and Maflex Premium, display markedly higher values (8.56–9.31 %). This trend is further supported by Maflex Premium Plus, whose soft/hard/soft architecture yields a low yield strain (5.27 %) despite the presence of a soft outer layer, suggesting that the high-modulus hTPU core constrains the overall deformation response and rendering the contribution of the sTPU outer layers negligible in terms of elastic modulus and yield strain.

The statistical grouping in Table 4.4 (shown graphically in Appendix D, Figure D.1) reveals statistically distinct groups for both elastic modulus and yield strain, as indicated by the corresponding statistical letters, confirming that material chemistry plays a critical role in the mechanical performance of the aligner [235]. The stiffest materials are the hTPU grades together with the PETG copolyester Duran (Zendura AT, Taglus, Tristar S and Duran, all with $E \approx$ 1214–1255 MPa), whereas the multilayer and PP materials are markedly more compliant (Essix C+, Maflex Premium and Zendura FLX, $E \approx$ 540–660 MPa).

From a clinical standpoint, the compliant high-strain group also shows the lowest yield stresses ($\sigma_y \approx$ 28 MPa), delivering lower forces and greater deformability at insertion, favouring patient comfort, while its extended elastic range helps sustain a controlled force over the range of tooth displacement [29], [236]. The stiffer PETG and hTPU grades, by

contrast, provide higher force output at the expense of a narrower elastic working range. The optimal material therefore depends on the balance sought between the biomechanical efficiency required for effective tooth movement and patient comfort.

Two toughness parameters were computed from the uniaxial tensile curves. The elastic toughness $Te = \sigma_y \varepsilon_y / 2$, also referred to in the materials science literature as the modulus of resilience [237], [238], quantifies the elastic strain energy stored and recoverable per unit volume, and is the parameter of primary clinical relevance for clear aligners, which are designed to operate exclusively within the elastic regime [29], [88]. Normalization by the elastic modulus yields the specific elastic toughness $Te/E$ [238] , which accounts simultaneously for the energy storage capacity and the stiffness of the material — and hence its propensity to generate discomfort upon insertion. The overall toughness T, defined as the area under the full stress-strain curve up to fracture [237], measures the total energy absorbed by the material before failure and provides an indication of resistance to crack initiation and propagation during clinical use [239]; higher values imply a greater capacity to sustain damage without catastrophic fracture.

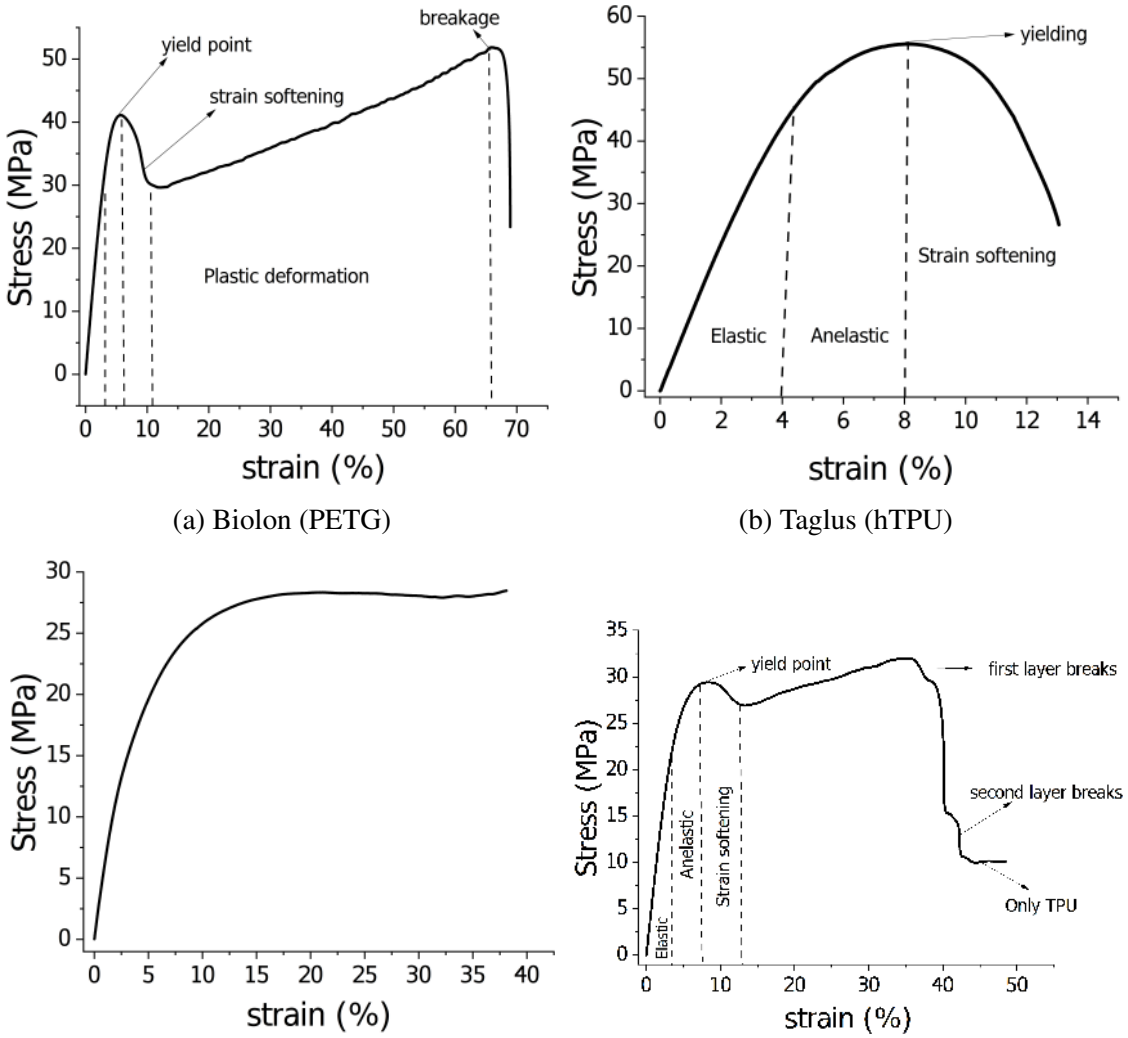
| Material | Chemistry | Te ($MJ/m^3$) | Te/E ($MJ/GPam^3$) | T ($MJ/m^3$) |
|---|---|---|---|---|
| Duran | PETG | 1.2 | 1.1 | 13.7 |
| Biolon | PETG | 1.3 | 1.1 | 26.7 |
| Leone | PETG | 1.0 | 1.1 | 19.3 |
| Tristar A | PETG | 1.0 | 0.9 | 20.2 |
| Prismlab C | PCTG | 1.6 | 1.7 | 15.6 |
| Tristar S | hTPU | 2.2 | 1.8 | 8.6 |
| Taglus | hTPU | 2.4 | 2.0 | 8.5 |
| Zendura AT | hTPU | 1.9 | 1.5 | 4.5 |
| Zendura FLX | ML | 1.3 | 2.0 | 13.7 |
| Maflex Premium | ML | 1.2 | 2.0 | 15.6 |
| Maflex Premium Plus | ML | 0.9 | 1.0 | 63.3 |
| Filtzwiesser | ML | 0.7 | 1.0 | 11.8 |
| Essix C+ | PP | 2.1 | 4.0 | 50.1 |

Table 4.5: Uniaxial tensile test values

Comparison of $Te$ and T across all materials (Table 4.5 and Figure 4.9) reveals a clear differentiation between material classes. The hTPU materials deform predominantly elastically, exhibiting high $Te$ but limited post-yield plasticity, resulting in near-brittle fracture behavior consistent with the well-documented susceptibility of high-hardness TPU to crack initiation and propagation [232], [234]. PETG and PCTG monolayers, by contrast, display extended plastic regimes and substantially higher overall toughness T reflecting their capacity to accommodate irreversible deformation prior to fracture. The multilayer configurations present overall toughness values comparable to those of the monolithic copolyesters, with the compliant sTPU interlayer effectively extending the elastic work-

ing range of the laminate relative to the corresponding hard layer alone.

Maflex Premium Plus presents a distinctive profile attributable to its inverted sTPU/hTPU/sTPU architecture. The soft outer layers are expected to provide crack-bridging at the device edges produced during laser cutting — a well-established toughening mechanism in layered polymer systems [239] — thereby mitigating the brittle fracture susceptibility that characterizes monolithic hTPU under the same processing conditions.




(a) Biolon (PETG)

(b) Taglus (hTPU)

(c) Essix C+ (PP)

(d) Zendura FLX (ML)

Figure 4.9: Stress-strain curves in as-received materials

As illustrated in Figure 4.9, the multilayer material Zendura FLX exhibits a sequential failure mechanism. The outer layers fracture successively, exposing the inner soft elastomeric core, which continues to elongate beyond the point of outer layer failure. A notable observation is that, upon unloading — that is, when the test is interrupted and the specimen is removed from the grips — the sample recovers its original dimensions, no matter whether the outer layers were broken beforehand. This elastic recovery was not observed in any other material tested, suggesting that the soft inner layer of Zendura

FLX behaves as a purely elastic component even after the fracture of both outer layers. This finding implies that either the outer copolyester layer in Zendura FLX and Maflex Premium possesses an intrinsically higher yield strain than conventional PETG or than the PCTG studied (Prismlab C), or that the combination of a hard copolyester with a soft elastomeric interlayer acts synergically to increase the effective yield strain beyond that of PETG or PCTG alone.

This hypothesis is further supported by the behavior of a specimen supplied by Filtzwiesser, which employs a structurally analogous PETG/sTPU/PETG trilayer architecture. The stress–strain response of this material closely resembles that of Zendura FLX, differing only in a marginally higher elastic modulus and a reduced yield strain of approximately 5 %. Given that the Filtzwiesser material uses PETG rather than PCTG as the outer layer, this comparison allows the higher yield strain observed in Zendura FLX and Maflex Premium to be attributed to the outer hard copolyester (PCTG) layer rather than to the soft interlayer.

Regarding the general shape of the stress–strain curves, nearly all materials display three characteristic regions: an initial linear elastic region, in which stress and strain are proportionally related through the elastic modulus; a pre-yield anelastic region, characterized by a progressive deviation from linearity; and a post-yield strain-softening region, in which stress decreases as deformation continues. The anelastic region is of particular clinical relevance, as anelastic deformation is time-dependent but fully recoverable given sufficient time, meaning that forces generated during aligner insertion may evolve over the course of wear.

As shown in Figure 4.9, the hTPU materials exhibit a markedly extended anelastic region compared to PETG, resulting in a progressive reduction of the effective stiffness at larger strains. This behavior is clinically advantageous, as it implies that orthodontic forces delivered by hTPU aligners decrease more gently with increasing deformation, reducing the risk of excessive loading. Furthermore, hTPU materials show limited plastic deformation capacity, with fracture occurring shortly after the yield point. PETG, by contrast, exhibits a pronounced post-yield plastic plateau. Zendura FLX displays intermediate behavior between these two extremes.

With respect to the yielding region, the breadth of the stress peak in the vicinity of the yield point carries direct clinical implications. In PETG, this region is narrow, meaning that small deviations in aligner fit or activation distance may produce disproportionate changes in the delivered force relative to the prescribed value. The hTPU and multilayer materials, by contrast, display a considerably broader yield region, conferring greater tolerance to dimensional inaccuracies and thus a more predictable force delivery under clinical conditions. The broadest region occurs for Zendura FLX, Maflex Premium, and Filtzwiesser multilayers, hence making them the most secure materials in this aspect.

It is also noteworthy that the stress–strain curves of Maflex Premium Plus closely resemble those of PETG, despite its trilayer sTPU/hTPU/sTPU architecture. Two factors

may contribute to this behavior. First, the hTPU core layer in this material may possess a lower hard-segment content than the monolithic hTPU materials, resulting in a reduced elastic modulus and yield strain that fall within the range characteristic of PETG. Second, the soft outer sTPU layers may partially attenuate the mechanical response of the stiff core, further shifting the overall behavior of the laminate toward that of a less rigid material, without, however, altering the fundamental shape of the stress–strain curve, which remains governed by the hTPU core.

Finally, the Essix C+ (PP) specimen exhibited a markedly different deformation behavior, characterized by the formation of a cold neck. A localized necking zone developed at the center of the specimen, becoming visibly white and opaque as strain-induced crystallization occurred within the striction. This whitened region subsequently propagated along the gage length as testing continued, and the specimen reached the displacement limit of the machine without fracture. As only a 30-minute test was performed on this material, the resulting curve — shown in Figure 4.9 — represents a partial stress–strain response.

Another feature that is worth noting is the transparency. Clear aligners must be practically invisible due to a highly transparent material. Transparency was measured using a UV-visible spectrophotometer. A plastic for clear aligners is defined as transparent if its transmittance is above 80% between 400 and 600 nm light wavenumber. [26], [221], [229], with the most important value in this range being the 550 nm wavenumber, as it is the value for the maximal sensitivity for the human eye [240], [241]. The literature on clear aligners focuses on the transmittance percentage of these materials, as more information can be obtained from the curves. For instance, the UVA and UVB protection of the material. The UVB protection is how much radiation the material blocks between 280 and 315 nm, and it has little importance in clear aligners because this type of radiation is almost fully absorbed by the atmosphere, and because plastics block this type of radiation naturally. UVA radiation (315-400 nm) is more damaging to plastics, which are usually protected against it. However, clear aligner plastics need to be highly transparent and biocompatible, and hence the protective additives must be in low proportions. For simplicity, only the UVA protection was analyzed. The range of acceptable values for large exposure of plastics is set to be above 40 % [242], [243] and between 45-65% for medical devices [243], but no clear range is actually defined for clear aligners, although some studies suggest that it must be over 40 % [244], [245].

Other important parameters are the color balance, the red-blue difference (RBD) and the yellowing index (YI). A transparent material should transmit all colors in the same manner; the color balance indicates the differences in light transmittance between all the visible color ranges in the spectra. For a highly transparent material, it must be near 0. The red-blue difference accounts for the physical homogeneity of the plastic and may indicate if small crystals in the polymer are scattering light. Acceptable values of RBD are between -5% and 3%, with the lower threshold indicating a blueish material and the upper threshold being the initial point of yellowing[246], [247], [248]. Finally, the yellowing

index indicates if the material has degraded by UV light or thermally, and hence shows a slight yellow color. For clear aligners, it must be below 3% for a perfectly transparent material and below 5% for a transparent material .[249], [250]. As we can see in Figure 4.10, almost all materials fulfill this requirement, except for Essix C+ (PP), which is less transparent even to the naked eye. The differences between that material and the others can be observed more clearly in table 4.6. Almost all the materials present an average transparency of between 85% and 92%. In the case of Essix C+, although its average transparency is over the threshold value for clear aligners, the YI is above the minimal value for perfect transparency.

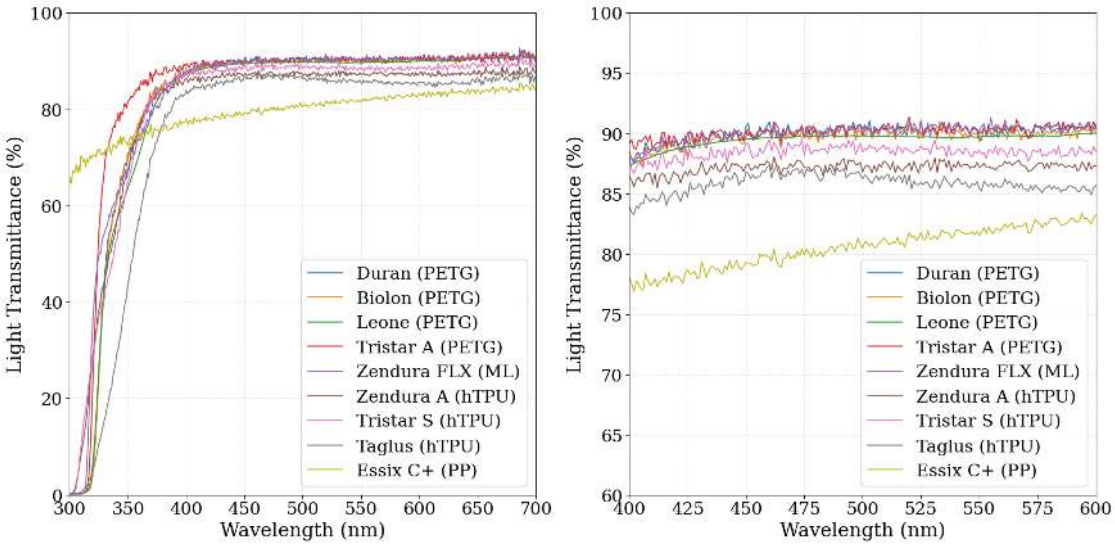



Figure 4.10: Light transmittance curves in as-received material

The reason for the lack of transparency in Essix C+ seems to be its high crystallinity.

| Material | Composition | AVG T (%) | $T_{550nm}$ | UVA Protection (%) | YI |
|---|---|---|---|---|---|
| Duran | PETG | 90.2 | 90.0 | 35.2 | 0.57 |
| Biolon | PETG | 90.0 | 89.9 | 35.3 | 0.76 |
| Leone | PETG | 89.8 | 89.7 | 37.6 | 0.9 |
| Tristar A | PETG | 88.6 | 88.3 | 34.6 | 0.52 |
| Prismlab C | PCTG | 92.7 | 92.8 | 23 | 0.47 |
| Tristar S | hTPU | 90.4 | 90.2 | 35.2 | 0.57 |
| Taglus | hTPU | 85.9 | 85.7 | 35.4 | -0.72 |
| Zendura AT | hTPU | 87.3 | 87.0 | 33.7 | 0.31 |
| Zendura FLX | ML | 90.3 | 90.3 | 32.1 | 0.87 |
| Essix C+ | PP | 81.6 | 81.8 | 26.1 | 4.92 |

Table 4.6: Transparency parameters in as-received materials

Vickers hardness was measured by stacking layers of plastics until reaching 6.5 mm in

thickness as required by the ISO 6507, which specifies that specimen thickness must be at least 1.5 times the mean indentation diagonal to prevent the test anvil from influencing the result [251]. Since individual aligner sheets (0.76-1.0 mm) fall well below this threshold, stacking was necessary; the constraint of using only chemically identical layers restricted measurements to one multilayer (Zendura FLX), three hTPU grades (Taglus, Zendura AT and Tristar S), and four PETG grades (Duran, Biolon, Leone and Tristar A).

Hardness is clinically relevant for two reasons: it governs resistance to surface wear and localized deformation during repeated insertion and removal cycles [229], and it is the primary descriptor used by thermoplastic suppliers to classify TPU grades — which span 60 Shore A to 90 Shore D depending on hard segment content. Measured Vickers values were converted to Shore D using standard conversion tables to enable direct comparison with supplier specifications.

| Material | HV0.025 | Shore D |
|---|---|---|
| Duran (PETG) | 8.5 ± 0.2 | 70 |
| Biolon (PETG) | 10.3 ± 0.4 | 74 |
| Leone (PETG) | 9.9 ± 0.2 | 74 |
| Tristar A (PETG) | 10.1 ± 0.3 | 74 |
| Tristar S (hTPU) | 12.9 ± 0.3 | 80 |
| Taglus (hTPU ) | 13.8 ± 0.9 | 82 |
| Zendura AT (hTPU) | 15.5 ± 1.1 | 85 |
| Zendura FLX (ML) | 8.9 ± 0.2 | 72 |
| Essix C+ (PP) | 5.9 ± 0.3 | 65 |

Table 4.7: Vickers hardness values ($n$ = 3)

Taking a look at Table 4.7, the differences between hTPU and PETG are clearly noticeable. hTPU is said to be more resistant to wear, mainly due to its higher hardness values. In Zendura FLX, the values obtained seem to be near those of PETG. Hence, it may be said that larger forces are needed to reach the TPU material in the inner core of Zendura FLX. Finally, Essix C+, as expected,is the softest material.

It seems that plastics with a shore hardness below 65 shore D cannot be used alone as clear aligner materials. In general, plastics with such low hardness show too low mechanical properties.

### 4.2. Tensile and 3PB at oral temperature

The uniaxial tensile and 3PB experiments were performed at 37 ºC on the DMTA equipment rather than on a universal testing machine, as the DMTA provides precise force-controlled loading at the low force magnitudes required for thin thermoplastic sheets, stable thermal conditioning through an actively regulated environmental chamber, and

continuity with the stress relaxation measurements subsequently carried out on the same instrument. Owing to instrument limitations, the maximum force attainable in both modes was 18 N. The tensile and flexural elastic moduli, together with the $T_g$ determined from the peak of the loss modulus, are summarized in Table 4.8.

| Material | $E_t$ (MPa) | $E_f$ (MPa) | $E_f/E_t$ | $T_g$ (°C) |
|---|---|---|---|---|
| Duran (PETG) | 1124 ± 113 | 2048 ± 182 | 1.82 | 88 |
| Taglus (hTPU) | 1498 ± 157 | 1940 ± 144 | 1.30 | 98 |
| Prismlab C (PCTG) | 973 ± 56 | 1591 ± 169 | 1.63 | 125 |
| Zendura FLX (ML) | 718 ± 16 | 1573 ± 95 | 2.19 | 118 |
| Filzwiesser (ML) | 743 ± 23 | 1961 ± 56 | 2.64 | 86 |
| Maflex Premium (ML) | 693 ± 18 | 1650 ± 206 | 2.38 | 124 |
| Maflex Premium Plus (ML) | 783 ± 39 | 838 ± 68 | 1.07 | 93 |

Table 4.8: Tensile ($E_t$) and flexural ($E_f$) elastic moduli and glass transition temperature of the aligner materials characterised at 37 ℃ (mean ± SD, $n = 5$).$T_g$ values determined from the position of the loss modulus peakin DMTA temperature sweeps.

The multilayer materials (Zendura FLX, Filzwiesser, and Maflex Premium) exhibit systematically lower tensile moduli than the single-polymer materials, while their flexural moduli remain comparable to those of PETG and PCTG. This discrepancy is captured by the $E_f/E_t$ ratio, which ranges from 1.07 (Maflex Premium Plus) to 2.64 (Filzwiesser). In homogeneous polymers this ratio typically falls between 1.2 and 1.7 [239], [252], [253], a range matched by all single-polymer materials in this study (hTPU: 1.30, PCTG: 1.63, PETG: 1.82). The elevated ratios of Zendura FLX (2.19), Filzwiesser (2.64), and Maflex Premium (2.38) indicate that their laminated architecture contributes to bending stiffness in a way not captured by the tensile modulus alone.

In 3PB the stress distribution is non-uniform across the thickness, with maximum tensile and compressive stresses at the outer surfaces and zero stress at the neutral plane (Figure 4.11). In a laminate combining a compliant elastomeric core and rigid outer sheets, the bending stiffness is therefore dominated by the stiff outer layers, whereas the tensile modulus reflects the effective homogeneous response of the whole stack. This explains why the multilayers appear disproportionately stiffer in bending than in tension, and has direct clinical relevance: the forces generated on insertion depend on the bending stiffness of the shell rather than on its uniaxial tensile response. The low $E_f/E_t$ ratio of Maflex Premium Plus (1.07) instead points to a more homogeneous layer architecture, discussed below.

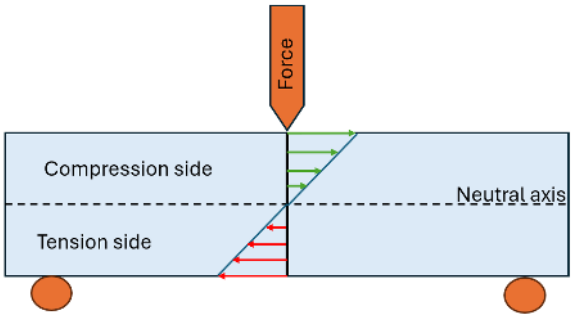
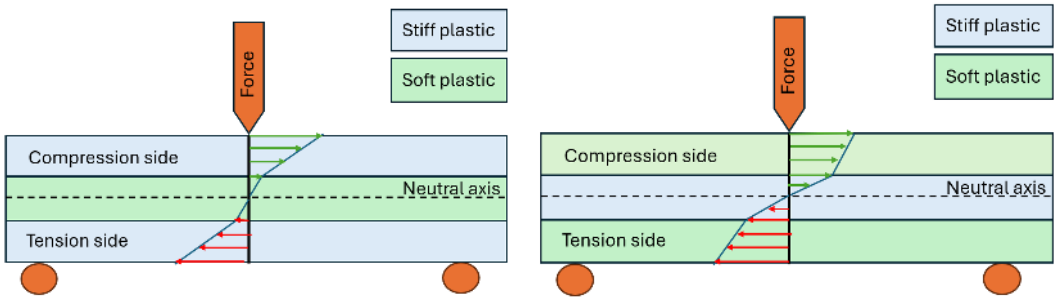



Figure 4.11: Stress profile in 3PB in a monolayer

A statistically significant global difference was detected across materials in both tensile ($H = 31.50$, $p < 0.001$) and flexural modulus ($H = 29.98$, $p < 0.001$), but no pairwise comparison reached significance after Bonferroni correction ($\alpha_{corr} = 0.0024$, 21 comparisons), consistent with the limited power of $n = 5$ specimens per group. The multilayer materials nonetheless showed markedly lower tensile variability than the single-polymer sheets, consistent with their more controlled layer-by-layer manufacturing.

The way a laminate carries a bending load depends on whether its core is soft or stiff (Figure 4.12). In the most common configuration (Case 1: stiff outer layers, soft core), realised here by Zendura FLX, Filzwiesser, and Maflex Premium, the soft core acts as a flexural spacer: it bears essentially no load and serves only to separate the stiff outer layers from the mid-plane, so that the entire bending load is carried by the outer layers, which lie furthest from the neutral axis where the flexural stress is greatest. Such a laminate appears stiffer in bending than in tension. In the opposite configuration (Case 2: soft outer layers, stiff core), realised by Maflex Premium Plus, the bending load is sustained by the inner core, so the laminate behaves as a thin monolayer of the core material. This is consistent with Table 4.8, where the tensile and flexural moduli of Maflex Premium Plus are almost equal ($E_f/E_t = 1.07$): the material effectively behaves as a very thin monolayer of hard hTPU.



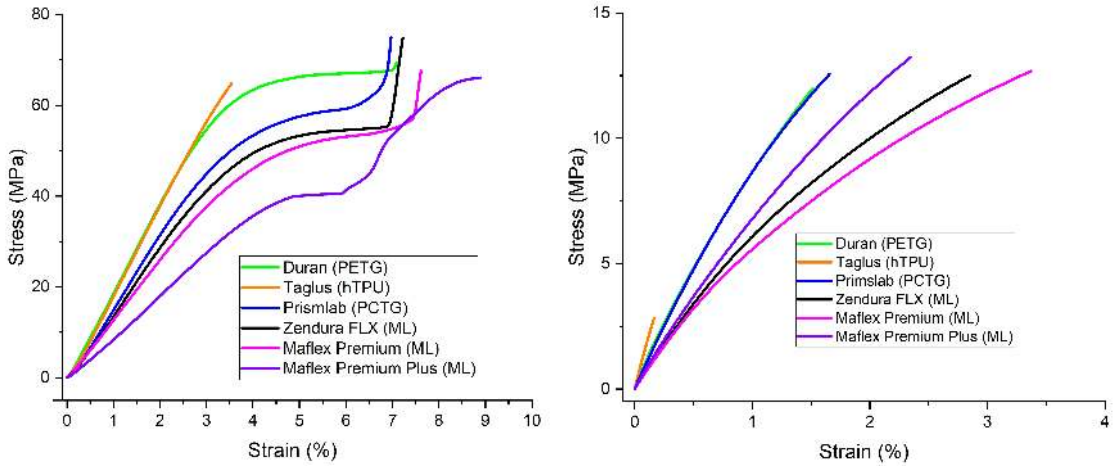

(a) Case 1 (b) Case 2

Figure 4.12: Stress profiles in 3PB in two multilayer types

The primary deformation mode in clear aligners is bending, through which teeth are

moved in translation; other movements (rotation, tipping, and crowding corrections) impose a biaxial stress state on the plastic. A lower tensile than flexural modulus ensures that most of the stored energy is directed into bending, narrowing the range of delivered forces relative to a material with comparable moduli in both modes.

Because bending preferentially probes the outer shell, the close match in flexural modulus and $T_g$ between the PCTG/sTPU/PCTG multilayers (Zendura FLX and Maflex Premium) and monolithic Prismlab C (PCTG) in Table 4.8 further supports the identification of PCTG as their outer layer. The Filzwiesser curves are omitted from the figures to avoid overcrowding and are reported only in Table 4.8.




(a) 3PB configuration

(b) Tensile configuration

Figure 4.13: Stress-strain experiments up to 18 N

As shown in Figure 4.13, mechanical plastification is reached early in all materials in 3PB, at around 4% strain, with similar slopes across materials. In the tensile configuration the curves are almost perfectly elastic, with only slight anelasticity. Taglus (hTPU) is a notable case, behaving as essentially perfectly elastic in both modes and exhibiting a high elastic modulus, so that only low strains are reached before the 18 N force limit of the equipment. Although previous studies report that clear aligners may be subjected to strains from 0.1% up to 2% or larger [143], [228], [229], with yielding beyond 2%, both bending and tensile configurations here sustained strains of up to 3% without mechanical yielding.

### 4.3. Stress relaxation in 3PB configuration

Stress relaxation experiments were conducted in both bending and tensile configurations. In each case, specimens were deformed to a predefined strain level $\varepsilon_0$, which was subsequently held constant for 3 hours while the resulting force and stress were continuously monitored. Upon completion of the relaxation period, a strain recovery phase was initi-

ated by reducing the applied force to 0.001 N and allowing the specimen to recover freely for 5 minutes. The complete protocol is shown in Figure 4.14.

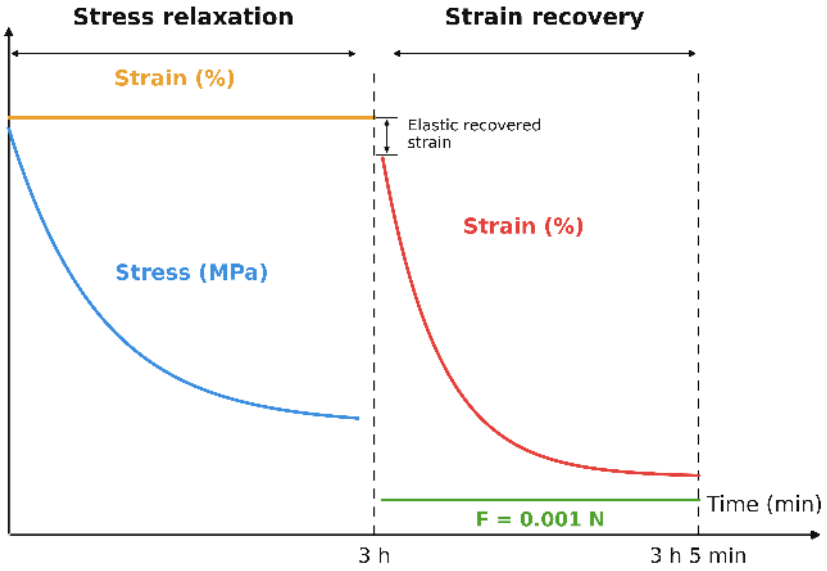



Figure 4.14: Stress relaxation and strain recovery experiment diagram

In the bending configuration, three strain levels were investigated: 2%, 4%, and 6%, each held constant for 3 hours and followed by a 5-minute recovery phase under a residual force of 0.001 N. The selection was motivated by both clinical and mechanical considerations. Clinically relevant bending deformations in clear aligner therapy have been reported in the 0.1-2% range, although larger strains occur in demanding scenarios, while strains exceeding 6% are generally discouraged as they may induce rapid mechanical degradation through plastic deformation. From a mechanical standpoint, the three levels represent a fully elastic regime (2%), a transitional regime (4%), and a condition within the post-yield plastic plateau (6%), as identifiable in Figure 4.13.

The relaxation modulus and strain recovery were quantified according to Equations 4.4 and 4.5, respectively:

$$E(\text{MPa}) = \frac{\sigma(t)}{\varepsilon_0} \tag{4.4}$$

$$\text{Strain Recovery}(\%) = \frac{\varepsilon_0 - \varepsilon(t)}{\varepsilon_0} \times 100 \tag{4.5}$$

where $\sigma(t)$ is the time-dependent stress, $\varepsilon_0$ the applied strain, and $\varepsilon(t)$ the strain recorded during the recovery phase.

The relaxation modulus curves are characterized by two parameters: the initial relaxation modulus ($E_0$), corresponding to the modulus at the earliest recorded time point, and the decay exponent $a$, which quantifies the rate at which the modulus decreases over time according to the power-law relationship $E(t) = E_0\, t^a$. A higher value of $a$ indicates a faster stress decay, reflecting a greater viscoelastic contribution to the mechanical response. The ratio $E(3h)/E(0)$ expresses the fraction of the initial modulus retained after

3 hours of relaxation, providing a normalized measure of stress decay that is independent of the absolute stiffness of each material. These parameters are used consistently in both the bending and tensile configurations.

Owing to the high similarity in the relaxation behaviour of Filzwiesser, Zendura FLX, and Maflex Premium, only the curves of the two latter multilayers are shown in the bending configuration, to avoid overcrowding the figures; the complete dataset for all materials is reported in the corresponding tables. In the tensile configuration, the curves of all multilayers are presented.

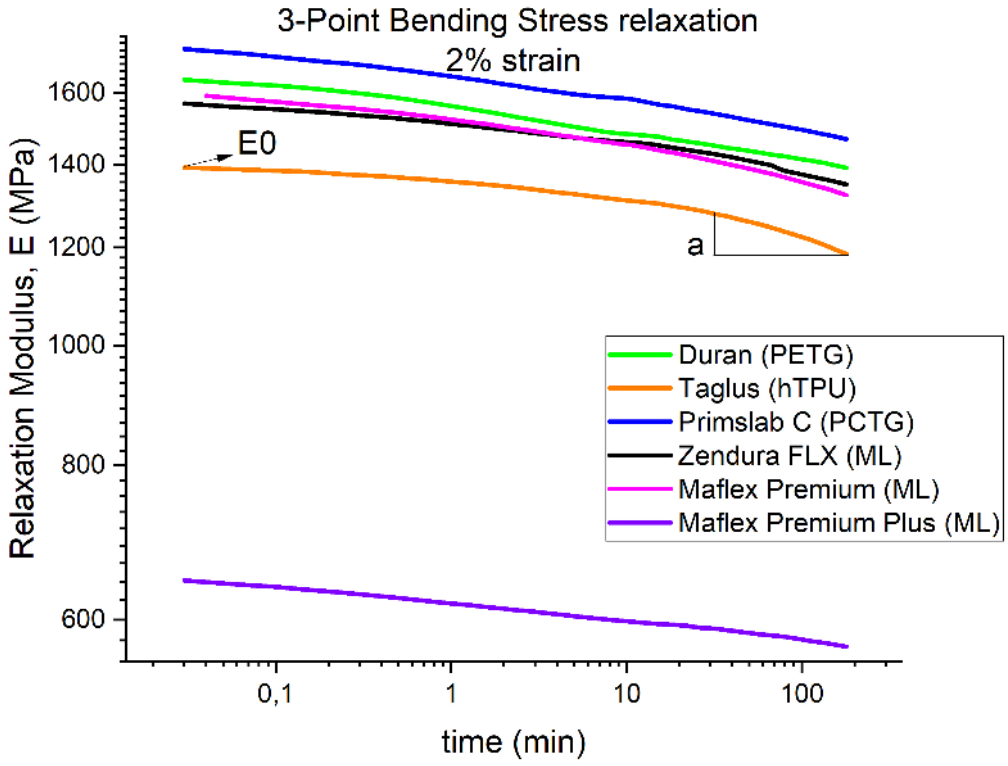



Figure 4.15: Stress relaxation curves at 2% strain in 3PB

At 2% strain in 3PB, all materials exhibit broadly similar relaxation behaviour, as shown by the convergence of the curves in Figure 4.15 and by the values in Table 4.9. The retained modulus after 3 hours ranges narrowly from 82 to 87% of the initial value, and strain recovery from 87 to 92%, across all materials. The multilayer materials display responses comparable to those of the monolithic PETG and PCTG sheets; hence, in bending at this strain level, the inner soft layer plays no discernible role in the mechanical response of the laminates. Maflex Premium Plus shows the highest strain recovery (92%) and the lowest $E_0$ (660 MPa), consistent with the effect of its inverted layer architecture on the bending response discussed in Section 4.2.

| Material | E0 (MPa) | $\varepsilon_0$ (%) | a | $E(3h)/E(0)$ (%) | Strain Recovery (%) |
|---|---|---|---|---|---|
| Duran (PETG) | 1640 | 2.0 | 0.02 | 84 | 88 |
| Taglus (hTPU) | 1392 | 2.0 | 0.04 | 84 | 87 |
| Prismlab C (PCTG) | 1770 | 2.0 | 0.02 | 83 | 88 |
| Zendura FLX | 1570 | 2.0 | 0.03 | 85 | 90 |
| Filtzwiesser (ML) | 1601 | 2.0 | 0.04 | 83 | 88 |
| Maflex Premium (ML) | 1620 | 2.0 | 0.04 | 82 | 88 |
| Maflex Premium Plus (ML) | 660 | 2.0 | 0.01 | 87 | 92 |

Table 4.9: Stress Relaxation and strain recovery parameters for 2% strain experiment in 3PB

At 4% and 6% strain, material-dependent differences become progressively more pronounced; the full results, tables, and material-by-material analysis are reported in Appendix F. In summary, monolithic PETG enters its plastic regime with increasing strain, losing both retained modulus and recovery, whereas PCTG retains superior recovery owing to its higher yield strain. Among the multilayers, the outer-layer copolyester dominates the response: the PETG-faced Filzwiesser mirrors monolithic PETG, while the PCTG-faced laminates (Zendura FLX and Maflex Premium) retain substantially more modulus and recover better, and Maflex Premium Plus behaves throughout as a thin hTPU monolayer. Monolithic Taglus (hTPU) could not be reliably characterised at 4% or 6% strain, as the 18 N force limit of the equipment was reached before the target deformation, highlighting an intrinsic limitation of 3PB for high-modulus aligner materials.

A general observation across all 3PB strain levels is that the contribution of the soft TPU interlayer to the measured mechanical response is inherently limited in this configuration: the stiff outer layers sustain the majority of the bending stress, while the soft core is subjected predominantly to shear rather than tensile or compressive loading. This geometry-dependent limitation, compounded by experimental artefacts; including specimen displacement within the supports and upward flotation of the free span during the relaxation phase; renders three-point bending poorly suited to the characterisation of multilayer aligner materials, particularly at higher strain levels where specimen stability cannot be guaranteed. These limitations motivated the adoption of tensile stress relaxation as the primary characterisation mode, described in the following section.

### 4.4. Stress relaxation in Tensile configuration

In the tensile configuration, substantial differences in the relaxation modulus are observed between materials, as shown in Figure 4.16. The monolithic materials display consistently higher $E_0$ than the multilayer systems, with Taglus (hTPU), Duran (PETG), and Prismlab C (PCTG) reaching 1220, 937, and 910 MPa respectively, compared with 625-640 MPa for Zendura FLX and Maflex Premium. A notable exception is Maflex Premium Plus,

whose $E_0$ of 1130 MPa is comparable to that of the monolithic materials, further supporting the dominant role of its hTPU core in governing the overall response.

At 0.5% strain the multilayers show both a lower retained modulus after 3 hours and lower strain recovery than the monolithic sheets (Table 4.10): setting aside the hTPU-cored Maflex Premium Plus, the laminates retain 29-42% of their modulus and recover 29-44% of the applied strain, against 33-60% retained and 33-56% recovery for the monolithic materials. This suggests that, even at low strain well within the elastic regime, the soft elastomeric interlayers introduce additional viscoelastic dissipation that accelerates relaxation and reduces recoverability. Maflex Premium Plus is again the exception, closely resembling Taglus (hTPU), retaining 50% and recovering 55%, against 60% and 56% for Taglus, reinforcing that its soft outer sTPU layers contribute negligibly and that its hTPU core governs the behaviour of the laminate across the full range of deformation.

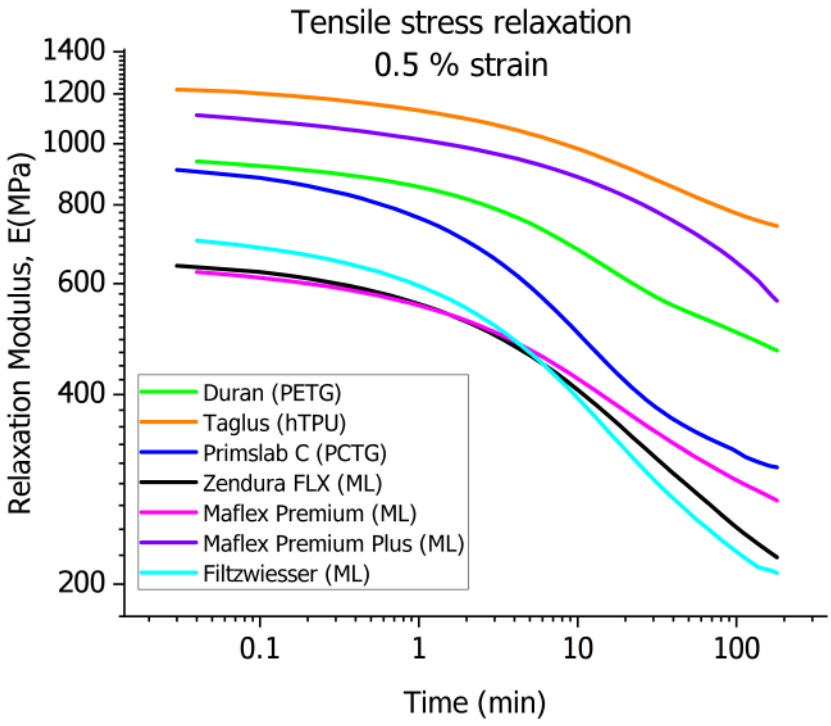



Figure 4.16: stress relaxation curves at 0.5 % strain in tensile configuration

| Material | E0 | $\varepsilon_0$ | a | $E(3h)/E(0)$ | Strain Recovery (%) |
|---|---|---|---|---|---|
| Duran (PETG) | 937 | 0.5 | 0.11 | 50 | 50 |
| Taglus (hTPU) | 1220 | 0.5 | 0.10 | 60 | 56 |
| Prismlab C (PCTG) | 910 | 0.5 | 0.16 | 33 | 33 |
| Zendura FLX | 640 | 0.5 | 0.2 | 33 | 38 |
| Filtzwiesser | 717 | 0.5 | 0.23 | 29 | 29 |
| Maflex Premium (ML) | 625 | 0.5 | 0.16 | 42 | 44 |
| Maflex Premium Plus (ML) | 1130 | 0.5 | 0.2 | 50 | 55 |

Table 4.10: Relaxation and strain recovery parameters for 0.5% strain experiment in tensile configuration

The intermediate 1.5% strain level, at which the same trends are seen developing, is reported in full in Appendix G.

At 2% applied strain the force ceiling was reached before the target deformation in four of the six materials (Figure 4.17, Table 4.11): Duran, Taglus, and Prismlab C attained only 1.04%, 1.12%, and 1.42% respectively, and Maflex Premium Plus 1.59%. As detailed in Appendix G.2, these shortfalls reflect the intrinsic stiffness of the materials rather than experimental artefacts. Valid relaxation parameters could therefore be extracted only for Zendura FLX and Maflex Premium, both of which reached the full 2% target; the data for Maflex Premium Plus are reported with caution given the 20% strain shortfall, while those for Duran, Taglus, and Prismlab C are considered unreliable at this level and are excluded from the following discussion.

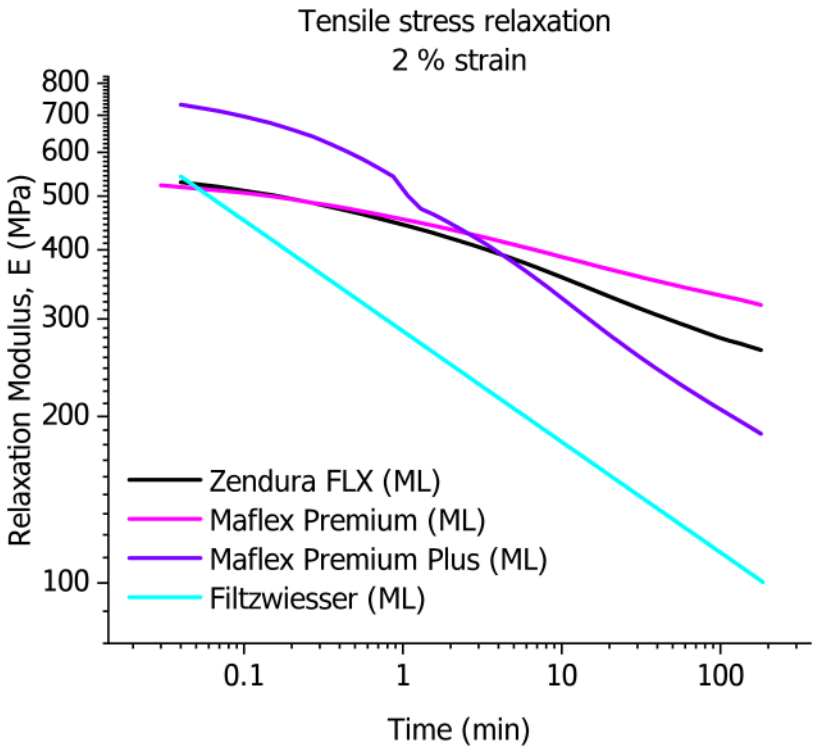



Figure 4.17: stress relaxation curves at 2 % strain in tensile configuration

Among the valid measurements, the initial moduli of Zendura FLX (544 MPa) and Maflex Premium (538 MPa) are nearly identical and substantially lower than their values at 0.5% and 1.5% strain, consistent with the progressive nonlinear softening of both materials as the deformation approaches and exceeds the yield region of their outer PCTG layers. Maflex Premium Plus, despite its reduced strain of 1.59%, exhibits a higher $E_0$ of 759 MPa, which may reflect the increasing load-bearing contribution of its hTPU core as the outer sTPU layers are strained beyond their linear range.

| Material | E0 (MPa) | $\varepsilon_0$ (%) | a | $E(3h)/E(0)$ (%) | Strain Recovery (%) |
|---|---|---|---|---|---|
| Duran (PETG) | 1096 | 1.05 | - | - | - |
| Taglus (hTPU) | 1065 | 1.12 | - | - | - |
| Prismlab C (PCTG) | 580 | 1.42 | - | - | - |
| Zendura FLX | 544 | 2.00 | 0.11 | 48 | 38 |
| Filtzwiesser (ML) | 540 | 1.98 | 0.2 | 29 | 25 |
| Maflex Premium (ML) | 538 | 2.00 | 0.07 | 60 | 53 |
| Maflex Premium Plus (ML) | 759 | 1.59 | 0.14 | 43 | 43 |

Table 4.11: Relaxation and strain recovery parameters for 2 % strain experiment in tensile configuration

Despite the near-identical $E_0$ of Zendura FLX and Maflex Premium, their long-term behaviour diverges substantially: Maflex Premium shows a lower decay exponent ($a$ =

0.07 versus 0.11) and retains 60% of its modulus after 3 hours, the highest value among the valid measurements, against 48% for Zendura FLX, thus delivering a more sustained mechanical stimulus over the wearing period. Both materials share the same PCTG/sTPU/PCTG architecture and identical layer thicknesses, as confirmed by optical microscopy, and both were strained to the full 2%; the difference in strain recovery (Maflex Premium 53% versus Zendura FLX 38%) therefore cannot be attributed to geometric factors and must reflect differences in the intrinsic viscoelastic properties of the constituent polymers; most plausibly in the elastic resilience of the sTPU core or the post-yield behaviour of the PCTG outer layers; which in turn depend on each manufacturer's formulation and processing. Maflex Premium Plus recovered 43% at $\varepsilon_0 = 1.59\%$; its outer sTPU layers, being more compliant than PCTG, may have remained closer to their elastic limit at this reduced strain, contributing more effectively to recovery than the plastified PCTG outer layers of the other two laminates. Across all materials, the recovery values at 2% strain are markedly lower than at 0.5%, reflecting the progressive accumulation of irreversible deformation as strain exceeds the elastic capacity of the constituent polymers, a finding with direct implications for the dimensional stability of aligners subjected to repeated or excessive activation.

One of the aims of this study was to evaluate whether multilayer PCTG/sTPU/PCTG materials, specifically Zendura FLX and Maflex Premium, generate lower initial forces than their monolithic counterparts while maintaining a slower force decay over time. The present data partially support this hypothesis. Regarding force magnitude, the PCTG/sTPU/PCTG materials consistently generated lower initial forces across all valid strain levels: at 0.5% strain, Zendura FLX and Maflex Premium produced $F_0$ values of 4.53 N and 4.56 N respectively, compared with 7.52 N for Duran (PETG) and 8.38 N for Taglus (hTPU); at higher strains, the monolithic materials systematically reached the 18 N equipment ceiling before achieving the target deformation, whereas the multilayers remained within the measurable range up to 2 % strain. This strongly suggests that the compliant sTPU core attenuates the force transmitted to the tooth, which is of direct clinical relevance given that forces substantially exceeding the 50-100 cN range considered optimal for orthodontic tooth movement [254] are associated with increased risk of root resorption and patient discomfort [255]. Regarding force decay, however, the data do not support the hypothesis of a slower relaxation rate in the multilayers: at 0.5% strain, Zendura FLX and Maflex Premium exhibited power-law exponents of $a = 0.20$ and $a = 0.16$ respectively, compared with $a = 0.11$ for Duran and $a = 0.10$ for Taglus, indicating that early-stage force decay was in fact more rapid in the multilayers. This is attributable to the viscoelastic sTPU core, which dissipates energy more readily than the stiffer PETG or hTPU matrices of the monolithic materials. The clinical implication is therefore nuanced: while PCTG/sTPU/PCTG multilayers deliver more physiologically appropriate initial forces, their faster relaxation means the delivered force diminishes more quickly over the wearing period, which may warrant careful consideration of aligner replacement intervals to maintain therapeutic efficacy.

### 4.5. Master curve

The master curve of each material at 37 °C was constructed in a tensile configuration by applying the time-temperature superposition principle to the DMTA frequency sweep data [142], [146], [256]. A representative example for Duran (PETG) is shown in Figure 4.18, illustrating the characteristic parameters extracted from each curve: the maximum storage modulus ($E'_{max}$), the minimum storage modulus ($E'_{min}$), the glass transition time ($t_g$), defined as the time at which the storage modulus decreases by 20 dB from its maximum value, and the onset of non-linear behavior ($t_L$), defined as the point at which the modulus drops by 2 dB when reading the curve from right to left. The mechanical parameters extracted from the master curves of all materials are reported in Table 4.12. The mathematical foundations of the time-temperature superposition principle and the physical interpretation of the master curve parameters are provided in Appendix A.

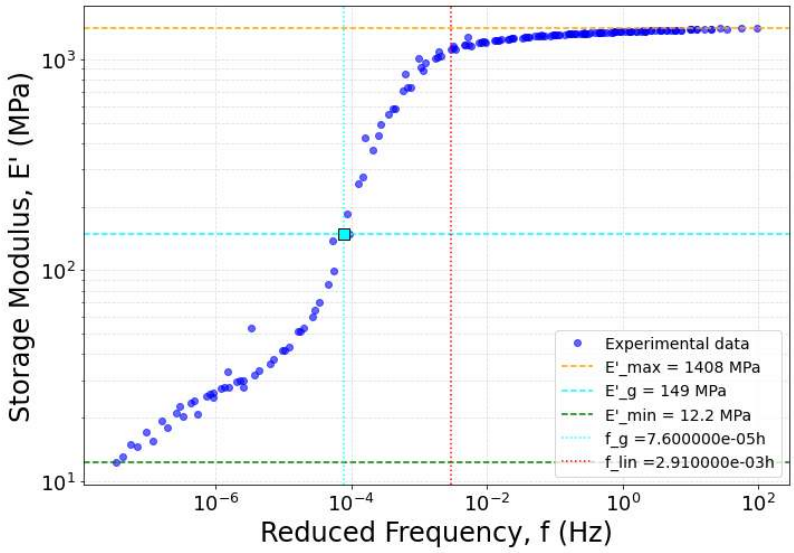



Figure 4.18: Duran (PETG) master curve.

Several clinically meaningful trends are apparent from Table 4.12. Monolithic PETG and hTPU materials exhibit the highest $E'_{max}$ values (1408 and 1761 MPa, respectively) but the shortest $t_g$ values (3.65 and 4.09 h), indicating that their glass transition — and thus the onset of substantial modulus loss — is reached within a few hours at body temperature, well before the recommended wearing period of 20-22 h per day has elapsed. Prismlab C (PCTG) shows an intermediate $t_g$ of 46.5 h, consistent with the higher glass transition temperature of PCTG relative to PETG. The multilayer materials exhibit substantially longer $t_g$ values despite their lower $E'_{max}$, with Maflex Premium reaching $t_g = 765\ h$ (around 32 days), the longest value in the dataset, suggesting that its force delivery remains within the linear viscoelastic regime throughout the full recommended wearing period. The general relationship between glass transition temperature and $t_g$ observed across materials is consistent with theoretical expectations: a higher Tg implies a greater thermal distance from body temperature, slowing the cooperative segmental relaxation responsible for the

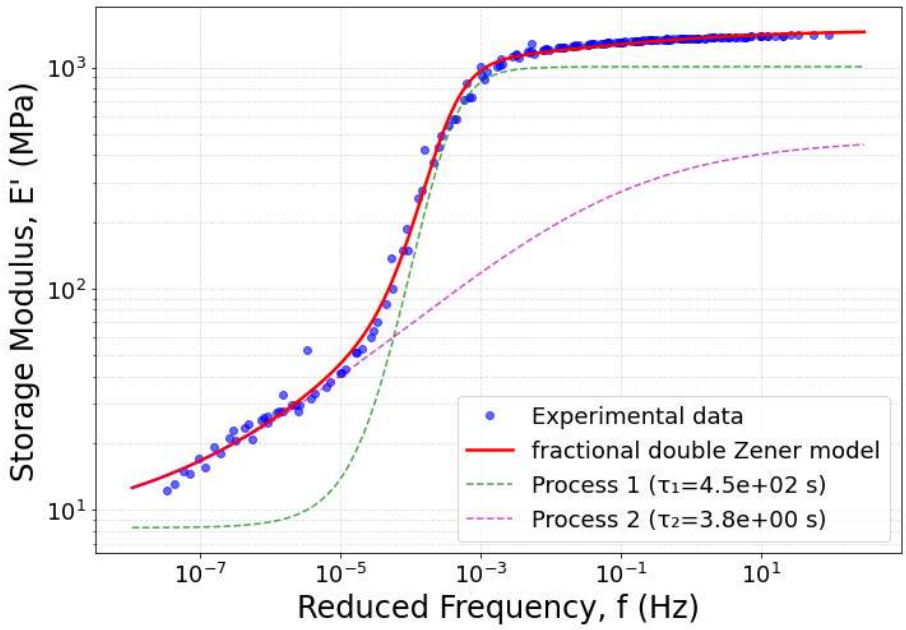
modulus drop. However, the scatter in the data indicates that additional factors, including multilayer architecture and the specific formulation of the sTPU interlayer, also influence the relaxation kinetics and preclude accurate quantitative prediction of $t_g$ from Tg alone.

| Material | $E'_{max}$ (MPa) | $E'_{min}$ (MPa) | $E'_g$ (MPa) | $t_L$ (h) | $t_g$ (h) |
|---|---|---|---|---|---|
| Duran (PETG) | 1408 | 12.2 | 149 | 0.09 | 3.65 |
| Taglus (hTPU ) | 1761 | 13.1 | 200 | 0.49 | 4.09 |
| Prismlab C (PCTG) | 1321 | 28.9 | 142 | 1.1 | 46.5 |
| Zendura FLX (ML) | 832 | 4.9 | 98 | 0.65 | 182 |
| Filtzwiesser (ML) | 975 | 2.2 | 91 | 0.5 | 155 |
| Maflex Premiun (ML) | 955 | 10.9 | 125 | 1.56 | 765 |
| Maflex Premium Plus (ML) | 968 | 12.4 | 105 | 0.42 | 117 |

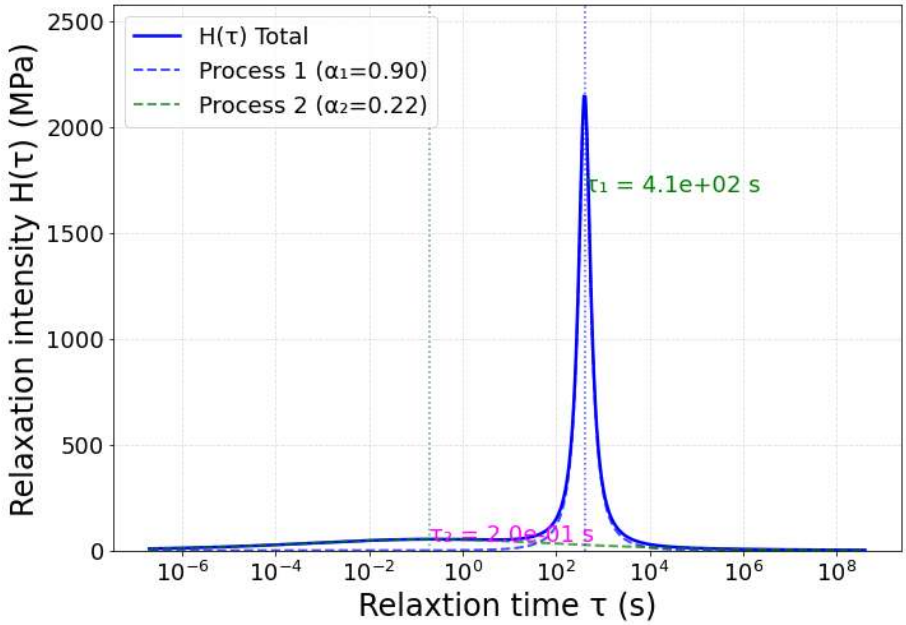
Table 4.12: Parameters obtained from the master curves in as-received materials

The clinical implications of the force delivery window are substantial. A material whose $t_g$ falls well below the prescribed wearing interval will have largely dissipated its stored elastic energy before the aligner is replaced, rendering it mechanically ineffective for much of the treatment stage. Conversely, a material with a wearing interval that is substantially exceeded sustains a therapeutically relevant force throughout the full stage, reducing the risk of treatment failure due to insufficient force. The data in Table 4.12 confirm that monolithic PETG and hTPU materials are mechanically active for only a few hours after insertion, whereas the multilayer materials, and particularly Maflex Premium, maintain their therapeutic force window across the entire clinically prescribed wearing period. This finding provides a mechanistic basis for the well-documented clinical superiority of multilayer aligner materials over their monolithic counterparts in terms of treatment predictability and patient compliance [29], [257].

Four fractional viscoelastic models were evaluated as candidates for fitting the experimental master curves: a simple Cole-Cole model, a fractional Zener model, a fractional double Zener (DZ) model, and a Havriliak-Negami (HN) model. The theoretical basis and selection criteria for these models are discussed in Appendix A. The Cole-Cole and fractional Zener models produced poor agreement with the experimental data and were discarded. Both the DZ and HN models achieved good fits across all materials; the DZ model was ultimately selected on the grounds of its superior balance between fitting accuracy, parameter parsimony, and physical interpretability. The DZ model fitting parameters for all materials are reported in Table 4.13, and their clinical interpretation is discussed below.




(a) master curve




(b) relaxation time spectra

Figure 4.19: Master curve and relaxation time spectra in Duran (PETG) at 37 ºC

The fitting parameters of the DZ model for all materials are reported in Table 4.13. Several clinically meaningful trends can be identified.

| Material | $E_r$ (MPa) | $E_1$ (MPa) | $\alpha_1$ | $\tau_1$ (h) | $E_2$ (MPa) | $\alpha_2$ | $\tau_2$ (s) |
|---|---|---|---|---|---|---|---|
| Duran (PETG) | 3.82 | 1096 | 0.89 | 0.11 | 461 | 0.22 | 0.2 |
| Taglus (hTPU) | 13.9 | 1190 | 0.60 | 4.7 | 500 | 0.4 | 170 |
| Prismlab C (PCTG) | 27 | 1030 | 0.84 | 1.4 | 252 | 0.41 | 34 |
| Zendura FLX (ML) | 5.8 | 578 | 0.76 | 5.0 | 225 | 0.42 | 114 |
| Filtzwiesser (ML) | 3.3 | 730 | 0.79 | 3.8 | 245 | 0.38 | 92 |
| Maflex Premium (ML) | 12.7 | 603 | 0.81 | 24.4 | 328 | 0.3 | 273 |
| Maflex Premium Plus (ML) | 12.3 | 673 | 0.78 | 2.7 | 257 | 0.48 | 67 |

Table 4.13: Master curve fitting parameters in as-received materials

The characteristic slow relaxation time ($\tau_1$) varied by more than two orders of magnitude across the dataset and is the parameter most directly relevant to clinical force delivery. Duran (PETG) exhibited the shortest $\tau_1$ (0.11 h), indicating that the force associated with the glass transition is largely dissipated within minutes of insertion. Taglus (hTPU) and Prismlab C (PCTG) showed intermediate values of 4.7 h and 1.4 h, respectively. Among the multilayer materials, Maflex Premium presented the longest $\tau_1$ in the dataset (24.4 h), implying that its slow relaxation process extends across the full recommended wearing period, a result consistent with its $t_g$ of 765 h and with the clinical observation that PCTG/sTPU/PCTG multilayers sustain more predictable forces over the wearing cycle than monolithic materials [29]. The breadth parameter $\alpha_1$ governs the shape of the slow relaxation process: Duran exhibited the highest value (0.89), consistent with a near-exponential and rapid force decay, while hTPU materials and multilayers showed lower values (0.60-0.81), indicating a more gradual and temporally distributed force loss. The fast relaxation parameters $\tau_2$ and $\alpha_2$ govern the insertion mechanics of the device: multilayer materials presented intermediate $\tau_2$ values of 67–273 s and moderate $\alpha_2$ values of 0.30-0.48, consistent with a mild initial tightness upon seating that resolves within minutes — a profile more comfortable than the abrupt high-force insertion characteristic of monolithic PETG ($\tau_2 = 0.2\ s$) and consistent with clinical reports of superior patient comfort with multilayer aligners [39].

## 4.6. Effect of thermoforming

### 4.6.1. Introduction

The mechanical and physical properties of thermoplastic aligner materials are not invariant with respect to the manufacturing process: thermoforming — the step in which flat polymer sheets are heated above their glass transition temperature and vacuum- or pressure-formed over a dental model — introduces thermal and mechanical stresses that can alter the microstructure of the material and, consequently, its functional properties. The extent and direction of these changes are material- and thickness-dependent, and the literature presents a somewhat heterogeneous picture, reflecting the diversity of materials,

thermoforming protocols, and characterisation methods employed across studies.

The most consistently reported effect of thermoforming is a reduction in elastic and flexural modulus, hardness, and glass transition temperature, accompanied by an increase in water absorption [26], [69], [171], [172], [258]. The reduction in elastic modulus is generally attributed to the thermal treatment exceeding the glass transition temperature of the material, which promotes chain mobility, microstructural reorganisation, and thickness reduction [26]. For PETG materials specifically, these effects have been quantified using 3PB and indentation tests on thermoformed specimens prepared over a standardised wedged mold, reporting reductions in flexural modulus of up to 88% in Duran (PETG) and 70% in Erkodur (PETG), and decreases in hardness of 22% and 7.6% respectively [171]. However, neither the direction nor the magnitude of thermoforming-induced changes is universal. The effect on hardness is particularly inconsistent: some studies report an increase in surface hardness after thermoforming for certain PETG materials, attributed to the conversion of amorphous regions into more ordered domains under thermal processing [259], while others report decreases [171], reflecting the sensitivity of this parameter to the specific thermoforming conditions employed. Similarly, the effect on elastic modulus is not uniformly negative: thermoforming has been reported to increase the elastic modulus of Duran (PETG) and Zendura A (hTPU) under certain conditions [172], [260], though it must be noted that in those studies thermoforming was performed in the absence of a forming mold, so that the sheets experienced only the thermal component of the process without the biaxial deformation and thickness reduction inherent to standard clinical fabrication. The increases observed under those conditions may therefore reflect thermal annealing effects — including relief of residual extrusion stresses and partial microstructural reorganization — rather than the combined thermo-mechanical response characteristic of actual aligner manufacture. Consistent with this interpretation, others [261] have studied the effect of mold geometry on the post-thermoforming mechanical response of Duran sheets and demonstrated that property changes are strongly dependent on the shape of the forming surface, with the largest reductions occurring for geometrically wedged molds [171], [261], and negligible changes when no mold or completely flat molds are used, a finding that retrospectively accounts for the discrepant results reported across studies employing different specimen preparation protocols.

Optical transparency is also affected by thermoforming, with the available evidence consistently indicating a decrease in slight transmittance following the thermal cycle [259], [260].This reduction has been linked primarily to the substantial increase in surface roughness induced by thermoforming, as the softened polymer conforms to and replicates the surface texture of the 3D-printed dental model during forming — reported to reach increases of over 1200% in average roughness (Ra) for PETG [260], which promotes light scattering at the material surface. Notably, no significant changes in the chemical structure of PETG have been detected after thermoforming by ATR-FTIR [260], suggesting that the observed property changes are primarily of physical and microstructural rather than chemical origin. It should be noted, however, that the existing evidence on the opti-

cal effects of thermoforming is limited in scope: studies have been conducted exclusively on flat thermoformed PETG sheets prepared in the absence of a forming mold [260], a condition that does not reproduce the biaxial deformation, thickness reduction, and surface contact with the mold that characterize actual clinical aligner fabrication. Other studies employ a forming wedged mold but remain restricted to PETG as the sole material under investigation [259]. In both cases, the range of materials examined is narrow, and multilayer architectures and TPU-based materials are notably absent from the optical characterization literature. A comparative study of the optical properties of a representative set of aligner materials in both the as-received and thermoformed-over-model states therefore represents a relevant contribution to the field.

The existing literature also lacks a comprehensive and systematic characterization of aligner materials before and after thermoforming across the full range of relevant analytical methods. While isolated studies have assessed individual properties such as Tg by DSC or DMTA; flexural modulus, or optical transparency in a limited number of PETG materials [171], [259], [260], no published work has simultaneously addressed the thermomechanical, calorimetry, optical, and thermal degradation behaviour of a representative set of aligner materials — including multilayer architectures — under controlled and standardize thermoforming conditions using wedged molds. In particular, the effect of thermoforming on thermal degradation behavior as assessed by Thermogravimetric Analysis (TGA) has not been previously reported in the aligner literature; TGA provides information on decomposition onset temperature, degradation kinetics, and residual mass that is complementary to calorimetry and mechanical methods and is of direct relevance for evaluating whether thermoforming induces any measurable reduction in the thermal stability of the processed material. It has been recommended that the properties of thermoplastic aligner materials should be characterized after thermoforming in order to reflect their clinically relevant state [259], and the present work responds to this recommendation by reporting a multi-method thermoforming analysis — encompassing tensile DMTA, DSC, optical transparency, and TGA — for the PETG, hTPU, PP, and a multilayer material subsets described above.

Although the primary aim of the present work was to characterize the as-received materials — which represent the starting point for the manufacturing process and provide material properties independent of the variability introduced by thermoforming — an analysis of thermoformed specimens was also conducted for two complementary purposes: to describe the materials in the state in which they are clinically deployed, and to evaluate whether the thermoforming cycle induces measurable changes in material structure or thermal stability that could inform the selection or optimization of processing parameters in thermoforming. Due to the unavailability of certain materials at the time this analysis was performed, the thermoforming characterization was restricted to a subset comprising the PETG group (Duran, Biolon, Leone, and Tristar A), the hTPU group (Tristar S, Taglus, and Zendura AT), PP (Essix C+), and a single multilayer material (Zendura FLX). The same material subset was used for the water sorption analysis described

in the following section.

### 4.6.2. Thermoforming effects materials and methods

Thermoforming was carried out in accordance with the specifications provided by each material manufacturer. In order to simulate the thermoforming conditions encountered in clinical aligner fabrication in a controlled and reproducible manner, a standardized model was designed to approximate the geometry of a canine tooth while allowing quantitative characterization of the resulting material deformation. This approach follows the methodology adopted by previous authors who employed simplified 3D-printed tooth-shaped molds to obtain thermoformed specimens with reproducible geometry and controlled deformation, avoiding the variability inherent in thermoforming over full dental arch models [171], [259]. The model consisted of two superimposed prisms: a rectangular base prism with a lateral length of 80 mm, a height of 10 mm, and a width of 8 mm, over which a triangular prism of identical width and lateral length was placed, with a height of 10 mm and an upper edge width of 2 mm. The resulting geometry — resembling a wedge-shaped or trapezoidal cross-section profile, as illustrated in Figure 4.20, produces a controlled and reproducible degree of material stretching during thermoforming that is representative of the deformation experienced by thermoplastic sheets when formed over the curved surfaces of a dental model, particularly in regions of pronounced curvature such as the buccal and incisal aspects of canine teeth. This approach ensures that the thermoformed specimens used for characterization reflect the microstructural and mechanical modifications induced by clinically relevant processing conditions, rather than those of flat, undeformed sheet material.

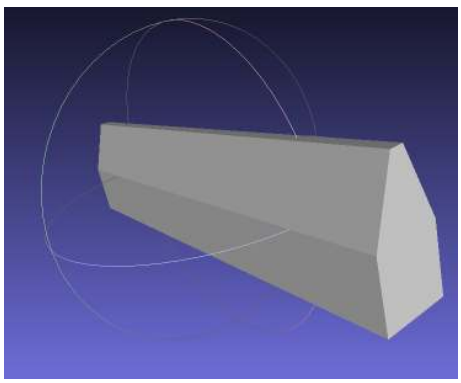


Figure 4.20: wedged mold designed for thermoforming analysis

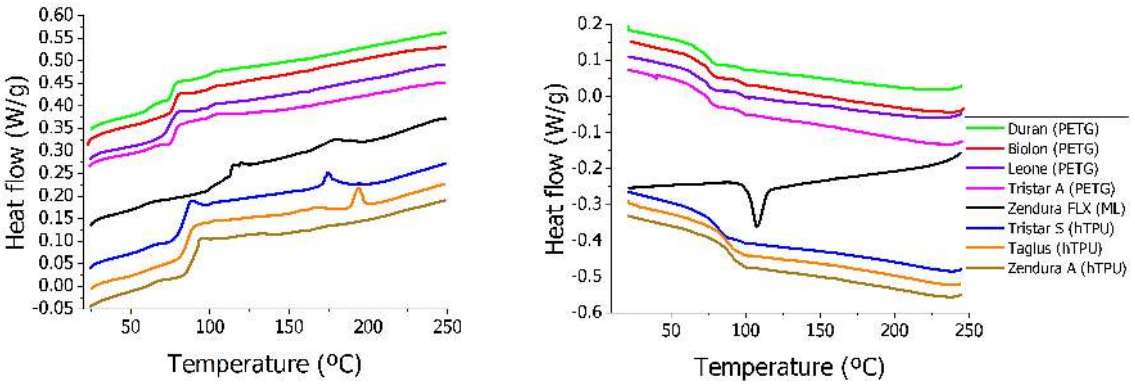
Following thermoforming, the lateral sides of the model were hand-cut to separate the thermoformed sheet from the mold. For the mechanical characterization by DMTA, specimens were preferentially taken from the lateral face of the upper triangular prism, which corresponds to the region of greatest curvature and therefore represents the most representative degree of material stretching. Samples were laser cut over this area. Specimens for DSC and TGA were extracted from the top and bottom faces of the prism,

where the material deformation during thermoforming is more uniform. Optical transparency measurements were performed on sections in which the upper and lower lateral faces remained joined, in order to cover the full sample window of the spectrophotometry equipment.

Uniaxial tensile testing was not performed on the thermoformed specimens due to the limited availability of certain materials and the difficulty in obtaining a sufficient number of specimens with adequate geometry from the thermoformed model. The effect of thermoforming on uniaxial tensile properties has been addressed in the existing literature for a limited set of PETG and hTPU materials [88], [172]. The characterization of tensile mechanical properties in the as-received state, reported in the preceding sections, is therefore considered the primary reference for the elastic and yield behavior of the materials in the present work, with the thermoforming analysis focused on the complementary methods described above.

### 4.6.3. Thermoforming effect results

First, the DSC traces were obtained using the same procedure as depicted previously in the as-received samples




(a) First heating (b) cooling

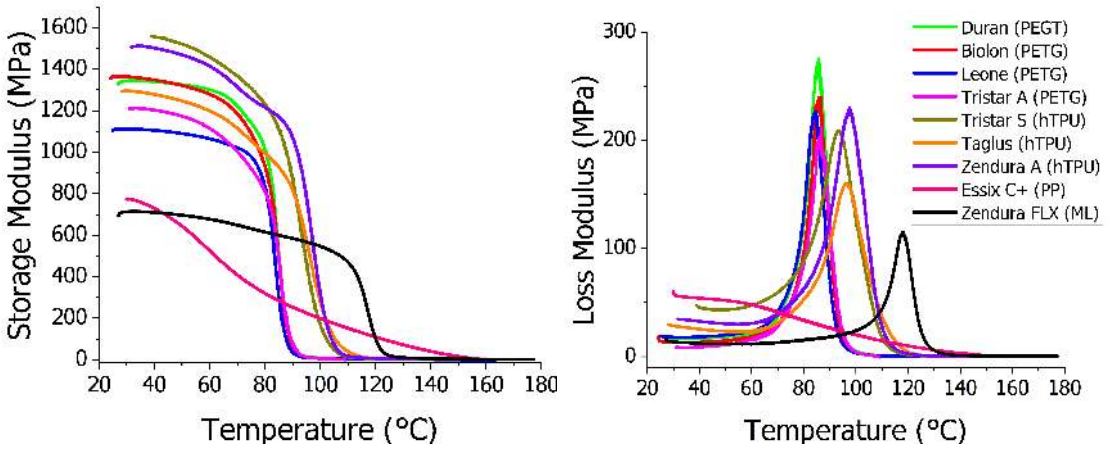
Figure 4.21: DSC traces after thermoforming

Comparison of the DSC curves before and after thermoforming (Figures 4.2 and 4.21) reveals two notable differences. First, the enthalpic relaxation peak associated with physical aging, visible in the as-received PETG materials as an endothermic overshoot near the trailing edge of the glass transition, is absent in all PETG specimens after thermoforming. Second, crystallization exotherms become more pronounced in several of the hTPU materials following thermoforming. No significant differences were observed in the DSC curves of the multilayer materials or of Essix C+ (PP) between the two conditions.

The disappearance of the physical aging peak in the thermoformed PETG samples is a direct consequence of the erasure of the thermal history during thermoforming. The

endothermic peak observed near the Tg in DSC heating scans of physically aged polymers (commonly referred to as relaxation enthalpy or enthalpic recovery) reflects the stored energy associated with the sub-Tg chain relaxation processes accumulated over time, and its magnitude is a measure of the extent of aging [262]. When the material is heated above its Tg during thermoforming, this stored history is erased, and the polymer chains are effectively reset to a disordered, non-relaxed state. The absence of the peak in the post-thermoforming DSC scans, therefore, confirms that the samples had not undergone sufficient room-temperature physical aging in the time elapsed between thermoforming and characterization, consistent with the well-established time-dependence of sub-Tg relaxation in amorphous thermoplastics [234], [263]. The presence or absence of crystallization exotherms in thermoplastic materials is strongly dependent on thermal history, and DSC results are not reproducible unless the thermal history of the sample is tightly controlled [262], [264] which further underscores the importance of characterizing aligner materials both before and after thermoforming.

An incidental but practically relevant observation was made during the laser cutting of thermoformed specimens from certain PETG batches: the material exhibited markedly increased stiffness and brittleness, to the extent that cutting specimens of adequate quality was not possible. This behavior is consistent with advanced physical aging, in which prolonged sub-Tg storage leads to densification of the amorphous phase, reduced free volume, and embrittlement of the polymer [263]. Technicians at Secret Aligner confirmed that this phenomenon is occasionally encountered with PETG batches that are old or have not been stored under the recommended conditions. Clear aligner thermoplastic sheets are typically supplied in sealed, light- and moisture-protected packaging and are intended to be used within a short time after opening or stored in controlled temperature conditions; improper storage or use beyond the recommended shelf life may therefore result in materials with altered mechanical properties that deviate from those of freshly processed sheets. Due to the limited availability of sufficiently aged material and time constraints, a systematic investigation of this aging effect fell outside the scope of the present thesis. Secret Aligner indicated that, as a matter of protocol, materials are stored in temperature-controlled cabinets and that opened batches are used within the stipulated time-frame or discarded upon expiration.




(a) Storage modulus

(b) Loss modulus

Figure 4.22: DMTA curves after thermoforming

Comparison of the DMTA curves before and after thermoforming (Figures 4.5 and 4.22) reveals generally modest but consistent differences across most materials. The DMTA parameters of the thermoformed materials are reported in Table 4.14, and a comparison of Tg and the slope of the storage modulus between 20 and 60 ºC before and after thermoforming is presented in Table 4.15.

| Material | Chemistry | E' at 37 ºC (MPa) | E" at 37 ºC (MPa) | Area ($GPa^2$) |
|---|---|---|---|---|
| Duran | PETG | 1343 | 16.7 | 0.23 |
| Biolon | PETG | 1355 | 13.2 | 0.22 |
| Leone | PETG | 1108 | 18.1 | 0.22 |
| Tristar A | PETG | 1178 | 15.8 | 0.17 |
| Tristar S | hTPU | 1493 | 40.1 | 0.15 |
| Taglus | hTPU | 1285 | 26.6 | 0.10 |
| Zendura A | hTPU | 1475 | 29.0 | 0.17 |
| Essix C+ | PP | 740 | 54.41 | - |
| Zendura FLX | ML | 712 | 12.6 | 0.04 |

Table 4.14: DMTA Parameters in thermoformed materials

Regarding the storage modulus at 37 °C, the effect of thermoforming was material-dependent, with no uniform trend across chemical families — a behavior consistent with the literature, where the direction of modulus change after thermoforming has been reported to depend on the specific material, its thickness, and the evaluation method employed [172], [259]. Among the PETG materials, Leone showed the largest reduction (-30%, from 1580 to 1108 MPa), followed by Duran and Taglus (-20%),while Tristar A was essentially unchanged (1182 to 1178 MPa). Lower reductions in storage modulus at oral temperature were obtained for hTPU materials; with reductions ranging between 10 % and 21%. Essix C+ (PP) showed a substantial reduction (-21%), consistent with

the sensitivity of semi-crystalline polymers to thermal processing above their softening range. Zendura FLX (ML) maintained the lowest storage modulus in both conditions (857 MPa as received, 712 MPa thermoformed, 17% reduction), reflecting the dominant contribution of its compliant sTPU interlayer.

The loss modulus E" at 37 °C showed a less consistent pattern: it increased after thermoforming in Duran, Tristar A, and Zendura FLX, while it decreased in Taglus (50.2 to 26.6 MPa), Zendura A (42.3 to 29.0 MPa), Tristar S (46.1 to 40.4 MPa), Leone (24.5 to 18.1 MPa), and Essix C+ (65.7 to 54.4 MPa). Changes in E" reflect alterations in the viscous dissipation mechanisms of the material, which are sensitive to modifications in chain mobility, free volume, and microstructural organization induced by thermoforming [263].

The Cole-Cole area decreased after thermoforming in most materials. The most pronounced reduction was observed in Taglus (-52%, from 0.21 to 0.10 $GPa^2$), followed by Tristar A (-29 %,from 0.24 to 0.17 $GPa^2$) and Zendura A (-19 %, from 0.21 to 0.17 $GPa^2$). The PETG materials showed more modest reductions of 10-21 %,while Zendura FLX exhibited a smaller absolute decrease (0.06 to 0.04 $GPa^2$).Tristar S was the only material whose Cole-Cole area remained essentially unchanged after thermoforming (0.18 $GPa^2$ in as received state and 0.15 $GPa^2$ after thermoforming), suggesting that its overall damping capacity is not significantly affected by the thermoforming thermal cycle. The systematic reduction in Cole-Cole area across the remaining materials may be interpreted as a consequence of the partial relaxation of thermally induced residual stresses and the reorganization of polymer chain entanglements during thermoforming, which reduces the density of chain segments available for viscous dissipation in the sub-Tg regime [234], [263]

| Material | Tg NT (ºC) | Tg T (ºC) | Slope NT (MPa/ºC) | Slope T (MPa/ºC) |
|---|---|---|---|---|
| Duran | 88.6 | 85.6 | -2.9 | -1.2 |
| Biolon | 85.6 | 85.9 | -3.1 | -2.3 |
| Leone | 85.6 | 84.4 | -2.8 | -1.5 |
| Tristar A | 85.1 | 87.4 | -2.7 | -3.3 |
| Tristar S | 96.1 | 93.1 | -2.9 | -3.8 |
| Taglus | 100.2 | 96.1 | -4.8 | -5.1 |
| Zendura A | 98.3 | 97.5 | -3.2 | -3.5 |
| Essix C+ | - | - | -10.6 | -8.3 |
| Zendura FLX | 118.3 | 121.1 | -1.9 | -1.6 |

Table 4.15: Comparison of DMTA data before and after thermoforming. T stands for thermoformed sample and NT for as-received (Not thermoformed) samples

With respect to Tg, thermoforming resulted in a decrease in all PETG and hTPU materials, with reductions ranging from 1 to 6.8 ºC for PETG and 3.0 to 4.1 ºC for hTPU. This attenuation of dynamic Tg after thermoforming in PETG aligners has been previ-

ously reported and interpreted as evidence of the thermomechanical degradation induced by the processing thermal cycle [171]. In contrast, Zendura FLX (ML) showed a slight increase in Tg after thermoforming (from 118.3 to 121.1 °C), which is noted as an observation without a definitive mechanistic interpretation; it may reflect partial orientation or strain-induced organization of the TPU hard segments during the stretching inherent to thermoforming, though this hypothesis would require further investigation to confirm.

The slope of the storage modulus at low temperature (20 ºC to 60 ºC) became less steep after thermoforming in most PETG materials (e.g. from -2.9 to -1.2 MPa/°C in Duran, from -3.1 to -2.3 MPa/°C in Biolon), suggesting a slightly more temperature-stable glassy response. In contrast, the hTPU materials showed a steeper slope after thermoforming (e.g., from -4.8 to -5.1 MPa/°C in Taglus), indicating greater temperature sensitivity of stiffness in the thermoformed state. Essix C+ (PP) exhibited a less steep slope after thermoforming (-10.6 to -8.3 MPa/°C), though its anomalously high absolute slope values reflect the semi-crystalline nature of PP and its broader thermal softening profile relative to the amorphous PETG and hTPU materials. Zendura FLX (ML) showed minimal change in slope (-1.9 to -1.6 MPa/°C), consistent with the dominance of the elastomeric sTPU component in governing its low-temperature mechanical response.

Talking about the van vGP curves after thermoforming, in PETG, almost the same curves as those in Figure 4.8 were obtained after thermoforming, suggesting minor changes in the material after this processing. In hTPU, on the other hand, the maximum phase angle decreased due to a reorganization of the soft segments. In some hTPU, a minimum phase angle is observed at 5º after thermoforming, implying high elasticity after the glass transition. This also suggests that, due to the reforming of the chemical structures, some trapped soft segments in the hard segment structure were released. In Zendura FLX and PETG, the only noticeable change between as-received and thermoformed material is a decrease in the $E^*$ value of the minimal phase angle, meaning there are fewer restrictions to deformation after the glass transition.

Regarding the transparency of plastics after thermoforming, it appears to increase in general, as can be seen by comparing Figure 4.23 and Figure 4.10. The largest increase in transmittance occurs in Essix C+ (PP), although this material still does not meet the minimum transparency requirements for clinical use. Prior to the transmittance measurements, ATR-FTIR spectra were collected for all thermoformed materials; no changes in chemical composition or functional group structure were detected relative to the as-received state, confirming that the thermoforming process does not induce any surface chemical degradation of any of the materials tested. This is consistent with previous reports showing that the chemical structure of thermoplastic aligner materials remains unaltered after thermoforming [260]. Initially, it was believed that the observed increase in transparency might be related to a decrease in crystallinity during thermoforming — an effect that could plausibly occur in PP and, to a lesser extent, in hTPU materials. However, the DSC traces collected after thermoforming do not support this interpretation. The increase in transmittance should instead be attributed to the reduction in material thickness inherent to the

thermoforming process and/or the rapid cooling applied after forming, which reduces the size of crystalline domains and thereby increases optical transparency [259], [260].

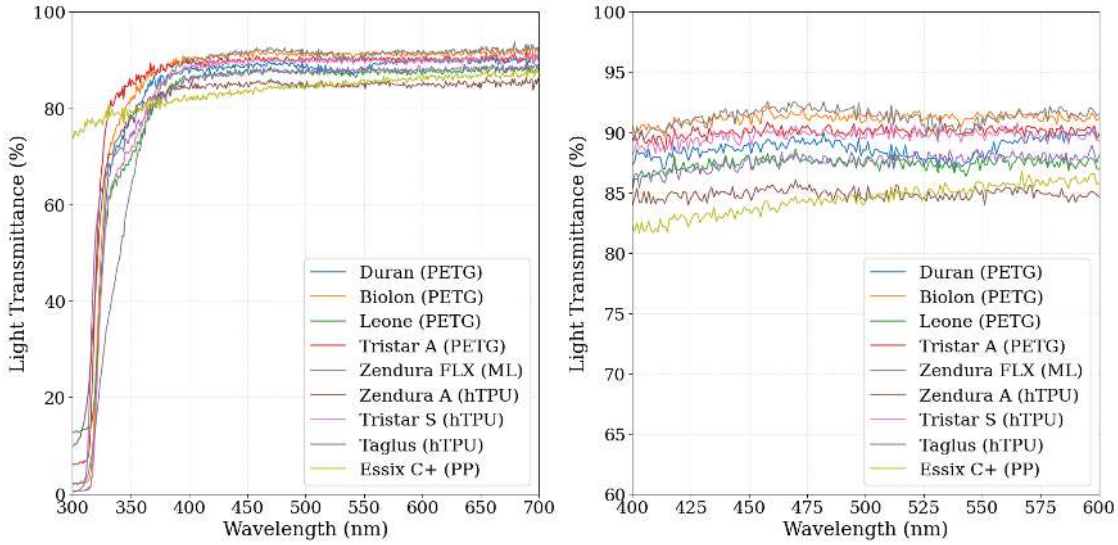



Figure 4.23: Light transmittance curves after thermoforming.

Despite meeting the commonly cited 80% transmittance threshold [26], [69], Essix C+ (PP) is clearly not transparent under clinical conditions, as is immediately apparent from the visual inspection of the thermoformed aligners shown in the figure 4.24: the PP aligner presents a distinctly translucent, whitish appearance compared to the fully transparent PETG aligner. This discrepancy between the numerical transmittance value and the perceived optical quality highlights a fundamental limitation of the 80% criterion: materials that marginally satisfy this threshold may still be visually translucent or opaque due to light scattering from crystalline domains rather than being genuinely transparent. The high crystallinity of Essix C+ (PP), as evidenced by its DSC traces, is the most likely cause of this behavior. On the basis of these observations, a minimum threshold of 85% transmittance is proposed as a more clinically appropriate criterion for clear aligner applications. Comparing Tables 4.16 and 4.6, the average transmittance in all the materials generally decreases by 1 to 3% after thermoforming, while the other parameters remain generally stable. These results are in line with those in the literature [259], [260]

| Material | Composition | AVG T (%) | $T_{550nm}$ | UVA Protection (%) | YI |
|---|---|---|---|---|---|
| Duran | PETG | 88.0 | 88.0 | 27.8 | 0.91 |
| Biolon | PETG | 89.0 | 88.9 | 24.4 | 0.23 |
| Leone | PETG | 85.0 | 85.2 | 31.5 | 0.37 |
| Tristar A | PETG | 88.1 | 88.0 | 19.9 | 0.19 |
| Tristar S | hTPU | 90.0 | 90.0 | 30.2 | 0.57 |
| Taglus | hTPU | 88.0 | 87.7 | 37.1 | 0.12 |
| Zendura AT | hTPU | 85.2 | 85.3 | 25.7 | 0.02 |
| Zendura FLX | ML | 88.5 | 85.3 | 26.1 | 0.70 |
| Essix C+ | PP | 80.1 | 84.5 | 20.2 | 3.17 |

Table 4.16: Transparency parameters in as-received materials

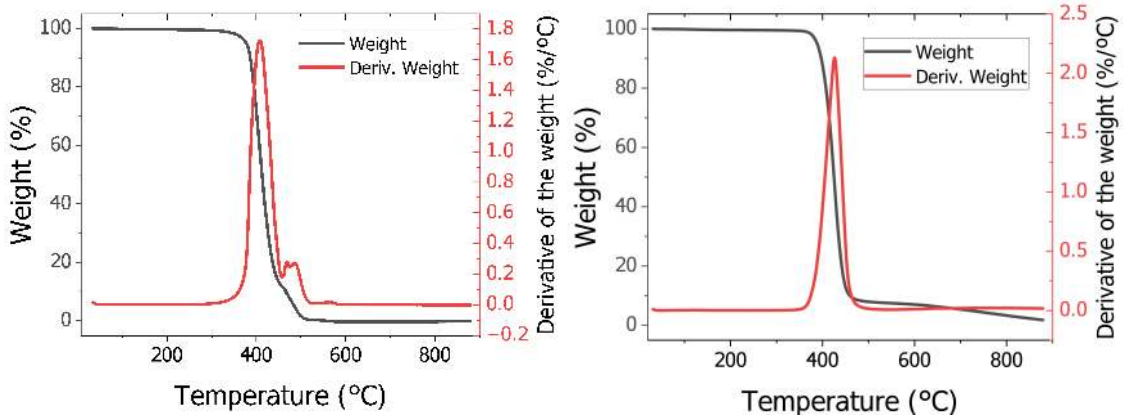


Figure 4.24: Thermoformed aligners. Left Essix C+ (PP) and right Duran (PETG)

The thermal stability of the plastics was analyzed by Thermogravimetric Analysis (TGA) in air and nitrogen atmospheres before and after thermoforming. During the clear aligner manufacturing process, the plastics are shaped by thermoforming and laser cutting, two processes that may thermally degrade the material. TGA measurements allow us to establish security limits for the thermoforming temperature and the laser cutting power.

Regarding thermoforming, this process is usually performed in low vacuum, and hence TGA curves in inert atmospheres are more significant. Laser cutting is usually performed in air, and hence the oxidative environment properties are more important.

In Figure 4.25 the TGA curves of Duran (PETG) can be seen in air and nitrogen atmospheres.



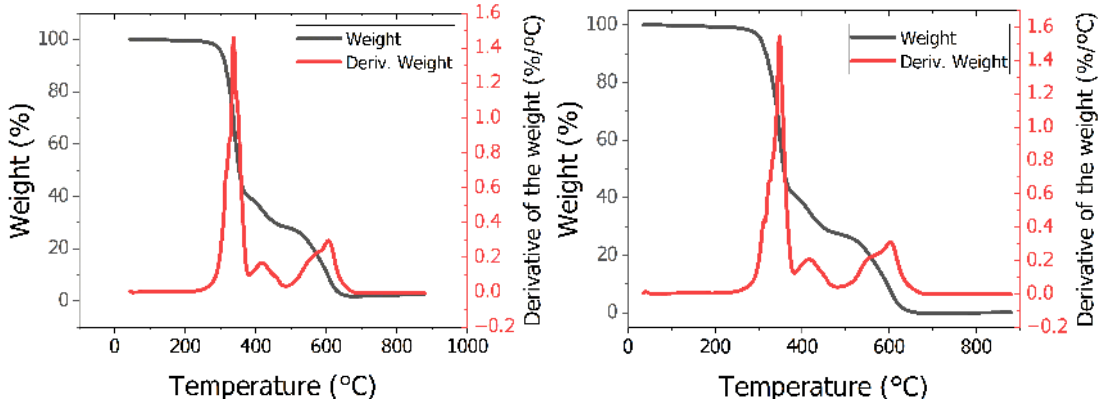

(a) Air (b) Nitrogen

Figure 4.25: TGA curves of Duran (PETG) in different atmospheres

As we can observe, the most noticeable difference in PETG between air and nitrogen atmospheres is the presence of secondary peaks in air after the main degradation. Those peaks have to be related to the charring of the material after degradation. In air, this char is not stable and is further degraded. This effect is common in plastic materials [237], [265], [266].

The level of additives present in the plastic can be determined by comparing the weight percentage at the end of the experiments. Burning a plastic in air leaves only inorganic residues, while burning a plastic in nitrogen also leaves organic charring[237], [266]. The inorganic residue corresponds to additives such as stabilizers or pigments. In clear aligners, the organic charring may be more important. When the devices are laser cut, the cut area is burned, and hence organic charring is deposited in the surrounding area [229].

The percentage of additives in most of the plastics was found to be relatively low in general, ranging from almost zero to 3%. Regarding the charring, it was observed that the charring in hTPU (2.5 - 6%) was higher than in PETG (0.15-3.8 %) and Essix C ($\simeq$ 0).




(a) Air (b) Nitrogen

Figure 4.26: TGA curves of Taglus (hTPU) in different atmospheres

Comparing Figure 4.25 with Figure 4.26, the thermal degradation of TPU is considerably more complex than that of PETG. The first degradation step corresponds to the decomposition of the hard segment urethane bonds, followed by degradation of the soft segment polyester polyol at 390-420 °C [267], [268], [269]. At higher temperatures, degradation of the aromatic structures within the hard segment takes place in a region referred to as the aromatic degradation region (430-600 °C). Degradation of charred material also occurs within this region and may partially obscure the analysis.

Valuable compositional information can be derived from the high-temperature degradation region of TPU thermograms. Commercial TPU formulations are typically synthesized using one of two primary diisocyanate types for the hard segment: MDI and TDI, both aromatic diisocyanates widely employed due to their favorable combination of low cost, high reactivity, ease of processing, and excellent mechanical properties [209], [267]. The aliphatic alternative, HDI, offers superior optical clarity, UV stability, and enhanced biocompatibility, making it particularly suitable for medical and optical applications [270], [271]; however, none of the monolayer TPU materials analyzed in the present study were identified as HDI-based by ATR-FTIR, and this diisocyanate type is therefore included here solely for contextual completeness.

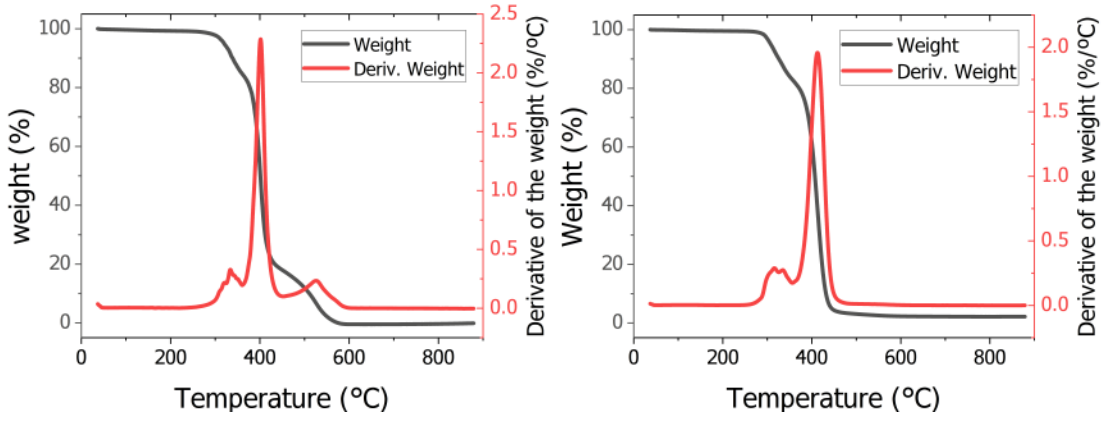
A key distinction in the thermal degradation behavior of MDI- and TDI-based TPU lies in their aromatic content. MDI-based TPU exhibits a characteristic degradation peak in the 420-480 °C range, attributed to the thermal decomposition of the aromatic rings present in the diphenylmethane structure [267], [272] — consistent with the MDI-based hard segment identified by ATR-FTIR for all monolayer TPU materials characterized in the present study. In MDI- and TDI-based TPUs, isocyanurate rings are less commonly present and are incorporated only deliberately to increase mechanical properties and slow relaxation processes while maintaining transparency and processability [273], [274], [275]. However, isocyanurate rings may also arise inadvertently during TPU processing as a consequence of excess diisocyanate, prolonged reaction times at elevated temperatures, or the presence of moisture during synthesis or extrusion. In such cases, the isocyanurate content is uncontrolled and typically results in a thermally stable product or acts as a crosslinker between chains, decreasing transparency and thermal processability while markedly increasing stiffness and brittleness [275], [276], [277]. These isocyanurate rings, formed through the trimerization of isocyanate groups, exhibit exceptional thermal stability and typically degrade in the 550–650 °C temperature range [269], [278], providing a diagnostic signature in TGA thermograms that can be used to identify their unintended presence in commercial formulations.

The hard segment content was estimated from DTG curve deconvolution, assuming that the area under each peak is proportional to the mass fraction of the corresponding component, as described in the equation 4.6 [267], [272]. This approach has been previously applied for quantitative analysis of multicomponent polymer systems [279], [280], although its specific application to TPU hard segment quantification through DTG peak deconvolution has been scarcely reported in the literature.

$$HS(\%) = \frac{A_{hard\ segment} + A_{Isocyanatures}}{(A_{hard\ segment} + A_{soft\ segment} + A_{Isocyanatures})} * 100 \quad (4.6)$$

It was possible to determine the exact composition of Taglus based on ATR-FTIR spectra and its TGA thermograms. Taglus is a polyester-based TPU in which the hard segments are composed of MDI and isocyanurate rings. The exact hard percentage in Taglus hTPU was found to be 87% taking into account the isocyanurate rings. In the case of the other commercial hTPU (Zendura AT and Tristar S), only one peak appears in the thermogram at 330 ºC for Zendura AT, indicating that this polyester-based TPU is almost fully composed of hard segments. It is very difficult to ascertain the specific chemical nature of the hard segment domain based only on its main degradation peak. In the case of Tristar S, the thermograms were similar to those of Taglus (figure 4.26), but the soft segment peak was absent. Hence, the chemical composition of Tristar S would be highly similar to that of Taglus, but with a higher hard percentage ratio. The hard segment percentage obtained by TGA is similar to the Degree of Phase Separation (DPS) obtained previously for Taglus, Zendura AT, and Tristar S.

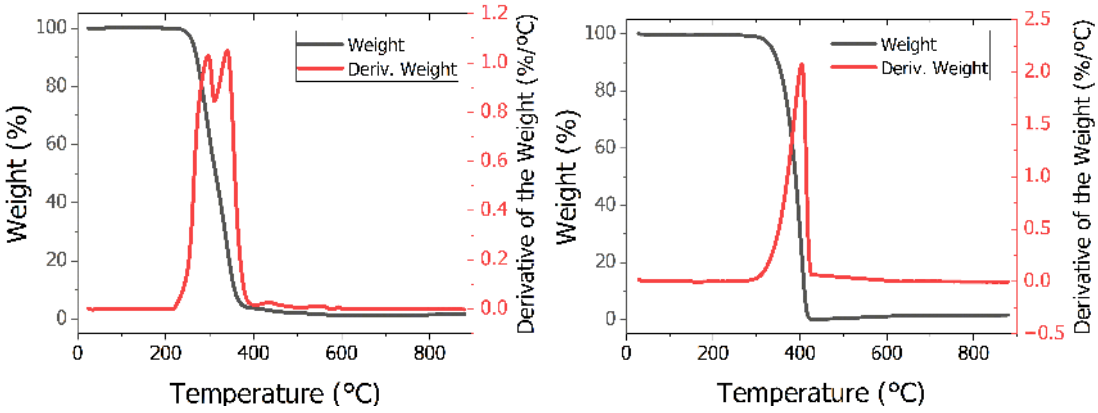
The characterization of the individual layers in Zendura FLX by thermogravimetric methods was not performed due to the fact that the film was clearly affected by the dissolution method and may lead to misleading results. Thermogravimetric characterization of the intact multilayer stack was nonetheless performed and is shown in Figure 4.27.




(a) Air (b) Nitrogen

Figure 4.27: TGA curves of Zendura FLX (ML) in different atmospheres

Regarding the TGA curves in multilayers, an example can be observed in Figure 4.27. Firstly, two degradation peaks appear. Both may be attributable to the TPU inner layer of the material. After this, the main degradation process in the plastic occurs at 400 ºC, with a similar nature to that of PETG. It can also be observed that there is a degradation peak of charring afterwards, as happened in PETG. From the curves, it can be noted that there is a lack of complex aromatic structures in the TPU of the multilayer. Hence, it may be supposed that it is a more flexible and softer TPU.




(a) Air (b) Nitrogen

Figure 4.28: TGA curves of Essix C+ (PP) in different atmospheres

Finally, Figure 4.28 presents the degradation curves of Essic C+ (PP). Under an oxidative atmosphere (air), molecular oxygen preferentially attacks the tertiary carbon atoms of polypropylene, initiating a chain auto-oxidation mechanism. This oxidative degradation pathway is complex and typically occurs at lower temperatures than pyrolysis under an inert atmosphere. In the initial stage, thermo-oxidative reactions lead to the formation of hydroperoxides, which subsequently decompose at approximately 293 °C, generating reactive radical species. The second degradation stage observed in air corresponds to the bulk decomposition of the polymer matrix. [281], [282]

These results indicate that PP is susceptible to oxidative degradation at relatively low temperatures in the presence of oxygen, which may represent a limitation for processes such as laser cutting. During laser processing, the localized high temperatures and exposure to air may promote radical formation at the edge. The persistence of such reactive species could potentially influence the surface chemistry and, consequently, the biocompatibility of the material.[283]

The activation energy ($E_a$) of the main thermal degradation process was obtained from the TGA curves by the Coats-Redfern method, whose derivation and fitting procedure are detailed in Appendix E. In Table 4.17, T5, T10, and T50 are the temperatures at which the weight drops for 5, 10, and 50% respectively; Tonset is the onset or starting temperature of the weight drop; Tmax is the temperature of the maximal point of the derivative of weight with respect to the temperature; and the last column reports the activation energy $E_a$ obtained as described above.

| Material | T5 (ºC) | T10 (ºC) | T50 (ºC) | Tmax (ºC) | Tonset (ºC) | $E_a$ (KJ/mol) |
|---|---|---|---|---|---|---|
| Duran (PETG) | 377 | 386 | 412 | 408 | 395 | 237 |
| Biolon (PETG) | 376 | 387 | 411 | 399 | 396 | 238 |
| Leone (PETG) | 327 | 381 | 411 | 407 | 391 | 235 |
| Tristar A (PETG) | 377 | 385 | 416 | 412 | 396 | 258 |
| Tristar S (hTPU) | 291 | 302 | 332 | 335 | 315 | 160 |
| Taglus (hTPU) | 302 | 314 | 353 | 337 | 327 | 142 |
| Zendura AT (hTPU) | 314 | 327 | 362 | 372 | 341 | 157 |
| Zendura FLX (ML) | 319 | 338 | 401 | 401 | 379 | 205 |
| Essix C (PP) | 262 | 272 | 313 | 339 | 283 | 81 |

Table 4.17: TGA results in as-received materials in oxidative atmosphere (air)

As it can be observed, the hTPU (Zendura AT, Tristar S, and Taglus) and the PP (Essix C+) degrade at a lower temperature than the PETG and the multilayer (Zendura FLX). Talking about the Zendura FLX multilayer, as may be observed in the Appendix E, Figure E.1, the degradation can be divided into two processes, one having $E_a$ of 205 KJ/mol and another one of 157 KJ/mol. The first one, showing similar values to PETG in $E_a$, may be attributable to a copolyester, whereas the second one can be identified as a degradation process of TPU origin.

Regarding the values in an inert atmosphere, the degradation is generally shifted towards higher temperatures, increasing between 10 and 40 ºC for all the temperature parameters in the materials. In Essix C+ (PP), the increase is even larger (80 ºC) due to the suppression of oxidative degradation in an inert atmosphere. Hence, an inert atmosphere may enlarge the thermoforming window of all the materials, especially for Essix C+ (PP).

Regarding the activation energy ($E_a$), this variable provides information about how difficult it is to scission the main chains of the polymer and hence provides information about the depolymerization. Nevertheless, it also measures the thermal stability of the product. The larger the $E_a$, the higher the thermal stability and the more independent the properties are from the temperature. The fact that PETG materials have larger $E_a$ than hTPU and PP is consistent with the lower slope of the DMTA storage modulus curve in the oral temperature range for PETG compared to the other materials (Table 4.2). It is also interesting to note that despite the degradation processes in Zendura FLX showing both lower $E_a$ than those of PETG materials, the thermal stability from DMTA is higher in Zendura FLX (ML) than in PETG.

The same procedure was performed on thermoformed samples, obtaining TGA curves in both atmospheres. No major differences were found in PETG (Duran, Biolon, Leone, and Tristar A) and ML (Zendura FLX) materials between thermoformed and as-received samples, implying that the thermoforming process does not alter or degrade the material.

In the case of Essix C+ (PP), all thermal parameters increased after thermoforming relative to the as-received state, including the activation energy. The TGA curves ob-

tained under both oxidative and inert atmospheres in the thermoformed material were highly similar to each other and closely resembled the curve recorded under nitrogen in the as-received material. The activation energies in the thermoformed state were likewise comparable to those obtained under inert atmosphere in the as-received material. This convergence suggests that the material underwent partial oxidation during thermoforming, consuming the most reactive tertiary carbon sites and thereby rendering the remaining polymer matrix more resistant to further thermo-oxidative degradation — effectively stabilizing its thermal behavior. Although thermoforming is performed under vacuum pressure, the initial heating of the sheet is carried out in air, which exposes thermally sensitive materials to an oxidative environment prior to forming. The absence of detectable changes in the ATR-FTIR spectra after thermoforming does not contradict this interpretation: the degree of oxidation is likely below the detection threshold of ATR-FTIR for carbonyl-containing oxidation products, which typically requires concentrations on the order of 5 % to produce discernible spectral features, and ATR-FTIR always underestimates the oxidation percentage and rate in polyolefins [284]. Meanwhile, TGA is sensitive to more subtle alterations in degradation kinetics and activation energy. For materials with high susceptibility to thermo-oxidative degradation, such as PP, thermoforming should ideally be performed under a fully controlled, non-oxidative atmosphere throughout the entire heating and forming cycle in order to preserve the original thermal and mechanical properties of the sheet.

The thermal parameters of the hTPU materials decreased slightly after thermoforming, consistent with partial degradation of both the soft and hard segments during processing. The absence of detectable changes in the ATR-FTIR spectra after thermoforming is consistent with this interpretation: the degree of degradation inferred from TGA is likely below the detection threshold of ATR-FTIR or may not occur in the surface of the material to a sufficient extent. The precise mechanism responsible — whether thermal chain scission, hydrolytic degradation of the ester linkages in the soft segment, or a combination of both — cannot be unambiguously determined from TGA data alone and would require complementary analysis using other techniques such as rheology or chemical analysis.

### 4.6.4. Thermoforming effect conclusions

In summary, thermoforming generally enhances the physical properties of the materials investigated, resulting in a modest reduction in stiffness and an improvement in optical transmittance in most cases. It should be noted, however, that the observed increase in transparency may be partly attributable to the reduction in material thickness inherent to the thermoforming process, rather than solely to changes in the polymer microstructure. The differences induced by thermoforming can be summarized as follows: a general decrease in storage modulus at oral temperature, a reduction in the area under the Cole-Cole curve, a slight increase in loss modulus at oral temperature, an improvement in optical transmittance, the disappearance of the enthalpic relaxation peak in PETG materials, and

the development of more pronounced crystallization exotherms in several hTPU materials. Regarding the Tg, the observed fluctuations are small and may be attributable to instrumental variability rather than genuine material changes. With respect to the slope of the storage modulus at low temperature, a decrease is observed in PETG and ML — indicating a less thermally dependent mechanical response — whereas hTPU materials show an increase, suggesting greater temperature sensitivity of stiffness in the thermoformed state.

These findings are broadly consistent with the softening, slight Tg reduction, and transparency changes previously reported for PETG materials following thermoforming [26], [171], [259], [260], and extend this characterization to materials not previously studied in this context, namely hTPU (Taglus, Zendura AT, and Tristar S), PP (Essix C+), and the ML Zendura FLX.

The most pronounced differences between the as-received and thermoformed states were observed by TGA in Essix C+ (PP) and, to a lesser extent, in the hTPU materials (Taglus, Zendura AT, and Tristar S). By contrast, the TGA parameters of the PETG and ML showed no appreciable changes after thermoforming, indicating that these materials are thermally stable under the processing conditions employed and that their degradation behavior is not meaningfully affected by the thermoforming cycle, as previously reported [260]. The results suggest that Essix C+ suffers a thermo-oxidative degradation process of relatively low intensity that was not detected by ATR-FTIR. Such degradation seems to render the material more resistant to further thermo-oxidative degradation. Regarding hTPU, it seems to suffer a partial degradation of the soft and hard segments of relatively low intensity as well, which was also not detected by spectroscopic means.

## 4.7. Commercial thermoformed multilayers characterization by nanoindentation

### 4.7.1. Introduction

A further consideration that emerged during the course of this thesis concerns the characterization of commercially manufactured aligners. The most widely used and clinically successful aligner brands supply only pre-fabricated, thermoformed devices and provide little or no information regarding the materials employed or the processing conditions applied during manufacture. According to available market data, approximately 70% of all clear aligners used globally in 2025 were marketed under the Invisalign brand (Align Technology), 12% under Spark, 10% under ClearCorrect, and the remaining 8% by smaller companies that source raw thermoplastic sheets from third-party suppliers such as Bay Materials, Maflex, Taglus, and Scheu Dental; and later thermoform the devices[285].

The mechanical characterisation of finished, clinically used aligners cannot be performed using the same methods applied to flat as-received sheets or to laboratory-thermoformed specimens. Finished aligners present curved, geometrically irregular surfaces that intro-

duce systematic errors in specimen preparation, and their relatively small dimensions preclude the extraction of samples in sufficient quantity or with the geometry required for conventional mechanical testing.

Nanoindentation was therefore selected as the most suitable technique for this purpose. The method employs a diamond indenter to probe the mechanical response of a material surface and the underlying volume by measuring the relationship between applied force and indenter displacement. The experiment proceeds in three sequential stages: during the loading phase, the indenter is driven into the surface under a controlled increasing force, and both load and displacement are continuously recorded; during the hold phase, the indenter is maintained at maximum penetration depth for a prescribed period to allow for creep and thermal drift stabilization; and during the unloading phase, the indenter is withdrawn from the surface under a controlled decreasing load, with the elastic recovery of the material captured in the unloading curve. The elastic modulus and hardness of the material are then extracted from the load-displacement data using established contact mechanics models [162].

It should be noted, however, that the standard interpretation of nanoindentation data rests on two idealizing assumptions: that the loading phase is governed exclusively by plastic deformation, and that the unloading phase is fully elastic. Neither assumption holds rigorously for viscoelastic materials such as polymers, in which the mechanical response is inherently rate- and time-dependent. As a consequence, the measured values of elastic modulus and hardness are sensitive to the indentation strain rate and to the duration of the hold phase, and direct quantitative comparison with moduli obtained by tensile testing or DMTA requires caution. As a rough guide, the tensile elastic modulus of a hard thermoplastic is commonly taken to be approximately one-third of the reduced modulus obtained by nanoindentation, though this approximation carries significant uncertainty and cannot be applied universally across material or test conditions.

A particular advantage of nanoindentation in the context of multilayer aligner materials is its ability to characterize each constituent layer individually, with spatial resolution sufficient to interrogate regions of interest independently of the surrounding layers and without the need for large or regularly shaped specimens. Of especial interest is the interlaminar region between adjacent layers: in this zone, the mechanical properties are expected to transition gradually between those of the bounding layers, and the spatial profile of modulus and hardness across the interface can provide insight into the degree of intermixing of the two polymers and, indirectly, into the quality of interfacial adhesion [286].

Although nanoindentation has been applied to characterize the bulk mechanical properties of aligner materials — notably to measure surface elastic modulus and hardness and to establish correlations with orthodontic force delivery [229] — and has also been employed on clinically retrieved aligners to quantify the mechanical changes induced by intraoral ageing [250], [287], no published study has, to the best of the author's knowl-

edge, employed this technique to resolve the mechanical properties of individual layers within multilayer aligner architectures or to characterise the interlaminar region between them. It is noteworthy that the aligner literature on instrumented nanoindentation is heavily skewed towards a single commercial brand: the overwhelming majority of studies have characterised Invisalign appliances, either in the as-manufactured thermoformed state or after intraoral retrieval [250], [287] while references to other thermoplastic materials — whether thermoformed or as-received sheet — remain scarce [229].

The existing nanoindentation studies in the aligner literature have therefore invariably treated the material as a homogeneous substrate, measuring average surface properties for comparative purposes, without exploiting the spatial resolution of the technique to probe layer-by-layer variation in cross-section. This stands in contrast to the broader polymer and composite science literature, where cross-sectional nanoindentation has been established as a powerful tool for the localised characterisation of constituent layers and interphase regions in multilayer polymer films and fibre-reinforced composites [286], [288], and where grid indentation techniques have been used to map mechanical property gradients across heterogeneous interfaces [288]. The present work therefore addresses both of these gaps simultaneously: it extends nanoindentation characterisation to a broader set of thermoplastic aligner materials beyond Invisalign, and represents the first systematic application of the technique to the cross-sectional characterisation of multilayer aligner architectures, providing localised mechanical data for each constituent layer and for the interfacial zone — information that is inaccessible by conventional bulk mechanical testing methods.

### 4.7.2. Materials and methods

Rectangular specimens were sectioned from the molar region of unused clear aligners supplied directly by Secret Aligner, sourced from different patients but never clinically deployed. Sample preparation, instrumentation, and the mathematical analysis of the load-displacement curves followed the protocols described in Section 3.6. The materials characterised are listed in Tables 4.18 and 4.19.

| Material | brand | composition | State |
|---|---|---|---|
| Invisalign | Align Technology | ML | Thermoformed |
| Spark | Ormco | ML | Thermoformed |
| Filtzwiesser | Filtzwiesser | ML | Thermoformed |
| Zendura FLX | Bay Materials | ML | Thermoformed |
| Zendura FLX | Bay materials | ML | As-received |
| Prismlab C | Prismlab | PCTG | Thermoformed |
| Prismlab C | Prismlab | PCTG | As-received |

Table 4.18: Commercial materials tested by nanoindentation

Raw constituent materials were also tested as reference standards, processed into plain 0.7 mm sheets using a heated press.

| Material | Composition | Shore hardness D |
|---|---|---|
| TPU85A | TPU | 33 |
| TPU93A | TPU | 41 |
| TPU52D | TPU | 52 |
| TPU70D | TPU | 70 |
| TPU84D | TPU | 84 |
| PETG | PETG | 75 |
| TritanMP100 | PCTG | 75 |

Table 4.19: Raw materials tested during nanoindentation. The shore hardness was obtained from the materials datasheets

The inclusion of raw sheet reference materials served two complementary purposes. First, it enabled accurate mechanical characterisation of the sTPU constituent layers without resorting to chemical dissolution, which risks altering the intrinsic polymer properties through plasticisation or chain scission. Reference data from pure constituent materials allows reliable attribution of the cross-sectional indentation response to each individual layer, particularly for the sTPU core, whose low modulus and high viscoelastic character make it the most mechanically distinctive and clinically relevant component of the multilayer stack. As-received PETG and PCTG sheets were also included to provide reference values for the hard outer layers and enable unambiguous identification of layer boundaries in the cross-sectional profiles.

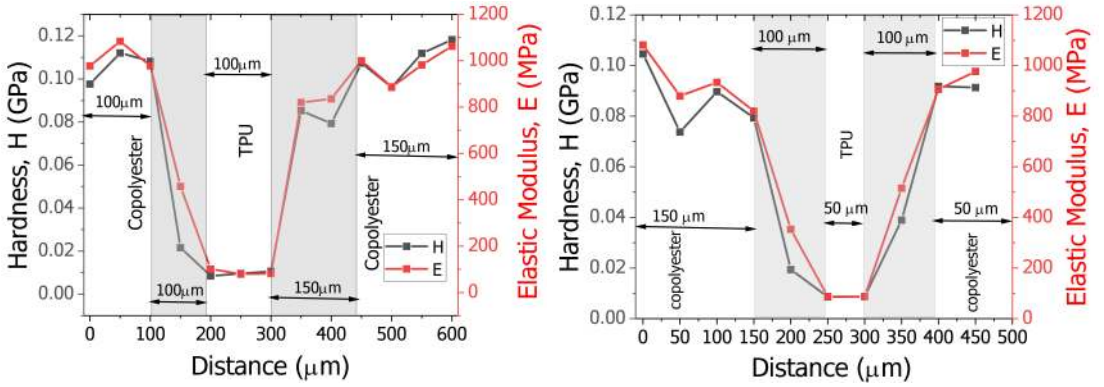
Second, the raw TPU series was used to establish an empirical relationship between nanoindentation-derived reduced modulus and Shore hardness across a range of TPU grades. Shore hardness is the primary mechanical descriptor used by thermoplastic sheet suppliers to classify soft polyurethane grades, yet it provides no direct information on the elastic modulus or viscoelastic character of the material. A quantitative hardness-modulus correlation allows the expected mechanical contribution of a given sTPU layer to be estimated directly from supplier datasheets, without the need for full mechanical characterisation at the material selection stage

### 4.7.3. Results on commercial materials

Prior to the nanoindentation experiments, ATR-FTIR spectra were acquired on the outer surface of the thermoformed specimens, restricted to the molar region in all cases. For Zendura FLX, Filtzwiesser, and Spark, the spectra were fully consistent with a copolyester chemistry of the PETG or PCTG type, confirming the identity of the outer layer. In the case of Invisalign, the spectrum was compatible with a polyester-based, MDI-type

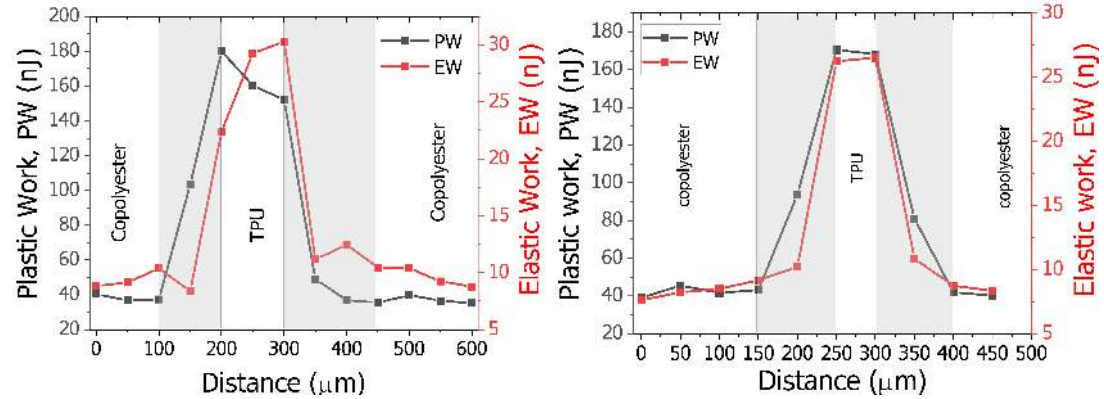
thermoplastic polyurethane, although the specific grade or Shore hardness of the material could not be determined from the spectroscopic data alone.

The multilayer materials were characterized in cross-section by performing indentation traverses perpendicular to the layer interfaces, enabling the construction of hardness and reduced modulus depth profiles across the full thickness of the specimen. The results for Zendura FLX are presented in Figure 4.31 as a representative example. It should be noted that the apparent layer thicknesses measured from the indentation profiles may not accurately reflect the true thicknesses of the constituent layers in the original sheet, as the thermoforming process induces non-uniform thinning across the aligner geometry, and the epoxy encapsulation and subsequent grinding may introduce a small degree of additional dimensional uncertainty in the cross-sectional preparation.




(a) As-received

(b) Thermoformed material




(c) As-received

(d) thermoformed material

Figure 4.29: Zendura FLX indentation before and after thermoforming

As shown in Figure 4.29, a transition zone of approximately 100 $\mu$m is observed at the interface between the sTPU core and the copolyester outer layers, over which the indentation modulus and hardness vary gradually between the values characteristic of each constituent. This interlaminar region is indicative of a degree of intermixing or interdiffusion between the two polymer phases at the interface, likely established during the co-extrusion process. Following thermoforming, the overall thickness of the laminate is reduced, as expected from the biaxial stretching and thinning inherent to the vacuum-

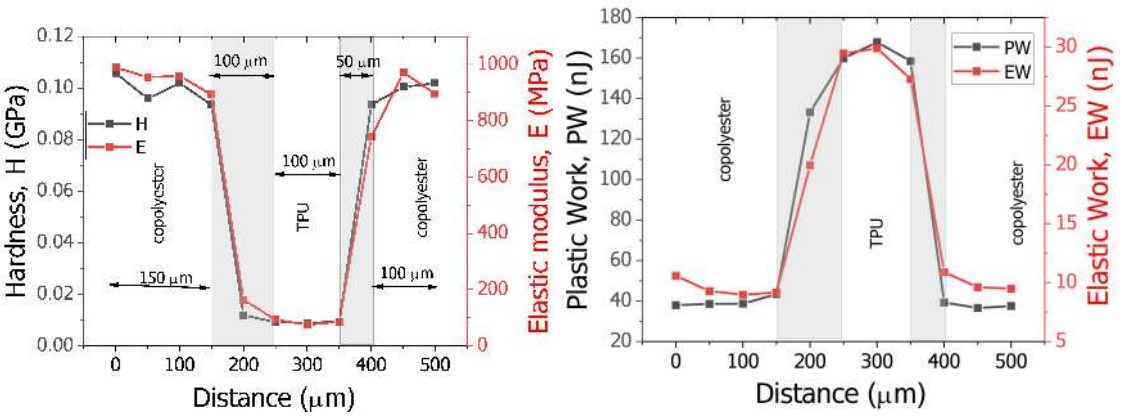
forming process. Notably, the interface between the two materials appears to become more sharply defined after thermoforming, suggesting that the thermal cycle does not promote further intermixing but may instead stabilise or consolidate the existing interfacial structure.

The nanoindentation parameters of the individual layers before and after thermoforming are summarised in Table 4.20 for Zendura FLX and Prismlab C (PCTG). For the outer copolyester layers of Zendura FLX, thermoforming induced a modest reduction in Vickers hardness from 11.3 to 9.4 HV and in elastic modulus from 1007 to 956 MPa, accompanied by a slight decrease in elastic energy and a marginal increase in plastic energy, consistent with the general softening trend observed in PETG and PCTG materials by DMTA previously depicted. The sTPU inner core showed negligible changes across all parameters — hardness remained essentially unchanged (0.98 against 0.92 HV) and elastic modulus varied by less than 3 % (87.7 against 89.9 MPa) — confirming that the thermal processing conditions employed do not alter the mechanical response of the elastomeric core.

| layer | State | HV (-) | Elastic modulus (MPa) | Elastic energy (nJ) | Plastic energy (nJ) |
|---|---|---|---|---|---|
| External | NT | 11.3 | 1007.3 | 9.15 | 36.3 |
| External | T | 9.4 | 956 | 8.4 | 42.1 |
| Inner | NT | 0.98 | 87.7 | 30.3 | 159.2 |
| Inner | T | 0.92 | 89.9 | 27.3 | 164.2 |
| Prismlab C | NT | 10.7 | 1000.2 | 8.8 | 37.9 |
| Prismlab C | T | 10.1 | 977.4 | 8.5 | 40.7 |

Table 4.20: Nanoindentation properties of the individual layers in Zendura FLX and Prismlab C in as-received (NT) and thermoformed (T) states. External are the outer layers in Zendura FLX and Inner are the inner core layer in Zendura FLX

Later, the mechanical properties of Spark, Invisalign, and Filtzwiesser in a thermoformed state were obtained.




(a) Hardness and elastic modulus

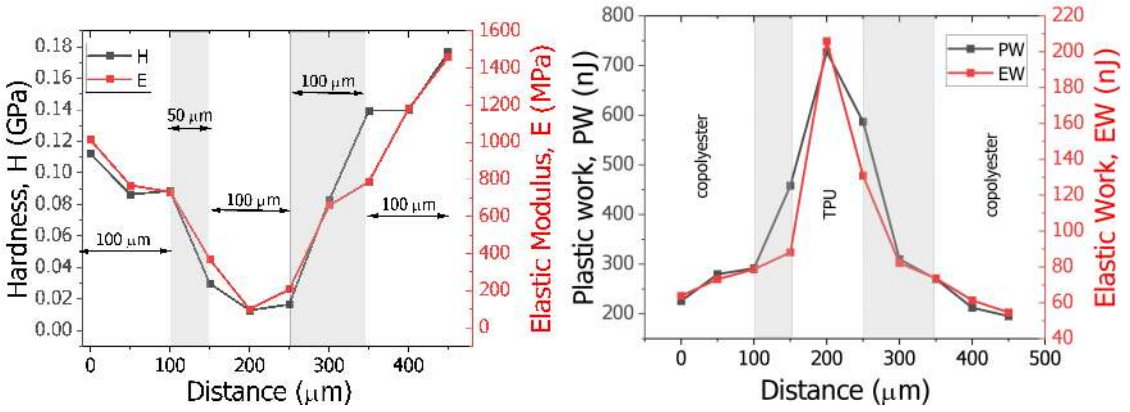
(b) Plastic and elastic energy

Figure 4.30: Indentation results along the cross-section of Spark clear aligner

A comparison of Figures 4.30 and 4.29 reveals a strong similarity in the layer-level mechanical properties of Spark and Zendura FLX: the outer copolyester layers and the sTPU inner core of both materials yield comparable values of hardness, elastic modulus, elastic energy, and plastic energy (Tables 4.20 and 4.21), suggesting a closely related or equivalent material chemistry in both cases. The most notable difference between the two profiles lies in the thickness of the interlaminar transition zone, which is substantially narrower in Spark than in Zendura FLX. This discrepancy is unlikely to reflect an intrinsic difference in the polymer chemistry or co-extrusion conditions of the two materials and is more plausibly attributable to differences in the thermoforming protocol — including mold geometry, heating time, or vacuum pressure — or to differences in the degree of biaxial deformation experienced by the interface region during forming over the dental model. Since only the thermoformed state was available for Spark, a direct comparison with the as-received condition analogous to that performed for Zendura FLX could not be carried out.

| Layer | State | HV (-) | Elastic modulus | Elastic energy (nJ) | Plastic energy (nJ) |
|---|---|---|---|---|---|
| External | T | 10.2 | 944.9 | 9.1 | 38.6 |
| Inner | T | 0.95 | 87.9 | 29.5 | 161.5 |

Table 4.21: Nanoindentation properties of the individual layers in Spark material in thermoformed (T) state




(a) Hardness and elastic modulus

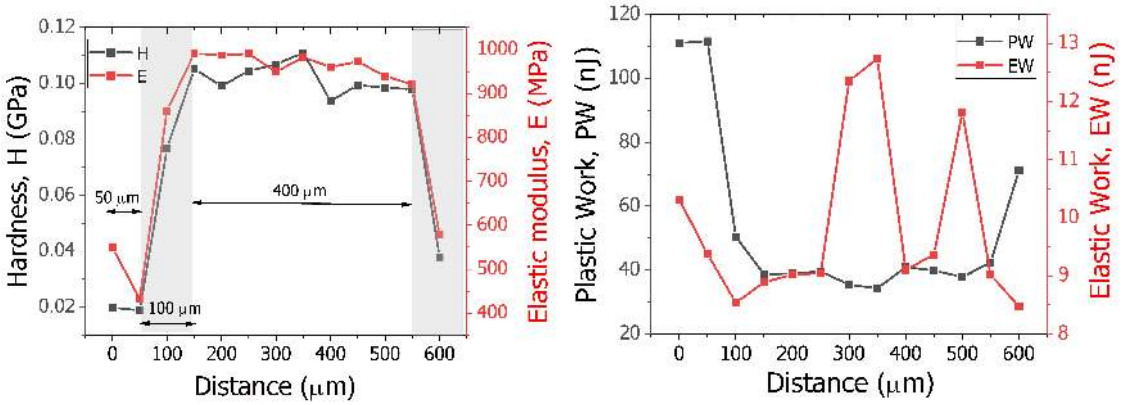
(b) Plastic and elastic energy

Figure 4.31: Indentation results along the cross-section of Filtzwiesser clear aligner

| Layer | State | HV (-) | Elastic modulus | Elastic energy (nJ) | Plastic energy (nJ) |
|---|---|---|---|---|---|
| External | T | 11.0 | 1154.1 | 7.7 | 40.1 |
| Inner | T | 0.14 | 3.93 | 82.9 | 184.7 |

Table 4.22: Nanoindentation properties of the individual layers in Filtzwiesser material in thermoformed (T) state

The nanoindentation results for Filtzwiesser (Figure 4.31 and Table 4.22) present a notably different profile from those of Zendura FLX and Spark. The outer layers exhibit a higher elastic modulus (1154 MPa) and a slightly lower elastic energy than the copolyester layers of the other multilayer materials, consistent with the manufacturer's specification that the hard outer layers of this product are fabricated from pure PETG rather than a copolyester such as PCTG. This is in agreement with the bulk characterisation results presented earlier in this work, where PETG was found to be stiffer than PCTG but to exhibit a narrower elastic range — a trade-off that is directly reflected in the higher modulus and reduced elastic energy storage capacity observed here at the layer level. The inner sTPU core of Filtzwiesser presents a markedly lower elastic modulus (3.93 MPa) and hardness (0.14 HV) than the corresponding core layers in Zendura FLX (87.7-89.9 MPa and 0.92-0.94 HV) and Spark (87.9 MPa and 0.95 HV), indicating that the elastomeric component employed in this architecture is substantially softer. This is consistent with a lower Shore hardness grade of the TPU used, and suggests that the Filtzwiesser multilayer is designed to deliver a considerably more compliant mechanical response than the other multilayer systems characterized in this work — a design choice with direct implications for the magnitude and distribution of orthodontic forces generated during clinical use.




(a) Hardness and elastic modulus

(b) Plastic and elastic energy

Figure 4.32: Indentation results along the cross-section of Invisalign clear aligner

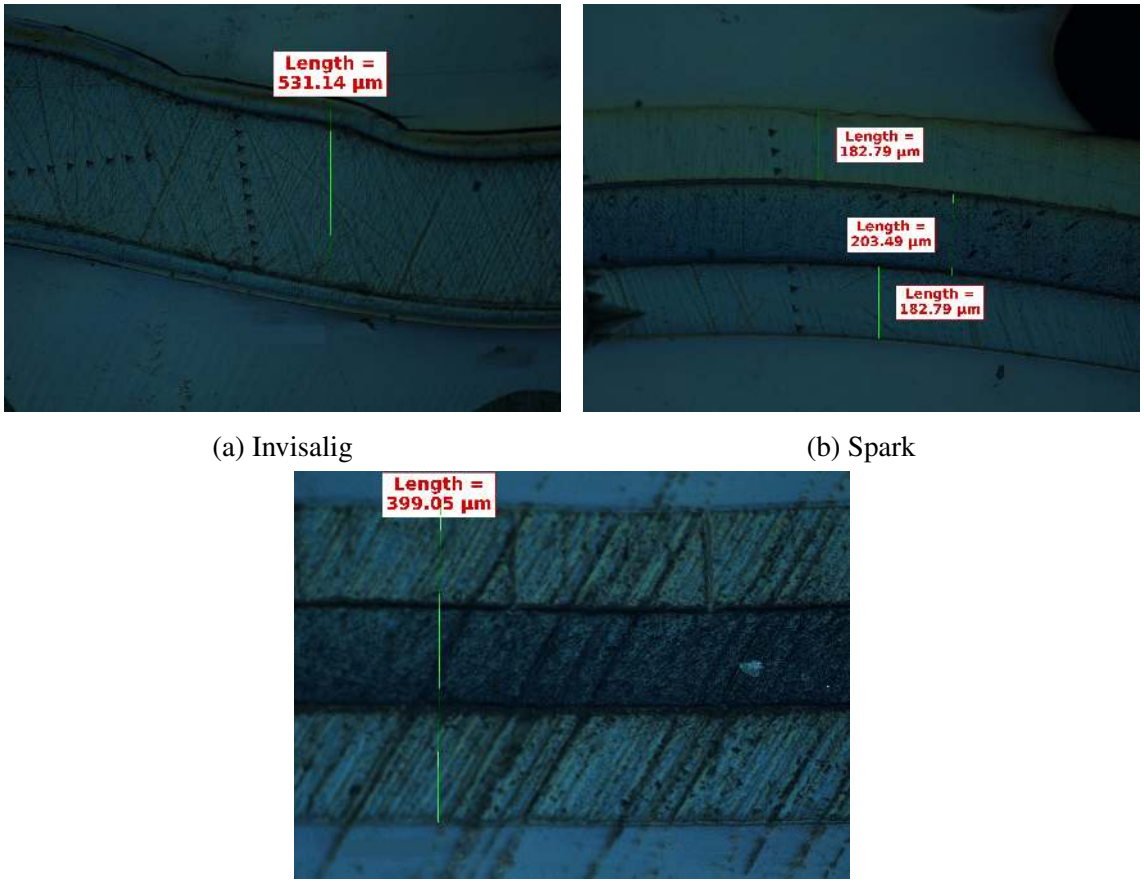
| Layer | State | HV (-) | Elastic modulus | Elastic energy (nJ) | Plastic energy (nJ) |
|---|---|---|---|---|---|
| External | T | 1.5 | 462.8 | 7.2 | 138.9 |
| Inner | T | 10.5 | 993.8 | 8.7 | 39.8 |

Table 4.23: Nanoindentation properties of the individual layers in Invisalign material in thermoformed (T) state

As shown in Figure 4.32 and Table 4.23, the multilayer architecture of Invisalign differs fundamentally from those of Zendura FLX, Spark, and Filtzwiesser. In contrast to the latter materials, which follow a hard-soft-hard arrangement with a compliant sTPU core sandwiched between stiff copolyester outer layers, Invisalign presents an inverted configuration: a hard, stiff inner core (elastic modulus 994 MPa, HV 10.5) enclosed by relatively soft and thin outer layers (elastic modulus 463 MPa, HV 1.5). This architecture is consistent with the ATR-FTIR identification of the outer surface as a polyester-based MDI-type TPU. surrounding a stiffer core that may be composed of either PETG, PCTG, or hTPU. It should be noted, however, that the mechanical values obtained for the outer layers are likely to be influenced by the proximity of the much stiffer inner core — a well-known substrate effect in nanoindentation of thin compliant films on rigid substrates [289], [290] — and may therefore overestimate the true modulus and hardness of the outer TPU layer. Given the limited thickness of the outer layers, a fully reliable and substrate-independent characterization of their intrinsic properties was not achievable under the indentation conditions employed.

Further evidence for this architectural inversion is provided by the cross-sectional images acquired after indentation (Figure 4.33). Residual indentation marks are clearly visible at the center of the Invisalign specimen, corresponding to the stiff inner core, whereas no permanent marks are discernible in the outer layers, consistent with the viscoelastic recovery characteristic of soft TPU under the applied loads. This behavior contrasts with Spark, where residual marks are confined to the hard outer copolyester layers and are ab-

sent in the soft inner core. The differential persistence of indentation marks thus provides a qualitative but visually compelling confirmation of the layer-level mechanical hierarchy in each multilayer architecture.




(a) Invisalig

(b) Spark

(c) Zendura FLX

Figure 4.33: Cross-section on thermoformed commercial multilayers

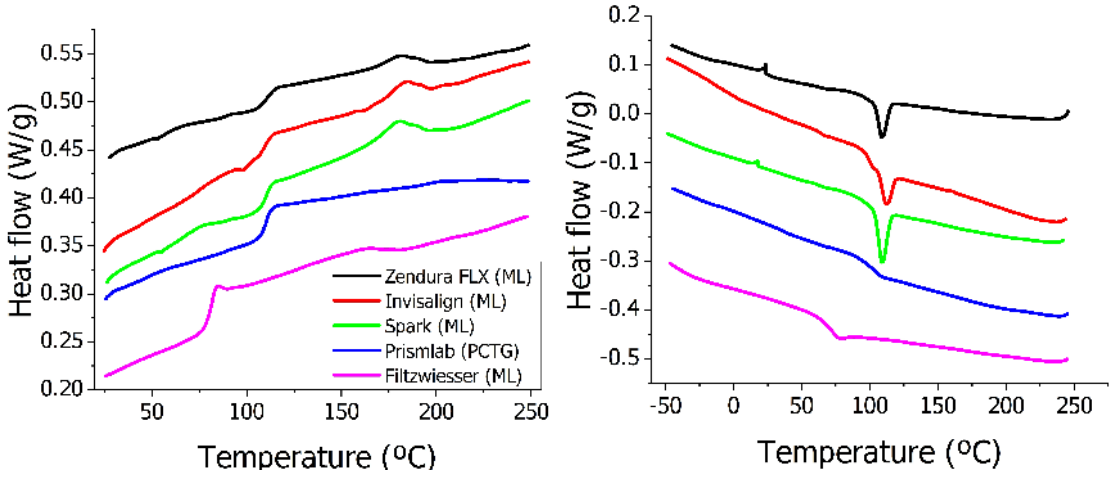
To provide a more complete thermomechanical characterization of the multilayer materials in their clinically relevant state, DSC traces were acquired directly from clear aligner specimens following the same procedure described previously. Specimens were extracted from the molar region of each aligner — the same location used for nanoindentation — in order to ensure spatial consistency across the two characterization methods. Only the traces obtained directly from the finished aligners are presented here (Figure 4.34), as these represent the material in its thermoformed state as clinically deployed.

Inspection of Figure 4.34 reveals a high degree of similarity in the thermal response across all materials, with each trace broadly consistent with the corresponding DSC profiles obtained from the thermoformed sheet specimens discussed previously (Figure 4.21). The principal distinction is observed in Filtzwiesser, where all thermal transitions — including the glass transition of the PETG outer layer and the melting and crystallization exotherms associated with the sTPU inner core — are shifted to lower temperatures rel-

ative to the other multilayer materials. This systematic depression of transition temperatures may reflect differences in the molecular weight distribution, degree of crystallinity, or specific grade of the constituent polymers employed in the Filtzwiesser architecture, and is consistent with the lower elastic modulus of its sTPU core identified by nanoindentation. It should also be mentioned that the appearance of an aging peak is typical in PETG-based materials.

In the multilayers incorporating PCTG outer layers, no aging peak was observed, and the crystallization peak has an sTPU origin. As may be observed in the traces of Prismlab C, no crystallization or aging peaks are observed in PCTG materials after thermoforming.

Of particular interest is the observation that, despite the fundamentally different layer architecture of Invisalign in contrast to Zendura FLX and Spark, its DSC trace is virtually indistinguishable from those of the latter two materials. The positions and relative magnitudes of the glass transition, melting endotherm, and crystallization exotherm are essentially coincident across the three products. This thermal equivalence raises the hypothesis that Invisalign, Zendura FLX, and Spark may share the same constituent polymer chemistries — a PCTG-type copolyester and a similar grade of MDI-based sTPU — but differ exclusively in the spatial arrangement of those components within the laminate architecture. If confirmed, this would imply that the distinct clinical force profiles associated with these products arise not from differences in material chemistry but solely from differences in layer configuration, a finding with significant implications for the rational design of next-generation multilayer aligner materials.



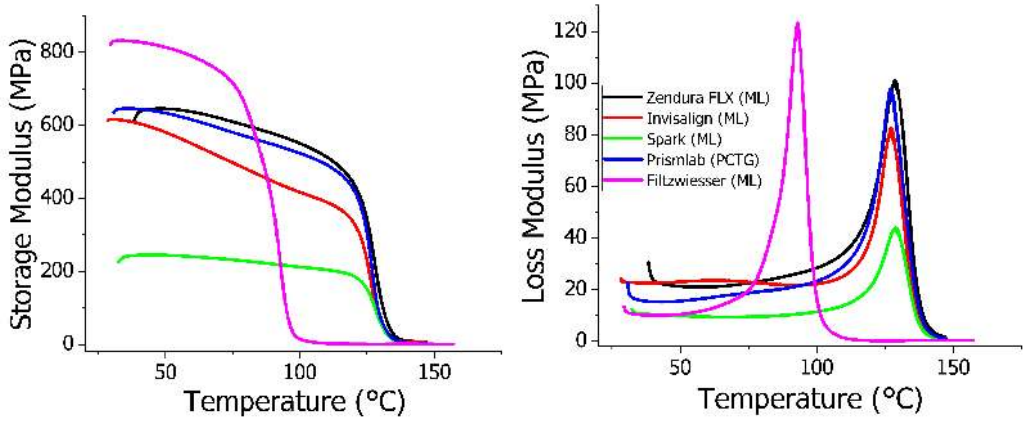

(a) First heating (b) Cooling

Figure 4.34: DSC traces on commercial clear aligners

The DMTA curves acquired directly from finished aligner specimens are presented in Figure 4.35. For several materials — notably Prismlab C, Zendura FLX, and Filtzwiesser — the storage modulus values at oral temperature differ substantially from those obtained on flat thermoformed specimens in the controlled characterization (see Figure 4.5). These discrepancies are attributable to the inherent geometric constraints imposed by the aligner

geometry rather than to genuine differences in material properties. Unlike the flat or near-flat specimens used in the as-received and controlled thermoforming characterization, finished aligners present curved, anatomically irregular surfaces whose geometry is dictated by the dental model over which they were formed. Also due to the small dimensions of some aligner samples, they had to be cut by hand instead of by laser. The non-planarity of the specimen, the restricted dimensions available for clamping, and the hand cutting introduce systematic errors in the measured bending response, compromising the reliability of the absolute modulus values obtained.

For the aligners thermoformed by Secret Aligner (Zendura FLX, Prismlab C, and Filtzwiesser), the dental model was selected deliberately to maximize the available flat area in the anterior incisor region, thereby improving specimen geometry for DMTA testing. This level of control was not possible for the commercially manufactured Invisalign and Spark aligners, whose geometry was fixed. By coincidence, the incisor dimensions of the Invisalign aligner proved sufficiently large to yield specimens of acceptable geometry, whereas this was not the case for Spark. This difference in specimen geometry is the most likely explanation for the large discrepancy in storage modulus at oral temperature observed between Zendura FLX and Spark — two materials whose nanoindentation and DSC data indicate nearly identical constituent polymer chemistries — and underscores the critical importance of specimen geometry standardization in the mechanical characterization of finished aligner appliances. In contrast, the Tg appears to be the parameter least sensitive to specimen geometry and preparation artifacts, as it is determined from the temperature position of the loss modulus peak, rather than from the absolute magnitude of any of the modulus. Accordingly, consistent Tg values were obtained across all specimens: Spark, Prismlab C, Zendura FLX, and Invisalign all exhibited a Tg at approximately 120 ºC, while Filtzwiesser presented a markedly lower Tg of approximately 85 ºC, further corroborating the distinct polymer chemistry of its outer layer identified by nanoindentation and DSC.




(a) Storage modulus (b) Storage modulus

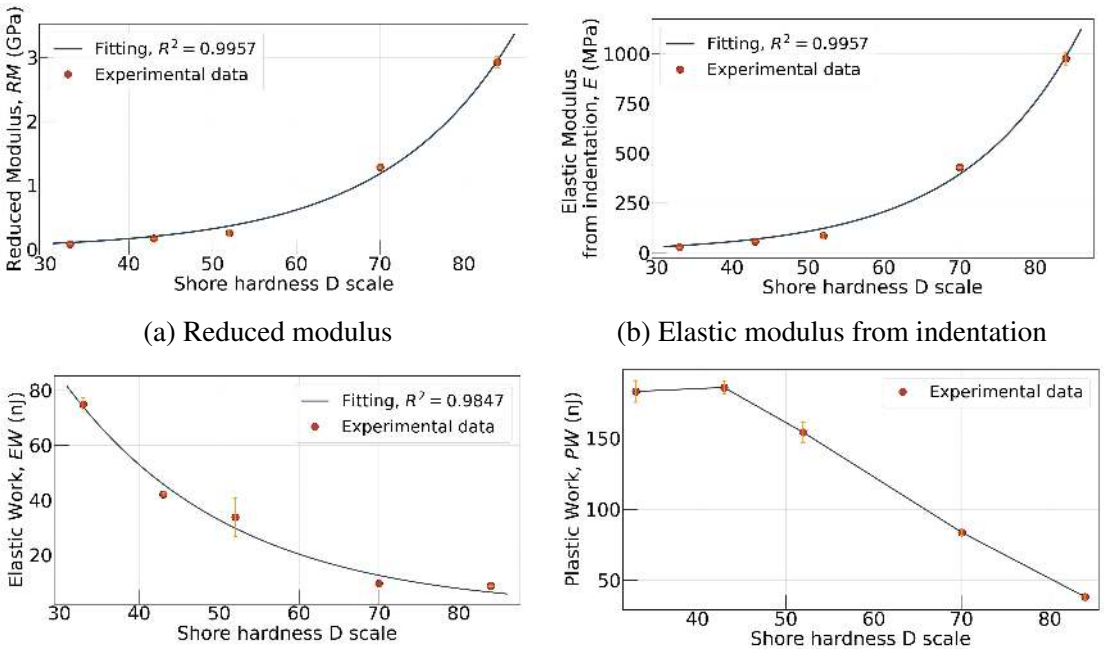
Figure 4.35: DMTA curves in commercial clear aligners

Transparency was attempted to be measured on these devices. However, the topography and small dimensions of the devices render the measurements too noisy to be analyzed. Hence, the characterization of materials directly from clear aligners by conventional methods (DMTA, tensile testing, and transparency) should be completely avoided.

### 4.7.4. Results on raw materials

To facilitate the unambiguous identification of the constituent layers in the multilayer aligner materials, raw sheet specimens of the relevant polymers were independently characterized by nanoindentation. For the hard outer layer candidates, the characterization was restricted to a representative commercial PETG, a hTPU (TPU 84D), and a PCTG (Tritan MP100). This restriction was justified on two grounds: first, the hard outer layer identity was already well supported by the ATR-FTIR and DSC data; and second, the nanoindentation-derived reduced moduli of PETG and PCTG grades are sufficiently similar that their differentiation by indentation alone is not reliable — consistent with their comparable elastic moduli measured by DMTA and uniaxial tensile testing reported in Chapter 4. For the soft inner layer candidates, a series of TPU materials spanning a range of Shore hardness values on the D scale were tested, as the large mechanical contrast between sTPU grades makes nanoindentation the most direct and discriminating technique for their identification and quantitative characterisation within the multilayer cross-section

A secondary but practically relevant objective of this characterization was to establish a quantitative relationship between the nanoindentation-derived mechanical parameters and the Shore hardness of the TPU materials, as reported in supplier data sheets, in order to provide a predictive tool for tailoring the mechanical properties of future multilayer architectures without the need for full nanoindentation characterization at the design stage. The resulting relationships are presented in Figure 4.36, where the indentation elastic modulus, reduced modulus, and elastic energy of the TPU materials are plotted as a function of their Shore D hardness. A clear and consistent dependence is observed across all three elastic parameters, with the relationship well described by the equation 4.7. This correlation confirms that Shore hardness, despite being a quasi-static surface property measured under standardized but largely empirical conditions, is a reliable proxy for the elastic character of TPU materials in the hardness range relevant to multilayer aligner design, and may therefore be used to guide material selection from commercially available grades.




(a) Reduced modulus

(b) Elastic modulus from indentation

(c) Elastic work

(d) Plastic work

Figure 4.36: Indentation properties of TPU

The reduced modulus, the elastic modulus, and the elastic released energy were fitted using the equation 4.7, in which A represents the fitted variable, $A_0$ represents the value of an ideal TPU only composed by soft segments, and b is the sensibility of the variable to the shore hardness variation.

$$A = A_0 * exp(b * H_D) \tag{4.7}$$

The exponential dependence of the elastic parameters on Shore hardness can be rationalized in terms of the microphase-separated structure of thermoplastic polyurethanes. A higher Shore hardness reflects a higher concentration of hard segments, which in isolation would be expected to increase the elastic modulus and elastic energy storage capacity in a roughly linear fashion. However, as the hard segment content increases, the hard segment domains become progressively more interconnected and thermodynamically stable, forming a more continuous and load-bearing network. It is this cooperative reinforcement of the hard segment microstructure [83], [291] — rather than a simple additive contribution of individual hard segments — that accounts for the observed exponential rather than linear dependence of the elastic properties on Shore hardness.

The fitted parameters of the equation 4.7 are as follows: for the elastic modulus, $A_0$ = 2.4 MPa and b = 0.07; for the released elastic energy $A_0$ = 307.4 nJ and b = -0.04. The negative exponent for elastic energy is consistent with the progressive transition from a predominantly elastic to a more viscoelastic and dissipative response as the hard segment content decreases. Extrapolating across the full commercial range of TPU grades, the fitted relationship predicts an elastic modulus spanning approximately from 3 MPa for the softest TPU available to 1000 MPa for the TPU 84D characterized here, a range of

nearly three orders of magnitude. This remarkable span highlights the versatility of the TPU chemistry as a design parameter for multilayer aligner architectures, enabling the mechanical response of the soft core layer to be tuned continuously across a wide window by simple selection of the appropriate Shore hardness grade.

The plastic energy recovered from the nanoindentation unloading curve provides an indirect measure of the capacity of the material to accommodate permanent deformation — that is, of its ductility in the indentation sense. Consistent with the general behavior of thermoplastic polymers, a monotonically decreasing relationship is observed between plastic energy and Shore hardness: as the material becomes harder and stiffer, its ability to undergo irreversible deformation diminishes, and the relationship is well approximated as linear over the hardness range investigated.

| Material | HV (-) | Elastic modulus (MPa) | elastic energy (nJ) | Plastic energy (nJ) |
|---|---|---|---|---|
| TPU85A | 0.6 | 26.9 | 74.9 | 183.0 |
| TPU93A | 0.7 | 56.0 | 42.1 | 185.8 |
| TPU52D | 0.9 | 85.0 | 33.8 | 154.2 |
| TPU70D | 1.0 | 95.5 | 9.7 | 83.6 |
| TPU84D | 10.7 | 975.3 | 8.8 | 38.2 |
| MP100 | 10.1 | 952.3 | 8.4 | 45.8 |
| PETG | 11.7 | 1220.5 | 7.5 | 37.7 |

Table 4.24: Indentation results on raw materials. HV stands for vickers hardness obtained from nanoindentation

As shown in Table 4.24 and Figure 4.36, the nanoindentation parameters of TPU materials vary substantially and systematically with Shore hardness grade, confirming that the elastic modulus, elastic energy, and hardness of the soft layer in a multilayer architecture can be tuned over a wide range through appropriate selection of the TPU grade. In contrast, the copolyester materials characterized here — PETG and PCTG (Tritan MP100) — exhibit very similar indentation properties, with limited scope for mechanical variation within this class of materials. The principal difference between the two is that PETG presents a higher elastic modulus and hardness (1221 MPa, HV 11.7) relative to PCTG (952 MPa, HV 10.1), while PCTG accommodates a somewhat greater degree of plastic deformation, consistent with its more compliant character identified in the bulk tensile and DMTA characterization presented earlier.

Applying the exponential relationship of the equation 4.7 inversely — that is, using the elastic modulus measured by nanoindentation on the cross-sectional layer profiles to back-calculate the equivalent Shore hardness — it is possible to identify the TPU grade of the soft constituent layers in each commercial multilayer material. The inner sTPU core of both Zendura FLX and Spark was identified as equivalent to a Shore D hardness of approximately 50, in both the as-received and thermoformed states for Zendura FLX and in the thermoformed state for Spark, consistent with the near-identical elastic moduli ob-

served for the inner layers of these two materials (Tables 4.20 and 4.21). In Filtzwiesser, the inner core yielded an equivalent Shore hardness of approximately 5 Shore D (corresponding to approximately 30 Shore A), confirming that this material incorporates a substantially softer elastomeric grade — more than an order of magnitude lower in modulus — than the sTPU used in Zendura FLX and Spark.

Finally, the outer TPU layers of Invisalign yielded an equivalent Shore hardness of approximately 75 D; however, as discussed above, this value must be interpreted with caution. The outer layers of Invisalign are comparatively thin, and the measured indentation response is likely to be confounded by the substrate effect of the surrounding epoxy resin and, more critically, by the proximity of the stiff inner thermoplastic core, both of which would tend to inflate the apparent modulus and hardness of the outer layer beyond its intrinsic values. The true Shore hardness of the outer TPU is therefore expected to be lower than the back-calculated estimate. Regarding the inner core of Invisalign, the nanoindentation results — which yield a modulus and hardness closely matching those of PCTG (Table 4.24) — together with the Tg of approximately 120 ºC identified by DMTA, which is characteristic of PCTG-type copolyesters, collectively suggest that the inner layer is most probably composed of a PCTG or PCTG-like copolyester rather than a standard PETG.

### 4.7.5. Conclusions on the commercial testing of materials by nanoindentation

The results presented in this section demonstrate that nanoindentation on polished cross-sections is a viable and informative method for the layer-level mechanical characterization of commercial multilayer aligner materials, enabling the unambiguous identification and quantification of each constituent layer without the need for chemical isolation or destructive separation. As expected from their respective chemistries, copolyesters and TPU materials exhibit markedly different mechanical signatures. The copolyester layers present elastic moduli in the range of 950-1200 MPa and Vickers hardness values of 9.5-11.5 HV, characteristic of glassy thermoplastics operating well below their glass transition temperature at the measurement conditions employed. The TPU layers, by contrast, span a substantially wider range — elastic moduli from approximately 4 to 460 MPa and hardness from 0.15 to 1.5 HV — reflecting the broad tunability of polyurethane mechanical properties through hard segment content. Despite their lower modulus, TPU layers exhibit a considerably greater capacity for elastic energy storage, with values of 7-82 nJ compared to 7-9 nJ for the copolyesters (PETG and PCTG respectively). The most pronounced difference between the two material classes, however, lies in plastic energy absorption: copolyesters dissipate 36-42 nJ per indentation cycle, whereas TPU materials absorb 160-180 nJ, confirming the far greater capacity of the elastomeric phase to accommodate irreversible deformation — a property directly relevant to the compliance and comfort of the aligner during clinical use.

Within the copolyester class, subtle but consistent differences are observed between

PETG and PCTG. PETG presents a higher elastic modulus and hardness but lower elastic and plastic energy relative to PCTG, indicating that PETG is stiffer but less capable of both storing and dissipating mechanical energy. From a clinical design perspective, PCTG therefore represents a more balanced choice for the hard outer layers of multilayer aligners, combining adequate stiffness with a greater capacity for energy management under the cyclic loading conditions of orthodontic use.

The characterization of raw sheet materials served primarily to establish a reference framework for identifying the TPU grade of the soft constituent layers in commercial multilayer products through comparison with a calibrated series of TPU materials of known Shore hardness. This approach identified the inner sTPU core of both Zendura FLX and Spark as equivalent to approximately Shore 50D, while the inner core of Filtzwiesser was found to correspond to a substantially softer grade of approximately Shore 80-85A — consistent with its markedly lower indentation modulus and the more compliant force profile expected from this architecture. Th e outer TPU layers of Invisalign were tentatively identified as equivalent to approximately Shore 70D, though this estimate carries greater uncertainty due to the substrate effect of the stiff inner core on the thin outer layer measurements. The inner core of Invisalign, whose indentation modulus and Tg are consistent with a PCTG-type copolyesters unlikely to be composed of an ultra-hard TPU grade: the hard segment-dominated TPU 84D presents a Tg of approximately 93 ºC, which is clearly distinguishable from the observed value and effectively rules out a high-hardness TPU as the inner core material. The combination of nanoindentation modulus and thermal data therefore points unambiguously to a PCTG or PCTG-like copolyester as the most probable composition of the Invisalign inner core.

A further contribution of the present work concerns the compositional characterization of Invisalign, the most widely used material in clear aligner therapy. A degree of confusion persists in the literature regarding its chemical nature: several studies describe Invisalign as a TPU-based material [26], [250], [287], [292], an assumption that is historically grounded in the composition of the first generation of Invisalign material (called EX30), which was a single layer polyurethane [26], [58], [292] and was in use from 1998 until 2013; or on surface spectroscopy data[250], [287]. However, this does not reflect the correct formulation of the Invisalign material in use nowadays (called Smart Track), which was introduced in 2013 [58]. The present results, obtained by combining cross-sectional nanoindentation and DMTA, unambiguously identify the architecture of SmartTrack as sTPU/PCTG/sTPU: a stiff PCTG-type copolyester inner core enclosed by thin soft TPU outer layers. This inverted architecture — relative to the PCTG/sTPU/PCTG configuration of most other commercial multilayer materials — has been independently corroborated by a recent study employing Raman microscopy on the cross-section of SmartTrack, which reported a polyester core with polyurethane outer skins [184]. The present work extends this finding by providing the first quantitative mechanical characterization of each individual layer and by identifying the inner core as a PCTG-type copolyester on the basis of both its indentation modulus and its Tg of nearly 120 ºC —

parameters that are fully consistent with PCTG and inconsistent with a high-hardness TPU, whose Tg would be expected at approximately 95 °C.

A secondary but practically significant finding of the raw material characterization is that the full mechanical response of a multilayer aligner architecture — including its stiffness, elastic energy storage, and plastic energy dissipation — can, in principle, be predicted and tuned based solely on the Shore hardness of the soft phase, using the exponential relationships established in equation 4.7. This provides a rational and experimentally grounded design tool for the development of next-generation multilayer aligner materials, enabling the target mechanical properties of the composite laminate to be mapped directly onto commercially available TPU grades without the need for iterative nanoindentation characterization at each design stage.

More generally, the layer-resolved characterization strategy presented here indicates that the thermomechanical behavior of multilayer aligner architectures are not determined solely by material chemistry, but can instead be deliberately engineered by design. By choosing component layers with known mechanical properties and tuning their relative thicknesses, the overall mechanical response of the laminate can be shaped in a predictable and controllable way. This, in turn, creates the possibility of a truly customized design framework for multilayer aligners, in which the target force profile, elastic recovery, and energy dissipation behavior are first specified in the clinical setting and then directly converted into a laminate architecture — defined by layer chemistry, Shore hardness, and thickness ratios — using the quantitative relationships derived in this study.

## 4.8. Effect of Water

### 4.8.1. Introduction

Clear aligners are exposed to a highly variable humidity environment during clinical use, with relative humidity (RH) ranging from approximately 100 % RH in a healthy oral cavity to as low as 30 % RH in patients with xerostomia (dry mouth), smokers, or in dry ambient conditions [293], [294]. Paradoxically, wearing a clear aligner has been reported to induce or exacerbate xerostomia [61], [62], [63], and the device is never fully immersed in saliva during normal use [295]. Although patients are instructed to remove their aligners while eating, drinking is commonly performed with the appliance in situ, and patients are routinely directed to rinse or clean the device with tap water [296], [297].

Water and water-based beverages are known to affect both the mechanical and chemical properties of clear aligner materials. Water absorption generally leads to a reduction in mechanical properties, most notably in stress relaxation behavior [71], [171], as water molecules act as a plasticizer by penetrating the polymer matrix and increasing the mobility of the polymer chains, thereby reducing the effective glass transition temperature and softening the material. The opposing effect — hygrothermal aging — is also well

documented in the aligner literature [143], [171], [298]. In this case, the water-induced enhancement of chain mobility allows the polymer chains to reorganize towards thermodynamically more stable conformations, resulting in a stiffer, more brittle, and more opaque material over time. Both effects can operate simultaneously, and their relative dominance depends on the material chemistry, the degree of water uptake, and the temperature and duration of exposure.

In terms of water absorption capacity, TPU-based materials are generally considered the most hygroscopic polymers in use for clear aligner fabrication, owing to the presence of highly polar urethane groups that readily form hydrogen bonds with water molecules, while PETG is regarded as among the least absorbing [71], [88], [299]. This consensus has, however, been recently challenged by studies reporting that certain TPU formulations exhibit among the lowest water absorption values of any dental polymer, surpassed only by PEEK [69], [300], [301], suggesting that the hygroscopic behavior of TPU is more grade- and formulation-dependent than previously assumed.

Despite the well-documented influence of thermoforming on the surface roughness and microstructure of aligner materials [259], [260], the effect of thermoforming on the subsequent water sorption behavior has not been systematically investigated across the range of material chemistries currently available. The only available data point is an incidental observation of a modest increase in water absorption in thermoformed PETG [259]; no equivalent data exist for hTPU, PP, or multilayer architectures. Since thermoforming substantially increases the surface roughness of aligner materials — with increases in Ra of over 1200% reported for PETG [260] — it may be expected to increase the effective surface area available for water contact and thereby accelerate water uptake kinetics, an effect that has not been previously characterized. The present work therefore characterizes water sorption in both the as-received and thermoformed states for all materials investigated, enabling a direct assessment of thermoforming-induced changes in hygroscopic behavior across material classes.

An additional motivation for this study arose from a practical observation by Secret Aligner during aligner manufacturing. Thermoplastic sheets for clear aligner fabrication are conventionally supplied as pre-cut sheets stored in sealed pouches protected from light and environmental exposure, with manufacturers recommending use within 15 minutes of opening and storage under controlled temperature and humidity conditions. Secret Aligner reported that certain PETG formulations that had not been properly protected during shipping or storage became impossible to cut by laser, exhibiting extreme brittleness — fracturing catastrophically into fragments upon contact, in a manner reminiscent of inorganic glass. This behavior, which was not observed in any other material class, is consistent with advanced physical aging of the PETG matrix driven by moisture-induced chain mobility during improper storage and provides a clinically and industrially relevant motivation for understanding the water sorption behavior of these materials under controlled conditions.

The aim of this section is to characterize the water sorption of commercial thermoplastic materials for clear aligner fabrication and to evaluate its effects on the thermal properties — principally the glass transition temperature — and the chemical structure of the materials, as assessed by DSC and ATR-FTIR respectively. DMTA characterization after water immersion was not performed: the absence of significant Tg shifts in the DSC traces of the immersed specimens — with the exception of Tristar S — indicated that the immersion-induced microstructural changes were of insufficient magnitude to be expected to produce detectable differences in the bulk thermomechanical response as probed by DMTA, rendering this analysis uninformative under the conditions employed.

### 4.8.2. Water absorption materials and methods

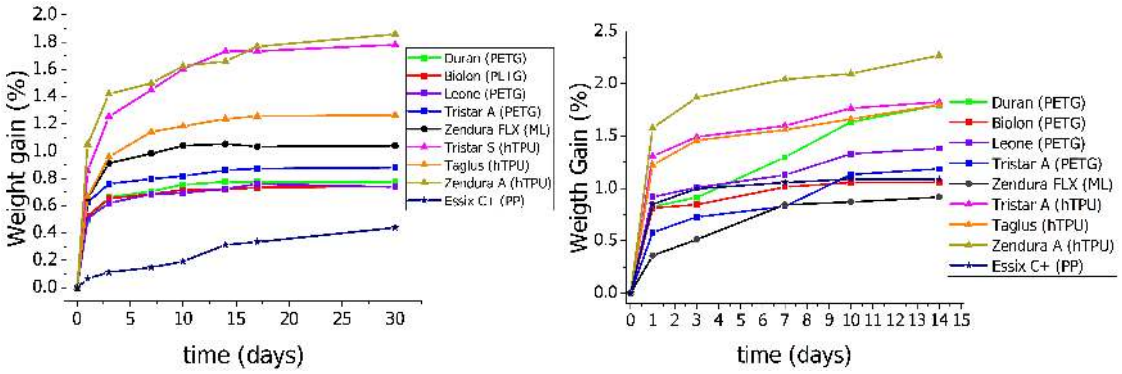
The materials evaluated in this section, the sample preparation protocols, and the complete experimental procedures for water absorption testing, post-immersion ATR-FTIR characterization, DSC, and DMTA are described in detail in Chapter 3, Section 3.7. Water absorption experiments were conducted on both as-received and thermoformed specimens.

### 4.8.3. Water absorption results

Figure 4.37 shows the mass uptake of all materials as a function of immersion time in distilled water at 37 °C, in both the as-received and thermoformed states. All materials exhibited a monotonically increasing mass gain that reached a well-defined saturation plateau after approximately 10 days, confirming that the observed weight increase reflects water absorption rather than the dissolution of the polymer matrix. The absorption curves were fitted using the fractional diffusion model of Ritger and Peppas [156], [157], whose physical basis and mathematical derivation are provided in Appendix A, sections A.1.3 and A.1.4:

$$M(t) = M_0 + \frac{\Delta M_\infty}{\Gamma(1+\alpha)} \left(\frac{t}{\tau}\right)^\alpha \tag{4.8}$$

The fitted parameters for all materials in both the as-received and thermoformed states are reported in Table 4.25. All materials presented $\alpha$ values between 0.09 and 0.65 and characteristic times ($\tau$) between 20 and 40 days, confirming their classification as low-permeability plastics with respect to water [302]. The physical interpretation of $\alpha$ and $\tau$ is discussed in detail in Appendix A, section A.1.4.




(a) As-received material

(b) Thermoformed materials

Figure 4.37: Water sorption in clear aligner materials

Among the as-received materials (Table 4.25), hTPU grades exhibited the highest saturation weight gains (1.26-1.85%), followed by the ML Zendura FLX (1.08%) and the PETG materials (0.74-0.89%). Essix C+ (PP) showed the lowest absorption in the as-received state (0.44%) but the largest relative increase after thermoforming, reaching 1.4%, a more than threefold increase attributable to the partial thermo-oxidative degradation identified by TGA during the thermoforming cycle, which increases the surface porosity and accessibility of the polymer matrix to water. Thermoforming systematically increased water uptake across all material classes (Table 4.25, consistent with the substantial increase in surface roughness induced by the thermoforming process [260], which exposes a greater effective surface area to water contact.

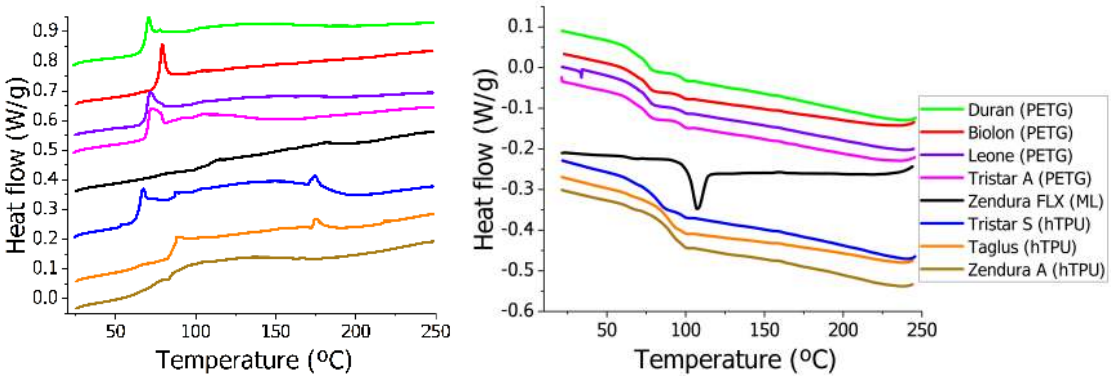
| Material | $\Delta M_\infty$ NT (%) | $\tau$ NT (days) | $\alpha$ NT | $\Delta M_\infty$ T (%) | $\tau$ T (days) | $\alpha$ T |
|---|---|---|---|---|---|---|
| Duran (PETG) | 0.74 | 27 | 0.12 | 1.51 | 21 | 0.34 |
| Biolon (PETG) | 0.75 | 32 | 0.09 | 1.10 | 20 | 0.13 |
| Leone (PETG) | 0.76 | 37 | 0.11 | 1.38 | 24 | 0.18 |
| Tristar A (PETG) | 0.89 | 34 | 0.09 | 1.18 | 24 | 0.25 |
| ZFLX (ML) | 1.08 | 35 | 0.12 | 1.12 | 16 | 0.35 |
| Tristar S (hTPU) | 1.76 | 33 | 0.21 | 1.81 | 24 | 0.12 |
| Taglus (hTPU) | 1.26 | 28 | 0.17 | 1.79 | 25 | 0.15 |
| Zendura A (hTPU) | 1.85 | 40 | 0.16 | 2.25 | 25 | 0.14 |
| Essix C+ (PP) | 0.44 | 35 | 0.65 | 1.4 | 24 | 0.6 |

Table 4.25: water sorption in as-received materials. NT stands for non-thermoformed and T for thermoformed

The results in Table 4.25 show that thermoforming affects the water sorption of the materials, accelerating the process. This may be due to the fact that the materials become softer after thermoforming. In general, hTPU materials absorb more water than PETG materials, but the least absorbent material is Essix C; however, it is also the material that changes the most after thermoforming. The increase in the maximal weight percentage

may be related to an increase in the surface roughness of the materials as well as the decrease in $\tau$.

However, in all cases, the water sorption in these materials may be considered surface-dominated or pseudofickian, with characteristic sorption times in the range of one month; hence, these plastics can be considered low-permeable plastics [302].

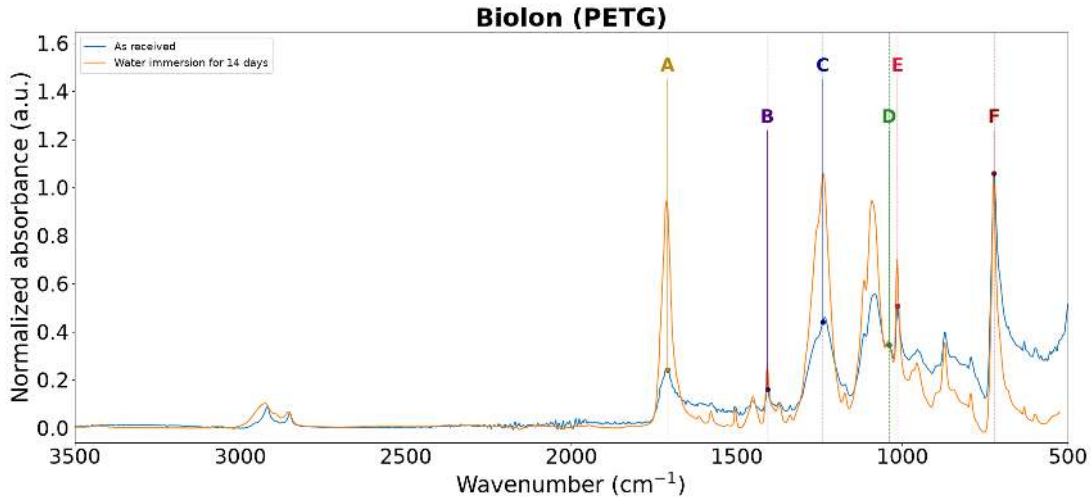



(a) Heating traces

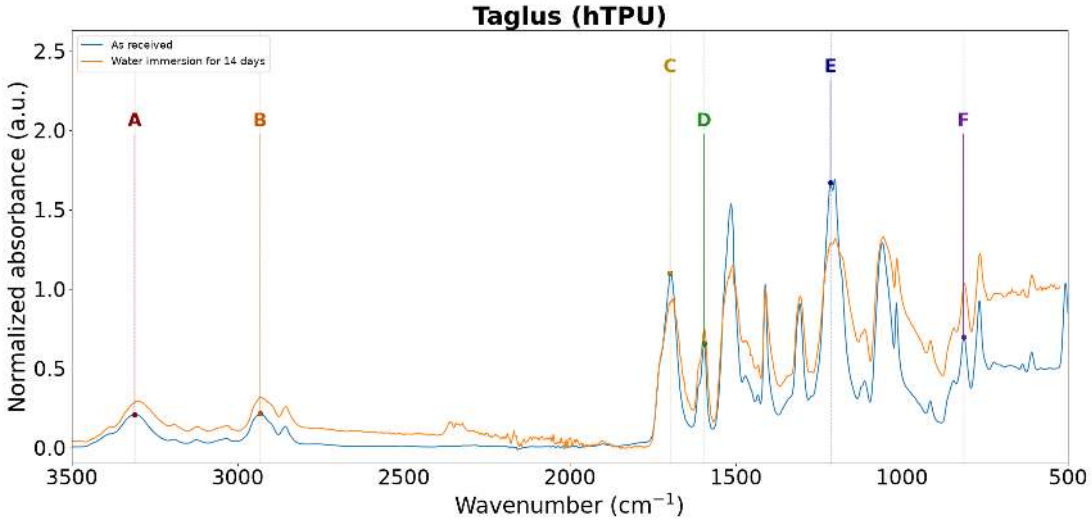
(b) Heating traces

Figure 4.38: DSC traces in as-received materials after 14 days of partial immersion in water at 37 ºC

DSC measurements on specimens immersed for 14 days (Figure 4.38) revealed a slight depression of the Tg of 2-5 °C in most materials, within the range of instrumental uncertainty and therefore not attributable with confidence to genuine plasticisation. A notable exception was Tristar S (hTPU), which exhibited a Tg depression of approximately 20 °C, substantially exceeding instrumental variability, consistent with partial hydrolytic degradation of the polyurethane matrix rather than reversible plasticisation. In PETG materials, the enthalpic relaxation peak associated with physical ageing reappeared after 14 days of immersion in both as-received and thermoformed specimens, confirming that intraoral immersion conditions reinitiate physical ageing within a single wearing period regardless of the thermal history of the material.




(a) PETG. Band assignments: **A** = 1709 cm$^{-1}$, C=O ester stretching; **B** = 1407 cm$^{-1}$, aromatic ring stretching; **C** = 1240 cm$^{-1}$, C-O-C asymmetric stretching; **D** = 1040 cm$^{-1}$, CHDM trans conformer; **E** = 1014 cm$^{-1}$, CHDM cis conformer; **F** = 722 cm$^{-1}$, C-H aromatic out-of-plane




(b) hTPU. Band assignments: **A** = 3310 cm$^{-1}$, N-H stretching (urethane); **B** = 2932 cm$^{-1}$, $CH_2$ stretching (soft segment); **C** = 1698 cm$^{-1}$, C=O H-bonded (ordered hard segment); **D** = 1596 cm$^{-1}$, Amide II / C=C MDI stretching; **E** = 1215 cm$^{-1}$, C-N stretching (urethane); **F** = 815 cm$^{-1}$, MDI para-substituted aromatic ring

Figure 4.39: ATR-FTIR after 14 days immersed in water at 37 ºC

ATR-FTIR analysis of immersed specimens (Figure 4.39) confirmed the absence of detectable backbone degradation in all materials except Tristar S, where a simultaneous decrease in both degree of phase separation and hard segment ordering was observed, consistent with the DSC evidence for partial hydrolytic chain scission in this material. In all remaining materials, the spectral changes observed after immersion; an increase in polyester band intensity in PETG materials, and an increase in hard segment ordering in Taglus and Zendura A; reflect physical reorganisation driven by water-induced plasticisation rather than chemical modification of the polymer backbone.

Regarding absorption dynamics, the apparent Fickian diffusion coefficient $D_{ap}$ estimated from the initial slope of the absorption curves (up to the third day) following the short-time solution to Fick's second law [302], was approximately $7 \times 10^{-8}$ $cm^2/s$ for the neat copolyester materials (Duran, Biolon, Leone and Tristar A), consistent with previously reported values for PETG and PET in water [143], [171]. The hTPU materials (Tristar S, Taglus and Zendura A) and the multilayer Zendura FLX showed higher values in the range of $1.2 - 1.9 \times 10^{-7}$ $cm^2/s$, approaching values characteristic of TPU and more hygroscopic thermoplastics [298], [303], indicating that the presence of urethane segments accelerates water uptake velocity relative to the neat copolyesters, independently of the total absorbed quantity. Essix C+ (PP) presented the lowest apparent diffusion coefficient ($3.1 \times 10^{-8}$ $cm^2/s$) consistent with its hydrophobic character, though its high $\alpha$ value (0.65) indicates that transport in this material is governed by a different mechanism than in the amorphous thermoplastics, as discussed in Appendix A.1.3. Thermoforming produced a consistent increase in $D_{ap}$ across all material classes, with the most pronounced effect observed in Essix C+ (PP) and the PETG materials, where $D_{ap}$ increased by factors of approximately 2-3 relative to the as-received state. This acceleration is attributed to the substantial increase in surface roughness induced by thermoforming [260], which increases the effective surface area available for water contact and thereby enhances the rate of surface-dominated sorption. In the hTPU materials, the increase in $D_{ap}$ after thermoforming was comparatively modest, consistent with their already higher baseline diffusivity and the limited additional surface area generated during thermoforming of these more compliant materials. The saturation weight gain $\Delta M_\infty$ also increased after thermoforming in all materials (Table 4.25), suggesting that thermoforming not only accelerates the kinetics of water uptake but also increases the total accessible free volume of the polymer matrix, likely through the partial relaxation of residual extrusion stresses and the microstructural reorganization associated with the thermoforming thermal cycle.

Taken together, these results confirm that all tested materials except Tristar S undergo reversible hygrothermal aging rather than irreversible chemical degradation under the experimental conditions, and that the dominant microstructural process during intraoral use is physical aging reinitiation in PETG materials and water-facilitated hard segment reorganization in hTPU materials. The substantially different behaviour of Tristar S, including a 20 °C Tg depression and spectroscopic evidence of hard segment disruption, warrants caution regarding its long-term mechanical stability during clinical use and suggests that its 14 day wearing protocol may need to be reconsidered.

### 4.8.4. Water absorption conclusions

The water sorption results are consistent with the general trends reported in the literature for clear aligner thermoplastics: hTPU-based materials exhibit substantially higher water absorption than PETG, which remains relatively dimensionally stable after 14 days of immersion [71], [88], [299]. The ML material Zendura FLX presents an intermediate

water uptake, consistent with the partial barrier role of its outer PETG layers in restricting water access to the inner sTPU core, as discussed above. With the notable exception of Tristar S — where a marked Tg depression of approximately 20 °C is attributed to partial hydrolytic degradation of the polyurethane matrix rather than reversible plasticization — no substantial depression of Tg was observed in any material after 14 days of immersion. This confirms that reversible hygrothermal aging, rather than irreversible chemical degradation, is the dominant process in the vast majority of the materials characterized under the immersion conditions employed. Hygrothermal aging was detected by DSC in all materials, manifesting as an increase in the enthalpic relaxation peak in PETG — reflecting reinitiation of physical aging from the thermoforming-reset baseline — and as more pronounced crystallization exotherms in several hTPU materials, reflecting water-facilitated hard segment reorganization. Thermoforming and hygrothermal aging act as sequential but independent processes: thermoforming resets the aging state of PETG to a near-zero baseline, but 14 days of intraoral immersion at 37 °C is sufficient to reinitiate physical aging to a level approaching that of the as-received material, indicating that the microstructural state of PETG aligners during clinical use is governed primarily by hygrothermal ageing rather than by the thermoforming history.

No significant changes in primary chemical structure were detected by ATR-FTIR in any of the materials following water immersion, confirming the absence of measurable backbone degradation within the 14-day immersion period. The spectral changes observed — an increase in the intensity of polyester surface bands in PETG and PCTG materials, and an increase in the degree of hard segment ordering in Taglus and Zendura A — are fully consistent with the physical reorganisation processes expected during hygrothermal ageing: water-induced plasticisation enhances local chain mobility, facilitating surface conformational ordering of polyester chains and lateral reorganisation of hard segment domains in TPU, without modifying the covalent backbone structure of either polymer class. The behaviour of Tristar S stands apart from this general pattern: the simultaneous decrease in degree of phase separation and hard segment ordering observed by ATR-FTIR, combined with the DSC-detected Tg depression, provides converging evidence for a qualitatively different and more severe degradation mechanism in this material — most probably hydrolytic chain scission of the polyurethane matrix — that distinguishes it clearly from the other hTPU materials characterised. This finding warrants caution regarding the long-term dimensional and mechanical stability of Tristar S under intraoral conditions, and suggests that its 14-day wearing protocol may need to be reconsidered in light of the degradation rate implied by these results.

The present work further demonstrates that thermoforming systematically modifies the water sorption behavior of aligner materials, an effect that has been largely overlooked in the literature. The mechanistic basis of this observation remains to be fully established; a plausible contributing factor is the substantial increase in surface roughness induced by thermoforming [260], which would increase the effective surface area available for water contact, though this was not directly measured in the present study. A systematic

investigation of the relationship between thermoforming-induced surface roughening and water uptake kinetics is therefore identified as a recommendation for future work.

## 4.9. Conclusions of the commercial material screening

The systematic characterization conducted in this chapter — encompassing thermal analysis by DSC and DMTA, mechanical testing by uniaxial tensile and three-point bending, chemical analysis by ATR-FTIR, optical characterization by UV-VIS spectroscopy, TGA, and cross-sectional nanoindentation, together with an assessment of the effects of thermoforming and water absorption — establishes a comprehensive reference framework for the physical and chemical properties of commercial thermoplastic aligner materials and provides a basis for rational material selection and development. The characterization confirms that virtually all materials currently in use are amorphous, transparent polymers — predominantly PETG, high-hardness MDI-based polyester TPU, or multilayer architectures combining both — engineered to meet a narrow window of optical, thermal, and mechanical requirements, with glass transition temperatures in the range 85–120 ℃ and elastic moduli between roughly 630 and 1300 MPa. Within this framework the multilayer materials occupy a distinctive position, combining a moderate elastic modulus with the highest elastic stored energy and elastic range — a combination not achievable by any monolithic material characterized, and which underpins their clinical advantage in sustained and controlled force delivery.

### 4.9.1. Chemical and thermal characterization

ATR-FTIR confirmed that all monolayer hTPU materials are polyester-based and MDI-type, with degrees of phase separation and ordering that meet or exceed literature values for high-performance polyester-MDI TPU, consistent with the well-developed microphase-separated microstructure that underpins their elastic recovery. Selective THF dissolution of Zendura FLX confirmed the copolyester nature of its outer layers and the MDI-based hard segment of its inner TPU. The CHDM content of the PETG materials, estimated from the DMTA glass transition temperatures, is consistent with the most widely used commercial grade, whereas the elevated $T_g$ of the outer copolyester layers of Zendura FLX and Maflex Premium requires the simultaneous presence of CHDM and CBDO, distinguishing them from standard PETG and identifying them as PCTG- or Tritan-type copolyesters.

### 4.9.2. Dynamic mechanical analysis and viscoelastic behaviour

The Cole-Cole and vGP representations revealed systematic differences in viscoelastic character between material classes: PETG exhibits the largest Cole-Cole areas and narrowest vGP phase-angle maxima, consistent with its homogeneous amorphous backbone

and near-exponential relaxation, whereas the multilayers present the smallest areas and broadest curves, reflecting the wider distribution of relaxation times introduced by their heterogeneous laminated architecture — a distribution that translates clinically into a more gradual and temporally distributed force decay. The bending-to-tensile modulus ratio identified a key mechanical advantage of the PCTG/sTPU/PCTG architecture: a substantially lower tensile modulus combined with a bending modulus comparable to that of monolithic PCTG, consistent with the role of the sTPU core as a flexural spacer. Since aligner forces are delivered primarily in bending, this stiffness anisotropy channels the stored energy preferentially into bending, yielding a more efficient and controllable force delivery.

### 4.9.3. Force delivery and stress relaxation

A central aim of this work was to evaluate whether multilayer PCTG/sTPU/PCTG materials generate lower initial forces than their monolithic counterparts while maintaining a slower force decay. The data support this hypothesis, with the evidence becoming progressively stronger at higher, more clinically representative strain levels. At 0.5% strain — the only level at which all materials could be fully characterized — the multilayers delivered markedly lower initial forces than the monolithic sheets, but their stress relaxation was broadly comparable, establishing no clear advantage in decay rate for either class. The picture changes at 1.5% and 2% strain: the monolithic materials systematically reached the 18 N equipment ceiling before the target deformation, whereas the multilayers remained within the measurable range up to 2% strain. This is itself mechanically informative, confirming that monolithic materials generate forces that may surpass physiological limits at moderate activation distances, leaving a narrower safety margin against dimensional inaccuracy.

The force decay behaviour shows a clinically significant, strain-dependent inversion. At low strain the multilayers exhibit faster early-stage decay than the monolithic materials, attributable to the viscoelastic sTPU core. As deformation increases toward clinically relevant levels, however, this relationship reverses: the decay of the multilayers slows while that of the monolithic materials accelerates with the onset of irreversible plastic deformation beyond their yield point. At higher strain the multilayers therefore deliver both lower initial forces and a slower, more gradual decay — a qualitatively different force–time relationship whose mechanical advantage is greatest precisely in the deformation range most relevant to clinical practice.

These findings translate into direct guidance for material selection by movement type. For simple movements involving low activation strains — arch expansion or constriction, mild spacing closure, minor rotations — monolithic PETG or hTPU may be preferable, as their higher initial modulus sustains a therapeutically relevant force over the wearing period at these low deformations. For complex movements requiring larger activation distances — significant bodily translation, torque correction, or large rotations — multilayer

PCTG/sTPU/PCTG materials are more appropriate, avoiding the excessive loads generated by monolithic materials at high strain and sustaining a moderate, gradually diminishing force across the wearing cycle. This distinction implies that the clinical protocol for multilayer aligners should not simply mirror that established for monolithic materials, but should be adapted — particularly with respect to activation distance and replacement interval — to exploit the distinctive force–time profile of the laminate architecture.

### 4.9.4. Thermoforming effects on the physical properties of the thermoplastics

Thermoforming generally reduced the storage modulus at oral temperature and the Cole-Cole area across all materials, consistent with partial relaxation of residual extrusion stresses and microstructural reorganization during the thermal cycle. The disappearance of the enthalpic relaxation peak in thermoformed PETG confirms the erasure of physical aging history, while the more pronounced crystallization exotherms in several hTPU materials reflect hard segment reorganization. These findings are consistent with the literature and extend the characterization to hTPU, PP, and multilayer materials not previously addressed. TGA revealed that the PETG and multilayer materials are thermally stable under the thermoforming conditions employed, whereas Essix C+ (PP) underwent partial thermo-oxidative degradation and the hTPU materials showed low-intensity partial degradation of both segments — effects detectable by TGA but below the threshold of ATR-FTIR, confirming the complementary sensitivity of the two techniques. Thermoforming and hygrothermal aging were found to act as independent sequential processes: thermoforming resets the aging state of PETG to a near-zero baseline, but a single wearing period of intraoral immersion is sufficient to reinitiate it, indicating that the in-service microstructural state of PETG aligners is governed primarily by hygrothermal aging rather than by thermoforming history.

#### Material suitability and exclusion criteria

Essix C+ (PP) is identified as unsuitable for clear aligner fabrication on multiple independent grounds: its semi-crystalline microstructure renders it optically translucent under clinical conditions despite marginally satisfying the commonly cited 80% transmittance criterion; its low elastic modulus and high loss modulus are more consistent with energy-absorbing applications such as bruxism or sports guards; and its thermoforming induces partial thermo-oxidative degradation. This case exposes the inadequacy of the 80% threshold, and on the basis of these findings a minimum transmittance of 85% and a requirement for fully amorphous character are proposed as necessary screening criteria. All other materials characterized satisfy the minimum requirements for functional use, but their clinical application is constrained by an important practical consideration: virtually all commercial planning software was developed and calibrated on the proprietary Invisalign material, so stiffer materials applied under the same activation parameters will generate higher initial forces. Multilayer materials, by delivering gentler and more sus-

tained forces, offer a mechanical profile inherently more compatible with existing planning software.

### 4.9.5. Nanoindentation of commercial multilayer aligners

Cross-sectional nanoindentation revealed that Zendura FLX and Spark share virtually identical layer-level mechanical properties and an equivalent inner TPU grade, consistent with a closely related material chemistry despite different commercial origins; Filtzwiesser employs a substantially softer inner TPU, delivering a markedly more compliant response; and Invisalign presents an inverted architecture — a stiff PCTG-type inner core surrounded by soft outer TPU layers — fundamentally different from the other multilayer systems. The combination of nanoindentation modulus and DMTA-derived $T_g$ enabled the unambiguous identification of the Invisalign core as PCTG, ruling out a high-hardness TPU alternative and illustrating a general methodological principle: the two techniques provide complementary information that enables compositional identification not achievable by either alone. The exponential relationship established between Shore hardness and nanoindentation-derived modulus across a calibrated TPU series provides a practical design tool, allowing the mechanical contribution of a given sTPU layer to be estimated directly from supplier data sheets. More broadly, this layer-resolved approach establishes that the thermomechanical properties of multilayer architectures are not fixed by material chemistry alone but can be engineered by design through the selection of layer chemistry, Shore hardness grade, and relative thickness.

### 4.9.6. Clinical and design implications

Four interdependent factors govern the outcome of clear aligner treatment (Figure 4.40): patient compliance, the clinician's experience and material knowledge, the planning software, and the thermoplastic material.

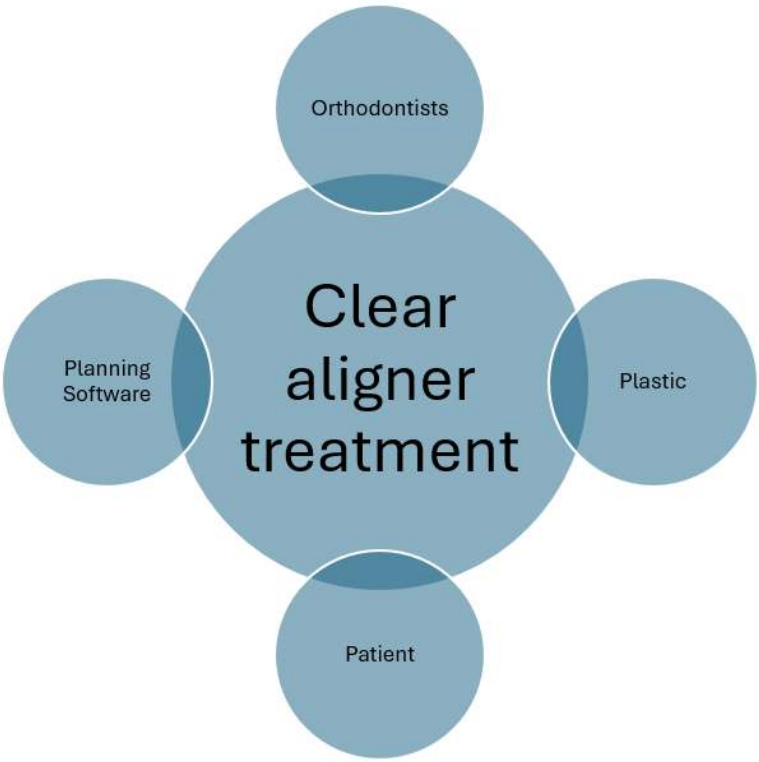



Figure 4.40: Key factors in clear aligner treatment planning

The relationship between material and software is particularly critical and currently underappreciated: since the software cannot readily be adapted to the material, it falls to the clinician to select and adapt the material to the software, the safest strategy being substitution with a material of closely matched mechanical properties. To address the absence of standardized minimum requirements or test protocols for clear aligner materials in the literature, a fast characterization pathway using Zendura FLX as a gold-standard reference is proposed (Figure 4.41); applied to a series of commercial materials submitted for evaluation by Secret Aligner, it enabled the identification of several viable alternatives with comparable or superior properties, including materials with a reduced elastic modulus.

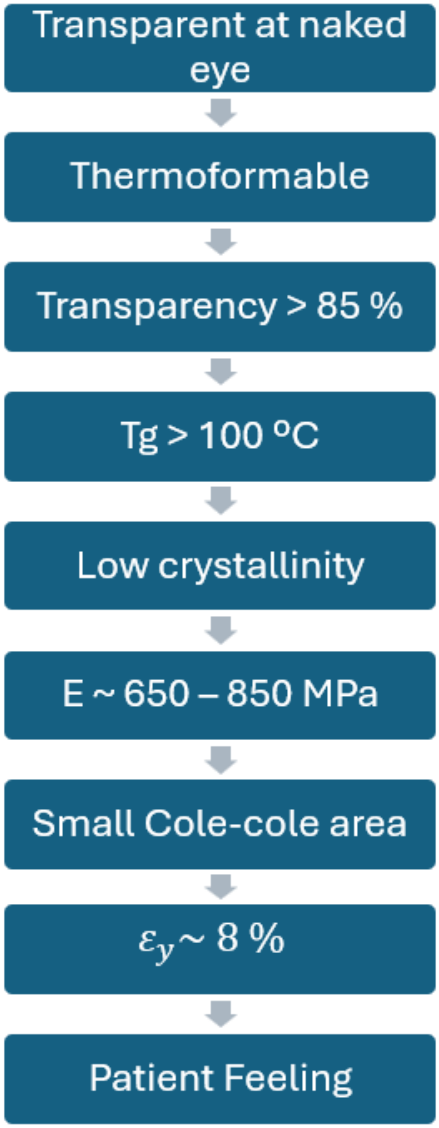



Figure 4.41: Fast pathway for clear aligner materials characterization

Finally, a strategy for the rational design of next-generation multilayer materials is proposed, based on the thermal co-bonding of PCTG and TPU layers of characterized properties: by exploiting the Shore hardness–modulus relationships established here, the thermomechanical response of the laminate can in principle be tailored to a prescribed clinical target, opening a pathway toward personalized aligner treatments matched to the biomechanical demands of individual patients or movement stages. The selective dissolution of the sTPU core in THF, which leaves the copolyester layers intact, further demonstrates that the constituent layers can be separated and recovered — a result of direct relevance to the recycling of the substantial thermoplastic scrap stream generated during manufacture, and developed further in the environmental analysis of Chapter 7.

# 5. HOMEMADE MULTILAYERS WITH TAILORED MECHANICAL PROPERTIES

### 5.0.1. Sourcing the raw materials

Building on the property requirements for hard and soft layer materials identified in the preceding chapters, a range of candidate thermoplastics was sourced from commercial suppliers for thermomechanical characterisation prior to multilayer fabrication. The hard layer candidates (Table 5.1) comprised commercial PETG, three grades of Tritan copolyester (MP100, MP150, and MP200), Polyethylene Naphthalate (PEN), and a PET-PEN copolymer, selected on the basis of their nominally higher Tg and VST relative to standard PETG. The soft layer candidates (Table 5.2) comprised EVA and a series of TPU grades — both polyether- and polyester-based — spanning the Shore 33D to Shore 70D hardness range, sourced to provide a broad compositional space for the design of multilayer architectures with tunable mechanical properties. Thin sheets of 0.7 mm thickness were produced from each material using a heated press to evaluate thermoformability and to provide specimens for thermomechanical characterisation by DSC and DMTA.

| Material | Composition | Hardness | VST (ºC) | source | Origin |
|---|---|---|---|---|---|
| PETG | PETG | 75 | 75 | Eolas prints | Spain |
| PEN | PEN | 85 | 125 | Indorama Ventures | USA |
| PENco | PEN-PET | 85 | 130 | Indorama Ventures | USA |
| MP100 | PCTG | 75 | 115 | Eastman | Netherlands |
| MP150 | PCTG | 75 | 110 | Eastman | Netherlands |
| MP200 | PCTG | 75 | 140 | Eastman | Netherlands |
| TPU Hardness + | TPU ether | 83 | 164 | Filament2Print | Spain |

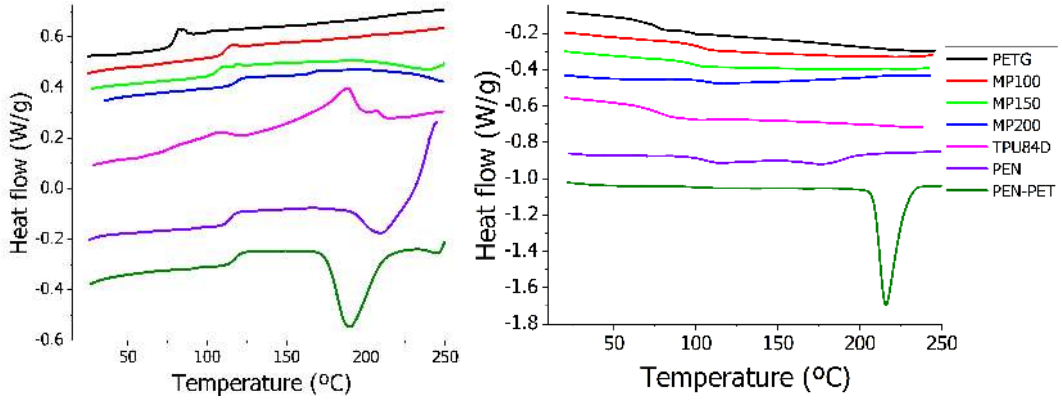
Table 5.1: List of hard materials tested in this chapter. Hardness is measured in shore D scale. VST stands for Vicat softening temperature

| Material | Composition | Hardness | VST (ºC) | source | Origin |
|---|---|---|---|---|---|
| EVA 1828 | EVA | 29 | 44 | Hanwha | South Korea |
| Laripur 9060 | TPU ether | 33 | 110 | COIM | Italy |
| TPU 93A | TPU ester | 41 | 165 | Eolas Prints | Spain |
| Estane-F95A | TPU ester | 46 | - | Filament2Print | Spain |
| Laripur 5260 | TPU ether | 52 | 120 | COIM | Italy |
| Estane-F98A | TPU ester | 60 | - | Filament2Print | Spain |
| Laripur 71D60 | TPU ether | 70 | 162 | COIM | Italy |

Table 5.2: List of soft materials tested in this chapter. Hardness is measured in shore D scale. VST stands for Vicat Softening temperature

### 5.0.2. Results on hard layers

The DSC traces of the hard layer candidate materials are shown in Figure 5.1. All selected thermoplastics except PETG exhibit a relatively high glass transition temperature, ranging from approximately 85°C in TPU 84D to 131°C in PET-PEN. However, a high degree of crystallisation or cold crystallisation was observed in TPU 84D and PEN-based materials respectively, which constitutes a critical processing limitation.



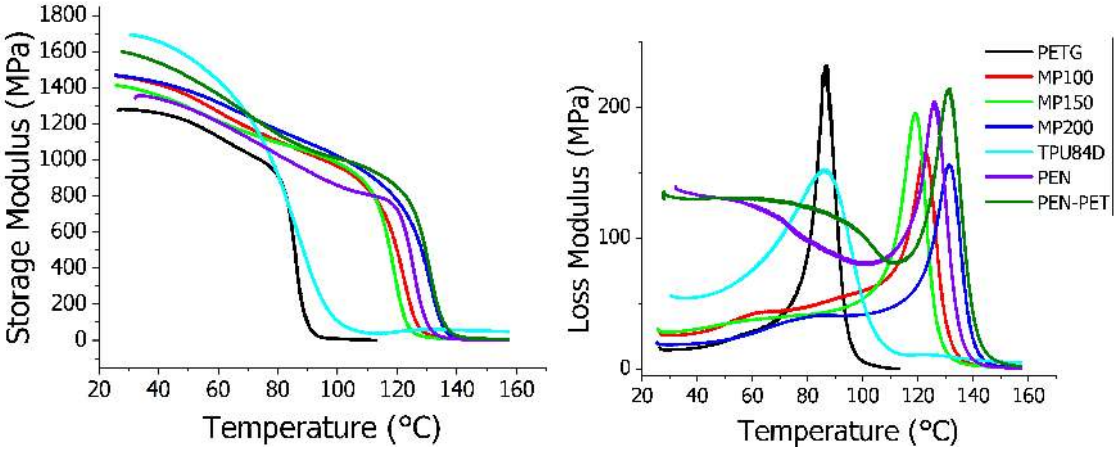

(a) First heating (b) Cooling

Figure 5.1: DSC traces on hard layers

Crystallisation in PEN and PET-PEN rendered the production of transparent layers impossible below 265°C. Attempts to obtain transparent films by heating PEN above its Tg (130°C) and below the onset of cold crystallisation (approximately 170°C) yielded either opaque crystalline layers at higher temperatures or insufficient material flow at lower temperatures. Conventional thermoforming of these materials consistently produced opaque, non-conforming parts; a fast thermoforming protocol of 10-15 s (reduced from 30-35 s) failed to prevent crystallisation while simultaneously producing insufficient material flow. PEN-based materials were therefore discarded on processing grounds, independently of their mechanical properties.

In the case of TPU 84D, the cold crystallisation observed in the first heating is almost negligible but becomes more significant in subsequent heating cycles, with both crystal populations — small unstable crystals and larger stable crystals — appearing to grow progressively at elevated temperatures, posing a risk during processing. In PETG, a clear physical ageing peak is observed — an undesirable feature for clear aligners as discussed in Chapter 4. In contrast, the Tritan copolyesters (MP100, MP150, and MP200) exhibit no ageing peak, no crystallisation, and a more suitable Tg, and thermoforming was conducted without issue.

The DMTA results in tensile configuration are shown in Figure 5.2 and Table 5.3. The secondary thermal transition ($T_\beta$) was obtained from cryogenic DMTA experiments at -120 ºC.




(a) Storage modulus (b) Loss modulus

Figure 5.2: DMTA curves on hard layers

| Material | E'37 (MPa) | E"37 (MPa) | Tg (ºC) | $T_\beta$ (ºC) |
|---|---|---|---|---|
| PETG | 1268 | 15.3 | 85.9 | -70 |
| MP100 | 1432 | 26.6 | 122.7 | -70 |
| MP150 | 1373 | 29.5 | 118.8 | -70 |
| MP200 | 1440 | 18.8 | 131.9 | -70 |
| TPU84D | 1672 | 54.2 | 86.1 | -75 |
| PEN | 1351 | 134.3 | 125.5 | 50 |
| PEN-PET | 1557 | 129.7 | 131.4 | 75 |

Table 5.3: Hard materials DMTA results

The storage moduli at oral temperature are broadly comparable across all materials and consistent with values reported for commercial clear aligner thermoplastics in Table 4.3. The most discriminating parameter is the loss modulus at 37°C: PEN and PET-PEN exhibit values of 134 and 130 MPa respectively — substantially higher than all other

candidates — indicating excessive viscoelastic dissipation that would produce rapid force decay and loss of device shape during wear. TPU 84D also presents an elevated loss modulus (54 MPa) and a Tg of only 86 °C, insufficient to ensure adequate force delivery throughout the wearing period. PETG and the Tritan copolyesters show intermediate loss moduli (15-30 MPa); the Tritan grades additionally present higher Tg values and no ageing peak, suggesting they would generate more comfortable insertion forces than PETG while retaining comparable shape stability. On these grounds, PETG, TPU 84D, and PEN-based materials were discarded.

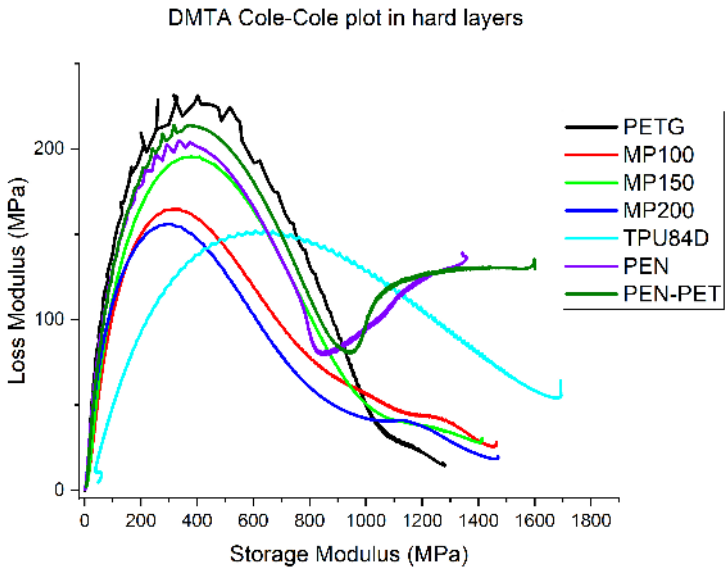



Figure 5.3: Cole-Cole curves from DMTA in soft layers

The Cole-Cole plots of the hard layers (Figure 5.3) confirm that all hard layer candidates exhibit areas under the curve comparable to those of commercial monolayer clear aligner materials, but substantially larger than those of commercial multilayer aligners. The vGP (Figure 5.4) reveal a notable absence of a phase angle minimum in all materials, including PETG — likely attributable to the broad molecular weight distribution of the 3D-printing grade used, which is formulated for high flowability rather than narrow polydispersity. The presence of plasticisers is excluded, as they would depress the Tg and reduce the maximum phase angle. Among the materials tested, TPU 84D tands out for its exceptionally low maximum phase angle (26°) and a narrow, complex vGP curve, consistent with a material of high elastic character but limited processability.

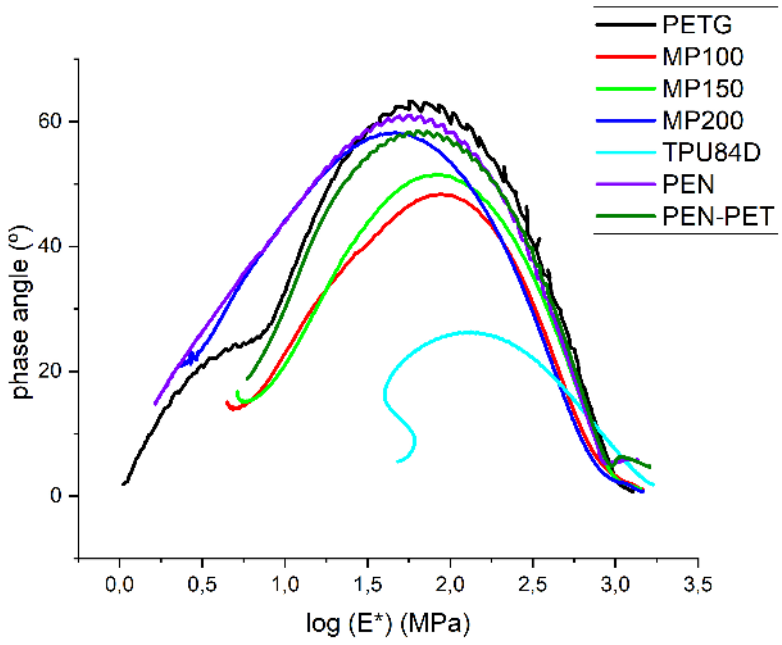



Figure 5.4: van Grup-Palmen curves of hard layers obtained from DMTA measurements

PEN and PET-PEN remain scientifically interesting candidates: both exhibit a secondary relaxation visible as a small additional semicircle in the Cole-Cole and DMTA loss modulus curves, attributable to localised low-amplitude chain motions that could enhance insertion comfort and reduce brittleness without compromising shape retention. However, the inability to process these materials into transparent amorphous layers under the current fabrication conditions renders their inclusion in this work impractical. From the hard layer screening, MP100, MP150, and MP200 were selected as the sole viable candidates, on the basis of their high Tg (118-132°C), absence of crystallisation, optical transparency, and thermoformability.

### 5.0.3. Results in soft layers

All soft layer candidates were transparent and thermoformable. Their DSC traces (Figure 5.5) reveal some degree of crystallisation in all materials, attributable to the hard segment phase of the TPU grades and to the polyethylene fraction of EVA. TPU 84D was included in this analysis despite its classification as a hard layer candidate, to provide a complete picture of how physicochemical properties evolve with increasing hardness.

The phase separation (DPS) and ordering degree (DOR) of the TPU materials were obtained from ATR-FTIR spectroscopy using Equations 4.1 and 4.2; results are reported in Table 5.4. All materials used MDI as the hard segment, as confirmed by ATR-FTIR. The hard segment concentration was obtained from the TGA derivative curve, and the degree of crystallinity was computed using the theoretical crystallisation enthalpy of pure MDI ($H_m^0 = 140\ J/g$ [304], [305]), as reported in Table 5.5. For computing the TPU

crystallinity ($\chi$), Equation 5.1 was used:

$$\chi\,(\%) = \frac{\Delta H_c}{(HS - I)\Delta H_m^0} \times 100 \tag{5.1}$$

where $\Delta H_c$ is the crystallisation enthalpy of the sample, $HS$ and $I$ are the hard segment and isocyanurate ring mass fractions obtained from TGA.

| Material | Hardness | DPS (%) | DOR (%) | $\Delta H_c$ (J/g) | Tc1 (ºC) | Tc2 (ºC) | TPU-type |
|---|---|---|---|---|---|---|---|
| TPU85A | 33 | 80 | 52 | 10.2 | 87 | - | ether |
| TPU93A | 41 | 87 | 60 | 11.2 | 97 | - | ester |
| TPU95A | 46 | 62 | 62 | 18.1 | 118 | 134 | ester |
| TPU52D | 52 | 79 | 79 | 15.1 | 111 | - | ether |
| TPU98A | 60 | 65 | 65 | 15.3 | 113 | 138 | ester |
| TPU70D | 70 | 99 | 88 | 17.2 | 120 | 142 | ether |
| TPU84D | 84 | 82 | 53 | 26.2 | - | - | ether |

Table 5.4: ATR-FTIR Characterization of all the TPU materials. The Crystallization enthalpy in the TPU 84D was actually computed using the melting peaks in the first heating. Tc1 refers to the low temperature crystallization peak and Tc2 to the high temperature. DPS and DOR were obtained based on the deconvolution of the ATR-FTIR spectra in the region 1600-1750 $cm^{-1}$ for TPUs. DPS stands for degree of phase separation, and DOR for ordering degree.

Upon heating, multiple melting processes are observed in nearly all materials, reflecting the coexistence of crystal populations of different sizes and thermal stability. The soft segments of TPU, which would be expected to crystallise in the 20-60 °C range, show no corresponding thermal event; the crystallising phase in all TPU grades studied is therefore attributed exclusively to the hard segments. The two crystallisation peaks observed in the cooling traces of several grades are consistent with the well-documented polymorphic behaviour of MDI/BDO-based hard segments, which can form two distinct crystalline morphologies (Type I and Type II) differing in their degree of phase separation and long-range order [306], [307]. Type I crystallites are disordered and associated with a higher degree of phase mixing; Type II crystallites exhibit stronger phase separation and greater structural perfection [307]. The soft segment phase, thermodynamically incompatible with the hard segment domains, acts as a structural defect that limits crystal growth and promotes the formation of smaller, less stable crystallites [307], [308], with each crystalline population contributing a distinct thermal event upon cooling.

In the case of hTPU grades, no crystallisation is observed in the as-received state (Figures 5.1 and 4.2), despite their high hard segment content. This apparent paradox is explained by the immobility of the hard segments at low temperatures: the high segment concentration traps them in a glassy state that prevents aggregation [85], [309]. Crystallisation therefore requires specific thermal activation, which accounts for the cold

crystallisation observed in the second heating cycle, the hygrothermal crystallisation seen in Figure 4.38, and the appearance of previously absent melting processes after thermoforming in Figure 4.21 [309].

Crystallisation is generally detrimental for clear aligner therapy: it increases material opacity, promotes stiffening and embrittlement, and complicates thermoforming. Small metastable crystals are particularly problematic, as their uncontrolled sizes and morphologies produce diffuse optical boundaries that increase haze; under intraoral conditions, they may grow further, rendering the device progressively more opaque, stiffer, and more brittle. Nonetheless, as shown in Table 5.5, the crystallinity of the TPU grades is relatively low (12-23 %), comparable to or lower than that of EVA (approximately 30 %).

| Material | Hardness | HS (%) | I (%) | $\chi$(%) |
|---|---|---|---|---|
| TPU85A | 33 | 41 | 0 | 18 |
| TPU93A | 41 | 61 | 0 | 13 |
| TPU95A | 45 | 51 | 4 | 23 |
| TPU52D | 52 | 66 | 0 | 16 |
| TPU98A | 60 | 70 | 13 | 13 |
| TPU70D | 70 | 68 | 0 | 18 |
| TPU84D | 84 | 73 | 22 | 36 |

Table 5.5: Crystallization analysis of the TPUs. HS stands for hard segment percentage and I stands for isocyanate ring percentage. Both percentages were obtained by the deconvolution of the TGA derivative curve.$\chi$ is the crystal percentage. The in-house Python script used for this analysis is openly available in the thesis repository

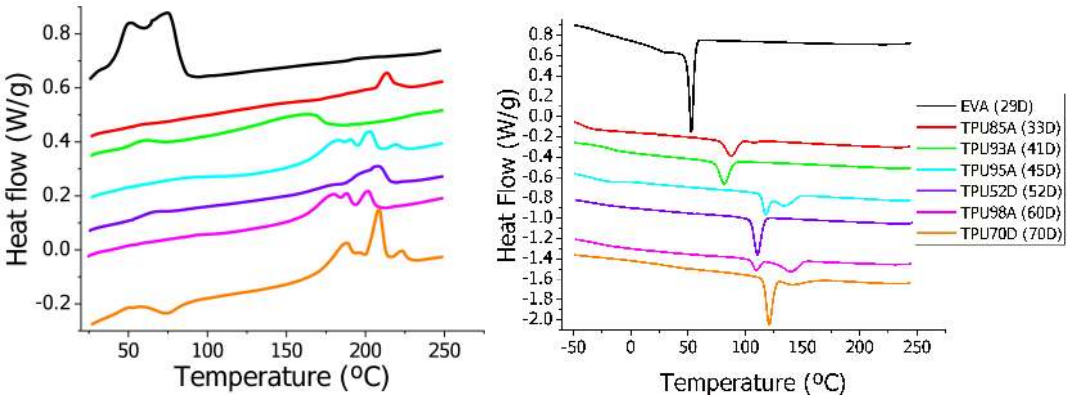
For EVA, the chemical composition was estimated from the experimentally determined Tg using Equation 5.2, which relates the copolymer Tg to the molar fraction of vinyl acetate (VA) based on the thermal properties of the constituent homopolymers [310]:

$$T_{g,EVA} = T_{g,PE} + 1.43 \cdot VA(mol(\%)) + 128.7 \cdot \exp(-0.028 \cdot VA(mol(\%)) \tag{5.2}$$

A VA content of 21% mol was obtained. At low VA content, the acetate units are sufficiently scarce to permit PE crystallisation while depressing the Tg; above approximately 40% mol VA, the copolymer becomes fully amorphous [310], [311]. The degree of crystallinity of the PE fraction was subsequently calculated from the crystallisation enthalpy measured in the DSC cooling scan ($\Delta H_c = 24.2\ J/g$, $T_c = 53$ ºC) using Equation 5.3 [312], [313]:

$$X_{PE} = \frac{\Delta H_c}{\Delta H_c^{\circ}(PE) \cdot (1 - w_{VA})} \times 100 \tag{5.3}$$

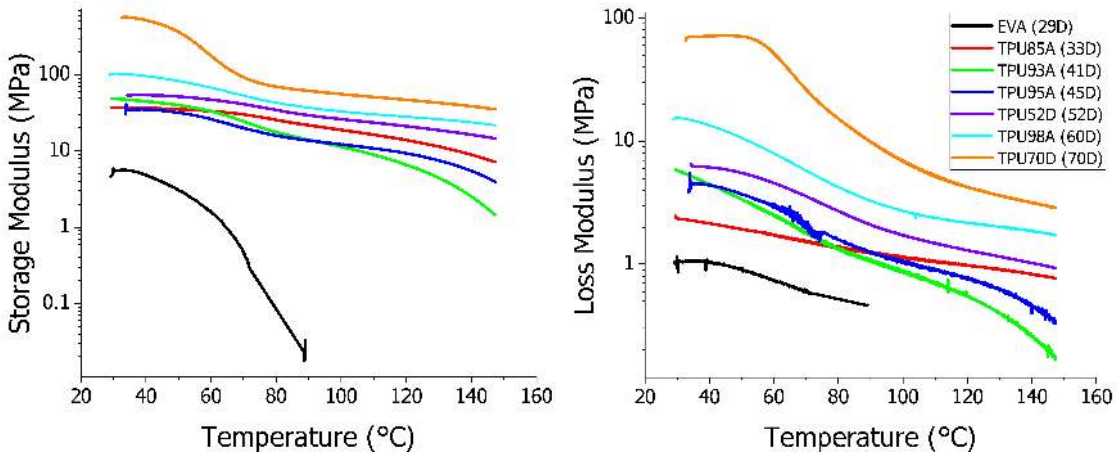
where $\Delta H_c(PE) = 293\ J/g$ [312], [314] is the theoretical crystallisation enthalpy of 100% crystalline PE amd $w_{VA} = 0.45$ is the weight fraction of VA. The resulting crystallinity of the PE phase is 14.9%, consistent with literature values for comparable VA contents [310], [311], [312], [313].




(a) First heating (b) First heating

Figure 5.5: DSC traces of soft layers

The DMTA curves of the soft layers are shown in Figure 5.6 and Table 5.6. All materials are above their Tg at room temperature.




(a) Storage modulus (b) Loss modulus

Figure 5.6: DMTA curves of soft layers

Cryogenic DMTA (from -120 to 160 °C) was performed on all grades; two distinct Tg were identified in the TPU materials, corresponding to the soft segment ($Tg_{ss}$) and hard segment ($Tg_{hs}$) phases respectively [183], [234]. The soft segment Tg appears independent of Shore hardness, being governed by the chemical nature of the soft segment rather than hard segment content [315], whereas $Tg_{hs}$ increases monotonically with hardness grade, consistent with the well-established relationship between hard segment content, phase separation, and viscoelastic response [307].

| Material | Hardness | E'37 (MPa) | E"37 (MPa) | $Tg_{hs}$ (℃) | $Tg_{ss}$ ( ℃) |
|---|---|---|---|---|---|
| EVA | 29 | 7.1 | 0.4 | -22 | - |
| TPU85A | 33 | 36.9 | 2.2 | -37 | -85 |
| TPU93A | 41 | 45.4 | 2.35 | -30 | - |
| TPU95A | 45 | 48.9 | 4.52 | -22 | -85 |
| TPU52D | 52 | 60.0 | 6.1 | -18 | -75 |
| TPU98A | 60 | 97.6 | 4.9 | -16 | -85 |
| TPU70D | 70 | 548.2 | 70.9 | 12 | -75 |

Table 5.6: Soft materials DMTA results at oral temperature.

The storage modulus scales monotonically with Shore hardness (Table 5.6 and Figure 5.7), reflecting the progressive stiffening of the hard segment network. The loss modulus follows the same trend up to TPU 70D but exhibits a slight decrease at TPU 84D, consistent with the known reduction in energy dissipation capacity at high hard segment crystallinity, where restricted chain mobility lowers $tan(\delta)$ despite the higher stiffness [307], [316]. Considering only the TPU data, a clear exponential relationship between storage modulus at oral temperature and Shore hardness is observed, as described by Equation 4.7. Tensile and bending experiments (Table 5.7) confirm this exponential dependence across measurement configurations, suggesting that Shore hardness alone provides a reliable predictor of the elastic modulus at oral temperature for soft layer material selection. The viscous component does not follow this trend: neither the plastic energy dissipation (Figure 4.36) nor the loss modulus values show a systematic dependence on Shore hardness, indicating that energy dissipation is governed by additional factors including soft segment chemistry and phase morphology.

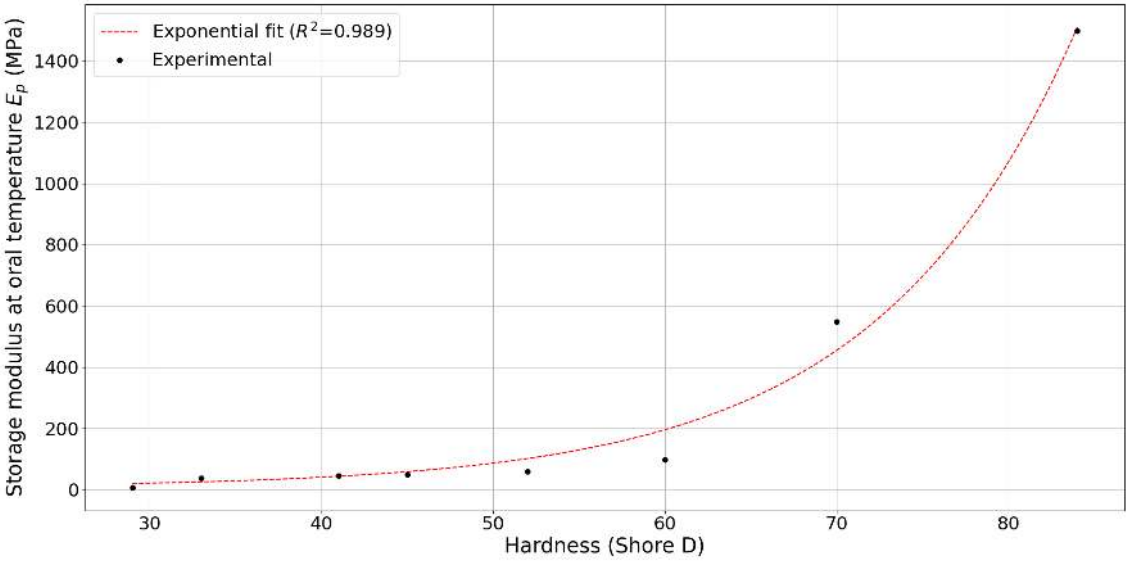



Figure 5.7: relationship between the Shore hardness of the soft layers and the storage modulus at oral temperature

| Material | Shore D hardness | E Tensile | E bending | Ratio (Eb/Et) |
|---|---|---|---|---|
| EVA | 29 | 9.15 | 17.57 | 1.91 |
| TPU85A | 33 | 15.28 | 41.57 | 2.71 |
| TPU52D | 52 | 34.75 | 115.46 | 3.32 |
| TPU70D | 70 | 130.37 | 459.14 | 3.52 |
| TPU84D | 84 | 1497.51 | 1939.76 | 1.30 |

Table 5.7: Elastic modulus in bending and tensile configuration in soft materials

As shown in Figure 5.8, all soft layer materials produce a nearly flat Cole-Cole response with negligible enclosed area, reflecting the post-transition rubbery behaviour of polymers whose Tg lie below room temperature. This contrasts markedly with the broad arcs observed in the hard layer candidates (Figure 5.3), and confirms that the soft layer contributes negligible viscoelastic dissipation to the multilayer composite in the clinical temperature range.

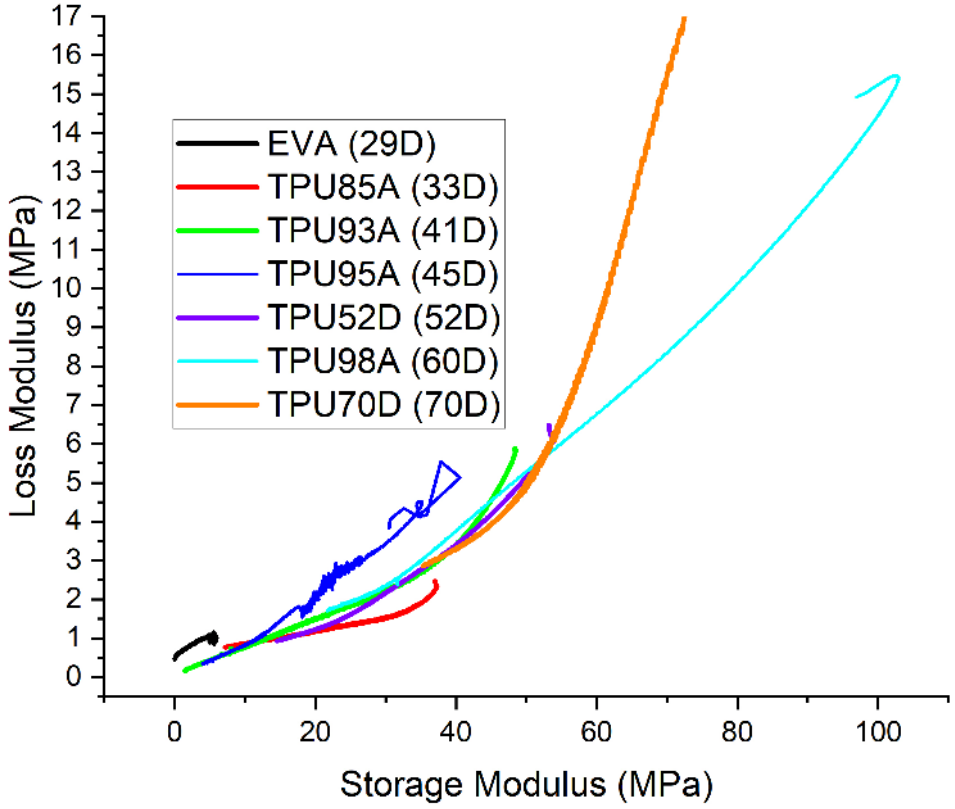



Figure 5.8: Cole-Cole curves of soft layer materials obtained from DMTA measurements.

Polyether TPU was preferred over polyester grades on several grounds: wider elasticity range at low temperatures, superior hydrolysis resistance, better resilience and strain

recovery, and a wider processing window with fewer extrusion-related degradation issues. EVA was included as a complementary soft layer candidate due to its mechanical properties comparable to the softest TPU grades and its ease of processing. Despite the absence of EVA-based multilayer aligners on the market — largely owing to incompatibility between its processing temperature and those of conventional hard layer materials — its inclusion broadens the compositional design space explored in this work. From this screening, three hard layers (MP100, MP150, MP200) and four soft layers (EVA, TPU 85A, TPU 52D, and TPU 70D) were selected for multilayer fabrication.

### 5.1. Results of the drying and shaping tests

Before extrusion, all of the materials were dried. At first, all materials were dried using a vacuum at 85 °C and 70 mBar for 7 h in the case of the hard layers and 3 h in the case of the soft layers. A yellowish decoloration was observed in all TPU materials, a clear sign of slight thermal degradation in those materials.

After drying, shaping tests were performed on all raw materials. The shaping tests consisted of melting the material to see if the melt could fill a square mold of 150 x 150 mm and 0.2 mm thick.

Hard layers filled the mold, and transparent monolayers were easily produced at the end of the cooling process with a thickness below 0.25 mm. Regarding soft layers, in these tests, only TPU materials were used. In this case, TPU85A and TPU52D were unable to fill the mold due to the high viscosity of the melts at molding temperature (190 °C). Only TPU70D was able to fill the mold and lead to a perfect and transparent monolayer. No yellowing was observed in most TPU monolayers after this processing.

Another molding test was performed that increased the molding time to 20 minutes on TPU materials. In those tests, a clear yellowing and bubbling of the materials was observed after processing. In the case of TPU70D, the coloration was white instead of yellow, possibly due to crystallization of the plastic.

### 5.2. Extrusion results

The hard layer extrusion process was relatively simple, and transparent sheets with the required thicknesses were obtained with almost no defects. The opposite happened with the soft layers extruded in the Center Català del Plàstic (Universidad Politécnica de Cataluña, Barcelona). During extrusions, 0.25 mm thick layers were difficult to obtain due to the low flowability of the plastics. In addition, heterogeneities and defects were common in TPU films. The most common defect was shark skin. Also, a clear yellowing was observed in most of the extruded TPU. During both extrusion processes, the TPU70D degraded in the machinery, blocking the screws and greatly increasing the torque; therefore, in both cases, the material was extracted, and the machines had to be completely cleaned

at high temperature. The degraded material was a white, viscous, and bubbling solution. Only a few meters could be obtained in the center Català del Plàstic before the material degraded, and even at the end of those meters, the degradation of the material was clear. Similar degradations were also observed in the two other types of TPU at the end of each extrusion, but to a lesser extent. Hence, since TPU70D is the one with a higher concentration of hard segments, it is reasonable to assume that degradation may be caused by the crystallization of hard segments

With respect to extrusions in ICTP (Instituto de ciencia y tecnología de polímeros, Madrid), the extruded TPU and EVA had better dimensional accuracy and fewer defects, but most of them presented a high degree of hazing. In TPU, a slight yellowish decoloration was also observed.

It was found that extruding EVA was not as complex as extruding TPU. It can also be done at lower temperatures, and the material does not degrade during extrusion and drying. Only one problem was encountered when EVA was spooled due to the autoadhesion of the plastic film.

To evaluate the directional dependence of mechanical properties after processing, tensile tests up to 18 N were performed on all materials, with specimens cut both in the machine direction (MD) and in the transverse direction (TD). The resulting data were analyzed using the anisotropy index (AI) (Equation 5.4), the anisotropy ratio (AR) (Equation 5.5), and a paired-sample t-test to assess whether statistically significant differences exist between the two directions.

For a perfectly isotropic material — one whose properties are independent of the sampling direction — the AR should approach unity and the AI should approach zero. Additionally, the p-value of the t-test should exceed 0.05, indicating a failure to reject the null hypothesis of equal means between the two directions.

$$AI = \frac{E_{md} - E_{td}}{E_{md} + E_{td}} \tag{5.4}$$

$$AR = E_{md}/E_{td} \tag{5.5}$$

| Material | $E_{md}$ (MPa) | $E_{td}$ (MPa) | AR | AI | P |
|---|---|---|---|---|---|
| MP100 | 836 ± 72 | 878 ± 100 | 1.05 | 0.024 | 0.133 |
| MP150 | 842 ± 180 | 997 ± 252 | 0.85 | -0.08 | 0.25 |
| MP200 | 898 ± 168 | 911 ± 209 | 0.99 | -0.007 | 0.85 |
| EVA | 9.78 ± 2.01 | 8.28 ± 1.83 | 1.18 | 0.083 | 0.07 |
| TPU85A | 14.76 ± 1.13 | 15.80 ± 3.07 | 0.93 | -0.03 | 0.5 |
| TPU52D | 31.42 ± 4.96 | 37.60 ± 7.65 | 1.19 | 0.089 | 0.07 |
| TPU70D | 117.25 ± 27.65 | 140.83 ± 29.24 | 0.83 | -0.09 | 0.05 |

Table 5.8: Anisotropy analysis on extruded materials

The data in table 5.8 confirm that no anisotropy or directionality is found, and no significant differences are observed between the sampling directions (P is greater than 0.05 in all cases) in any of the materials.

## 5.3. Multilayer manufacturing and characterization

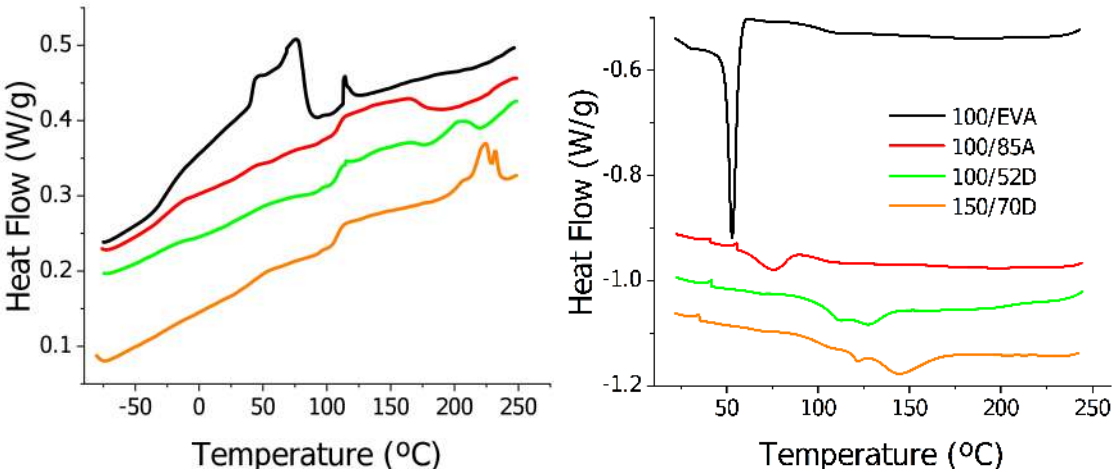
After extrusion, different types of multilayers were manufactured using the extruded materials. Most of the analysis was performed on 5 cm x 5 cm squared multilayers, but 120 cm x 120 cm multilayers were processed to test thermoformability and to try to manufacture real clear aligners.

The forming process consisted first of cutting large enough samples of each layer needed by hand and introducing all layers into the hot press, which was previously heated to the required molding temperature. Hence, the molding was performed by applying a slight pressure at the required temperature for 2 minutes. The molding time was kept low to avoid as much TPU degradation as possible, as well as bubbling and yellowing of the materials.

Different molding temperatures and pressing conditions were studied until a multilayer with good and tough bonding between layers, and with the required dimensions, was obtained. In the case of TPU-based multilayers, the molding temperature used was the same as the corresponding hard layer molding temperature (220 °C for MP100 and MP150, and 230 °C for MP200). In the case of EVA-based materials, the molding temperature was 180 ºC. Temperatures below these lead to easy debonding of the layers or to multilayers that were easily peeled. The final pressure for all multilayer processing was 200 KPa. Higher pressure values lead to unwanted material flow of the soft layers and to too-thin multilayers. The layer blending occurred using too high pressures (> 800 KPa) and/or too high temperatures.

### 5.3.1. Thermal properties

First, the multilayers were thermally characterized using DSC as described previously. In figure 5.9 only multilayers containing MP100 copolyesters are displayed for simplicity. Multilayers using other copolyesters show the same type of DSC traces, with the unique difference of a slightly higher Tg in MP200.




(a) First heating

(b) Cooling

Figure 5.9: DSC traces of MP100 multilayers

In figure 5.9, it can be seen that the glass transition and crystallization processes observed in the individual layers are also present in the multilayers without significant changes.

The main thermal events in multilayers based on EVA are first melting peaks in heating at 48 °C and 70 °C from the EVA origin, followed by the glass transition of the corresponding hard layer at 120 or 130 °C. In cooling, the only thermal event is EVA crystallization at 65 °C.

Regarding the multilayers based on TPU, the first thermal event in heating is the Tg of the hard layer. After that, a complex melting range of crystals appears between 180 and 240 ºC. In cooling, only crystallization of those crystals occurs.

The thermal events observed in the multilayer DSC traces can be attributed to the individual constituent layers, with no apparent shift in peak positions relative to the neat materials, suggesting that the layers remain thermally distinct without significant intermixing. Also the absence of new thermal events in the multilayer discard the formation of new interphases during processing.

Slight differences can be observed in the position of the crystallization peaks of the multilayers with respect to those of the original soft layers (Table 5.9). In multilayers using EVA as the soft layer, the crystallization peak position remains unchanged compared to the neat EVA, and the only notable difference lies in the crystallization enthalpy, which decreases proportionally to the EVA weight fraction in the multilayer structure. This confirms that the EVA layer crystallizes independently, without interference from the adjacent layers.

TPU-based multilayers show a broadly similar behavior, although with some notable exceptions. In the case of TPU 85A, the crystallization peak shifts from 87 °C in the neat material to 76 °C in all the multilayers analyzed. This downward shift suggests that the

presence of the hard layers hinders the crystallization kinetics of TPU 85A, likely due to a confinement effect at the interface that restricts chain mobility and delays nucleation, resulting in crystallization occurring at lower temperatures. This effect is particularly pronounced for TPU 85A given its low hardness and high chain flexibility, which make it more susceptible to interfacial constraints imposed by the hard layers.

In the case of TPU 52D, the behavior depends on the specific hard layer grade used. In the MP100 multilayer, the main crystallization peak is retained at 111 °C but a secondary peak appears at 126 °C, suggesting the presence of two crystallizing populations: one corresponding to bulk-like crystallization and another associated with chains near the hard layer interface that crystallize under different conditions, possibly induced by nucleating interactions with the MP100 surface. This splitting is not observed in the MP150 and MP200 multilayers, which display only the main crystallization peak at 111 °C. This difference points to a specific interfacial interaction between TPU 52D and MP100 that is not replicated with the other Tritan grades, despite their structural similarities, suggesting that subtle differences in the chemical composition or molecular architecture of the Tritan copolyesters influence the nucleation behavior of the adjacent TPU layer.

The weight fraction of the soft layer, $X$, was estimated from the ratio of the crystallization enthalpy of the multilayer to that of the neat soft layer material, following the standard calorimetric mixing rule in equation 5.6. [312], [314], [317]

$$X(\%) = \frac{\Delta H_c^{Multilayer}}{\Delta H_c^{Soft\ layer}} * 100 \tag{5.6}$$

| Material | $\Delta H_c$ (J/g) | Tc1 (ºC) | Tc2 (ºC) | X (%) |
|---|---|---|---|---|
| 100/EVA | 6.9 | 53 | - | 29 |
| 100/TPU85A | 2.8 | 74 | - | 28 |
| 100/TPU52D | 5.4 | 126 | 111 | 35 |
| 100/TPU70D | 5.9 | 144 | 120 | 35 |
| 150/EVA | 7.2 | 53 | - | 29 |
| 150/TPU85A | 2.9 | 74 | - | 28 |
| 150/TPU52D | 4.8 | 113 | - | 32 |
| 150/TPU70D | 6.2 | 153 | 120 | 36 |
| 200/EVA | 6.9 | 53 | - | 28 |
| 200/TPU85A | 3.0 | 54 | - | 29 |
| 200/TPU52D | 6.5 | 112 | - | 38 |
| 200/TPU70D | 6.4 | 147 | 121 | 38 |

Table 5.9: Crystallization enthalpy ($\Delta H_c$), crystallization peak temperatures (Tc1 and Tc2), and soft layer weight fraction (X) obtained from DSC cooling scans for all multilayer configurations. X was calculated as the ratio of the crystallization enthalpy of the multilayer to that of the neat soft layer material. A dash indicates the absence of a second crystallization peak

The weight fraction of the soft layer X in the table 5.9, estimated from the ratio of the crystallization enthalpy of the multilayer to that of the neat soft layer material, yields values ranging from 28% to 38% in all multilayer configurations (Table 5.9). These values are in good agreement with the theoretical soft layer fraction of 33% expected for a symmetric trilayer structure with equal layer thicknesses, confirming that the layer composition is consistent across all manufactured multilayers. The small deviations from the theoretical value are within the experimental uncertainty associated with DSC enthalpy measurements and minor thickness variations inherent to the multilayer manufacturing process and do not indicate any systematic compositional deviation.

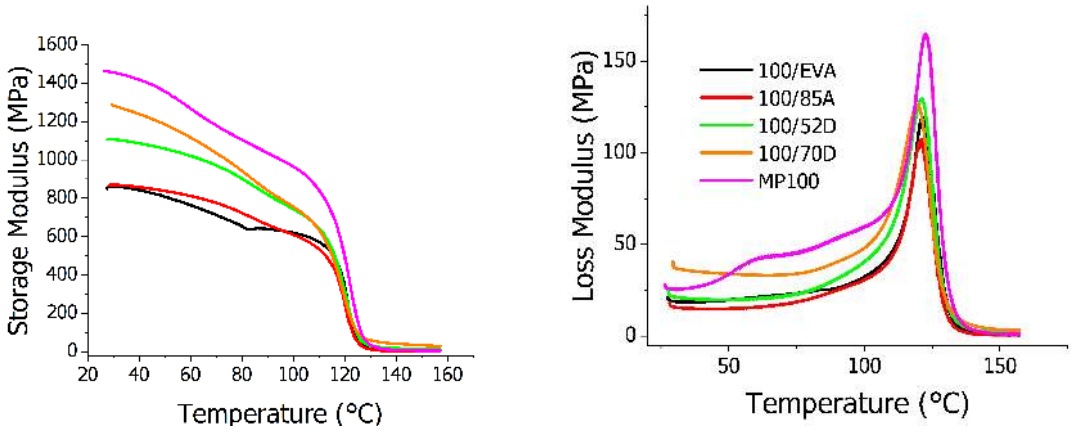
### 5.3.2. Thermomechanical properties

The thermomechanical properties were studied by DMTA in a tensile configuration between room temperature and 160 °C. The figure 5.10 shows representative storage and loss modulus curves for the series of multilayers manufactured using MP100 as the hard layer with different soft layer grades. Several features are immediately apparent. The storage modulus at oral temperature decreases systematically with decreasing soft layer hardness, from the neat MP100 monolayer (1432 MPa) down to the MP100/EVA configuration (850 MPa), confirming the tunability of the composite stiffness through soft layer selection. The glass transition, evidenced by the sharp drop in $E'$ and the corresponding peak in $E''$ at approximately 120 °C, is governed exclusively by the hard layer chemistry and remains essentially unchanged across all multilayer configurations, consistent with

the thermal independence of the constituent layers established by DSC previously.

A notable feature of the EVA-containing multilayers is a discontinuity in the storage modulus curve at approximately 80 °C, coinciding with the melting of the EVA crystalline phase identified by DSC (see figure 5.9). This transition, which manifests as an abrupt softening superimposed on the gradual glassy-to-rubbery decay of the hard layer, was observed consistently in all multilayer configurations incorporating EVA as the soft layer and is absent in TPU-based multilayers, whose crystalline melting occurs above 100 °C and therefore overlaps with the onset of the hard layer glass transition. It is interesting to note that the melting of the crystalline phase in EVA was not observed in DMTA experiments performed over neat EVA (see figure 5.6). This happens because, at room temperature, the EVA is already in a rubbery state, and the melting of the crystalline phase results in a decrease in modulus too small to be observed. In a multilayer, this effect is amplified, but it also reflects the glass transition in the interphase of the multilayer. While there are crystals in EVA, they mechanically anchor the interphase between the EVA and the copolyester and hence provide a load-bearing phase. However, once the crystals are molten, the EVA behaves like a viscous fluid, which transmits load by friction, reducing the effective rigidity of the multilayer. A similar effect may occur in TPU-based multilayers, but since the melting of the TPU crystals happens in the range of the glass transition of the hard layers, that effect is not observable.

The loss modulus curves reveal an additional low-temperature shoulder between 30 and 80 °C in the MP100/TPU 70D multilayer, attributed to the glass transition of the TPU 70D hard segment phase ($T_g^{hs} \approx 12$ °C, Table 5.6), which is the only soft layer grade with a hard segment $T_g$ within the measured temperature window. This feature is absent in the other multilayer configurations, whose soft segment relaxations occur well below room temperature.




(a) Storage modulus

(b) Loss modulus

Figure 5.10: DMTA curves of homemade multilayers manufactured using MP100 as hard layer and different soft layers

As shown in the table 5.10, the storage modulus at oral temperature increases sys-

tematically with the hardness of the soft layer, ranging from approximately 850-860 MPa for the softest grades (EVA and TPU 85A) to values approaching those of the neat hard layer for TPU 70D (1194-1253 MPa). The loss modulus follows a similar trend, increasing with soft layer hardness, consistent with the greater energy dissipation capacity of harder TPU grades at oral temperature, as discussed in the commercial material screening. As expected, the glass transition temperature is governed exclusively by the hard layer chemistry and remains essentially unaffected by the nature or hardness of the soft layer, retaining the values characteristic of each Tritan grade (approximately 120 °C for MP100 and MP150 and 131 °C for MP200).

| Hard layer | Soft layer | E'37 (MPa) | E"37 (MPa) | Tg (ºC) | A ($GPa$) |
|---|---|---|---|---|---|
| MP100 | - | 1432 | 26.5 | 122.7 | 0.110 |
| MP100 | EVA | 850 | 18.6 | 122.7 | 0.046 |
| MP100 | TPU85A | 862.6 | 14.9 | 120.8 | 0.043 |
| MP100 | TPU52D | 1093 | 20.22 | 121.3 | 0.064 |
| MP100 | TPU70D | 1253 | 35.4 | 118.9 | 0.060 |
| MP150 | - | 1373 | 29.5 | 118.8 | 0.132 |
| MP150 | EVA | 856 | 18.4 | 118.5 | 0.048 |
| MP150 | TPU85A | 883 | 15.4 | 118.7 | 0.045 |
| MP150 | TPU52D | 909.7 | 25.3 | 118.8 | 0.057 |
| MP150 | TPU70D | 1198 | 35.7 | 121.6 | 0.056 |
| MP200 | - | 1440 | 18.8 | 131.1 | 0.102 |
| MP200 | EVA | 1091 | 18.9 | 131.1 | 0.043 |
| MP200 | TPU85A | 927 | 15.4 | 131.0 | 0.042 |
| MP200 | TPU52D | 954 | 26.6 | 130.7 | 0.052 |
| MP200 | TPU70D | 1194 | 29.1 | 130.44 | 0.051 |
| Commercial | - | 918 | 16.04 | 122.6 | 0.061 |

Table 5.10: DMTA values of manufactured multilayers

The storage modulus of the multilayer can be approximated by an iso-strain rule of mixtures, as depicted in the equation 5.7. [214], [215]

$$E'_{\text{multilayer}} \approx \phi_{\text{hard}} E'_{\text{hard}} + \phi_{\text{soft}} E'_{\text{soft}} \tag{5.7}$$

where $\phi_{\text{hard}}$ and $\phi_{\text{soft}}$ are the volume fractions of the hard and soft layers, respectively, both equal to 0.67 and 0.33 for the symmetric trilayer architectures manufactured here. This relationship implies that the storage modulus of the composite can be predicted and tuned in a controlled manner simply by selecting the appropriate soft layer grade, without the need for iterative fabrication and characterization. It is important to distinguish these volume fractions from the weight fraction $X$ determined independently by DSC (Equation 5.6): whereas $X$ quantifies the mass contribution of the soft layer to the total laminate—as

inferred from the ratio of crystallization enthalpies—$\phi_{\text{soft}}$ describes its volumetric contribution, which is the physically relevant quantity for any mechanical mixing rule. The two are related through the layer densities:

$$\phi_{\text{soft}} = \frac{X/\rho_{\text{soft}}}{X/\rho_{\text{soft}} + (1 - X)/\rho_{\text{hard}}} \tag{5.8}$$

where $\rho_{\text{hard}} \approx 1.27\ \text{g cm}^{-3}$ for the Tritan copolyester layers and $\rho_{\text{soft}}$ ranges from 0.95 $\text{g cm}^{-3}$ (EVA) to 1.22 $\text{g cm}^{-3}$ (TPU 70D). Because the hard layer is systematically denser than the soft layer, the soft phase occupies a slightly larger fraction of the total volume than it contributes by mass, i.e. $\phi_{\text{soft}} > X$. Substituting the DSC-derived weight fractions into Equation 5.8 and using the resulting volume fractions in the mixing rule does not improve the predictive accuracy relative to the nominal value $\phi_{\text{soft}} = 0.33$, confirming that the small deviations of $X$ from the nominal 33% reflect experimental scatter in the DSC enthalpy measurements rather than genuine compositional variability, and that the nominal volume fraction is the most reliable input for the mechanical model.

The agreement between this prediction and the experimental data therefore establishes a rational design framework for future multilayer aligner materials, in which a target storage modulus at oral temperature is translated directly into a soft layer Shore hardness through the exponential relationship derived in Section 4.7 and the mixing rule of the equation 5.7. The validity of the iso-strain mixing rule (Equation 5.7) was assessed by comparing the predicted and experimental storage moduli at 37 °C for all twelve multilayer configurations. The predicted values, computed using $\phi_{\text{hard}} = 0.67$ and $\phi_{\text{soft}} = 0.33$ together with the experimentally determined moduli of the neat constituent layers (Tables 5.6 and 5.3), deviate from the experimental results by no more than 13%, with a mean absolute error of approximately 8%. This level of agreement is satisfactory for a polymeric multilayer system and confirms that the iso-strain rule of mixtures provides a reliable first-order design tool for predicting the storage modulus of these laminates from the properties of their constituent layers alone, without the need for iterative fabrication.

The iso-strain rule of mixtures consistently overestimates the experimental storage modulus, with predicted values exceeding the measured ones in all twelve configurations. The iso-strain model supposes that all the layers undergo ideally the same deformation [214], [215]. In practice, several factors cause the effective modulus to fall below this bound. The first problem is that the soft layer gathers a disproportionate fraction of the total deformation, violating the iso-strain hypothesis and reducing the effective stiffness of the laminate. Second, the interlayer transition region of around 100 $\mu m$ identified before (Section 4.7) constitutes a region of intermediate mechanical properties that behaves more like the softer constituent under tensile loading, effectively reducing the load-bearing cross-section of the hard layer. Third, the thermal pressing cycle required to consolidate the layers may induce localized softening at the hard-layer surface in contact with the molten soft layer, reducing its effective modulus relative to that of the neat extruded sheet used as input to the model. Notwithstanding these limitations, the mean absolute error of 7.8% confirms that the iso-strain model provides a reliable first-order

design tool, and the systematic overestimation implies that the true modulus will always lie slightly below the predicted value.

A more refined prediction could in principle be obtained by incorporating the interphase as an explicit third phase in the mixing rule, with a modulus equal to the geometric mean of the two constituent layers [318]. Preliminary calculations show that fitting a single interphase volume fraction $\phi_i$ reduces the mean absolute error from 7.8% to 6.0%, with an optimal value of $\phi_i = 0.098$ corresponding to a mechanically active interphase of approximately 37 $\mu$m per interface. This value represents the equivalent homogeneous interphase thickness that best reproduces the observed deviation from Voigt behavior, and is substantially smaller than the compositional interphase width of $\sim$100 $\mu$m identified by nanoindentation, consistent with a gradual modulus profile in which only the central fraction of the transition zone deviates appreciably from the adjacent bulk values. However, a rigorous implementation of this three-phase model would require dedicated characterization of the interphase modulus profile - for instance through high-resolution nanoindentation across the cross-section of each multilayer configuration - which was beyond the scope of the present work. Given that the improvement over the simple Voigt model is modest and that the Voigt error of 7.8% is already acceptable for design purposes, the iso-strain rule of mixtures is retained as the preferred predictive tool.

It should be noted that the iso-strain rule cannot be extended to the loss modulus. The storage modulus quantifies the elastic energy stored reversibly per deformation cycle and is governed by the volume-weighted contributions of the constituent phases, which is the physical basis of the mixing rule. The loss modulus, by contrast, quantifies irreversible energy dissipation and depends not only on the intrinsic damping capacity of each phase but also on the spatial distribution of stress and strain fields within the laminate, the viscoelastic coupling between phases at the interface, and the frequency- and temperature-dependent relaxation mechanisms of each constituent - contributions that are inherently non-additive and cannot be reconstructed from the neat-layer $E''$ values alone.

Regarding the Cole-Cole curves the arc width increases systematically with the soft layer hardness as may be observed in the figure 5.11. MP100/EVA and MP100/TPU 85A exhibit the narrowest arcs, while MP100/TPU 52D and MP100/TPU 70D approach the monolayer more closely, consistent with the higher rubbery modulus of the stiffer grades and the smaller reduction in Cole-Cole area reported in the table 5.10. The position of the loss modulus peak along the $E'$ axis - equivalent to the storage modulus at the midpoint of the glass transition - is also displaced towards lower values in the multilayers relative to the monolayer, reflecting the compression of the accessible modulus range rather than any shift in the glass transition temperature itself, which remains unchanged as confirmed by the temperature-domain data (Table 5.10).It must be noted the appearance of a discontinuity in the EVA-based multilayers due to the melting of the crystalline phase in EVA.

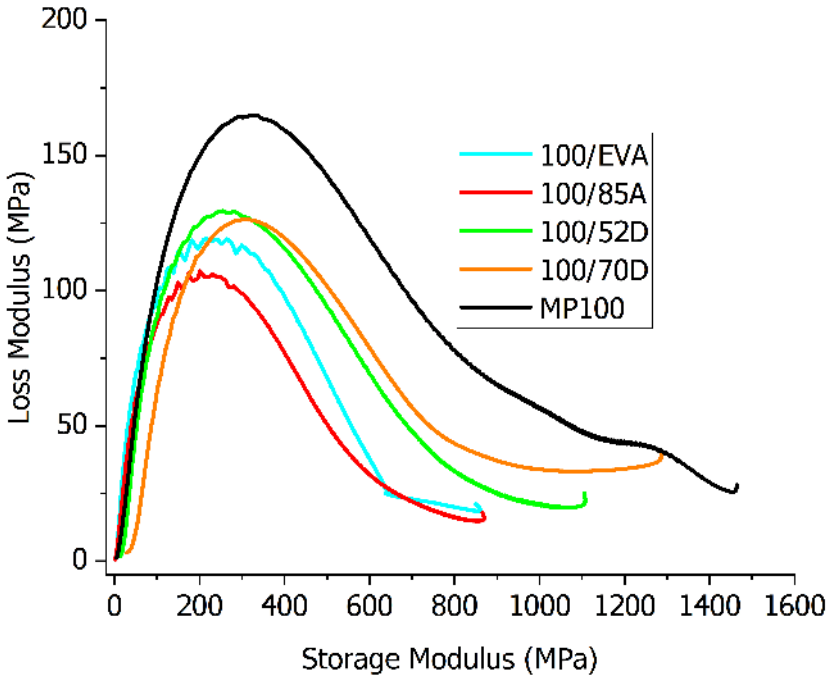



Figure 5.11: Cole-Cole curves of homemade multilayers using MP100 as hard layer

The Cole-Cole areas of the homemade multilayers (0.041-0.064 $GPa^2$) fall within the same range as the commercial multilayer reference materials (0.030-0.060 $GPa^2$, Table 4.3), with an identical mean value of 0.050 $GPa^2$, confirming that the lamination process produces a viscoelastic relaxation spectrum comparable to that of state-of-the-art commercial aligners. All homemade multilayer areas are substantially below those of the corresponding neat hard layers (0.10-0.13 $GPa^2$), consistent with the compression of $\Delta E'$ induced by the compliant inner layer, as described for the commercial multilayers in Chapter 4.

The magnitude of this compression depends systematically on the soft layer hardness. Softer grades (EVA and TPU 85A, Shore hardness 29-33) reduce the hard layer Cole-Cole area by approximately 60-62%, whereas stiffer grades (TPU 52D and TPU 70D, Shore hardness 52-70) produce a smaller reduction of 51-52%. This trend reflects the degree to which the compliant layer elevates the rubbery plateau modulus and thereby compresses the dynamic range of the relaxation transition: a softer inner layer imposes a lower rubbery modulus floor, broadening the accessible $\Delta E'$ range and increasing the area, while a stiffer inner layer restricts the compression to a narrower band. In clinical terms, multilayers incorporating the softest grades (EVA, TPU 85A) exhibit the smallest Cole-Cole areas and therefore the most stable force delivery profile over the wear cycle, whereas those with TPU 70D approach the larger areas characteristic of the neat hard layer, suggesting a somewhat greater force decay between insertion and removal - though still substantially lower than any monolithic material characterized in Chapter 4.

### 5.3.3. Tensile testing

All uniaxial tensile characterization of the homemade multilayers was performed at room temperature using a universal testing machine under displacement control, allowing the full stress–strain curve to be recorded up to final rupture and enabling the determination of the elastic modulus, yield stress, yield strain, and elastic toughness reported above. Although the elastic modulus of thermoplastic aligner materials decreases slightly between room temperature and oral temperature (37°C), this difference is small relative to the overall modulus range of the materials studied. Room temperature testing was therefore preferred over intraoral temperature testing for this characterization, as it provides a richer mechanical dataset: the 37 °C measurements performed on the commercial materials (see Table 4.8) were conducted in a DMTA instrument under a force ramp of up to 18 N, which precludes reaching the yield point and does not allow the post-yield behavior or the fracture response to be characterized. The room temperature tensile data therefore provide complementary and more complete mechanical information than the DMTA-based measurements, while remaining representative of the relative ranking of materials in terms of stiffness and elastic deformability.

The uniaxial tensile parameters of the homemade multilayer series are reported in the Table 5.11. The elastic modulus of the neat copolyester monolayers (996.2, 964.1, and 1002.3 MPa for MP100, MP150, and MP200, respectively) is comparable to that of the commercial PETG materials reported in Table 4.4 (941-1155 MPa), confirming that the Tritan copolyester grades studied here fall within the expected stiffness range for hard aligner thermoplastics. Upon incorporation of a soft interlayer, the elastic modulus decreases substantially in all configurations, with the magnitude of the reduction governed by the hardness of the soft layer grade. Within the MP100 series, the elastic modulus ranges from 623.3 MPa (100/EVA) to 695 MPa (100/70D), placing all multilayer configurations within the range of commercial multilayer materials (618-838 MPa, Table 4.4). Analogous trends are observed in the MP150 and MP200 series, where the elastic modulus increases monotonically with soft layer hardness from 622.5 to 789.3 MPa and from 625.3 to 750.3 MPa respectively. These values are consistent with the iso-strain mixing rule validated in the previous section, confirming that the elastic modulus of the composite under quasi-static tensile loading is governed by the same volume-weighted contribution of the constituent layers as the dynamic storage modulus.

| Material | E (MPa) | $\sigma_y$ (MPa) | $\varepsilon_y$ (%) | $T_e$ $(MJ/m)$ | $T_e/E$ $(MJ/GPam)$ | statistical Group |
|---|---|---|---|---|---|---|
| MP100 | 996.2 ±48.9 | 35.6 ±3.2 | 6.9 ±1.3 | 1.234 | 1.238 | d |
| 100/EVA | 623.3 ±64.4 | 27.4 ±2.4 | 7.2 ±0.8 | 0.986 | 1.583 | a |
| 100/85A | 660.3 ±89.2 | 29.7 ±3.1 | 7.4 ±1.0 | 1.099 | 1.664 | a |
| 100/52D | 675.8 ±36.1 | 28.9 ±3.0 | 7.8 ±1.0 | 1.127 | 1.668 | a |
| 100/70D | 695.0 ±71.7 | 22.3 ±4.5 | 5.4 ±1.2 | 0.602 | 0.866 | b |
| MP150 | 964.1 ±86.3 | 34.9 ±6.9 | 5.9 ±1.4 | 1.030 | 1.068 | d |
| 150/EVA | 622.5 ±59.9 | 24.8 ±1.8 | 6.3 ±0.3 | 0.781 | 1.255 | a |
| 150/85A | 694.4 ±49.5 | 27.4 ±3.9 | 6.6 ±1.6 | 0.904 | 1.302 | b |
| 150/52D | 720.9 ±65.3 | 30.8 ±4.2 | 6.8 ±1.0 | 1.047 | 1.453 | b |
| 150/70D | 789.3 ±36.8 | 32.6 ±3.3 | 5.26 ±1.2 | 0.857 | 1.086 | c |
| MP200 | 1002.3 ±45.1 | 42.3 ±1.9 | 7.5 ±0.9 | 1.586 | 1.583 | d |
| 200/EVA | 625.3 ±41.1 | 27.8 ±1.4 | 7.9 ±0.7 | 1.098 | 1.756 | a |
| 200/85A | 667.2 ±62.4 | 29.6 ±2.3 | 8.1 ±0.5 | 1.199 | 1.797 | a |
| 200/52D | 692.3 ±47.8 | 31.5 ±4.1 | 8.2 ±1.4 | 1.292 | 1.866 | b |
| 200/70D | 750.3 ±91.3 | 33.2 ±8.4 | 6.4 ±1.2 | 1.062 | 1.416 | c |

Table 5.11: Uniaxial tensile parameters of homemade multilayer materials. Different letters indicate statistically significant differences in elastic modulus (Tukey HSD, $\alpha$ = 0.05).

The yield stress $\sigma_y$ follows a less consistent trend with soft layer hardness. In the MP100 and MP150 series, $\sigma_y$ decreases for the softest grades (EVA and 85A) relative to the monolayer, then increases progressively with TPU hardness up to 52D before dropping sharply at 70D, a behaviour attributable to the embrittling effect of the rigid TPU 70D hard segment domains, which restrict plastic deformation and reduce the yield stress despite the higher elastic modulus. The MP200 series shows a more monotonic increase of $\sigma_y$ with soft layer hardness (27.8 to 33.2 MPa), consistent with the higher intrinsic stiffness of the MP200 copolyester providing a more dominant load-bearing contribution across all TPU grades.

Statistical analysis confirmed that the three monolayer grades are not significantly different from one another in elastic modulus (one-way ANOVA, F = 0.71, p = 0.51), indicating that the choice of Tritan grade does not affect the static elastic modulus, a result consistent with their similar backbone chemistry and comparable Tg values. The global ANOVA nonetheless confirmed highly significant differences across all materials (F = 28.70, p «< 0.001), and Tukey HSD post-hoc analysis revealed a consistent grouping structure: all monolayers form a distinct high-modulus group (letter d, 958-1002 MPa), statistically separated from all multilayer configurations. Within the multilayer groups, EVA and 85A grades are statistically equivalent in the MP100 and MP200 series (letter a, 597-668 MPa), while 52D occupies an intermediate position and 70D consistently forms its own higher-modulus subgroup (letters b-c, 708-788 MPa). The MP150 series shows the clearest progression, with all four soft layer grades forming statistically distinct sub-

groups (EVA: a, 85A-52D: b, 70D: c), confirming a monotonic and statistically significant increase of elastic modulus with TPU hardness in this series.

The yield strain $\varepsilon_y$, is the parameter most directly relevant to clinical safety, as it defines the maximum elastic deformation the aligner can sustain without permanent plastic deformation. Within the MP100 series, $\varepsilon_y$ increases progressively from 6.93 % (neat MP100) and 7.8 % (100/52D), before dropping to 5.4 % (100/70D), the lowst value in the MP100 group, anf comparable to PETG materials (5.01-5.70 % Table 4.4) This sharp reduction in $\varepsilon_y$, at 70D confirms that the rigid TPU 70D hard segment phase constrains the deformation capacity of the laminate, rendering its elastic working range comparable to that of a monolithic polyester despite its lower elastic modulus. A qualitatively identical pattern is observed in the MP150 series (5.26 % for 150/70D) and the MP200 series (6.4 % for 200/70D), confirming that this behaviour is independent of the hard layer copolyester grade and is an intrinsic consequence of the TPU 70D microstructure. In contrast, multilayers incorporating EVA, TPU85A or TPU52D consistenly show $\varepsilon_y$ values exceeding those of the neat monolayers and comparable to or above those of commercial multilayer materials incorporating PCTG outer layers (8.56-9.31% for Zendura FLX and Maflex Premium, Table 4.4). The MP200 series in particular shows the highest values across all soft layer grades (7.9-8.2% for EVA, 85A and 52D), suggesting that the combination of the higher-Tg MP200 copolyester with a compliant interlayer produces a particularly favourable elastic working range. It must be acknowledged, however, that the fabrication route employed here (Extrussion and thermopressing joining) may introduce surface and interfacial defects not present in commercially produced multilayer sheets, where continuous co-extrusion or lamination under precisely controlled conditions ensures superior interfacial adhesion and surface finish. In particular, the hot-plate pressing step may generate localized thickness variations, air bubbles, residual stress gradients, and interfacial imperfections that act as preferential crack initiation sites under tensile loading. Since yield strain is particularly sensitive to such stress concentrations, the $\varepsilon_y$ values reported here for the homemade multilayers should be regarded as conservative lower-bound estimates of the intrinsic elastic deformability of these material combinations, and are expected to improve upon optimization of the consolidation process.

The elastic toughness $T_e$ and the normalized elastic toughness ($T_e/E$), of the homemade multilayer series are as well reported in the table 5.11. These parameters are of primary clinical relevance, as clear aligners are designed to operate exclusively within the elastic regime: $T_e$ quantifies the elastic strain energy that the device can store and release per unit volume, while $T_e/E$ normalises this capacity by the stiffness of the material, providing a composite index of elastic efficiency that accounts simultaneously for the force delivered and the comfort of insertion.

The neat copolyester monolayers present $T_e$ values of 1.234, 1.030 and 1.586 $MJ/m$ for MP100, MP150 and MP200 respectively, comparable to the PETG commercial materials (1.04-1.28 $MJ/m$, Table 4.5) but below the hTPU range (1.85-2.39 $MJ/m$). However, when normalised by the elastic modulus, the $T_e/E$ values of the monolayers (1.068-1.583

$MJ/GPam$ ) are intermediate between those of PETG (0.91-1.15 $MJ/GPam$) and hTPU (1.52-1.96 $MJ/GPam$), reflecting the higher-Tg and correspondingly greater backbone rigidity of the Tritan copolyester grades relative to standard PETG.

Upon incorporation of a soft interlayer, $T_e/E$ increases substantially in all configurations except those incorporating TPU 70D. Within the MP100 series, $T_e/E$ rises from 1.24 $MJ/GPam$ (monolayer) to 1.583 $MJ/GPam$ (100/EVA), 1.664 $MJ/GPam$ (100/85A) and 1.668 $MJ/GPam$ (100/52D), before dropping sharply to 0.866 $MJ/GPam$ for 100/70D — the lowest value in the entire dataset and below that of any commercial material except Filtzwiesser (1.02 $MJ/GPam$) and Maflex Premium Plus (1.01 $MJ/GPam$). This sharp reduction confirms that TPU 70D compromises the elastic efficiency of the laminate to a clinically significant degree, despite its higher contribution to the elastic modulus. Qualitatively identical trends are observed in the MP150 and MP200 series, where $T_e/E$ peaks at the 52D grade (1.453 and 1.866 respectively) before declining at 70D (1.086 and 1.416).

The MP200 series presents the highest $T_e/E$ values across all soft layer grades, with 200/52D achieving the maximum in the entire dataset (1.866 $MJ/GPam$), exceeding all commercial multilayer materials (1.01-1.97 $MJ/GPam$, Table 4.5) and approaching the values of the best-performing commercial hTPU materials (Taglus: 1.96 $MJ/GPam$, Tristar S: 1.75 $MJ/GPam$). This result is particularly significant because the MP200-based multilayers achieve this elastic efficiency at a substantially lower elastic modulus (625-750 MPa) than monolithic hTPU (1213-1255 MPa), implying that they can store and release a comparable amount of elastic energy per unit stiffness while exerting considerably lower absolute forces on the dental arch, a combination that is not achievable with any monolithic material chemistry currently available.

Across all three copolyester grades, multilayers incorporating EVA, TPU 85A and TPU 52D consistently outperform their neat monolayer counterparts in $T_e/E$, confirming that the incorporation of a compliant interlayer is an effective strategy for improving elastic efficiency independently of the hard layer chemistry. TPU 70D represents the exception to this trend and should therefore be regarded as suboptimal from an elastic efficiency standpoint, despite its higher modulus contribution — a conclusion that is consistent with the reduced yield strain observed for all 70D configurations and with the embrittling effect of the rigid hard segment domains discussed previously.

The shape of the stress-strain curves provides additional insight into the deformation mechanisms of the homemade multilayer series, and a representative example for the MP100 series is shown in the figure 5.12. The neat copolyester monolayers exhibit a response characteristic of commercial glassy thermoplastics such as PETG and PCTG: a well-defined linear elastic region, a sharp and well-localised yielding peak, and a subsequent plastic flow regime extending to final rupture. This behaviour is consistent with the response of amorphous copolyesters operating well below their glass transition temperature at room temperature.

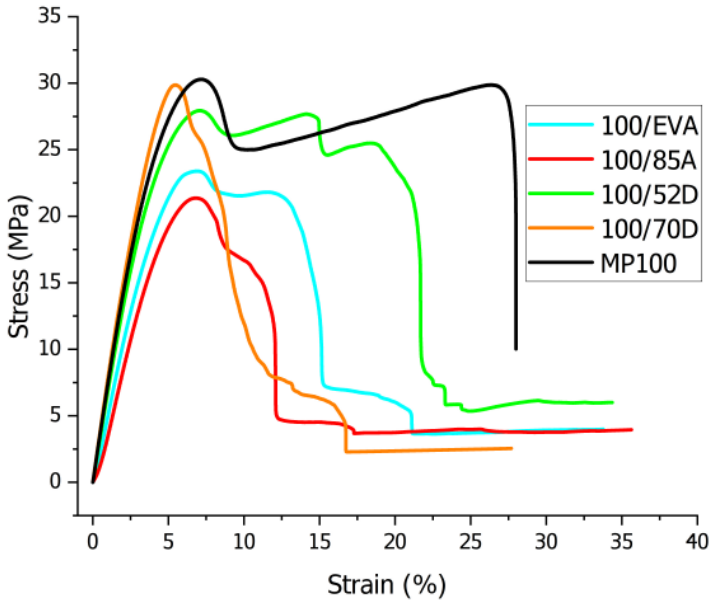



Figure 5.12: stress-strain curves of all the 100 series multilayers

The homemade multilayer configurations reproduce qualitatively the stress-strain response observed in commercial multilayer aligner materials previously (see figure 4.9). The curve is characterized by a well-defined elastic region, followed by a relatively broad and diffuse yielding zone — in contrast to the sharp yield peak of the monolayers — and a short plastic regime. Beyond this point, a process of sequential layer rupture occurs: the outer rigid copolyester layers fracture progressively rather than simultaneously, each failure event manifesting as a discrete load drop in the stress–strain curve. The tail of the curve represents the mechanical response of the intact elastomeric inner layer alone, which — by virtue of its high extensibility and low modulus — continues to sustain load after the complete fracture of both copolyester outer layers, acting as a crack-bridging ligament that prevents catastrophic failure and confers a degree of damage tolerance not present in any monolithic material. This sequential failure mechanism is clinically relevant: it implies that even after partial structural failure of the hard outer layers, the device retains a residual load-bearing capacity through the intact elastomeric core, potentially maintaining some degree of mechanical function and preventing sharp-edged fracture fragments that could injure the soft tissues of the oral cavity

### 5.3.4. Master curves of homemade multilayers

The master curves of all homemade multilayer configurations at 37 °C were constructed following the same time-temperature superposition protocol applied to the commercial materials in Chapter 4. The extracted parameters and DZ model fitting results are reported in Tables 5.12 and 5.13 respectively. The theoretical basis of the master curve analysis

and the DZ model is provided in Appendix A.

| Material | $E'_{max}$ (MPa) | $E'_{min}$ (MPa) | $E'_g$ (MPa) | $t_{lin}$ (h) | $t_g$ (h) |
|---|---|---|---|---|---|
| MP100 | 1315 | 3.76 | 141 | 2.85 | 595 |
| 100/85A | 846 | 21 | 105 | 1.68 | 790 |
| 100/52D | 909 | 5.1 | 104 | 1.42 | 540 |
| 100/70D | 1150 | 25 | 115 | 1.35 | 425 |
| MP150 | 1310 | 18.9 | 145 | 0.60 | 167 |
| 150/85A | 685 | 10.2 | 80 | 0.42 | 171 |
| 150/52D | 745 | 16.2 | 85 | 0.43 | 51 |
| 150/70D | 1050 | 20.2 | 82 | 0.31 | 32 |
| MP200 | 1401 | 10.6 | 124 | 5.3 | 1260 |
| 200/85A | 832 | 5.5 | 76.8 | 1.3 | 2300 |
| 200/52D | 950 | 27.1 | 114.2 | 1.2 | 1160 |
| 200/70D | 1190 | 25.2 | 114.8 | 0.95 | 1008 |

Table 5.12: TTS parameters of homemade multulayers

Several systematic trends are immediately apparent. The maximum storage modulus $E'_{max}$ decreases consistently upon incorporation of a soft layer relative to the neat hard layer monolayer, with the magnitude of the reduction governed by the TPU hardness grade, consistent with the iso-strain mixing rule validated for the static modulus. The rubbery modulus $E'_{min}$ increases systematically with TPU hardness within each series, reflecting the higher plateau modulus of stiffer TPU grades and their greater contribution to the composite rubbery floor.

The glass transition time $t_g$ is the parameter most directly relevant to clinical force delivery. Within the MP100 series, $t_g$ decreases monotonically with increasing TPU hardness, from 790 h for 100/85A, which exceeds the neat MP100 monolayer (595 h), to 425 h for 100/70D. This behavior reflects two competing effects of the soft interlayer: at low TPU hardness, the compliant layer retards cooperative segmental motion in the hard layer through mechanical coupling at the interface, extending $t_g$ beyond the monolayer value; as TPU hardness increases, the higher rubbery modulus of the soft layer elevates the composite floor modulus and accelerates the apparent onset of the modulus drop. A qualitatively identical pattern is observed in the MP150 and MP200 series. The MP200 series yields by far the largest $t_g$ values in the entire dataset: even the stiffest configuration (200/70D, $t_g = 1008\ h$) substantially exceeds the longest $t_g$ reported among commercial materials (765 h for Maflex Premium), and 200/85A reaches $t_g = 2300\ h$ (approximately 96 days), confirming that the combination of the high-Tg MP200 copolyester with a compliant interlayer extends the therapeutic force window far beyond what is achievable with any currently available commercial material.

Substantial differences in $t_g$ are also observed between the MP100 and MP150 series despite their similar glass transition temperatures ($Tg \simeq 120\ C$), suggesting that factors

beyond Tg alone, including molecular weight distribution and chain architecture, govern the absolute relaxation kinetics of these materials.

| Material | $E_r$ (MPa) | $E_1$ (MPa) | $\alpha_1$ | $\tau_1$ (h) | $E_2$ (MPa) | $\alpha_2$ | $\tau_2$ (s) |
|---|---|---|---|---|---|---|---|
| MP100 | 4.4 | 842.2 | 0.78 | 21.0 | 427.1 | 0.55 | 900 |
| 100/85A | 5.2 | 524.7 | 0.74 | 16.3 | 294.2 | 0.34 | 648 |
| 100/52D | 20.3 | 607.0 | 0.78 | 13.7 | 360.1 | 0.42 | 540 |
| 100/70D | 22.5 | 745.3 | 0.77 | 10.3 | 385.9 | 0.55 | 468 |
| MP150 | 5.43 | 941.0 | 0.73 | 2.6 | 422.9 | 0.23 | 180 |
| 150/85A | 9.4 | 436.3 | 0.72 | 3.7 | 239.6 | 0.38 | 180 |
| 150/52D | 18.6 | 469.3 | 0.84 | 1.6 | 257.8 | 0.47 | 180 |
| 150/70D | 24.6 | 720.9 | 0.78 | 1.3 | 305.3 | 0.51 | 216 |
| MP200 | 6.4 | 783.7 | 0.77 | 38.0 | 289.9 | 0.45 | 180 |
| 200/85A | 9.8 | 479.9 | 0.77 | 58.0 | 342.4 | 0.49 | 1008 |
| 200/52D | 16.5 | 575.1 | 0.75 | 48.0 | 360.3 | 0.34 | 936 |
| 200/70D | 25.3 | 763.7 | 0.76 | 30.0 | 402.3 | 0.51 | 720 |

Table 5.13: DZ model fitting parameters of homemade multilayer master curves

The rubbery modulus $E_r$ increases systematically with TPU hardness within each series, consistent with the higher plateau modulus of stiffer grades. The characteristic slow relaxation time $\tau_1$, the parameter most directly relevant to sustained force delivery, follows a clear and clinically significant pattern. Within the MP100 series, $\tau_1$ decreases monotonically with increasing TPU hardness, from 16.3 h (100/85A) to 10.3 h (100/70D), all below the neat MP100 monolayer value (21.0 h). The MP200 series presents the longest $\tau_1$ values in the entire dataset: 58.0 h (200/85A), 48.0 h (200/52D) and 30.0 h (200/70D), all substantially exceeding the longest $\tau_1$ reported among commercial materials (24.4 h for Maflex Premium). These values imply that the slow relaxation process in MP200-based multilayers extends well beyond the full recommended wearing period of 20-22 h per day, confirming the exceptional force delivery characteristics of this material series.

The breadth parameter $\alpha_1$ ranges from 0.72 to 0.84 across all homemade multilayer configurations, narrower and systematically lower than monolithic PETG (0.89) but comparable to commercial multilayer materials (0.76-0.81), confirming that the lamination process produces a more broadly distributed slow relaxation process than monolithic thermoplastics regardless of the specific copolyester or TPU grade employed. No systematic trend with TPU hardness is apparent for $\alpha_1$, confirming that the shape of the slow relaxation process is governed primarily by the hard layer chemistry rather than by the soft interlayer.

The fast relaxation parameters $\tau_2$ and $\alpha_2$ govern insertion comfort. The MP200 series presents substantially longer $\tau_2$ values (720-1008 s) than the MP150 series (180-216 s), suggesting that local segmental mobility is not determined solely by Tg, but depends also on additional molecular characteristics of the copolyester. Across all series, the combina-

tion of intermediate $\tau_2$ values of the order of minutes and moderate $\alpha_2$ values (0.23-0.55) is consistent with a gentle and progressive adaptation to the dental arch upon seating, a profile comparable to that of state-of-the-art commercial multilayer aligners and substantially more comfortable than the abrupt insertion feel characteristic of monolithic PETG.

Taken together, the DZ model parameters reveal two interdependent design levers for the homemade multilayer series: the hard layer copolyester grade governs the timescale and shape of the slow relaxation process through its Tg and molecular architecture, while the soft layer TPU grade modulates the absolute force magnitude and the rubbery floor modulus. The combination of these two levers — independently tunable through material selection alone — provides a rational and experimentally validated basis for the design of multilayer aligner materials with prescribed clinical force delivery profiles and represents one of the central methodological contributions of the present thesis.

### 5.3.5. Stress relaxation curves

Stress relaxation experiments in tensile configuration were performed on all homemade multilayer materials at 37 °C, following the same protocol applied to the commercial materials. In the interest of experimental efficiency, only the 0.5% strain condition was evaluated for the homemade series, whereas the commercial materials were characterized at three strain levels (0.5, 1.5, and 2%). Each experiment consisted of an initial thermal equilibration of the DMTA chamber at 37 °C, followed by a step deformation to 0.5% strain held constant for 3 h while force and stress were continuously monitored. Upon completion of the relaxation period, the applied force was reduced to 0.001 N, and the samples were allowed to recover freely at 37 °C for 5 minutes under strain monitoring, enabling assessment of the elastic recovery capacity of each material following the relaxation cycle.

| Material | $E_0$ (MPa) | a (-) | $E(3h)/E_0$ (%) | Strain recovery (%) |
|---|---|---|---|---|
| MP100 | 1075 | 0.21 | 17.5 | 43 |
| 100/EVA | 535 | 0.08 | 63 | 67 |
| 100/85A | 798 | 0.06 | 63 | 62 |
| 100/52D | 828 | 0.11 | 47 | 48 |
| 100/70D | 1011 | 0.14 | 44 | 44 |
| MP150 | 972 | 0.25 | 27 | 28 |
| 150/EVA | 489 | 0.08 | 64 | 62 |
| 150/85A | 685 | 0.10 | 55 | 56 |
| 150/52D | 763 | 0.13 | 51 | 53 |
| 150/70D | 963 | 0.14 | 48 | 47 |
| MP200 | 1252 | 0.36 | 19.2 | 20 |
| 200/EVA | 682 | 0.09 | 50 | 63 |
| 200/85A | 767 | 0.12 | 52 | 62 |
| 200/52D | 859 | 0.15 | 56 | 59 |
| 200/70D | 968 | 0.21 | 40 | 45 |

Table 5.14: Stress relaxation parameters for homemade multilayers

The stress relaxation parameters extracted from the 3-hour tensile experiments at 37 °C are reported in Table 5.14. The initial relaxation modulus $E_0$ follows the same trends as the dynamics storage modulus reported in the table 5.10 and the elastic modulus at room temperature in the table 5.11. It increases monotonically with TPU hardness within each series and is substantially higher in the neat monolayers than in any multilayer configuration, confirming the consistency between dynamic and quasi-static mechanical characterization, and at the same time the iso-strain rule of mixtures.

Regarding the relaxation rate parameter (a), it is defined by the power-law decay $E(t) \sim E_0 t^{-a}$. The parameter is obtained by fitting the log(E) against log(t) between 10 minutes and 3h, a time window dominated by the slow cooperative relaxation process associated with the glass transition. The a parameter quantifies the rate at which the material loses its load-bearing capacity at body temperature. The initial transient decay within the first 10 minutes, governed primarily by the fast local segmental relaxation process, is excluded from the fit as it does not follow the same power-law behaviour as the subsequent long-time regime.

Within each multilayer series, a decreases substantially upon incorporation of a soft interlayer and increases progressively with TPU hardness, ranging from 0.06-0.09 for EVA and TPU85A to 0.14-0.21 for TPU70D. The softest grades therefore produce the most gradual and temporally distributed force decay, consistent with their lower $\alpha_2$ values in the DZ model. Among the neat monolayers, a increases markedly with the glass transition temperature of the copolyester: from 0.21 for MP100 to 0.25 for MP150 and 0.36 for MP200. This counterintuitive result — that the highest-Tg material relaxes most

rapidly in terms of the power-law exponent — is consistent with the master curve data: at 37 °C, the greater thermal distance from Tg in MP200 places the measurement deeper in the sub-glassy regime, where the fractional viscoelastic exponent a reflects the cumulative contribution of multiple relaxation processes distributed over a very wide time window, producing a steeper apparent power-law decay despite the longer absolute relaxation times.

The percentage of modulus retained after 3 hours, $E(3h)/E_0$, provides the most clinically direct index of force stability over the wearing period. The neat monolayers retain only 17.5% (MP100), 27% (MP150) and 19.2% (MP200) of their initial modulus after 3 hours — values substantially lower than any multilayer configuration — confirming that monolithic copolyesters undergo rapid and nearly complete stress relaxation well within the first hours of wear, consistent with the very short $\tau_1$ values reported in the master curve analysis in contrast, all multilayer configurations retain between 40 and 64% of their initial modulus at 3 hours, representing a two- to threefold improvement in force retention relative to the corresponding monolayer. Within each series $E(3h)/E_0$ decreases monotonically with increasing TPU hardness: the softest grades (EVA and TPU 85A) consistently retain 55-64% of their initial modulus, while TPU 70D retains 40-48%. This trend confirms that softer interlayers not only reduce the absolute force delivered but also stabilize it more effectively over the wearing period — a combination that is clinically advantageous for sustained and comfortable orthodontic force delivery. The MP200 series presents an interesting observation: despite having the highest values among all multilayer groups, 200/52D retains 56% of its modulus at 3 hours — comparable to the softest grades in the MP100 and MP150 series — suggesting that the broader relaxation distribution of MP200 redistributes force decay more evenly across the measurement window even at intermediate TPU hardnesses.

The strain recovery after 5 minutes at near-zero load provides a complementary index of elastic reversibility following the relaxation cycle. The neat monolayers show markedly poor recovery: MP100 recovers only 43 % MP150 28%, and MP200 20% of the applied strain — confirming that the 0.5% strain imposed over 3 hours induces substantial irreversible creep in the monolithic copolyesters, particularly in MP200 where the combination of the highest a and lowest recovery indicates the greatest permanent set among all materials tested. All multilayer configurations recover substantially more strain than their corresponding monolayers, with EVA and TPU 85A grades consistently recovering 56–67% — values approaching those reported for commercial multilayer materials. The strain recovery values are generally consistent with $E(3h)/E_0$, confirming that materials retaining more force also recover more deformation elastically. The slight discrepancies between the two parameters — most notably in the MP200 series, where 200/EVA and 200/85A show strain recovery values (63% and 62%) exceeding their $E(3h)/E_0$ ratios (50% and 52%) — suggest that these configurations exhibit a degree of delayed elastic recovery during the 5-minute unloading period that is not captured by the modulus retention ratio alone, consistent with the broad relaxation time distribution and long $\tau_1$ values

characteristic of MP200-based laminates.

## 5.4. Discussion of the homemade multilayer production

The results presented in this chapter demonstrate that multilayer thermoplastic laminates with controlled and predictable mechanical properties can be successfully manufactured through a semi-industrial extrusion and hot-plate pressing route, using Tritan copolyesters as hard layers and polyether TPU or EVA as compliant interlayers. The practical viability of this fabrication route is confirmed by the thermoformed clear aligners shown in Figures 5.13 and 5.14, produced from homemade multilayer sheets within the scope of this project, which demonstrate that the manufactured laminates are suitable for direct clinical device fabrication.

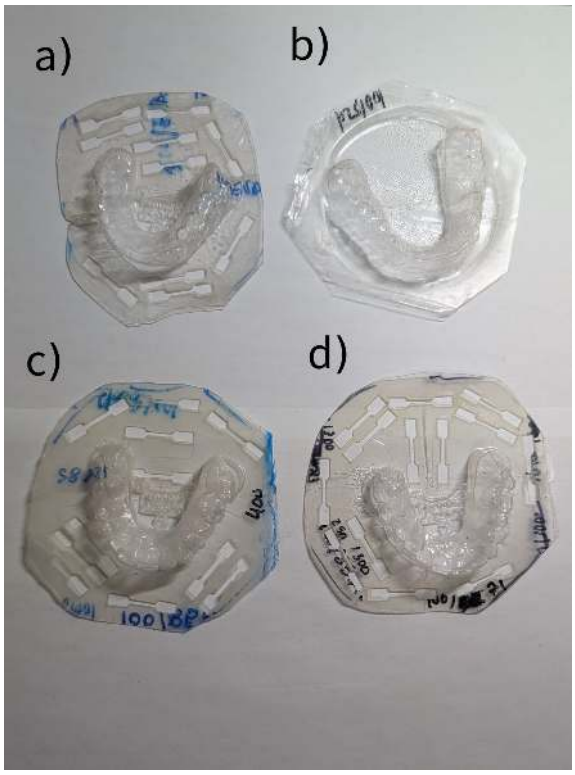



Figure 5.13: Overview of four homemade thermoformed multilayer aligners: a) MP100/EVA, b) MP100/52D, c) MP100/85A, and d) MP100/70D.

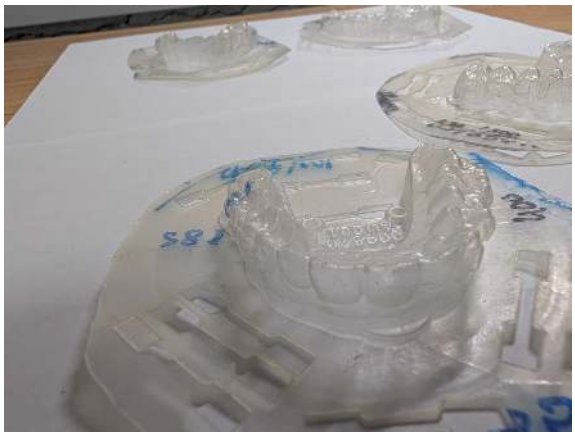


Figure 5.14: Close-up view of a MP100/85A aligner

### 5.4.1. Material selection and processability

The screening of candidate hard layer materials identified the Tritan copolyester grades (MP100, MP150, MP200) as the only viable candidates, based on their high and stable glass transition temperatures (120-131 °C), fully amorphous character, absence of physical aging peaks, and processability by extrusion and thermoforming without crystallization-related opacity. PEN- and PET-PEN-based materials, despite their attractive combination of high Tg and secondary transition, which could in principle provide enhanced damping and reduced brittleness, were discarded due to the impossibility of obtaining transparent amorphous films under any of the processing conditions explored. TPU 84D was similarly discarded due to its crystallization behavior and elevated loss modulus at oral temperature, which would compromise the shape retention of the device during clinical use. Among the soft layer candidates, polyether TPU grades and EVA were selected over polyester TPU on the grounds of hydrolysis resistance, processing stability, and supply chain reliability. TPU70D was retained in the experimental matrix but consistently identified as suboptimal: it was the only grade to cause severe thermal degradation and screw blocking during extrusion, and its mechanical characterization revealed that it compromises both yield strain and normalized elastic toughness $T_e/E$ relative to TPU52D, without providing a commensurate improvement in any clinically relevant parameter.

A particularly promising direction opened by the present results is the use of EVA as an alternative soft layer to TPU. To the best of our knowledge, no commercial or under development multilayer aligner material currently incorporates EVA as a soft layer. The mechanical results reported here suggest that EVA-based multilayers could represent a clinically relevant alternative, particularly for patients with high mechanical sensitivity — for whom the lower modulus of EVA relative to TPU85A may reduce insertion discomfort — or for patients with known sensitivity or allergy to TPU components. From a processing standpoint, EVA presents several practical advantages over TPU: it is more readily extruded into thin films with fewer defects, does not require the extended drying times or specialized equipment imposed by TPU, and enables multilayer consolidation

at substantially lower pressing temperatures (180°C versus 220°C for TPU-based configurations), reducing both energy consumption and the risk of thermal degradation at the layer interfaces. However, the elastic moduli of the EVA grade characterized in this study were relatively low — below those of even the softest TPU grade (TPU85A), which limits the contribution of the EVA layer to the composite modulus and may constrain the range of target stiffnesses achievable with EVA-based laminates. More broadly, the successful fabrication and characterization of EVA-based multilayers demonstrates that the design space for the soft interlayer is not limited to TPUand opens the door to the systematic exploration of other elastomeric materials as soft layer candidates, including EVA grades with higher vinyl acetate content or alternative polyolefin elastomers, potentially enabling further tuning of the mechanical and biological response of the laminate beyond what is achievable with the TPU grades currently available.

### 5.4.2. Comparison with commercial multilayer materials

The homemade multilayers compare favorably with commercial multilayer aligner materials across all characterization domains. In terms of dynamic storage modulus at oral temperature, the softest homemade configurations (EVA and TPU 85A grades, 685-927 MPa) fall within the range of commercial multilayer materials (618- 857 MPa for Zendura FLX and Maflex Premium), while the stiffer grades (TPU52D and TPU70D) approach or exceed the upper bound of the commercial range. The Cole-Cole arc areas of the homemade multilayers (0.041-0.064 $GPa$) are identical in mean value and range to those of the commercial reference materials (0.030-0.060 $GPa$) , confirming that the lamination process produces a viscoelastic relaxation spectrum comparable to that of state-of-the-art commercial aligners, despite the fundamentally different manufacturing route.

The most significant advantage of the homemade multilayers over commercial materials lies in their viscoelastic performance at body temperature. The MP200 series in particular produces $\tau_1$ and $t_g$ values that substantially exceed those of any commercial material surveyed: the longest $\tau_1$ in the commercial dataset is 24.4 h (Maflex Premium), whereas 200/85A achieves $\tau_1$=58.0 h and $t_g$=2300 h, approximately three times the clinical wear cycle of 14 days. Even the stiffest MP200 configuration (200/70D) presents $t_g$=1008 h, exceeding Maflex Premium by a factor of 1.3. These results confirm that the synergistic combination of a high-Tg copolyester (MP200,Tg = 130 ºC) with a compliant elastomeric interlayer produces a therapeutic force window that is qualitatively beyond the reach of any currently available commercial material, regardless of its architecture.

In terms of elastic efficiency, the MP200/52D configuration achieves the highest $T_e/E$ in the entire dataset (1.866 $MJ/GPam$), approaching the values of the best-performing commercial hTPU materials (Taglus 1.96 $MJ/GPam$, and Tristar S 1.75 $MJ/GPam$) while operating at a substantially lower elastic modulus (692 MPa against 1213-1255 MPa). This combination — high elastic energy storage per unit stiffness at a lower absolute modulus — is not achievable with any monolithic material chemistry and represents

a genuinely novel performance envelope accessible only through the multilayer architecture. The stress relaxation data further confirm this advantage: all multilayer configurations retain 40-64% of their initial modulus after 3 hours at 37 °C, compared to 17-27% for the neat copolyester monolayers, with the softest TPU grades (EVA and 85A) consistently achieving the highest retention values regardless of the hard layer chemistry.

The normalized elastic toughness of the homemade multilayers also compares favorably with commercial materials. Configurations incorporating EVA, TPU85A and TPU52D consistently outperform their monolayer counterparts and are comparable to or exceed the commercial multilayer range (1.01-1.97 $MJ/GPam$), confirming that the lamination process does not compromise the elastic efficiency of the hard layer but systematically enhances it through the introduction of the compliant interlayer. The only exception is TPU70D, which reduces $T_e/E$ below the commercial range in all three copolyester series — a finding that, combined with its processing difficulties, provides a strong basis for its exclusion from any future scale-up.

### 5.4.3. Design framework

A key outcome of this chapter is the establishment of a rational quantitative design framework for multilayer aligner materials. The iso-strain rule of mixtures (mean absolute error 7.8%) enables the storage modulus of the laminate at oral temperature to be predicted directly from the known properties of the constituent layers, without the need for iterative fabrication and characterization of intermediate compositions. For TPU-based soft layers, this framework can be further simplified: if the Shore hardness of the TPU falls within the experimental range covered in this study (33-84 Shore D) its elastic modulus at oral temperature can be estimated from the exponential relationship validated here across four independent measurement methods (nanoindentation, DMTA, tensile, and bending), using only the Shore hardness value reported in the material datasheet — a parameter that is universally available, reliable, and requires no additional experimentation. This means that, in practice, a target composite modulus can be translated directly into a soft layer Shore hardness selection without characterizing the soft layer independently, reducing the material development cycle to a single fabrication and validation step rather than an iterative experimental process. The systematic overestimation by the Voigt model — attributable to the mechanically active interphase, the disproportionate deformation of the soft layer, and localized softening during hot-plate pressing — is consistent across all configurations, implying that the true modulus will reliably fall slightly below the predicted value and providing a conservative and safe design bound. Taken together, these results establish a predictive materials design workflow in which the elastic properties of a multilayer aligner can be specified in advance from constituent layer data alone, enabling rational material selection for a given clinical target modulus.

The DZ model parameters provide an additional predictive layer: $\tau_1$ and $E_r$ scale systematically with TPU hardness and copolyester Tg, enabling the therapeutic force window

and residual force to be estimated from material selection alone. The finding that $\alpha_1$ is governed primarily by the hard layer chemistry rather than the soft layer grade simplifies the design space further: the shape of the slow relaxation process can be independently controlled through the choice of copolyester, decoupled from the modulus and $\tau_1$ tuning provided by the soft layer.

## 5.5. Conclusions of the homemade multilayer production

The following conclusions can be drawn from the results presented in this chapter:

1. The main finding of this thesis is that multilayer thermoplastic sheets for clear aligners with tailored elastic properties can be fabricated by semi-industrial means — extrusion of individual layers followed by hot-plate consolidation — starting from widely available commercial thermoplastics in pellet form. This is of direct strategic relevance for a medium-sized company such as Secret Aligner: the capital investment and production infrastructure required are substantially lower than those of industrial co-extrusion, the volume can be scaled to the demand of a mid-sized producer, and the material inputs are commercially accessible without dependence on proprietary polymers. The results therefore establish a technically and economically viable route to in-house multilayer production that does not require the capabilities of large multinational suppliers.

2. The Tritan copolyester grades MP100, MP150 and MP200 were identified as the most suitable hard layer candidates, on the basis of their high glass transition temperatures (120–131 °C), fully amorphous character, absence of physical aging under normal processing and storage conditions, and processability by extrusion and thermoforming. PEN-, PET-PEN- and TPU84D-based materials were discarded due to crystallization-related opacity and processing incompatibilities.

3. Among the soft layer candidates, polyether TPU grades (85A and 52D) and EVA were identified as the most viable options. TPU70D was consistently suboptimal across all characterization domains: it caused severe thermal degradation during extrusion and reduced both yield strain and normalized elastic toughness relative to TPU52D without a commensurate improvement in any clinically relevant parameter.

4. Thermal characterization confirmed that the constituent layers remain thermally and compositionally distinct after hot-plate pressing, with no evidence of intermixing or new interphase formation, and that the soft layer weight fraction is consistent with the nominal trilayer geometry across all configurations, confirming the compositional reproducibility of the process.

5. The storage modulus at oral temperature is accurately predicted by the iso-strain rule of mixtures (mean absolute error below 8%), establishing a rational quantitative

design framework. For TPU-based soft layers, the elastic modulus can be estimated directly from the Shore hardness reported in the material datasheet, so that a target composite modulus can be translated into a soft layer selection without additional experimental characterization.

6. The Cole-Cole arc areas of the homemade multilayers are equivalent in mean value and range to those of commercial multilayer reference materials, confirming that the lamination process produces a viscoelastic relaxation spectrum comparable to that of state-of-the-art commercial aligners.

7. The normalized elastic toughness $T_e/E$ was systematically improved by the incorporation of a compliant interlayer for all soft layer grades except TPU70D. The MP200/52D configuration achieved the highest $T_e/E$ in the entire dataset, approaching the values of the best commercial hTPU materials at a substantially lower elastic modulus — a combination not achievable with any monolithic material chemistry.

8. The MP200 series presents the most clinically favourable viscoelastic performance of all materials tested. Its characteristic slow relaxation time and glass transition timescale substantially exceed the values of any commercial multilayer material, including the best-performing commercial reference (Maflex Premium), confirming that the synergistic combination of a high-$T_g$ copolyester with a compliant interlayer extends the therapeutic force window beyond what is currently available commercially.

9. Stress relaxation at 37 °C confirmed that all multilayer configurations retain a substantially larger fraction of their initial modulus after 3 hours than the corresponding neat copolyester monolayers, with the softest TPU grades and EVA achieving the highest force retention and elastic recovery.

10. EVA represents a promising alternative soft layer to TPU, offering easier processability, lower pressing temperatures, and potential clinical advantages for mechanically sensitive or TPU-allergic patients. To the best of our knowledge, no commercial multilayer aligner material currently incorporates EVA as a soft layer, and the present results identify this as an unexplored direction worthy of further investigation.

11. The primary limitations of the current fabrication route — air entrapment at layer interfaces, surface irregularities, and the absence of optical transparency quantification — must be addressed through process optimization before clinical translation, as detailed in the limitations and future work of Chapter 8.

# 6. BIOCOMPATIBILITY AND INTRAORAL STABILITY OF HOMEMADE MULTILAYER ALIGNER MATERIALS

## 6.1. Introduction

Clear aligner thermoplastics are in prolonged and intimate contact with the oral mucosa, gingival tissues, and saliva throughout the treatment period, making biocompatibility a fundamental requirement alongside mechanical performance [26], [39], [104]. Although copolyesters such as PETG are generally considered to exhibit low or negligible cytotoxicity, the biocompatibility of TPU, a material of increasing relevance in clear aligner therapy, remains controversial, with reports of slight to moderate cytotoxic effects depending on the specific formulation [103], [104], [105], [319]. Thermoforming has been reported to further compromise cytocompatibility, likely through the release of low-molecular-weight leachables generated during thermal processing, increased surface area and reactivity, or thermally induced structural changes in the polymer [104], [105], [319]. The biocompatibility of multilayer materials, which combine copolyester and elastomeric phases in intimate contact, has received comparatively little attention in the literature[103], [104].

Beyond cytotoxicity, the inflammatory and oxidative potential of aligner materials represents an equally important but insufficiently characterized dimension of their biological performance. Regulated oxidative stress plays a central role in orthodontic tooth movement and bone remodeling, arising from the combined contribution of mechanical forces applied to the periodontium and the presence of foreign materials in the oral environment [320], [321], [322]. Reactive oxygen species (ROS) generated by aligner plastics — through microplastics release, leaching of degradation by-products, or surface interactions with oral fluids — may amplify the physiological inflammatory response and compromise periodontal tissue integrity [323], [324], [325]. The oxidative and pro-inflammatory potential of clear aligner plastics specifically remains insufficiently characterized [326], [327]. Metallic appliances have been shown to produce ROS through Fenton reactions, with an initial peak that normalizes after approximately seven days [328], [329], [330], [331], [332], but equivalent data for thermoplastic aligner materials are lacking.

The measurement of biochemical markers such as cytokines and enzymes has been proposed as a tool for orthodontic treatment monitoring and force control, as discussed in Chapter 1 (see Figure 1.7) [27], [28], [333], [334]. However, their interpretation requires careful consideration: elevated marker levels in the periodontal environment are not specific to mechanical loading and may equally reflect material-induced cytotoxicity, microplastic release, or leaching of degradation by-products from the aligner [326], [327]. Failure to account for the material contribution could therefore lead to erroneous clinical conclusions. No study has yet characterized the release of biochemical damage markers specifically associated with the use of clear aligner thermoplastics, and the contribution

of material-induced effects to the overall periodontal inflammatory response during treatment remains unresolved.

Cases of severe local irritation reactions have also been reported in association with specific polyurethane formulations in contact with the oral epithelium [43], [106], [108], and Secret Aligner has reported no biological adverse effects or allergic reactions associated with the clinical use of either Zendura FLX or Maflex Premium — providing a relevant clinical safety baseline against which the homemade materials can be contextualized. A further factor of clinical relevance is the stability of aligner materials under simulated intraoral conditions: water absorption, plasticization, and hydrolytic degradation during prolonged immersion in saliva may alter the physicochemical and mechanical properties of the thermoplastic, potentially compromising the force delivery profile of the device over the clinical wear cycle [74].

This chapter reports the biocompatibility and intraoral stability characterization of the homemade multilayer materials developed in Chapter 5, alongside their monolayer constituents and the commercial reference material Maflex Premium, which is currently employed by Secret Aligner in clinical practice. The characterization comprises indirect and direct cytotoxicity assessment using human gingival fibroblasts (hGFs) following ISO 10993-5 guidelines [164], quantification of reactive oxygen species (ROS) in material extracts, cytokine profiling by ELISA, and evaluation of artificial saliva absorption and its effects on the thermomechanical, spectroscopic, and thermal properties of the materials after 17 days of immersion. The commercial reference material Maflex Premium was included as a benchmark against which the performance of the homemade multilayers is evaluated, given its current use in clinical practice by Secret Aligner.

Prior to biocompatibility testing, a subset of materials was selected from the full experimental matrix characterized in Chapter 5. Biological testing is inherently resource-intensive in terms of both time and cost; the number of materials was therefore reduced to those configurations whose mechanical behaviour most closely approaches that of current commercial multilayer aligners, ensuring that the biocompatibility data are generated on the most clinically relevant candidates.

From the hard layer series, MP100 and MP200 were selected as representative copolyester grades. MP150 was excluded for two reasons: first, its stress relaxation performance was systematically inferior to that of MP100 despite sharing a similar glass transition temperature, offering no clear mechanical advantage to justify its inclusion; and second, MP150 is a specialized extrusion grade whose formulation may incorporate processing aids or additives not present in the standard Tritan grades, introducing an additional and poorly characterized source of variability in the biological response that would complicate the interpretation of the biocompatibility results. MP100 and MP200 together span the full range of glass transition temperatures represented by the Tritan series (120 °C and 130 °C respectively), providing representative coverage of the hard layer design space with the minimum number of materials.

From the soft layer series, TPU 85A and TPU 52D were selected as the most clinically relevant candidates. Both grades consistently delivered the best stress relaxation performance in the homemade multilayer series, combining the highest force retention at 3 hours with the best elastic recovery, and both fall within the elastic modulus range identified as optimal for multilayer aligner applications. TPU 70D was excluded on multiple grounds: its processing by extrusion was severely compromised by thermal degradation, yellowing and bubble formation, resulting in films with frequent surface and interfacial defects that would introduce uncontrolled variability in the biological response; and the mechanical characterization demonstrated that it does not provide a commensurate improvement in elastic modulus or stress relaxation relative to TPU 52D to justify the additional processing complexity. EVA was excluded from the biocompatibility study due to the inability to secure a reliable large-scale supply from the original material provider, precluding the preparation of sufficient sample quantities for the planned biological assays. The commercial multilayer Maflex Premium was included as a comparative reference, as it is the material currently employed by Secret Aligner in clinical practice and therefore represents the benchmark against which the homemade multilayers must be evaluated. Secret Aligner has reported no biological adverse effects or allergic reactions associated with its clinical use, providing a relevant safety baseline for contextualising the present results.

## 6.2. Materials and methods

The materials evaluated in this chapter, the sample preparation protocols, and the complete experimental procedures for all biological and physicochemical assays — including ATR-FTIR characterization of culture medium extracts and recovered plastic samples, water and artificial saliva absorption testing with post-immersion DSC and DMTA characterization, indirect and direct cytotoxicity assessment, reactive oxygen species quantification, and cytokine profiling by ELISA — are described in detail in Chapter 3, Sections 3.9-3.9.9. The specific combinations of materials and assays performed in the present chapter are summarized in Table 6.1.

| Material | Indirect | Direct | ROS | ELISA | Saliva |
|---|---|---|---|---|---|
| MP100 | ✓ | ✓ | ✓ | ✓ | ✓ |
| MP200 | ✓ | ✓ | ✓ | ✓ | ✓ |
| TPU 85A | ✓ | ✓ | ✓ | ✓ | – |
| TPU 52D | ✓ | ✓ | ✓ | ✓ | – |
| 100/85A | – | ✓ | ✓ | ✓ | ✓ |
| 100/52D | – | ✓ | ✓ | ✓ | ✓ |
| 200/85A | – | ✓ | ✓ | ✓ | ✓ |
| 200/52D | – | ✓ | ✓ | ✓ | ✓ |
| Maflex Premium | ✓ | ✓ | ✓ | ✓ | ✓ |

Table 6.1: Materials evaluated in each biological and physicochemical assay performed in Chapter 6. Checkmarks indicate assays performed; dashes indicate assays not conducted for that material.

## 6.3. Results of the biocompatibility and intraoral stability of homemade multilayers

### Analysis of culture medium extracts

ATR-FTIR spectra of culture medium conditioned by immersion of each material for 1, 3, and 7 days at 37 °C were compared against freshly prepared medium to detect leachable compounds released from the plastics. As shown in Figure 6.1, the differences between conditioned and non-conditioned media were minimal across all materials and time points. The most notable spectral changes were a slight increase in band intensity at 1044, 1090, 2360, and 2342 $cm^{-1}$

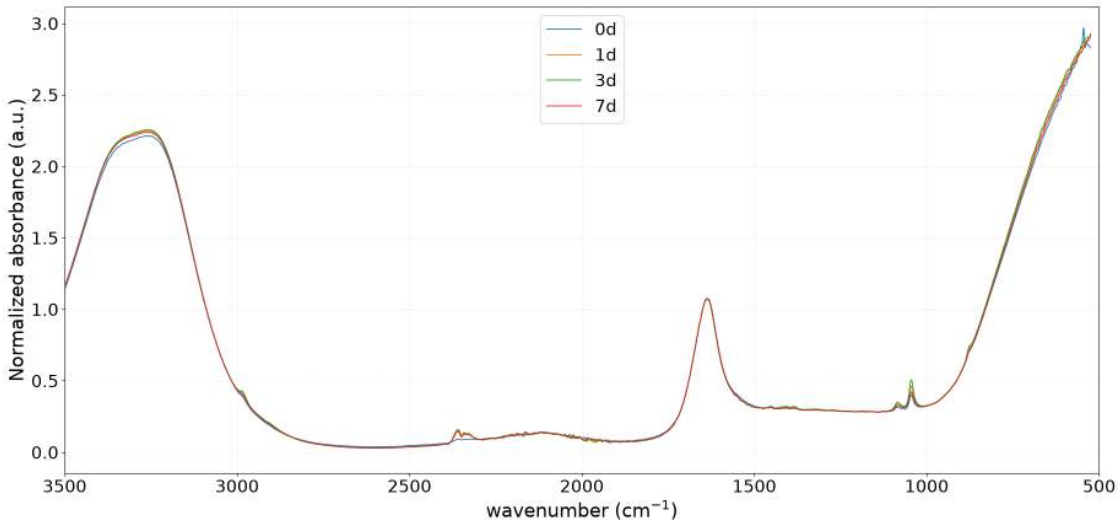



Figure 6.1: ATR-FTIR spectra of culture media exposed to MP100 copolyester for 1,3 and 7 days

The bands at 1044 and 1090 $cm^{-1}$ are attributed to C – O stretching vibrations. In copolyesters, these signals may reflect the release of low-molecular-weight diol species originating from CHDM or TMCD units along the backbone [189], [335]; in polyether TPU materials, the same bands may arise from the diffusion of poly(tetramethylene ether glycol) (PTMEG) oligomers from the soft segment phase or the chain extender into the medium [336]. The bands at 2360 and 2342 $cm^{-1}$ are characteristic of dissolved $CO_2$ and are indicative of a slight acidification of the medium during incubation [336], [337]. The low intensity of all observed differences — particularly of the 1044 and 1090 $cm^{-1}$ bands, indicates that leaching of compounds from any of the tested materials is minimal under the experimental conditions employed and is unlikely to account for any significant biological response in the subsequent cytotoxicity assays.

**Analysis of recovered plastic samples**

ATR-FTIR spectra of plastic samples recovered after the indirect cytotoxicity assay were compared against non-exposed controls of the same material. No significant spectral changes were observed in any of the materials tested: differences in band intensities were negligible and attributable to minor variations in contact pressure between the sample and the ATR crystal rather than to any material degradation or surface modification induced by immersion in cell culture medium. This confirms that 72 h exposure to culture medium at 37°C does not produce detectable physicochemical alterations in the surface chemistry of any of the materials studied. Equally, no significant spectral changes were observed in any of the materials following 17 days of immersion in artificial saliva at 37°C, confirming that prolonged exposure to simulated intraoral conditions does not induce detectable surface degradation or chemical modification in any of the thermoplastic materials evaluated.

### 6.3.1. Indirect cytotoxicity

Indirect cytotoxicity assessment using material-conditioned extracts generated over 1, 3, and 7 days at 37 °C demonstrated high mitochondrial activity for all tested monolayer materials across all time points (see figure 6.2). Mitochondrial activity values ranged from 84.7% to 100% relative to the untreated control, well above the 70% viability threshold established by ISO 10993-5 [164] at all time points and for all materials tested. No material exhibited a reduction in cell viability below the cytotoxicity threshold at any of the three conditioning periods evaluated.

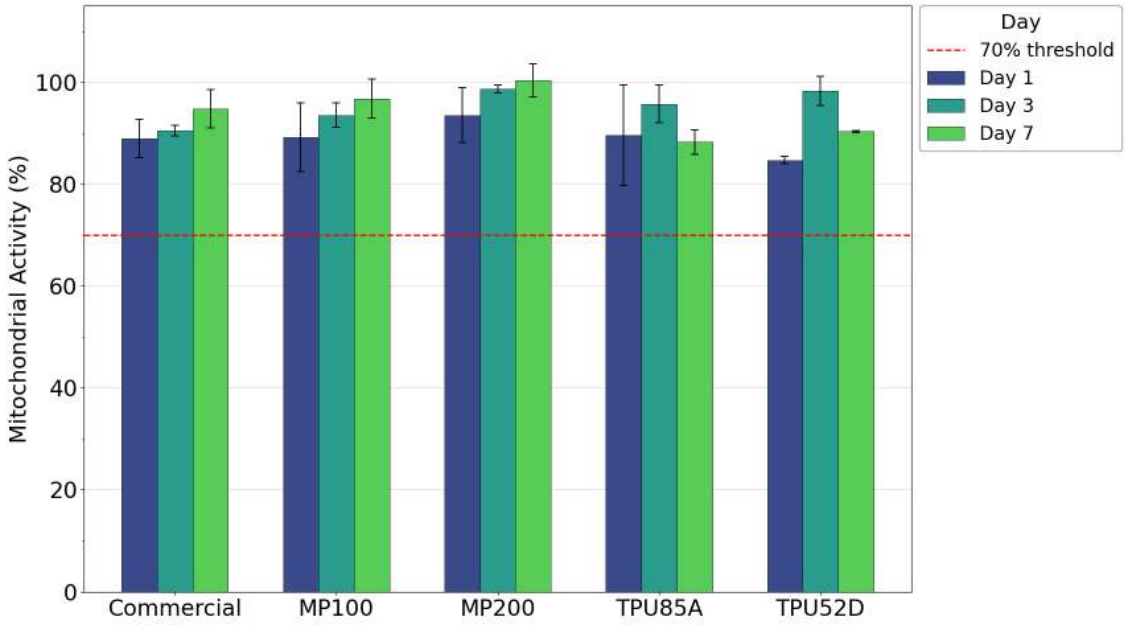



Figure 6.2: Mitochondrial activity of the monolayers and the commercial (Maflex Premium) materials expressed as % of the positive control (P.Control) by indirect manners at 1,3, and 7 days

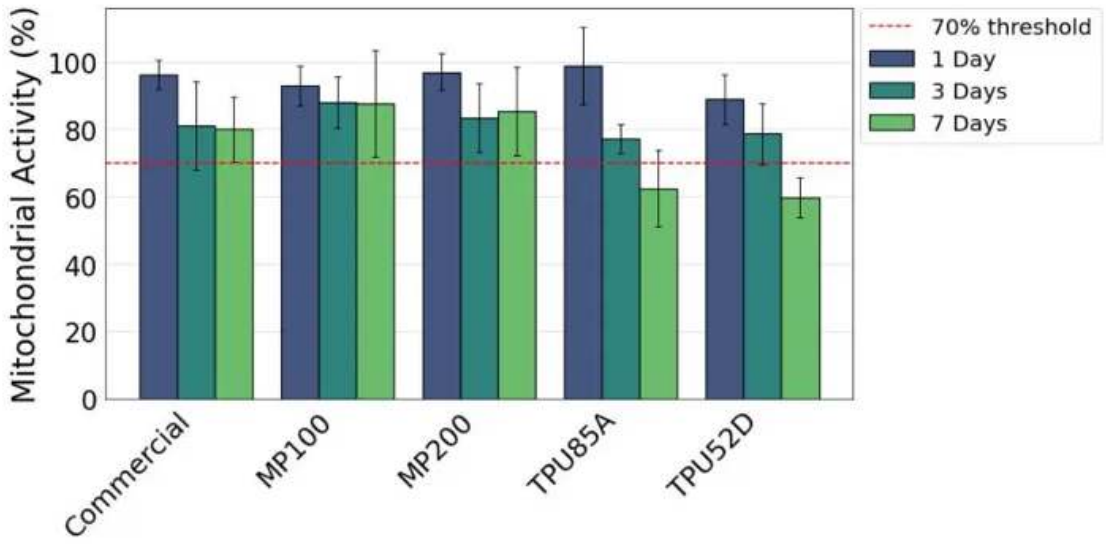
All copolyester-based materials (MP100, MP200, and Maflex Premium) showed stable or slightly increasing mitochondrial activity over time, indistinguishable from the untreated control. TPU-based materials (TPU85A and TPU52D) likewise remained above the viability threshold throughout the conditioning period. These results confirm the absence of cytotoxic leachables from any of the tested materials under the experimental conditions employed and are consistent with the ATR-FTIR analysis of the conditioned extracts.

### 6.3.2. Direct cytotoxicity and cell-material interaction

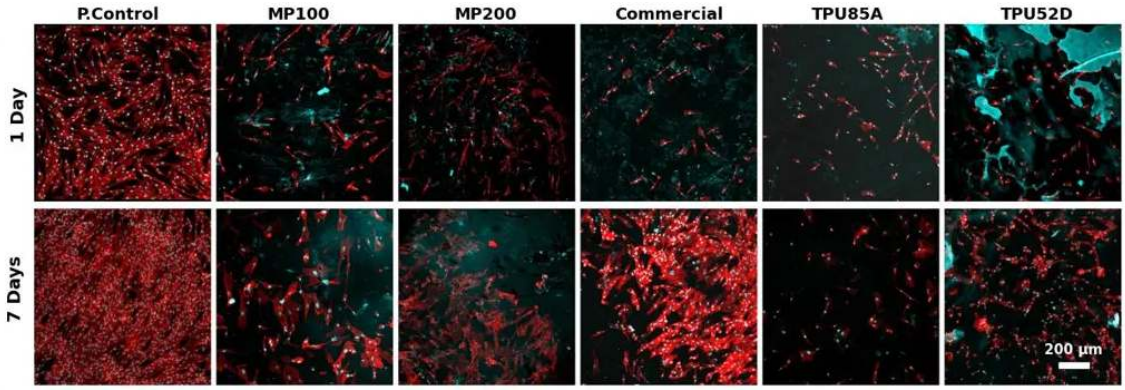
Direct cytotoxicity assessment of monolayer materials by direct culture of hGFs on material surfaces demonstrated mitochondrial activity values ranging from 60% to 99% relative to the untreated control across all time points and materials (see figure 6.3). Copolyester-based materials (MP100 and MP200) and the commercial reference Maflex Premium maintained mitochondrial activity above the 70% viability threshold at all time points, confirming their cytocompatibility under direct contact conditions. In contrast, TPU-based materials (TPU 85A and TPU 52D) exhibited a reduction in mitochondrial activity below the 70% threshold at day 7, suggesting a progressive substrate-dependent modulation of cellular metabolic activity under prolonged direct contact.

Cell membrane integrity was confirmed by LDH release quantification at day 7, with cytotoxicity values below 1.4% for all monolayer materials and no statistically significant differences relative to the spontaneous control ($p > 0.05$). This confirms that the reduction

in mitochondrial activity observed for TPU materials reflects a metabolic or proliferative response to the substrate rather than necrotic cell death or loss of membrane integrity.




(a) Mitochondrial activity of hGF cultured directly over monolayer materials and the commercial (Maflex Premium) at 1,3, and 7 days, normalized to the positive cytocompatibilty control (P.Control). The red dashed line indicates the 70 % cell viability threshold established by the ISO 10993-5

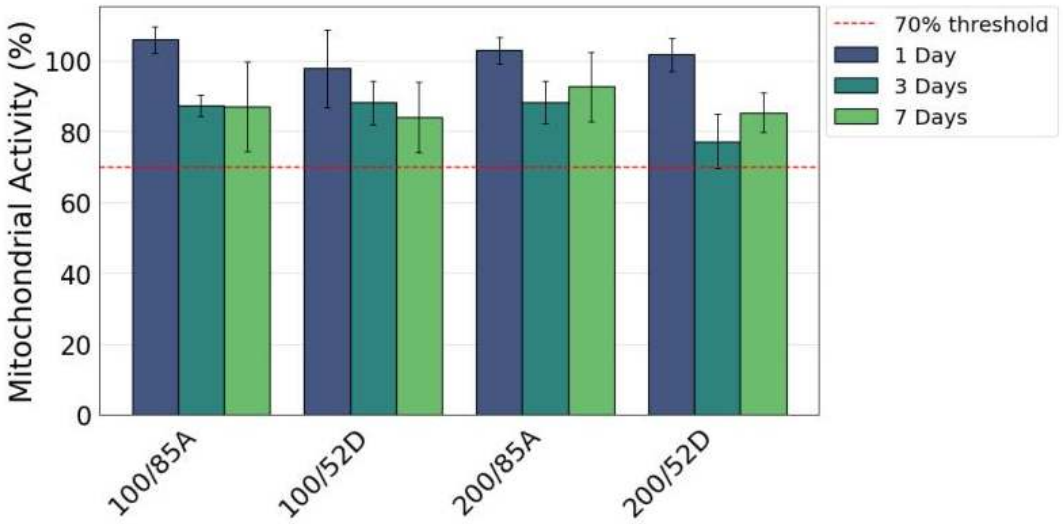



(b) Confocal fluorescence images of hGF cultured over monolayer and commercial material for 1 and 7 days. Actin cytoskeleton is stained with Alexa Fluor 660 phalloidin (red) and cell nuclei with DAPI (cyan).

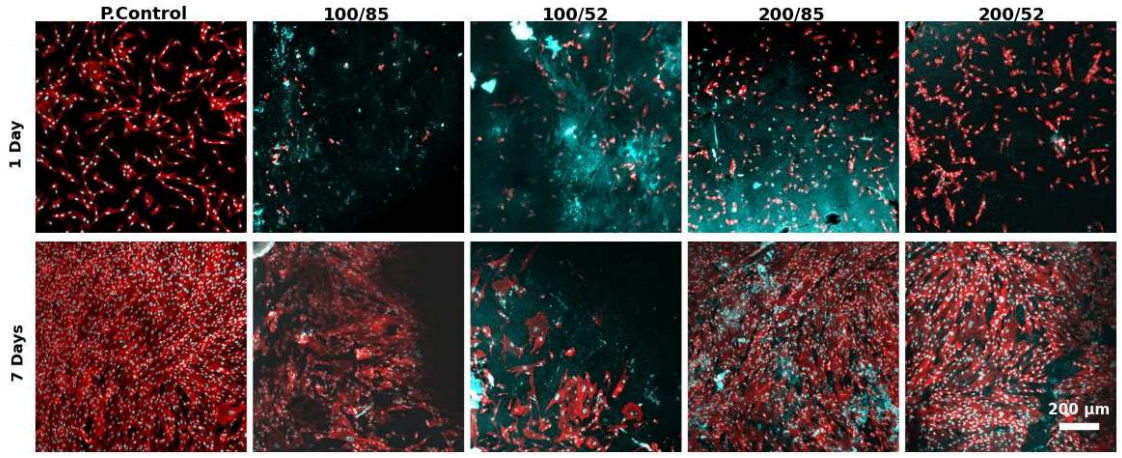
Figure 6.3: Direct-contact cytocompatibility of human gingival fibroblasts (hGF) cultured on the monolayer materials and the commercial multilayer (P.Control)

Confocal imaging at days 1 and 7 (see Figure 6.3) revealed clear differences in cell morphology and density depending on the substrate. Cells on copolyester and commercial reference surfaces displayed a typical fibroblastic morphology, well-spread, elongated, and highly confluent by day 7, despite appearing smaller and less numerous at day 1, indicating progressive adaptation and proliferation over the culture period. In contrast, cells on TPU substrates exhibited a rounded, poorly spread morphology with reduced

cytoskeletal organization from day 1 onward, which persisted throughout the culture period and is consistent with impaired integrin-mediated adhesion on the TPU surface[338], [339].




(a) Mitochondrial activity of hGF cultured directly over homemade multilayers at 1,3 and 7 days, normalized to the positive cytocompatibilty control (P.Control). The red dashed line indicates the 70 % cell viability threshold established by the ISO 10993-5.




(b) Confocal fluorescence images of hGF cultured over monolayer and commercial material for 1 and 7 days. Actin cytoskeleton is stained with Alexa Fluor 660 phalloidin (red) and cell nuclei with DAPI (cyan)

Figure 6.4: Direct-contact cytocompatibility of human gingival fibroblasts (hGF) cultured on the homemade multilayer materials.

Homemade multilayer materials exhibited proliferation behavior comparable to that of the copolyester monolayers and the commercial reference across all time points (see Figure 6.4). LDH cytotoxicity values remained below 2.4% for all multilayer configurations, with no statistically significant differences relative to the spontaneous control ($p >$

0.05), confirming the absence of membrane damage and the good biocompatibility of all homemade multilayer combinations under direct contact conditions.

Confocal imaging of hGFs cultured on homemade multilayer surfaces at days 1 and 7 (Figure 6.4) revealed cell morphology and proliferation behavior comparable to those observed on copolyester monolayers and the commercial reference material.

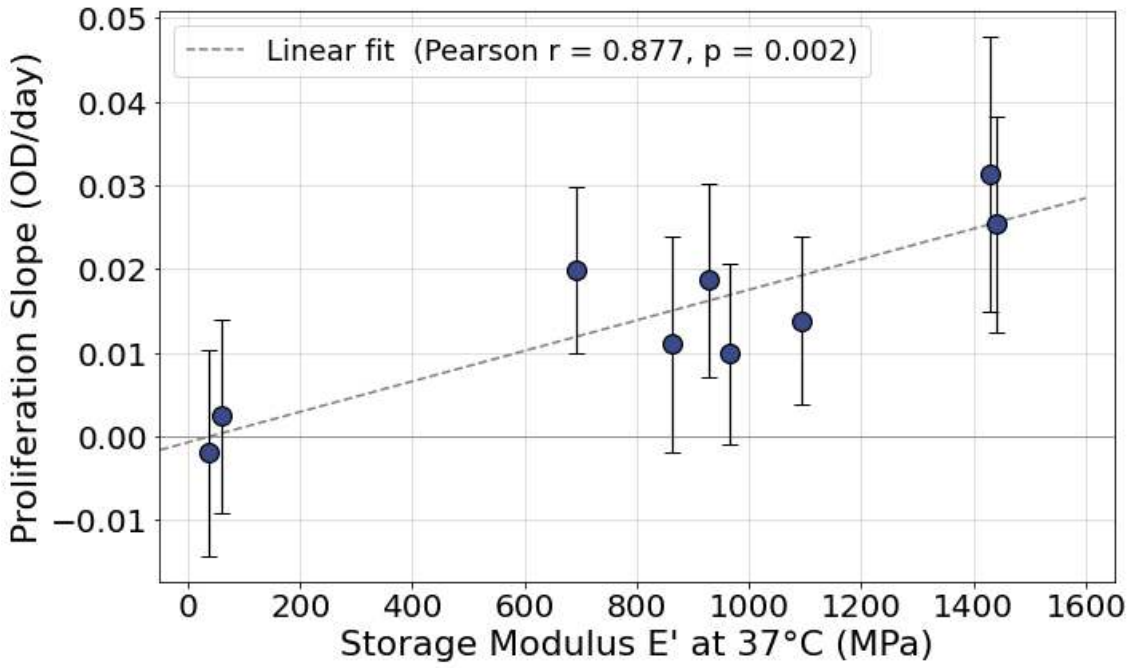



Figure 6.5: Correlation between the storage modulus E' at 37 ºC and the proliferation slope d(OD)/dt. Relation calculated as the linear rate in meean optical density of the direct cell culture experiments between day 1 and 7

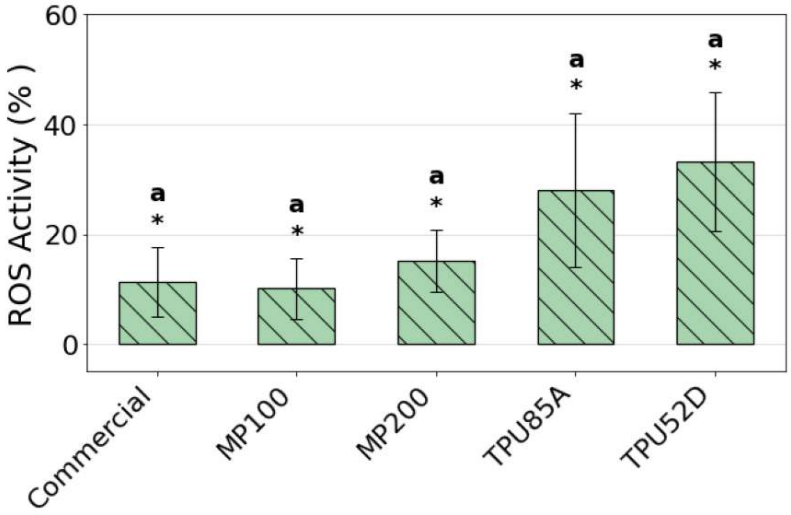
Analysis of proliferation dynamics across all tested materials — monolayers and multilayers combined — revealed a strong positive correlation between substrate stiffness and hGF proliferative activity (see Figure 6.5). Cell proliferation was quantified as the slope of the optical density signal between day 1 and day 7 of direct contact culture, expressed in OD/day. This parameter represents the rate of change in mitochondrial metabolic activity over time and serves as a proxy for proliferation rate: a steeper positive slope indicates faster proliferation, while a slope close to zero or negative indicates slower growth or stagnation. When this proliferation slope was plotted against the dynamic storage modulus ($E'$) measured at 37 °C by DMTA, a strong and statistically significant positive correlation was found (Pearson's r = 0.877, p = 0.002, see figure 6.5), indicating that hGFs proliferated faster on stiffer substrates. This finding is consistent with the well-established mechanosensitive behavior of fibroblasts, which sense and respond to substrate rigidity through integrin-mediated focal adhesion formation and downstream mechanotransduction pathways, including YAP/TAZ signaling [338], [339], [340], [341], [342]. The positive correlation between stiffness and proliferation observed here across a range of thermoplastic materials spanning nearly two orders of magnitude in elastic modulus (7-

1440 MPa) suggests that the mechanical properties of the aligner material may influence not only force delivery but also the biological response of the gingival tissues in contact with the device, with stiffer materials supporting a more active fibroblast proliferative environment at the tissue-device interface.

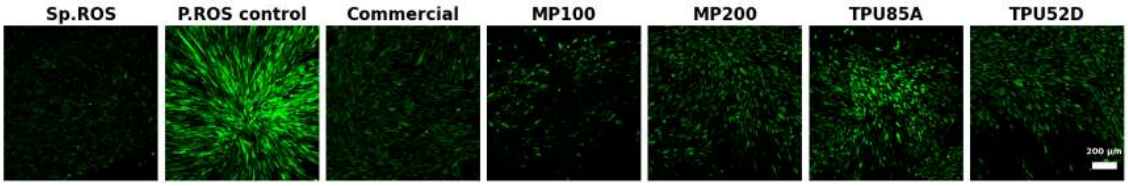
### 6.3.3. Reactive oxygen species (ROS)

All tested monolayer materials induced intracellular ROS levels substantially below those of the positive control (Figure 6.6), with values ranging from approximately 11% (MP100) to 33% (TPU 52D) relative to the positive control. Despite these low absolute values, statistically significant differences were detected relative to the spontaneous ROS control for all materials ($p>0.05$), indicating that all tested materials elicit a measurable but modest oxidative response above the cellular basal level. No statistically significant differences were found among the five monolayer materials (Tukey HSD, all sharing letter "a"), suggesting a broadly comparable oxidative potential across all monolayer configurations. A tendency towards higher ROS induction was observed for TPU-based materials (TPU 85A and TPU 52D) relative to copolyesters (MP100 and MP200) and the commercial reference, consistent with the known capacity of polyurethane degradation products to contribute to oxidative stress [343], [344], [345]; however, this difference did not reach statistical significance within the present sample size.

.




(a) ROS activity expressed as a percentage of the positive control (pyocyanin-treated cells). Asterisks indicate statistically significant differences relative to the spontaneous ROS level (Student's t test *p<0.05, **p<0.01,***p<0.001,****p<0.0001); letters indicate homogeneous subgroups determined by Tukey's HSD post hoc test (groups sharing the same letter are not significantly different from each other).

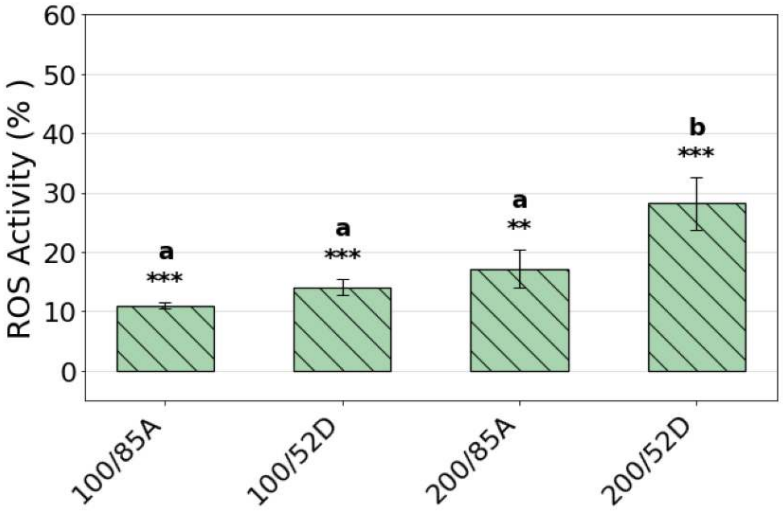



(b) Representative confocal fluorescence images of hGFs stained with H2DCFDA following 24 h exposure to each material extract. Green fluorescence intensity is proportional to intracellular ROS levels. Sp.ROS: spontaneous ROS level of unexposed hGFs; P.ROS control: positive ROS control (pyocyanin-treated cells).

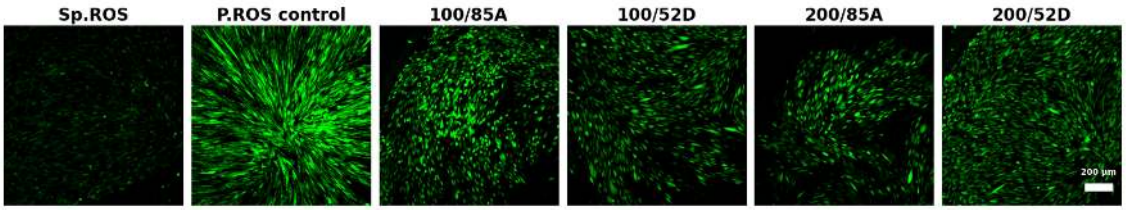
Figure 6.6: Intracellular reactive oxygen species (ROS) activity in human gingival fibroblasts (hGFs) exposed to material-conditioned extracts from monolayer materials and the commercial reference (Maflex Premium)

ROS activity in the homemade multilayer configurations followed a clear material-dependent pattern (Figure 6.7). Materials incorporating MP100 as the outer layer (100/85A and 100/52D) showed the lowest values (approximately 11% and 14% respectively), while those based on MP200 (200/85A and 200/52D) exhibited higher activity (approximately 17% and 28%). Statistically, 200/52D showed significantly higher ROS induction compared to 100/85A ($p<0.001$), while the remaining three multilayer configurations were statistically comparable to each other. All multilayer materials showed statistically significant differences relative to the spontaneous ROS control ($p<0.05$), consistent with the pattern observed for the monolayer materials. The higher ROS values observed in MP200-

based multilayers relative to MP100-based configurations may reflect the greater surface area and reactivity of the MP200 copolyester under the experimental conditions, though the absolute values remain well below those associated with pathological oxidative stress.




(a) ROS activity expressed as a percentage of the positive control (pyocyanin-treated cells). Asterisks indicate statistically significant differences relative to the spontaneous ROS level (Student‘s t test *p<0.05, **p<0.01,***p<0.001,****p<0.0001); letters indicate homogeneous subgroups determined by Tukey’s HSD post hoc test (groups sharing the same letter are not significantly different from each other).




(b) Representative confocal fluorescence images of hGFs stained with H2DCFDA following 24 h exposure to each material extract. Green fluorescence intensity is proportional to intracellular ROS levels. Sp.ROS: spontaneous ROS level of unexposed hGFs; P.ROS control: positive ROS control (pyocyanin-treated cells).

Figure 6.7: Intracellular reactive oxygen species (ROS) activity in human gingival fibroblasts (hGFs) exposed to material-conditioned extracts from homemade multilayers.

Optical microscopy images acquired before and after H2DCFDA staining confirmed that cell morphology, density and adhesion were preserved following dye addition across all conditions and materials, validating the staining procedure and excluding dye-induced cytotoxic artefacts as a confounding factor in the fluorescence measurements.

### 6.3.4. Cytokine quantification (ELISA)

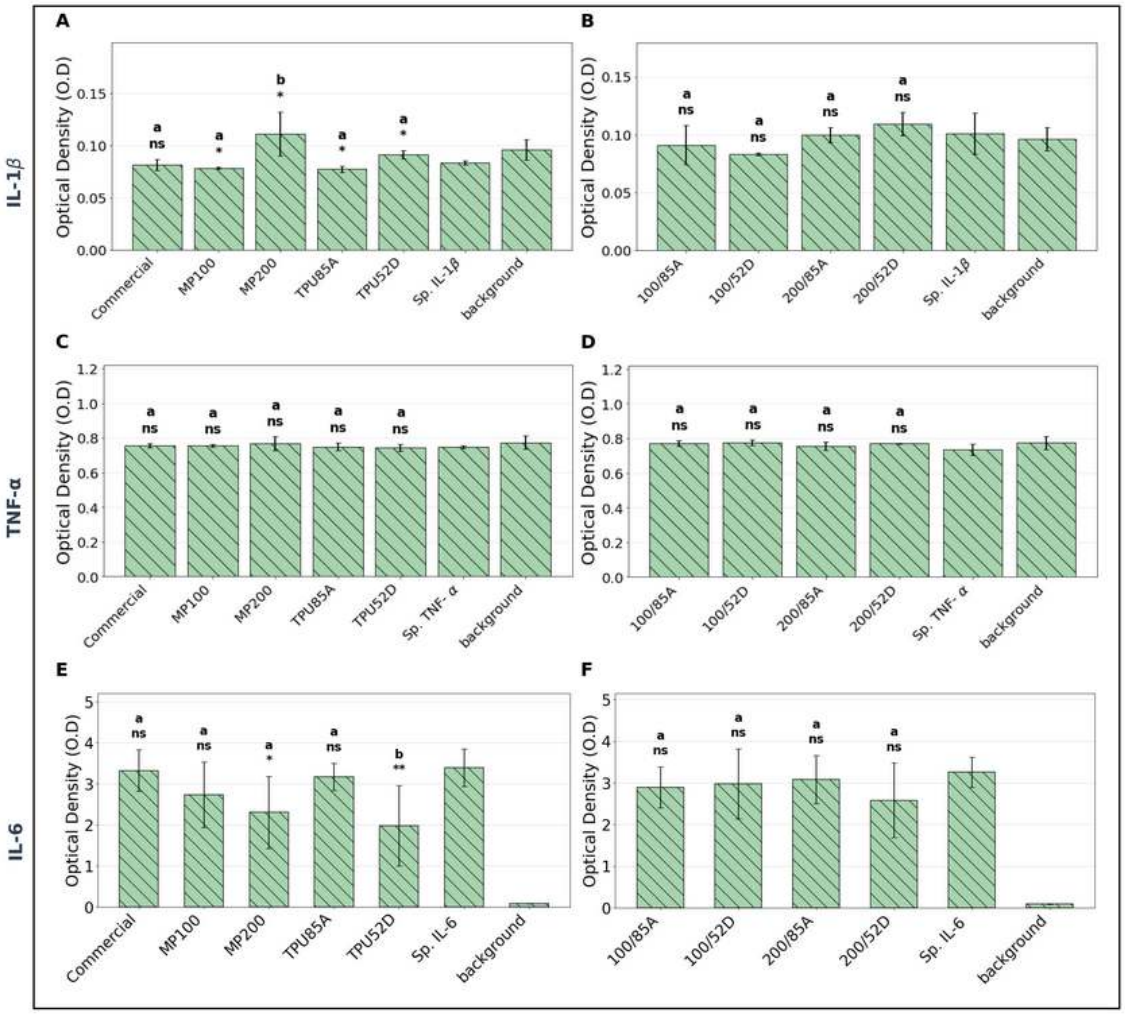



Figure 6.8: IL-1$\beta$, TNF-$\alpha$ and IL-6 secretion by HGF cells after direct contact with tested biomaterials. Optical density (O.D.) values measured by ELISA for interleukin-1$\beta$ ($IL-1\beta$,A-B), tumour necrosis factor- $\alpha$ ($TNF-\alpha$, C–D) and interleukin-6 (IL-6, E–F) in monolayer (A, C,E) and multilayer (B, D, F) configurations. Sp. $IL-1\beta$, Sp. $TNF-\alpha$ and Sp. IL-6 represent the spontaneous cytokine release from untreated cells. Data are expressed as mean ± SD (n = 3 biological replicates). Different letters indicate statistically significant differences among materials (Tukey HSD, $p < 0.05$); ns: not significant vs. spontaneous control (Student's t- test); *$p < 0.05$, **$p < 0.01$ vs. spontaneous control.

As shown in Figure 6.8 (panels A-B), the spontaneous $IL-1\beta$ signal was indistinguishable from the background absorbance of cell-free medium, and all material-conditioned supernatants yielded equivalent optical densities, with no statistically significant differences relative to the spontaneous control (all groups: ns, $p > 0.05$). All values were below the lower limit of quantification of the assay (LLOQ), indicating that none of the tested materials induced detectable $IL-1\beta$ secretion under direct contact conditions.

Analogous results were obtained for $TNF-\alpha$ (Figure 6.8, panels C-D): all tested conditions yielded optical densities indistinguishable from background, with no statistically significant differences between groups and all values below the LLOQ. These findings are consistent with the established biology of both cytokines in unstimulated gingival fibroblasts, which do not constitutively secrete significant quantities of $IL-1\beta$ or $TNF-\alpha$ in the absence of a pro-inflammatory stimulus [346], [347], [348]. The absence of detectable secretion of either cytokine confirms that none of the tested materials induces an acute pro-inflammatory response in gingival fibroblasts under the experimental conditions employed.

In contrast to $IL-1\beta$ and $TNF-\alpha$ IL-6 was readily detectable in all conditions, with optical density values substantially exceeding the upper limit of quantification of the assay (ULOQ = 150 pg/mL), consistent with the constitutive expression of IL-6 by healthy gingival fibroblasts [349], [350]. As shown in Figure 6.8 (panels E-F), no statistically significant differences in IL-6 optical density were detected among multilayer configurations (all groups sharing letter "a", $p> 0.05$). In the monolayer group, TPU 52D induced a statistically significant reduction in IL-6 optical density relative to the spontaneous control and to most other materials (letter "b", $p<0.01$), while MP200 also showed a modest but significant decrease ($p<0.05$). These reductions suggest a mild inhibitory effect of specific material surface chemistries on constitutive IL-6 secretion, possibly mediated by modulation of basal $NF-\kappa B$ autocrine signalling [351]. Importantly, no material induced an increase in IL-6 above spontaneous levels, which would be indicative of a pro-inflammatory response, confirming the absence of material-induced inflammation across all tested configurations.

### 6.3.5. Artificial saliva absorption and intraoral stability

All materials tested showed relatively low artificial saliva absorption after 17 days of immersion at 37 °C (Figure 6.9), with total weight gains ranging from 0.76% (MP100) to 1.30% (200/85A). Homemade multilayers showed absorption behavior comparable to that of the commercial reference material, and the nature of the TPU inner layer exerted a clear influence on the absorption response: multilayers incorporating the softer TPU 85A grade consistently showed higher water uptake than those incorporating TPU 52D, consistent with the greater hydrophilicity of the softer polyether segment.

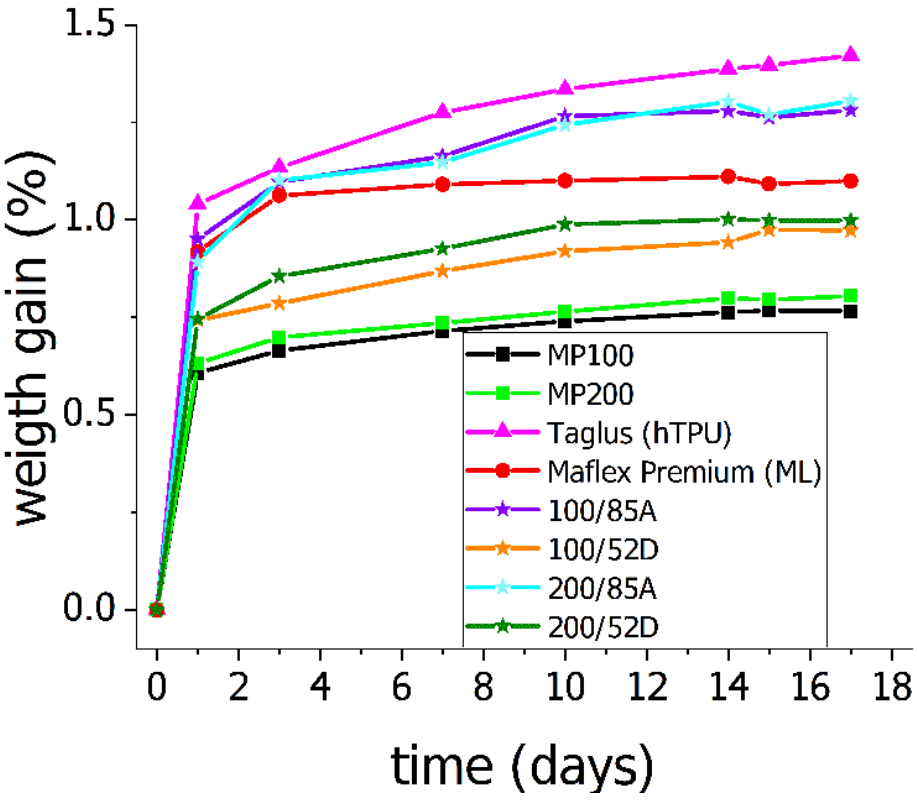



Figure 6.9: Weight gain 24 of tested plastics immersed in artificial saliva at 37 ºC

The absorption kinetics were described using the fractional diffusion model of Ritger and Peppas [156], [157], whose physical basis and mathematical derivation are provided in Appendix A, section A.1.4:

$$M(t) = M_0 + \frac{\Delta M_\infty}{\Gamma(1+\alpha)} \left(\frac{t}{\tau}\right)^\alpha \tag{6.1}$$

All tested materials showed $\alpha$ values between 0.08 and 0.13 and characteristic times $\tau$ between 25 and 32 days, indicating predominantly superficial sorption with a characteristic diffusion timescale of approximately one month, consistent with their classification as low-permeability plastics with respect to water [302]. The neat copolyesters MP100 and MP200 presented lower $\alpha$ values and longer $\tau$ than the commercial material and the homemade multilayers, consistent with their lower saturation weight gain (approximately 0.8% for the neat copolyesters versus 1.1% for the commercial reference and 1.0-1.3% for the homemade multilayers). The apparent diffusion coefficient $D_{ap}$ of the neat copolyesters was approximately $7 \times 10^{-8}\ cm^2/s$, consistent with previously reported values for similar materials such as PETG and PET in water [352], [353]. The commercial reference and the homemade multilayers showed higher diffusion coefficients in the range of $1.06 - 1.75 \times 10^{-7}\ cm/s$, indicating that the presence of the TPU inner layer accelerates water uptake velocity independently of the total absorbed quantity [298], [302], [303].

DSC measurements on immersed samples revealed a slight depression of the glass

transition temperature of 3-8 °C relative to non-immersed controls across all materials, attributable to the plasticization of the polymer matrix by absorbed water. In the case of MP100 and its multilayers, an additional melting process was detected at approximately 240 °C in the first heating scan, accompanied by a crystallization exotherm at 180 °C in the cooling trace — thermal events absent in the non-immersed controls and not observed in MP200 or its multilayers, suggesting that water absorption promotes a degree of hygrothermal crystallization in MP100-based materials under the experimental conditions [354], [355].

DMTA measurements performed on the immersed surface of samples after 15 days confirmed the plasticization observed by DSC. No significant change in Tg was detected by DMTA for any material. The storage modulus at 37 °C decreased by approximately 100 MPa in most materials following immersion, reflecting a modest but consistent softening of the polymer matrix. A notable exception was the commercial reference material (Maflex Premium), whose storage modulus at 37 °C increased substantially after immersion, from 690 MPa in the as-received condition to 1225 MPa after 15 days, an increase of approximately 600 MPa that may be attributed to hygrothermal crystallization or physical aging of the sTPU outer layers under the immersion conditions. The loss modulus at oral temperature showed more pronounced changes: in the neat copolyesters MP100 and MP200, $E''$ approximately doubled after immersion (from 15 and 18 MPa to 38 and 35 MPa respectively), while in the homemade multilayers, the increase was more modest (1-7 MPa). The commercial reference showed an intermediate increase of approximately 15 MPa.

ATR-FTIR spectra of plastic samples recovered after 17 days of immersion in artificial saliva showed no significant spectral changes relative to non-immersed controls in any of the materials tested, confirming the absence of detectable surface degradation or chemical modification under the experimental conditions.

Taken together, these results indicate that the absorption of artificial saliva by all tested materials is low (0.76-1.30% weight gain after 17 days) and produces only modest changes in mechanical properties, with no significant shift in glass transition temperature detectable by either DSC or DMTA. The absence of bulk Tg depression, combined with the superficial nature of the absorption process ($\alpha << 0.5$), confirms that no significant bulk hydrolysis of the polymer matrix occurs under the experimental conditions, and that the mechanical effects of water uptake are confined to the surface region of the material.

### 6.4. Biocompatibility and oral stability discussions

The ATR-FTIR characterization confirmed the chemical identity of all tested materials prior to biological testing and validated that ethylene oxide sterilization did not introduce detectable surface modifications, consistent with previous reports [356], [357]. Analysis of culture media conditioned by each material for 1, 3, and 7 days revealed only negligible

spectral changes at all time points, with the low intensity of the 1044 and 1090 $cm^{-1}$ bands supporting minimal leaching from all materials across the full conditioning period. The absence of spectral changes in recovered plastic samples after both cell culture immersion and 17 days in artificial saliva confirms that no bulk degradation occurred under any of the tested conditions, supporting the chemical stability of all materials under conditions representative of intraoral use.

The thermomechanical characterization reported in Chapter 5 demonstrated that the homemade multilayers exhibit mechanical properties intermediate between those of the stiff copolyester monolayers and the compliant TPU layers, confirming that the multilayer architecture enables rational tuning of the mechanical response of the device. Water sorption values (0.76–1.30% after 17 days) were low across all materials and consistent with a predominantly superficial absorption mechanism, as confirmed by the fractional calculus model ( $\alpha$ = 0.08-0.13, $\tau$ = 25-35 days) [156], [157], [358], [359]. The absence of Tg shifts and degradation-related FTIR peaks after immersion confirms that water sorption induces superficial plasticization rather than bulk hydrolysis, supporting the chemical and mechanical durability of all tested materials. The notable increase in storage modulus observed for the commercial reference material (Maflex Premium) after 15 days of immersion — from 690 MPa to 1225 MPa — is an unexpected finding that may reflect hygrothermal crystallization of the sTPU outer layers or physical aging under the immersion conditions, and warrants further investigation.

All materials satisfied the cytocompatibility criteria established by ISO 10993-5 [164]. The reduction in mitochondrial activity observed for TPU-based materials at day 7 under direct contact conditions is more appropriately attributed to a slower proliferation rate relative to the tissue culture plastic control than to overt cytotoxicity: normalization against a highly proliferative reference substrate inherently penalizes materials that support more moderate but physiologically normal cell growth [360]. LDH results confirmed preserved membrane integrity across all conditions, with values remaining comparable to spontaneous release and well below the ISO 10993-5 cytotoxicity threshold. It should be noted that LDH data should be interpreted with caution given the known susceptibility of LDH enzyme to denaturation under prolonged Triton X-100 exposure, which may affect the performance of the positive control [189], [361], [362]; however, this limitation does not affect the interpretation of the experimental groups, which consistently showed negligible LDH release.

The strong positive correlation between hGF proliferation rate and substrate storage modulus (Pearson's r = 0.877, p = 0.002) is consistent with the well-established mechanosensitive behaviour of fibroblasts [338], [339], [341], [342]. Differences in protein adsorption — particularly for TPU materials, whose higher surface hydrophobicity results in a distinct protein corona composition relative to copolyesters [363], [364], [365] — likely influenced integrin-mediated signaling and the activation of proliferative pathways including MAPK/PI3K and mechanosensitive YAP/TAZ signalling [338], [339], [366], [367], [368]. From a clinical perspective, this finding suggests that the mechan-

ical properties of the aligner material may influence not only the orthodontic force delivery profile but also the biological response of the gingival tissues in contact with the device. Stiffer materials appear to support a more active fibroblast proliferative environment, which could be relevant to periodontal tissue remodeling during tooth movement; however, this relationship requires validation in more physiologically representative experimental models before clinical conclusions can be drawn.

All tested materials induced low intracellular ROS levels (10-33% of the positive control), consistent with a non-pathological adaptive oxidative response that does not indicate oxidative stress-mediated cytotoxicity [326], [369]. The slightly higher ROS levels observed for TPU-based materials may reflect differences in leachable composition relative to copolyesters [370], [371], but these differences did not reach statistical significance and remain well within the range of physiological oxidative activity. The material-dependent pattern observed in the multilayer group — with MP200-based configurations showing higher ROS activity than MP100-based ones — may reflect differences in surface chemistry or leachable profile between the two copolyester grades, though the absolute values were low in all cases.

No pro-inflammatory cytokine response was detected under any tested condition. The absence of detectable $IL-1\beta$ and $TNF-\alpha$ is consistent with the requirement for exogenous pro-inflammatory stimulation for their production in gingival fibroblasts [372], [373], [374], [375], [376], [377], [378], and confirms that none of the tested materials activates an innate inflammatory pathway under direct contact conditions. Basal IL-6 secretion was constitutive across all conditions, consistent with its established homeostatic role in gingival tissue [348], [349], [350]. The modest reduction in IL-6 observed for TPU52D and MP200 monolayers may indicate subtle modulation of fibroblast homeostatic signaling by surface-associated leachables [351], but does not alter the overall biocompatibility assessment given the absence of elevated pro-inflammatory cytokines or reduced cell viability in any condition.

Collectively, these results confirm the chemical stability and cytocompatibility of the homemade multilayers under simulated intraoral conditions, on a par with the commercial reference; the integrated mechanistic interpretation is presented in the conclusions below.

### 6.5. Conclusions of the biological testing

The biological and physicochemical characterization presented in this chapter can be integrated into a single mechanistic model of the cell–material interface (Figure 6.10), which distinguishes a mechanotransduction-driven modulation of proliferation from a leachate-driven, sub-pathological oxidative response, both occurring in the absence of membrane damage or pro-inflammatory signalling. From this model, the following conclusions can be drawn:

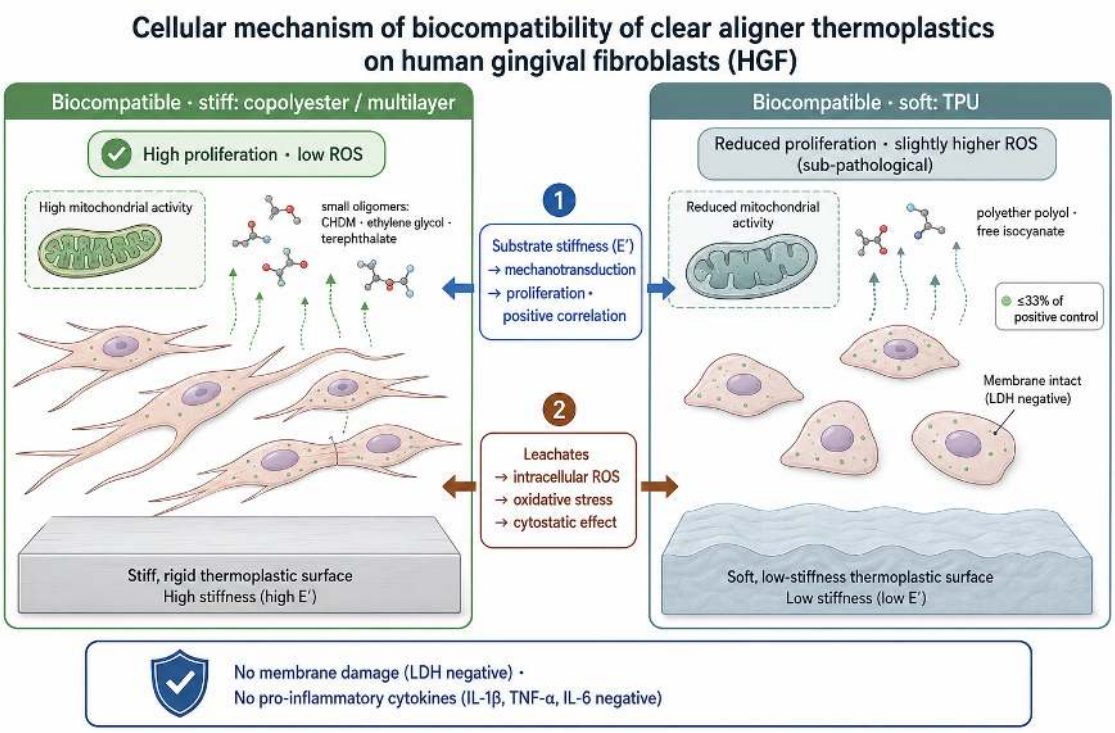



Figure 6.10: Proposed cellular mechanism governing the interaction of clear aligner thermoplastics with human gingival fibroblasts. Figure generated with the assistance of ChatGPT (OpenAI, 2026)

1. ATR-FTIR analysis of culture media conditioned by each material for 1, 3, and 7 days confirmed minimal leaching from all tested materials, with only low-intensity spectral changes attributable to diol species, PTMEG oligomers, and dissolved $CO_2$. No spectral changes were detected in recovered plastic samples after cell culture immersion or after 17 days in artificial saliva, confirming the chemical stability of all materials under conditions representative of intraoral use.

2. All tested materials — the homemade copolyester monolayers (MP100 and MP200), TPU monolayers (TPU 85A and TPU 52D), the homemade multilayer configurations, and the commercial reference Maflex Premium — satisfied the cytocompatibility criteria of ISO 10993-5 in both indirect and direct cytotoxicity assays. The reduction in mitochondrial activity observed for TPU-based materials at day 7 under direct contact reflects a substrate-dependent modulation of proliferation rate rather than cytotoxic cell death, as confirmed by the absence of LDH release above spontaneous levels.

3. All tested materials induced low intracellular ROS levels, indicating a non-pathological adaptive oxidative response. No material induced detectable secretion of $IL\text{-}1\beta$ or $TNF\text{-}\alpha$, and none elevated IL-6 above constitutive basal levels, confirming the absence of a material-induced pro-inflammatory response under direct contact conditions.

4. A strong positive correlation was found between substrate storage modulus at 37 °C and hGF proliferation rate, suggesting that the mechanical properties of the aligner material may influence the biological response of the gingival tissues at the tissue–device interface beyond their role in orthodontic force delivery. This mechanobiological relationship warrants further investigation in more physiologically representative models.

5. Water absorption after 17 days of immersion in artificial saliva was low for all materials, with kinetics consistent with predominantly superficial sorption. No bulk hydrolysis was detected by DSC or DMTA and no significant shift in glass transition temperature was observed, confirming the intraoral stability of all tested materials over clinically relevant timescales.

6. The homemade multilayer configurations demonstrated biocompatibility and intraoral stability comparable to the commercial reference Maflex Premium across all assays performed. Combined with the superior viscoelastic performance documented in Chapter 5, these results establish the homemade multilayers — particularly the MP200-based configurations — as mechanically and biologically viable candidates for next-generation aligner materials.

7. Several dimensions of the biological safety of these materials remain uncharacterized, including their potential for microplastic release under clinically representative conditions, their endocrine-disrupting activity, and their inflammatory potential in the presence of immune cells, as detailed in the limitations and future work of Chapter 8.

# 7. ECONOMICAL AND ENVIRONMENTAL ANALYSIS OF THE INDUSTRIAL PRODUCTION OF CLEAR ALIGNERS

## 7.1. Computer simulations

### 7.1.1. Introduction

Computational simulation has emerged as a fundamental tool in dental biomechanics, enabling precise analysis of the forces, deformations, and displacements that occur during orthodontic tooth movement [379], [380], [381]. In the specific context of clear aligners, planning software constitutes a critical component of the clinical workflow, and its quality largely determines treatment efficacy.

Currently, the market offers two broad categories of planning software. The first relies entirely on the clinician's expertise, with the professional manually defining the sequence of tooth movements [382]. The second corresponds to proprietary systems with algorithms based on clinical datasets, the most representative example being the software developed by Align Technology for the Invisalign system [382]. Most commercially available solutions combine elements of both categories, yet share a fundamental limitation: they do not explicitly model the physics of the tooth movement process [381], [383]. As a consequence, the relationship between aligner material stiffness, activation distance, and the forces actually delivered to the tooth remains poorly characterized at the individual patient level.

Historically, orthodontic biomechanics has relied on two-dimensional analytical models derived from beam theory and simplified force diagrams to predict tooth movement and force transmission [8], [384]. These models, while computationally tractable, neglect the three-dimensional geometry of the tooth crown and root, the heterogeneous distribution of stresses within the periodontal ligament, and the complex contact mechanics at the aligner-tooth interface. Despite growing awareness of their limitations, two-dimensional conceptual frameworks remain in widespread clinical use, and many orthodontists continue to apply force prescriptions derived from planar mechanics to inherently three-dimensional clinical scenarios [8]. Furthermore, the optimal force levels for orthodontic tooth movement remain poorly established: a systematic review identified a high degree of uncertainty regarding the appropriate force magnitude for bodily tooth movement, with forces between 50 and 100 cN appearing broadly optimal, but the evidence base remains weak [385]. This uncertainty is compounded in clear aligner therapy, where the forces delivered depend simultaneously on material stiffness, shell thickness, activation distance, and patient anatomy—variables that interact in ways that two-dimensional models cannot capture.

In contrast, the Finite element method (FEM) offers a more scientifically rigorous

three-dimensional framework. FEM is a well-established numerical technique in structural engineering and biomechanics that solves partial differential equations over complex geometries by discretising them into smaller elements [379]. In the orthodontic field, FEM has been employed to quantify forces and moments acting on teeth, to analyze stress distribution within the periodontal ligament and alveolar bone, and to optimize attachment design and activation distances [65], [99], [380], [386]. These studies have established that thermoplastic aligner materials with moderate elastic moduli can generate clinically effective force systems provided appropriate attachments are incorporated [65], and that multilayer aligner designs produce more selective tooth displacement than monolayer equivalents [109]. However, the published FEM literature on clear aligner biomechanics presents several persistent methodological limitations that compromise reproducibility and clinical interpretability.

First, the overwhelming majority of published FEM studies model only a single tooth or a small subset of teeth rather than the complete dental arch [66], [387], [388], [389], [390]. This simplification is motivated primarily by computational cost - single-tooth models require orders of magnitude fewer elements than full arch models - but introduces systematic errors in the predicted force distribution, as the mechanical response of the aligner is governed by its global geometry and the simultaneous engagement of the full dentition [99], [383], [391]. The increase in computational power and the availability of open-source solvers capable of handling large mesh sizes now make full arch FEM analyses tractable on standard hardware, reducing the justification for single-tooth simplifications.

Second, in a significant proportion of published FEM studies, the elastic moduli and Poisson's ratios used as simulation inputs are adopted from generic polymer databases or from values reported in earlier works, without independent experimental characterization of the specific materials under investigation [66], [387], [388], [389], [390]. This opacity undermines both reproducibility and the physical fidelity of the simulated force fields. Furthermore, the elastic moduli used in most published FEM studies are determined at room temperature (approximately 22-25 ºC) rather than at body temperature. The elastic modulus of thermoplastic aligner materials decreases significantly at intraoral temperature (37 ºC) compared to room temperature [143], [171], meaning that simulations using room-temperature moduli systematically overestimate aligner stiffness and the predicted contact forces relative to the clinical scenario.

Third, the simulation tools used in most published studies are commercial finite element packages - ANSYS [390], [392], [393], Abaqus [394], [395], [396], or COMSOL multyphysics [397] - whose licensing costs preclude adoption by research groups without institutional access, and whose proprietary nature limits auditability and extensibility [383]. To the authors' knowledge, no published FEM study of clear aligner biomechanics has been implemented using open-source software, making the present work the first to demonstrate a complete open-source simulation pipeline for this application.

Fourth, a significant proportion of published FEM studies impose tooth movement as a prescribed displacement applied directly to the tooth crown, or apply point forces of arbitrary magnitude to the aligner or tooth surface without linking the force magnitude to the geometric and material properties of the system [66], [387], [388]. This decoupling between loading and material stiffness means that the simulated force field cannot be directly related to the clinical activation distance encoded in the aligner design, limiting the clinical interpretability of the results. Moreover, in all published FEM studies of clear aligners known to the author, the aligner geometry is generated digitally as an offset surface over the target dental model rather than acquired from a physical specimen, meaning that the post-thermoforming geometry of the aligner - which differs from the nominal CAD surface due to material thinning and springback effects [398] - has not been incorporated into any published simulation to date.

Applying FEM to the design and planning of clear aligners offers substantial advantages over current approaches. First, FEM allows the explicit incorporation of the mechanical properties of the thermoplastic aligner material - Young's modulus, Poisson's ratio, viscoelastic behaviour - enabling planning to be adapted to the specific material used, reversing the current logic in which material selection is dictated by the software [399], [400]. Second, patient-specific biomechanical properties can be included: bone density, periodontal ligament stiffness, and enamel characteristics obtained from CBCT, opening the door to the design of force-personalized aligners [39], [99]. Third, the availability of free and open-source FEM codes (such as FEniCSx or CalculiX) makes the development of open orthodontic planning software technically feasible, free from the proprietary constraints that currently dominate the field [401].

The present chapter addresses these limitations through the development of an open-source finite element simulation framework for clear aligner biomechanics, implemented in Python using FEniCSx [401]. The framework incorporates the following methodological advances. First, the elastic moduli of four of the five aligner materials simulated - PETG, hTPU, PCTG, and a multilayer thermoplastic - were determined experimentally at 37 ºC by the authors in the preceding chapters of this thesis through quasi-static uniaxial tensile and three-point bending tests, establishing a direct link between material characterization and simulation input. Second, aligner insertion is modeled through a geometric activation mechanism in which each node of the aligner shell is displaced radially outward by the prescribed activation distance prior to contact analysis, so that the contact forces emerge naturally from the elastic response of the shell rather than being imposed externally. Third, a material-scaled contact force formulation is introduced in which the nodal contact force scales linearly with the elastic modulus of the aligner material relative to a reference material (PETG), ensuring that stiffer materials generate proportionally higher contact forces under identical geometric activation - consistent with linear elastic theory and experimental observations [261], [400]. Fourth, the complete simulation pipeline is implemented in freely available open-source software and is publicly available, ensuring full reproducibility.

The framework is validated across three levels of geometric complexity: a cubic surrogate geometry for numerical verification, an isolated anatomical crown representing the upper right first molar, and a complete lower dental arch — the latter representing, to the authors' knowledge, one of the first full arch clear aligner FEM analyses implemented in open-source software. Five aligner materials spanning the full stiffness range of thermoplastics currently used in clinical practice are compared under identical loading conditions, and results are evaluated against physiologically admissible ranges established in the literature [65], [386], [400].

### 7.1.2. Computational Framework

The finite element simulation framework was implemented in Python using FEniCSx [401]. The complete source code, including mesh preparation scripts, solver configuration, post-processing routines, and example input geometries, is publicly available at https://github.com/Ozyman1998/Thesis/tree/main/fem. Only the key methodological choices are summarised in Figure 7.1; the reader is referred to the repository for full implementation details.

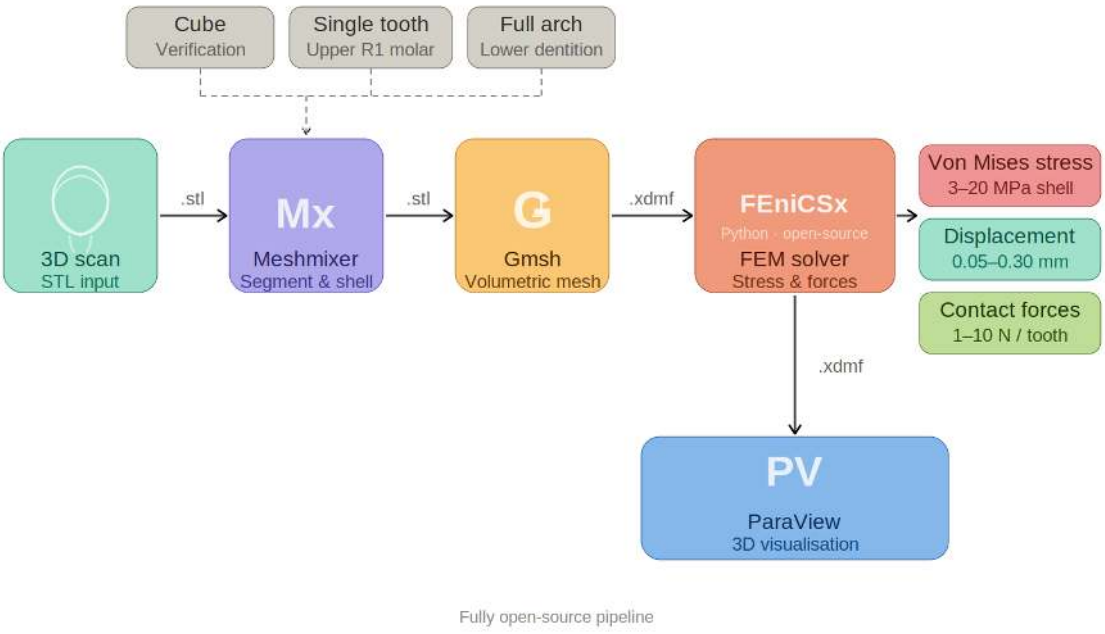



Figure 7.1: Open-source simulation pipeline for clear aligner biomechanics. Figure generated with the assistance of Claude (Anthropic, 2026).

### 7.1.3. Material Properties

Both the aligner shell and the tooth were modeled as linear elastic, isotropic, and homogeneous materials, which is a standard assumption in small-displacement FEM analyzes of orthodontic appliances [383], [402]. Five aligner materials were simulated, covering the full stiffness range of thermoplastic materials currently used in clear aligner fabrication.

The elastic moduli of four of the five materials simulated— ML (Zendura FLX, $E$ = 720 $MPa$), PCTG (Prismlab C+, $E$ = 990 $MPa$), PETG (Duran, $E$ = 1124 $MPa$), and hTPU (Taglus, $E$ = 1497 $MPa$)—were determined experimentally by the authors at 37 ºC through quasi-static uniaxial tensile testing and three-point bending, as described in the Chapter 4, section 4.2 and summarized in Table 4.8. The tensile modulus was adopted as the FEM input parameter in all cases, as it constitutes a direct measure of the uniaxial elastic response consistent with the isotropic linear elastic constitutive model employed and avoids the geometric dependence and systematic overestimation associated with three-point flexural moduli in thin thermoplastic sheets [261]. The use of moduli determined at body temperature rather than at room temperature represents a methodological advance over the majority of published FEM studies of clear aligners, which assign room temperature values that systematically overestimate aligner stiffness under clinical conditions [143], [171].

Polycarbonate (PC) ($E$ = 2300 MPa) was included as an upper-bound reference material representing the maximum stiffness end of the thermoplastic spectrum relevant to orthodontic applications [26], [39], [172]. No PC aligner specimens were available for mechanical testing within the scope of this thesis, and its use as a clear aligner material remains marginal in the clinical literature [88], [172]; it is therefore included as a limiting case rather than as a clinically representative scenario, and its elastic modulus was adopted from standard polycarbonate material data [403].

The Poisson's ratios were adopted from the polymer mechanics literature. For PETG and PCTG, $\nu$ = 0.33 was adopted following Elkholy et al. [261], with the same value assumed for PCTG given the structural and chemical similarity of the two copolyester materials; for hTPU, $\nu$ = 0.45 was taken from experimental digital image correlation measurements on TPU [404]; and for PC, $\nu$ = 0.37 was adopted from standard polycarbonate material data [403]. For the ML material, no specific value is consistently reported in the literature, with values ranging from 0.35 to 0.45 [287], [394], [395], [405];an intermediate value of $\nu$ = 0.40 was therefore selected as a representative estimate. The experimental determination of Poisson's ratio for all five materials through digital image correlation during tensile testing is identified as a priority for future characterization work. The material properties used in the FEM simulations are summarised in Table 7.1.

| Material | Trade name | E (MPa) | $\nu$ (-) | E source | $\nu$ source |
|---|---|---|---|---|---|
| Multilayer | Zendura FLX | 720 | 0.40 | This work | [287], [394], [395], [405] |
| PCTG | Prismlab C+ | 990 | 0.33 | This work | [99] |
| PETG | Duran | 1124 | 0.33 | This work | [99] |
| hTPU | Taglus | 1497 | 0.45 | This work | [404] |
| PC | — | 2300 | 0.37 | [403] | [403] |
| Tooth | — | 18000 | 0.30 | [65], [99], [383] | [65], [99], [383] |

Table 7.1: Material properties used in the finite element simulations. Elastic moduli for ML, PCTG, PETG, and hTPU determined experimentally at 37 ºC in this work (mean values, $n = 5$, see Table 4.8); PC and tooth moduli from literature.

The tooth was modeled as a single homogeneous elastic body with properties representative of dental enamel ($E = 18000 MPa$, $\nu = 0.30$), following the values adopted in published FEM studies of clear aligner biomechanics [65], [99], [383]. This is a simplification of the actual heterogeneous dental anatomy, which comprises enamel ($E \simeq$ 40-80 $GPa$), dentine ($E \simeq$ 15-20 $GPa$), pulp, periodontal ligament, and alveolar bone [65], [406]. The single-material approximation is adopted for consistency with the majority of the published FEM literature on clear aligners and is justified by the fact that the tooth is modeled as a near-rigid body relative to the aligner shell; its internal stress distribution is of secondary interest compared to the contact forces and aligner deformation fields that are the primary outputs of the simulation.

The constitutive behavior of the aligner was governed by the generalized Hooke's law for linear elastic solids. The stress tensor $\boldsymbol{\sigma}$ was computed from the strain tensor $\boldsymbol{\varepsilon}$ following the equation 7.1

$$\boldsymbol{\sigma} = \lambda \, \mathrm{tr}(\boldsymbol{\varepsilon}) \, \mathbf{I} + 2\mu \boldsymbol{\varepsilon} \tag{7.1}$$

where $\lambda$ and $\mu$ are the Lamé parameters defined in the equations 7.2 and 7.3 respectively.

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \tag{7.2}$$

$$\mu = \frac{E}{2(1+\nu)} \tag{7.3}$$

and the linearized strain tensor is defined by equation 7.4

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) \tag{7.4}$$

The von Mises equivalent stress, used to evaluate the risk of material yielding and to compare results against clinically reported stress ranges [65], [386], [400], was computed using the equation 7.5

$$\sigma_{\mathrm{vm}} = \sqrt{\frac{3}{2}\, \mathbf{s} : \mathbf{s}} \tag{7.5}$$

where $\mathbf{s} = \boldsymbol{\sigma} - \frac{1}{3}\mathrm{tr}(\boldsymbol{\sigma})\mathbf{I}$ is the deviatoric stress tensor.

### 7.1.4. Boundary Conditions

To simulate the clinical constraint imposed by the patient's gingival tissues and the anatomical limits of aligner seating, the upper and lower margins of the aligner shell — corresponding to the gingival and occlusal borders, respectively — were assigned homogeneous Dirichlet boundary conditions, fixing all displacement components to zero at nodes located within the outermost 10% of the shell's vertical extent at both ends. This bilateral constraint prevents rigid body motion and reflects the physical reality that the aligner is held in place by friction and anatomical retention at its margins during the insertion process.

The interior of the shell was left free to deform in response to the contact forces generated at the aligner–tooth interface.

### 7.1.5. Geometric Activation

Prior to the contact analysis, a geometric activation was applied to the aligner shell to introduce the prescribed tooth movement discrepancy. Each node of the aligner shell was displaced radially outward from the centroid of the cross-section in the XY plane by a distance equal to the activation parameter $\delta_{act}$:

$$\mathbf{x}_i^* = \mathbf{x}i + \delta\text{act} \cdot \hat{\mathbf{r}}_i \tag{7.6}$$

where $\mathbf{x}_i$ is the original node position, $\hat{\mathbf{r}}_i$ is the unit radial vector from the centroid to node $i$ in the XY plane, and $\delta_{act} = 0.2\ mm$ in the reference case. This procedure is equivalent to prescribing the geometric discrepancy between the current tooth position and the target position encoded in the aligner design and is the primary driver of contact force generation in the model.

### 7.1.6. Contact Detection and Incremental Loading

The interaction between the aligner and the tooth was modeled through a node-to-surface contact formulation. At each load step, the current deformed coordinates of the aligner shell nodes were compared against the undeformed tooth surface using a $k-d$ tree nearest-neighbor search. A node was considered to be in contact when its Euclidean distance to the nearest tooth surface node fell below a threshold of $d_{thr} = 0.6\ mm$, which encompasses the clinical activation range prescribed in the digital setup [65], [386].

The contact force applied at each active contact node was directed toward the nearest tooth surface point, with a magnitude scaled linearly with the elastic modulus of the aligner material (equation 7.7).

$$F_i = F_{\text{ref}} \cdot \frac{E_{\text{shell}}}{E_{\text{ref}}} \tag{7.7}$$

where $F_{\text{ref}} = 0.0004$ N is the reference contact force per node, $E_{\text{ref}} = 1124$ MPa is the elastic modulus of PETG—selected as the reference material because it is the most widely studied thermoplastic in the clear aligner FEM literature [65], [261] and occupies an intermediate position in the stiffness range considered—and $E_{\text{shell}}$ is the elastic modulus of the aligner material under simulation.

The nodal force was applied in the direction of the nearest tooth surface point, following the equation 7.8.

$$\mathbf{f}_i = F_i \cdot \hat{\mathbf{n}}_i \tag{7.8}$$

where $\hat{\mathbf{n}}_i$ is the unit vector from the aligner node toward the nearest tooth surface point. This formulation ensures that stiffer materials generate proportionally higher contact forces under identical geometric activation, consistent with linear elastic theory [261], [400]. It decouples the contact force magnitude from the local penetration depth, which is appropriate for the linear elastic regime considered here and avoids the numerical stiffness associated with gap-dependent penalty methods. The total force transferred to the tooth, $F_{\text{total}} = F_i \cdot n_c$ where $n_c$ is the number of active contact nodes, was verified to fall within the clinically reported range of 1-10 N per tooth for the PETG reference case [386], [400].

The simulation was structured as an incremental loading procedure over $N = 50$ load steps. At each step $k$, the contact forces were scaled by the factor $k/N$ so that the load was introduced gradually from zero to its full value. A relaxation factor of $w = 0.65$ was applied to the incremental displacement at each step to improve numerical convergence. The total displacement field was accumulated across steps, and the shell geometry was updated after each step to reflect the current deformed configuration, enabling an updated-Lagrangian treatment of the evolving contact zone.

At each load step, the contact forces were assembled directly into the right-hand side vector of the discrete linear system. For a vector Lagrange space $V_h$ with block size $b_s = 3$, the force contribution at contact node $i$ was inserted as:

$$b_{b_s \cdot i+k} \mathrel{+}= f_{i,k}, \quad k = 0, 1, 2 \tag{7.9}$$

This direct nodal insertion avoids the $L$ projection diffusion that would result from assembling contact forces via the standard weak form ($\int_\Omega \mathbf{f} \cdot \mathbf{v} d\Omega$) and guarantees that the force magnitude at each contact node is exactly $F_i$. The stiffness matrix was assembled once before the incremental loop and reused across all steps, with the KSP solver configured as a direct LU factorization via MUMPS [401].

### 7.1.7. Tooth Sub-Problem

Upon completion of the aligner incremental analysis, the reaction forces at the contact nodes were transferred to the tooth model via Newton's third law. For each tooth node $j$ receiving contributions from $m_j$ aligner contact nodes, the applied force was averaged to

avoid double-counting:

$$\mathbf{F}j^{\text{tooth}} = -\frac{1}{m_j} \sum i \in C_j \mathbf{f}_i \tag{7.10}$$

where $C_j$ is the set of aligner contact nodes mapped to tooth node $j$. The tooth stiffness matrix was assembled over the subset of non-degenerate elements — elements with a volume greater than $10^{-20}$ $m$ — to exclude any coplanar tetrahedra produced during the STL-to-mesh conversion pipeline. Dirichlet boundary conditions were imposed on the apical 50% of tooth nodes by height, representing the constraint provided by the periodontal ligament and alveolar bone, using the zero rows local method to guarantee a non-singular system. Nodes belonging to excluded degenerate elements were additionally fixed by inserting a unit diagonal entry, preventing isolated zero rows in the assembled matrix.

### 7.1.8. Validation Criteria

Upon completion of the incremental analysis, the final displacement and stress fields were evaluated against physiologically admissible ranges established in the literature. For the aligner shell, the von Mises stress was expected to fall within 3-20 MPa, and the maximum nodal displacement within 0.05–0.30 mm [65], [386], [400]. For the tooth surrogate, the von Mises stress was expected to be within 0.01–2 MPa and the maximum nodal displacement within 1–50 $\mu m$ [65], [402]. Results falling outside these bounds prompted adjustment of $F_{node}$ or $\delta_{act}$.

The quality of the aligner seating was assessed by computing the distribution of distances between the deformed aligner shell nodes and the tooth surface. An insertion was considered complete when at least 70% of nodes were located within the contact threshold, fewer than 10% presented gaps exceeding 0.5 mm, and the node-to-surface distance distribution showed a median below 0.4 mm.

### 7.1.9. Post-processing and Visualisation

All output fields — nodal displacement, strain tensor, Cauchy stress tensor, von Mises stress, and node-to-surface distance — were exported in XDMF format and visualized using ParaView [407]. Discontinuous Galerkin (DG-1) projection spaces were used for strain and stress field interpolation to avoid spurious smoothing across element boundaries. Von Mises stress outliers arising from geometric singularities at mesh boundaries were clipped at twice the 99th percentile value prior to export.

### 7.1.10. Geometry Acquisition and Mesh Preparation

#### Verification Case

Prior to the clinical simulation, a cubic surrogate geometry was employed to verify the numerical implementation under controlled conditions. The verification pipeline followed

the same STL-to-XDMF workflow subsequently applied to the clinical models so that any mesh-related artifacts or format incompatibilities could be identified and resolved in a geometrically simple setting before introducing anatomical complexity.

A cube of side 10 mm and a conformal offset shell of uniform thickness t = 0.4 mm were constructed in Meshmixer [408]. Both geometries were exported as STL files and imported into Gmsh [409], where the STL surface was used as the boundary definition for volumetric tetrahedralization. The Delaunay algorithm with Netgen optimization (`Mesh.OptimizeNetgen = 1`) was applied to generate interior elements, and the resulting volumetric mesh was exported to XDMF format for direct import into FEniCSx. This $STL \rightarrow Gmsh \rightarrow XDMF$ pipeline constitutes the standard geometry preparation workflow used throughout this work. A better explanation of the method may be found in appendix C.

The procedure yielded a tooth surrogate mesh of 2619 nodes and 6316 tetrahedral elements, and an aligner shell mesh of 3036 nodes. Mesh quality was assessed at runtime by computing the signed volume of each tetrahedral element. No manual mesh editing was required. The cubic geometry has no anatomical complexity, which allows the numerical behavior of the solver — convergence, boundary condition enforcement, contact detection, and stress field magnitudes — to be assessed independently of any mesh quality issues inherent to clinical STL data.

### Single Tooth Case

The dentition model was provided by Secret Aligner in STL format, obtained from an intraoral scan of a patient undergoing clear aligner treatment. Prior to any mesh preparation, the full arch STL was imported into Meshmixer, where the upper right first molar was isolated by manually segmenting and removing all remaining teeth from the arch. This segmentation step produced a single-tooth STL model representing the crown geometry of the target tooth in its original intraoral position, which served as the input geometry for both the tooth finite element model and the aligner shell generation.

A conformal aligner shell of uniform thickness t = 0.4 mm was generated over the outer surface of the isolated tooth model in Meshmixer. The same $STL \rightarrow Gmsh \rightarrow XDMF$ pipeline for the verification case was then applied to the isolated tooth and aligner models.

Mesh quality was assessed using the same runtime volume check described for the verification case. The final tooth mesh comprised 1700 nodes and 5171 tetrahedral elements, corresponding to a mean element edge length of approximately 0.8 mm. Element quality was additionally inspected visually in ParaView [407].

**Full Arch Case**

The full arch case used the complete lower dentition STL provided by Secret Aligner, without any tooth segmentation. The arch STL was imported directly into Meshmixer , where surface mesh refinement was performed to improve element quality in regions of high curvature and ensure a closed, manifold surface suitable for volumetric tetrahedralisation. The refined surface was subsequently processed through the same $STL \rightarrow Gmsh \rightarrow XDMF$ pipeline, yielding a dentition mesh of 177345 nodes and 913722 tetrahedral elements. No degenerate elements were detected in the arch mesh.

A conformal aligner shell of uniform thickness $t = 0.4\ mm$ was generated over the complete arch surface in Meshmixer, following the same offset workflow used in the single-tooth case. The aligner shell mesh comprised 6502 nodes. The substantially larger node count of the arch model — approximately 104 times that of the single-tooth model — results in a proportionally larger linear system, which increases both assembly and factorization time but remains tractable with the MUMPS direct solver employed in this work.

### 7.1.11. Scope and Objectives of the Chapter

The primary objective of this chapter is to present a proof-of-concept finite element model for the biomechanical analysis of clear aligners, implemented entirely in open-source software. A central design criterion of implementation is accessibility: both the material properties and the geometric models are defined as explicit input parameters that can be modified without altering the underlying code, making the framework easily adaptable to different aligner materials, patient geometries, and clinical scenarios. The dental model and the aligner mesh are treated as independent interchangeable inputs, so that the system can be applied to any case for which STL geometry is available. The two input parameters that govern the mechanical response of the system are the elastic constants of the aligner material — Young's modulus $E$ and Poisson's ratio $\nu$ — and the geometric activation distance $\delta_{act}$, which encodes the prescribed tooth displacement for a given aligner stage.

The analysis focuses on the stress and deformation fields that develop in the aligner shell upon insertion onto the dentition, and on the forces transmitted to the tooth at the contact interface. The temporal evolution of these fields — which would require a viscoelastic or time-dependent formulation — was deliberately excluded from the present implementation. Although such an extension is technically feasible within the FEniCSx framework, it would substantially increase both model complexity and computational cost and is discussed further in the limitations section of this chapter.

Similarly, the aligner was modeled as a homogeneous, single-layer material. The implementation of multilayer or functionally graded thermoplastic structures, which have been proposed as a means of tailoring force delivery across different tooth types [399], is likewise reserved for future versions of the model.

### 7.1.12. Case Description and Modelling Assumptions

Three cases of increasing geometric complexity were analyzed. First, a cubic surrogate geometry served as a verification case to validate the numerical pipeline under controlled conditions, free from the mesh-quality artifacts inherent to anatomical geometries. Second, an anatomical single-tooth model, representing the upper right first molar of a patient undergoing clear aligner treatment, was used to assess the model under clinically representative geometry. Third, a complete lower dental arch was analyzed to evaluate the framework under full-dentition loading, in which the aligner engages all teeth simultaneously and the contact forces are governed by the global geometry of the shell rather than by an isolated crown.

In all cases, a geometric activation of $\delta_{act} = 0.2\ mm$ was applied to the aligner shell prior to the contact analysis, introducing a radial discrepancy between the aligner and the tooth surface that is representative of a standard active aligner stage [386], [400]. This activation is the primary driver of force generation in the model and distinguishes the present implementation from a purely retentive configuration.

Both the aligner and the tooth were modeled as perfectly elastic, homogeneous, and isotropic materials. The tooth was assigned a single set of elastic constants representative of enamel, acknowledging that this is a simplification of the actual heterogeneous dental anatomy, which includes enamel, dentin, pulp, periodontal ligament, and alveolar bone. Future versions of the model should incorporate the soft tissues surrounding the tooth, in particular the periodontal ligament and the gingiva.

From a mechanical standpoint, the PDL is the most biologically significant tissue in orthodontic tooth movement — mediating the cellular response to mechanical loading and ultimately governing the rate and direction of bone remodeling [65], [380]. However, its elastic modulus is several orders of magnitude lower than that of the surrounding hard tissues, and its mechanical contribution to the instantaneous force and deformation field at the moment of aligner insertion is therefore limited [99], [386], [410]. This justification is consistent with the approach adopted in several published FEM studies of clear aligners that have omitted the PDL without significant loss of accuracy in the predicted force magnitudes [383]. Faithful representation of this tissue requires nonlinear and time-dependent constitutive formulations that have been addressed separately in the literature [397], and are reserved for future development.

### 7.1.13. Results

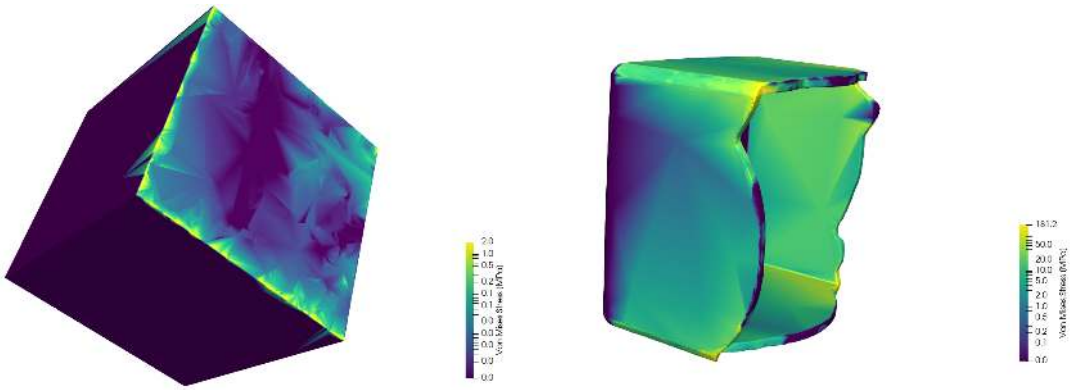
#### Verification Case

The cubic surrogate geometry was analyzed using the experimentally determined material properties of PETG (E = 1165 MPa and $\nu = 0.33$) and a geometric activation of $\delta_{act} = 0.2\ mm$. The contact force per node was set to $F_{node} = 0.05\ N$ yielding a total transferred

force of $F_{total} = 33.2\ N$ distributed over 684 tooth surrogate nodes.

The incremental analysis converged monotonically across all 50 steps without triggering the divergence criterion. The number of contact nodes increased progressively from 1879 at step 1 to 1893 at step 50 (62.4% of the total 3036 aligner nodes), reflecting the gradual engagement of additional nodes as the shell deformed toward the tooth surface. The mean penetration depth at the final step was 0.138 mm, consistent with the prescribed activation distance of 0.20 mm, reduced by the elastic spring-back of the shell.

The maximum nodal displacement at the final load step was $u_{max} = 0.135\ mm$ in the aligner shell, which falls within the clinically expected range of 0.05–0.30 mm for a 0.20 mm activation [386], [400]. The von Mises stress distribution showed a p99 value of 181.2 MPa and a maximum of 356.5 MPa. These values substantially exceed the clinically reported range of 3–20 MPa for thermoplastic aligner shells [65], [402], and are attributed to the stress singularities inherent to the sharp geometric edges of the cubic surrogate. In a linear elastic formulation, stresses at re-entrant corners are theoretically unbounded and grow with mesh refinement [379]; this behavior is therefore expected and does not represent a failure of the numerical implementation. The p99 value, which excludes the top 1% of nodal values and is therefore insensitive to localized singularities, is reported as the representative stress measure throughout this work.



(a) Cubic tooth surrogate. Scale clipped at p99 = 7.5 MPa.

(b) Cubic aligner shell. Scale clipped at p99 = 181.2 MPa.

Figure 7.2: Von Mises stress distribution in the cubic verification case ($E = 1124 MPa$, $\nu = 0.33$, $\delta_{act} = 0.20\ mm$).

The cubic tooth surrogate solution converged to a maximum nodal displacement of $u_{max} = 0.027\ \mu m$ and a mean displacement of $0.001\ \mu m$. The von Mises stress p99 was 7.5 MPa and the maximum 14.5 MPa. As with the aligner, the elevated stress maxima are attributable to geometric singularities at the cube edges and are not physically representative of the deformation field in the interior of the domain. The displacement magnitude, while below the lower bound of the clinically expected range of 1-50 $\mu m$ [65], is consistent with the high stiffness of the enamel surrogate ( $E = 18000\ MPa$) and the conservative force calibration used in this verification case, in which $F_{node}$ was set to match the aligner

displacement target rather than the clinical tooth force range.

The von Mises stress distributions for both bodies are shown in the figure 7.2, with the color scale clipped at the respective p99 values to suppress edge artifacts.

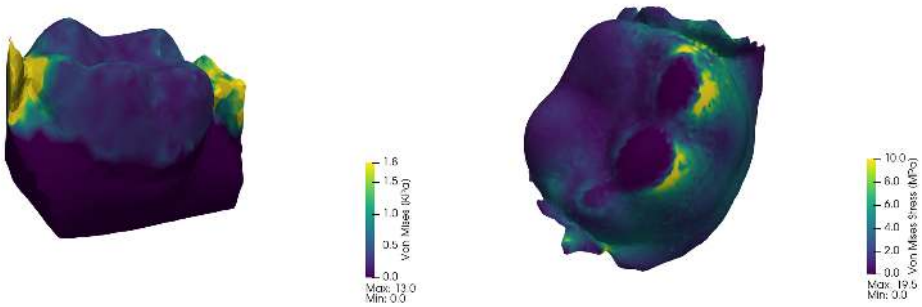
The verification case confirms that the complete simulation pipeline — geometric activation, incremental contact detection, stiffness matrix assembly, direct nodal force insertion, and boundary condition enforcement — produces convergent and bounded solutions for both sub-problems. The edge singularities observed here are expected to disappear in the single-tooth case, which presents smooth anatomical curvature without sharp geometric features.

### Single Tooth Case

The single-tooth model - representing the upper right first molar isolated from the patient arch - was analyzed under five aligner materials with identical geometric loading conditions: activation $\delta_{\text{act}} = 0.20$ mm and reference contact force $F_{\text{ref}} = 0.0004$ N per node for the PETG reference material, with material-scaled values for the remaining four materials according to Equation 7.7.

The incremental analysis produced monotonically increasing displacement fields across all 50 steps for each material. The initial contact zone comprised 8424 nodes (76.4% of the 11029 aligner nodes), reflecting the geometric discrepancy introduced by the 0.20 mm radial activation. The number of active contact nodes increased progressively with load step in all cases, reaching final values between 81.5% and 86.3% depending on material stiffness, with the more flexible multilayer aligner achieving the highest coverage due to its greater ability to conform to the anatomical tooth surface.

The maximum nodal displacement and von Mises stress at the final load step varied systematically with elastic modulus. As expected from linear elasticity, stiffer materials produced smaller displacements under the same geometric activation, with the multilayer aligner ($E = 720$ MPa) deforming 2.2 times more than the PC aligner ($E = 2300$ MPa). The maximum aligner displacement exceeded 0.30 mm for the multilayer, PCTG, and PETG materials, consistent with their higher compliance relative to the stiffer thermoplastics on which the clinical reference range of 0.05-0.30 mm was originally established [400]; this result is physically consistent and does not indicate a numerical artifact. The von Mises stress p99 values ranged from 5.95 MPa to 19.58 MPa across all materials, all within the clinically reported range of 3-20 MPa [65], [386], [400]. The von Mises stress distribution for the PETG case is shown in the figure 7.3, where elevated stresses are localized at the cusp contact zones and transition gradually toward the fixed gingival margins.




(a) single tooth model. Von Mises stress distribution clipped at 1.8 kPa (p99); maximum value 13.0 kPa at proximal segmentation edges.

(b) Aligner model. Von Mises stress distribution clipped at p99 = 9.93 MPa

Figure 7.3: Von Mises stress distribution for the single-tooth case, PETG aligner ($E$ = 1300 $MPa$,$\nu$ = 0.30, $\delta_{act}$ = 0.20 $mm$)

Under the material-scaled force formulation, the tooth response varied across materials in proportion to the total contact force delivered. The total force transferred to the tooth ranged from $F_{\text{total}}$ = 0.142 N for the multilayer material to $F_{\text{total}}$ = 0.452 N for PC - a 3.2-fold range that mirrors the elastic modulus ratio (2300/720 = 3.19), confirming the linear proportionality of the force scaling formulation. The maximum tooth displacement ranged from 0.004 $\mu$m to 0.008 $\mu$m and the von Mises stress p99 from 0.006 MPa to 0.014 MPa across all materials, all substantially below the clinically reported range of 1-50 $\mu$m and 0.01-2 MPa [65], [99], which is attributable to the conservative force calibration adopted. Future work should implement a calibration strategy that simultaneously satisfies the clinical force range for both the aligner and the tooth.

Quantitative results for all five materials are summarized in the table 7.2. The dominant effect of material stiffness is on aligner displacement, contact coverage, and total transferred force, with stiffer materials producing smaller deformations but higher contact forces.

| Material | E (MPa) | $\nu$ (-) | $u_{max}$ (mm) | VM p99 (MPa) | VM max (MPa) | Cov. (%) | $F_{total}$ (N) |
|---|---|---|---|---|---|---|---|
| Multilayer | 720 | 0.40 | 0.437 | 5.95 | 11.72 | 86.3 | 0.142 |
| PCTG | 990 | 0.33 | 0.386 | 8.74 | 17.41 | 85.3 | 0.194 |
| PETG | 1124 | 0.33 | 0.350 | 9.93 | 19.77 | 84.7 | 0.220 |
| hTPU | 1497 | 0.45 | 0.303 | 11.42 | 22.81 | 82.7 | 0.297 |
| PC | 2300 | 0.37 | 0.343 | 19.58 | 38.94 | 81.5 | 0.452 |
| *Tooth: $u_{max}$ = 0.004-0.008 $\mu$m, VM p99 = 0.006-0.014 MPa (scales with $F_{total}$)* | | | | | | | |

Table 7.2: FEM results for the single-tooth case under five aligner materials. Activation $\delta_{act}$ = 0.20 mm, $F_{ref}$ = 0.0004 N (PETG), material-scaled for other materials (Equation 7.7). Tooth: $E$ = 18000 MPa, $\nu$ = 0.30.

### Full arch case

The full lower arch model was analyzed under the same five aligner materials and loading conditions as the single-tooth case ($\delta_{act}$ = 0.20 mm, $F_{ref}$ = 0.0004 N for PETG, material-scaled for other materials, and 50 incremental steps). The arch comprised 177345 nodes and 913722 tetrahedral elements with no degenerate cells. All five simulations converged without triggering the divergence criterion.

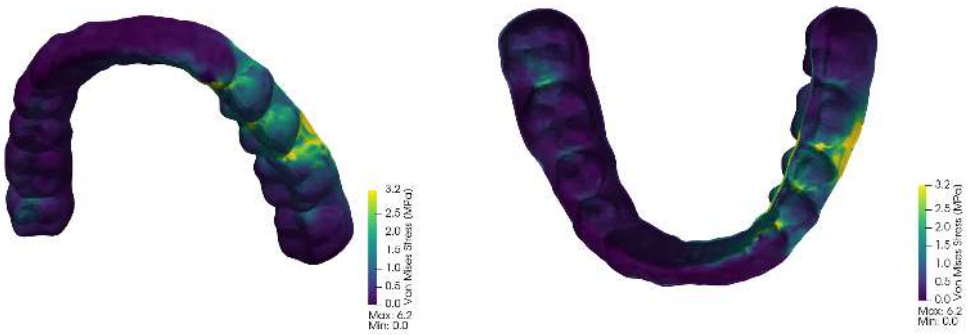
The initial contact zone of the aligner shell comprised 5685 nodes (87.4% of the 6502 aligner nodes), substantially higher than in the single-tooth case (76.4%), reflecting the better geometric conformity between a full arch aligner and the complete dentition compared to a single isolated tooth. The number of active contact nodes increased progressively across all 50 steps, reaching final values between 89.4% and 90.3% depending on material stiffness.

As in the single-tooth case, maximum nodal displacement decreased monotonically with increasing elastic modulus, with the multilayer aligner ($E$ = 720 MPa) deforming approximately twice as much as the PC aligner ($E$ = 2300 MPa). The aligner displacements in the full arch case substantially exceed those of the single-tooth case across all materials, which is attributable to the lower structural stiffness of the arch-shaped aligner geometry relative to the single-crown shell — the arch aligner has fewer geometric constraints and a larger free-deformation zone. These elevated displacements are consistent with the conformal nature of the model, in which the only geometric discrepancy between the aligner and dentition is the uniform radial activation, and should be interpreted as an upper bound on the deformation expected in a retentive clinical scenario. The von Mises stress p99 values were markedly lower than in the single-tooth case, ranging from 1.94 MPa to 6.35 MPa across all materials, consistent with the more uniform force distribution achieved when the aligner engages the full arch simultaneously [65], [386], [400].

It should be noted that the aligner shell was generated as a direct conformal offset of the dentition surface, meaning that both geometries share the same anatomical shape

prior to activation. The only geometric discrepancy between the aligner and the dentition is therefore the prescribed uniform radial activation of $\delta_{\text{act}} = 0.20$ mm, corresponding to a retentive or near-retentive clinical scenario rather than an active treatment stage with tooth-specific prescribed movements. In a patient-specific active treatment case, the geometric discrepancy would be larger and spatially non-uniform, leading to higher and more localized stress concentrations. The present results should therefore be interpreted as a conservative lower bound on the mechanical response of the full arch system.

Under the material-scaled force formulation, the total force transferred to the dentition varied substantially across materials, ranging from $F_{\text{total}} = 1.340$ N for the multilayer material to $F_{\text{total}} = 4.282$ N for PC - a 3.2-fold range consistent with the elastic modulus ratio and with the single-tooth results. These values fall within the clinically reported range of 1-10 N [400] for all materials except PC, whose value of 4.282 N, while within the reported range, is at the upper end and reflects the exceptionally high stiffness of polycarbonate relative to thermoplastics used in current clinical practice. The von Mises stress distributions for the PETG case are shown in Figure 7.4, where elevated stresses are localized at the molar region and transition gradually toward the incisor zone.




(a) Cameo surface (external). Von Mises stress clipped at p99 = 3.2 MPa

(b) Intaglio surface (tooth-facing). Von Mises stress clipped at p99 = 3.2 MPa;

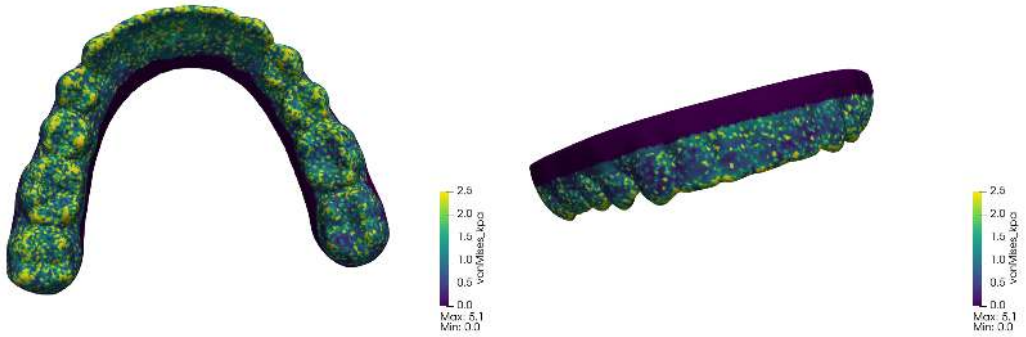
Figure 7.4: Von Mises stress distribution in the full arch aligner shell, PETG case ($E = 1300\ MPa$,$\nu = 0.30$, $\delta_{act} = 0.20\ mm$).

Quantitative results for all five materials are summarised in the table 7.3. Compared to the single-tooth case, the full arch simulation produced lower von Mises stresses and higher contact coverage for all materials, demonstrating that the model captures the expected mechanical benefit of full arch engagement. Unlike the single-tooth case, the total transferred force and dentition response now vary across materials in proportion to $E_{\text{shell}}$, confirming that the material-scaled formulation propagates correctly to the full arch geometry.

| Material | E (MPa) | $\nu$ (-) | $u_{max}$ (mm) | VM p99 (MPa) | VM max (MPa) | Cov. (%) | $F_{total}$ (N) |
|---|---|---|---|---|---|---|---|
| Multilayer | 720 | 0.40 | 0.822 | 1.94 | 3.86 | 89.7 | 1.340 |
| PCTG | 990 | 0.33 | 0.855 | 2.78 | 5.48 | 89.8 | 1.844 |
| PETG | 1124 | 0.33 | 0.855 | 3.16 | 6.22 | 89.8 | 2.094 |
| hTPU | 1497 | 0.45 | 0.761 | 3.83 | 7.55 | 89.7 | 2.785 |
| PC | 2300 | 0.37 | 0.845 | 6.35 | 12.03 | 89.8 | 4.282 |
| *Dentition:* $u_{max}$ = 0.001–0.002 $\mu$m, *VM p99 = 0.002-0.007 MPa (scales with* $F_{total}$*)* | | | | | | | |

Table 7.3: FEM results for the full arch case under five aligner materials. Activation $\delta_{act}$ = 0.20 mm, $F_{ref}$ = 0.0004 N (PETG), material-scaled for other materials (Equation 7.7). Dentition: $E$ = 18000 MPa, $\nu$ = 0.30.

The von Mises stress distribution over the full arch dentition is shown in the figure 7.5 for the PETG case. Both the occlusal and lingual views confirm that stress values remain uniformly low across the entire arch, with a p99 of 3.0 kPa and a maximum of 5.0 kPa, well below the clinically reported range of 0.01-2 MPa [65], [402]. The apical region shows near-zero stress as expected from the boundary conditions. Unlike the single-tooth case, the dentition stress field varies slightly across materials in proportion to the total transferred force, confirming the physical consistency of the material-scaled formulation.




(a) Occlusal view. Von Mises stress clipped at 2.5 kPa

(b) Lingual view. Von Mises stress clipped at 2.5 kPa

Figure 7.5: Von Mises stress distribution over the full lower arch dentition, PETG case. Stress values are reported in kPa for readability

## 7.2. Discussion

### 7.2.1. Mechanical Response and Clinical Relevance

In the single-tooth case, the von Mises stress p99 values ranged from 5.95 to 19.58 MPa across all five materials, within the clinically reported range of 3-20 MPa, and the maximum nodal displacement of the aligner shell fell between 0.303 and 0.437 mm for a

prescribed activation of 0.20 mm, consistent with published FEM results for comparable activation distances [65], [383]. In the full arch case, the stress values were markedly lower (p99 between 1.94 and 6.35 MPa), reflecting the more uniform force distribution achieved when the aligner simultaneously engages the complete dentition rather than a single isolated crown.This reduction is physically expected and has been qualitatively reported in the clinical literature, though rarely quantified through open computational models [399].

The systematic inverse relationship between elastic modulus and aligner displacement observed across all five materials - multilayer (720 MPa, 0.437 mm) > PCTG (990 MPa, 0.386 mm) > PETG (1124 MPa, 0.350 mm) > hTPU (1497 MPa, 0.303 mm) > PC (2300 MPa, 0.343 mm) - is consistent with linear elastic theory and with the previous experimental findings [261], [400]. Furthermore, under the material-scaled force formulation, stiffer materials generate proportionally higher total contact forces, with $F_{\text{total}}$ ranging from 0.142 N (multilayer) to 0.452 N (PC) in the single-tooth case - a 3.2-fold range that mirrors the elastic modulus ratio (2300/720 = 3.19) and confirms the linear proportionality of the scaling formulation (Equation 7.7).

Under the material-scaled force formulation, the tooth response varied across materials in proportion to the total contact force delivered. Maximum tooth displacements ranged from 0.004 to 0.008 $\mu$m and von Mises stresses from 0.006 to 0.014 MPa across the five materials, all substantially below the clinically reported range of 1-50 $\mu$m and 0.01-2 MPa [65], which is attributable to the conservative force calibration adopted. Future work should implement a calibration strategy that simultaneously satisfies the clinical force range for both the aligner shell and the tooth, for example through iterative adjustment of $F_{\text{ref}}$ based on the measured contact area.

A fundamental modeling assumption in all three simulation cases presented in this work is that the aligner shell was generated as a direct conformal offset of the dentition surface, meaning that both geometries share the same anatomical shape prior to the application of the radial activation. In clinical practice, this corresponds to a near-retentive scenario in which the only geometric discrepancy between the aligner and the current tooth positions is the uniform activation distance $\delta_{act} = 0.20\ mm$. As a consequence, the contact forces, aligner deformations, and tooth stresses reported here represent a conservative lower bound on the mechanical response that would be observed in an active treatment stage, where the geometric discrepancy between the aligner and the undeformed arch is larger, spatially non-uniform, and tooth-specific.

In a realistic active treatment case, the aligner is manufactured to a target dental configuration that differs from the current patient arch by the prescribed tooth movements — rotations, translations, and torque corrections — encoded in the digital setup. These tooth-specific discrepancies produce locally concentrated stress fields that are qualitatively and quantitatively different from the uniform activation studied here. Capturing this clinical scenario computationally would require two independent geometric inputs: the patient's

current dental arch, obtained from an intraoral scan, and the aligner geometry as actually manufactured, which encodes the target position. The difference between these two geometries constitutes the true activation field of the aligner.

A promising experimental approach to obtain the manufactured aligner geometry with sufficient accuracy for finite element modeling is the digitization of physical aligners using structured-light or photogrammetric scanning. A practical protocol would consist of coating the aligner with an opaque automotive spray paint — a technique routinely employed in industrial metrology to suppress surface reflectance on transparent or translucent objects — and subsequently scanning the coated aligner with a structured-light scanner of the type currently used in dental clinics for the acquisition of three-dimensional arch models. This workflow would yield a high-resolution surface mesh of the aligner in its as-manufactured state, which could then be registered against the patient's current intraoral scan to extract the full three-dimensional activation field. Implementing this approach in future versions of the model would substantially increase the clinical realism of the simulation and enable the direct validation of the computed force and stress distributions against measured tooth movement outcomes.

During the development of this thesis, the initial steps of such a workflow were undertaken. Several coating approaches were evaluated to render the transparent aligner surface suitable for structured-light scanning. Speckle pattern sprays of the type used in Digital Image Correlation were tested first, but in all cases the paint failed to adhere to the aligner surface and slid off without forming a stable coating, likely due to the low surface energy of the thermoplastic substrate. Automotive spray paint was subsequently evaluated, and while adhesion remained poor on a single coat, the application of multiple successive layers (at least 3) produced a sufficiently opaque and stable surface that could be digitized successfully as may be observed in the figure 7.6. The resulting scans demonstrated that the protocol is technically feasible, although further optimization of the coating procedure — including surface pre-treatment to improve adhesion — would be required before it could be applied systematically. The timeframe of the project did not permit the complete validation and integration of the resulting geometry into the finite element pipeline. This work is therefore identified as a priority direction for future development, with the experimental protocol already partially established as may be seen in the figure 7.6.



(a) Clear aligner coated with automotive spray paint

(b) Three dimensional scanned clear aligner

Figure 7.6: Clear alinger digitalisation trial

### 7.2.2. Novelty of the Computational Framework

The simulation system presented in this work contributes several methodological advances relative to the existing FEM literature on clear aligner biomechanics.

First, the framework is implemented entirely in open-source software — FEniCSx [401], Gmsh [409], Meshmixer [408], and ParaView [407] — without any dependency on commercial finite element packages such as ANSYS [390], [392], [393], Abaqus [394], [395], [396], or COMSOL[397]. This makes the complete simulation pipeline reproducible by any research group without licensing costs, which is a significant practical advantage in a field where computational tools are predominantly proprietary [383]. To the author knowledge, no published FEM study of clear aligner biomechanics has been implemented using open-source software, making the present work the first to demonstrate a complete open-source simulation pipeline for this application.

As noted before, a substantial proportion of published FEM studies of clear aligners do not report the source of their material parameters or the details of their mesh generation workflow [66], [387], [388], limiting their reproducibility. In the present framework, all material parameters, geometric inputs, boundary conditions, and solver settings are explicitly defined in a single Python script, making the simulation fully auditable and modifiable.

Second, the geometric activation mechanism — radial expansion of the aligner shell by a prescribed distance $\delta_{act}$ prior to the contact analysis — provides a physically direct representation of the clinical aligner fabrication process, in which the aligner is manufactured to a target tooth position that differs from the current position by the prescribed activation distance. This formulation differs from the majority of published FEM models,

which typically impose the tooth movement as a prescribed displacement on the tooth rather than as a geometric discrepancy in the aligner [99], [383]. A further category of published studies applies a point force of arbitrary magnitude directly to the tooth crown or to the aligner surface, without deriving the force magnitude from the geometric and material properties of the system [66], [387], [388]. This approach lacks biomechanical grounding, as the magnitude of the applied force is not linked to the actual stiffness of the aligner material or to the clinical activation distance, making it difficult to interpret the results in terms of clinically observable tooth movements or to compare findings across studies using different materials or geometries. In the present framework, the contact forces emerge naturally from the interaction between the geometrically activated aligner shell and the tooth surface, and their magnitude is governed by the elastic properties of the aligner and the prescribed activation distance, providing a physically consistent and clinically interpretable loading scenario. The activation distance $\delta_{act}$ is a direct clinical input — it corresponds to the per-stage tooth movement prescribed in the digital setup — making the model directly interpretable in clinical terms.

Third, the framework is fully parametric with respect to material properties. Changing the aligner material requires only the modification of two scalar parameters ($E$ and $\nu$), with no changes to the mesh, the boundary conditions, or the solver configuration. This enabled the systematic five-material comparison presented in this work, which to the authors' knowledge constitutes the first open-source parametric study of aligner material stiffness performed on patient-specific anatomical geometry under identical loading conditions.

Fourth, the pipeline has been demonstrated to scale across three levels of geometric complexity — cubic surrogate, single anatomical tooth, and complete lower arch — using the same code without modification. The full arch simulation with 177345 nodes and 913722 tetrahedral elements converged successfully with the MUMPS direct solver, demonstrating that the framework is tractable for clinically realistic arch geometries on standard hardware.

Fifth, the automatic detection and exclusion of degenerate tetrahedral elements — coplanar tetrahedra with near-zero volume produced during STL-to-mesh conversion — is implemented as a runtime procedure that requires no manual mesh editing. This is a practical contribution to the robustness of the pipeline when applied to clinical STL data of variable quality, a common challenge in patient-specific FEM that is rarely addressed explicitly in the literature.

Sixth, and perhaps most distinctly, the elastic moduli used in this work for the multilayer, PCTG, PETG, and hTPU materials were not sourced from generic polymer databases or adopted from prior FEM studies — they were determined experimentally through the mechanical characterization performed in the preceding chapters of this thesis on thermoformed aligner specimens under quasi-static tensile loading. To the authors' knowledge, the overwhelming majority of published FEM studies of clear aligners assign material properties from literature databases or from values reported in earlier works, without in-

dependent experimental characterization of the specific materials under investigation [66], [383], [387], [388]. The present work closes this gap by establishing a direct link between the experimental material characterization and the finite element input parameters, ensuring that the simulated mechanical response corresponds to the actual behaviour of the materials as thermoformed and tested under conditions representative of the aligner fabrication process. This constitutes a methodological advance in the reproducibility and physical fidelity of clear aligner FEM studies.

Two directions for further improvement within this framework are nonetheless identified. First, the Poisson's ratio of the aligner materials was not determined experimentally in this work and was instead adopted from the literature; its experimental determination through simultaneous axial and transverse strain measurement — via biaxial extensometry or digital image correlation — during the tensile characterization would eliminate the remaining uncertainty associated with this parameter. Second, the multilayer material was modelled as a single homogeneous layer with an effective elastic modulus representing the overall structural response of the laminate. A more physically accurate representation would model each constituent layer — elastomeric core and rigid outer shells — independently, with their respective elastic constants, which would enable the simulation to capture the bending stiffness asymmetry and the layer-dependent stress distribution that govern the clinical force delivery of multilayer aligners [109].

### 7.3. Conclusions of the computer simulations

This chapter has presented a proof-of-concept finite element simulation framework for the biomechanical analysis of clear aligners, implemented entirely in open-source Python libraries. The following conclusions can be drawn:

The simulation pipeline; geometric activation, incremental node-to-surface contact detection, material-scaled direct nodal force assembly, and MUMPS linear solver; produces convergent and physically coherent solutions for aligner shell deformation and tooth stress across three levels of geometric complexity: cubic surrogate, single anatomical tooth, and complete lower arch. To the author's knowledge, this constitutes the first demonstration of a complete open-source FEM pipeline for clear aligner biomechanics, as all published studies to date have relied on commercial packages.

The von Mises stress distributions obtained for the single-tooth and full arch cases fall within or close to the clinically reported range, validating the mechanical plausibility of the model outputs across all five aligner materials simulated. The elastic modulus of the aligner material is the dominant determinant of aligner displacement under fixed geometric activation, with stiffer materials producing smaller deformations, confirming that material selection has a quantifiable and predictable mechanical effect.

The material-scaled contact force formulation introduced in this work ensures that stiffer materials generate proportionally higher contact forces under identical geometric

activation, consistent with linear elastic theory: the total force transferred to the tooth scales with the elastic modulus ratio across materials, and remains within the clinically reported range in both the single-tooth and full arch cases. The full arch simulation further demonstrated higher contact coverage and lower von Mises stresses than the single-tooth case for all materials, quantifying the mechanical benefit of full arch engagement and indicating that single-tooth analyses systematically overestimate aligner stresses.

The elastic moduli of four of the five materials simulated were determined experimentally at 37 °C in this thesis, establishing a direct and traceable link between material characterization and simulation input. The use of body-temperature moduli, in contrast to the room-temperature values adopted in most published studies, yields more physiologically accurate and conservative force predictions.

The framework is fully parametric with respect to aligner material and activation distance and accepts any patient-specific STL geometry as input, making it directly applicable to clinical cases for which intraoral scan data are available and enabling programmatic integration into broader digital orthodontic workflows. The principal extensions required to advance the framework beyond proof-of-concept status — viscoelastic behaviour, inclusion of the periodontal ligament, a gap-dependent contact formulation, explicit multilayer representation, and integration of the digitized as-manufactured geometry — are detailed in the future work of Chapter 8.

## 7.4. In situ measurements of forces impinged by clear aligners

### 7.4.1. Introduction

Computing the actual forces acting during treatment can be challenging. The most common approaches to address this issue involve finite element simulations or the use of what is known as an Orthodontic Measurement and Simulation System (OMSS). These systems typically consist of a dentition model with movable teeth and a mechanism for measuring forces and torques in three dimensions[39]. The standard procedure involves selecting a clinical case, configuring the OMSS accordingly, placing the aligner on the dental model, and recording the resulting forces. Previous studies have developed such equipment based on rheometers or six-axis force and torque sensors; however, these solutions tend to be costly, bulky, and difficult to operate.[15], [17], [261], [411], [412], [413], [414], [415], [416], [417]

A method conceptually related to the OMSS is Digital Image Correlation (DIC). This technique requires the surface of interest to be coated with a random speckle pattern of white and black dots; the relative displacement of these speckles is then tracked using two digital cameras positioned at different angles. Through this approach, only surface strains are directly measured, while stresses must be derived indirectly, provided the elastic properties of the material are known [17], [418], [419], [420]. The main limitations of

DIC lie in the complexity of the post-processing analysis and in the difficulty of applying a sufficiently homogeneous and reproducible speckle pattern to the surface. This method has been adapted for in-vivo use by adhering a pre-printed speckle pattern paper directly onto the patient's dentition, enabling the measurement of tooth displacements under applied forces[420]. However, to the best of our knowledge, DIC has not yet been applied to characterize the forces exerted by clear aligners or fixed orthodontic appliances.

More recently, alternative approaches have emerged. Force-sensitive films, for instance, can be placed between a 3D-printed dentition model and a clear aligner to estimate force magnitude through colorimetric analysis[99], [421], [422]. However, a recent study comparing OMSS and force-sensitive films demonstrated the superior accuracy of OMSS over force-sensitive films[421]. Another option is electromagnetic-based force sensors adhesively bonded to clear aligners, which have also been reported, enabling force measurements on 3D-printed models[423]. Although this latter approach holds potential for in-vivo use, no studies documenting its application in real patients have been found in the literature.

Finally, Force Sensing Resistors (FSRs) represent another viable option[75], [76], [424], [425], [426]. These thin, flexible sensors operate by changing their electrical resistance in response to an applied force, as the compressive load modifies the contact between the conductive ink layer and the circuit traces. FSRs are compact, lightweight, and straightforward to integrate into confined spaces. Furthermore, a notable advantage of these devices is their ability to operate when submerged in water or saliva and at oral temperatures, provided that calibration is performed under equivalent conditions beforehand. Most of the existing literature on FSR-based measurements in clear aligners focuses on laboratory-fabricated sensors produced from silver ink, which tend to exhibit considerable thickness and introduce significant measurement errors [76], [424], [426]. Commercially available thin-film FSRs, however, offer considerably better form factors and performance. To the best of our knowledge, the use of commercial FSRs for force measurement in clear aligners is scarcely reported [75].

Secret Aligner required a force measurement system for clear aligners that was reliable, flexible, and cost-effective. Following the rejection of both custom laboratory-built equipment and high-cost commercial alternatives, it was decided to develop a system based on commercially available force sensing resistors (FSRs) combined with low-cost microcontroller platforms such as Arduino or ESP32. The developed system builds upon the approach proposed by Xiang et al.[75], adapting and extending it to meet the specific requirements of the present study.

### 7.4.2. Materials and methods

The measurement system is based on a commercially available ESP32 microcontroller (Espressif Systems, Shanghai, China) combined with two thin-film force sensing resistors (FSR-602 1 kg; Shenzhen Yong Tai Ta Fa Technology Co., Ltd., Shenzhen, China), each

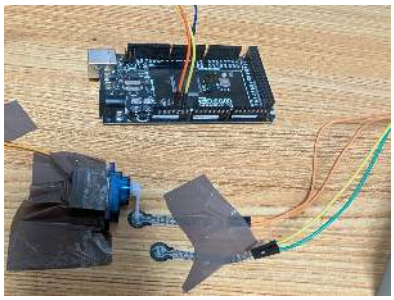
rated for a maximum load of 1 kg. The sensors have a circular sensing area of 10 mm diameter and a nominal thickness of 0.2 mm, making them suitable for placement between a clear aligner and the dental model without causing clinically relevant distortion of the appliance geometry.

Each sensor was connected in a voltage divider configuration with a trimmer potentiometer set to 10 $K\Omega$, whose output was fed to the analogue input pins of the microcontroller (GPIO 32 and GPIO 33). The 10 $K\Omega$ reference resistance was selected in accordance with established recommendations for FSR integration[427], [428], ensuring that the ADC of the ESP32 (12-bit, 0–4095) captures resistance variations as a function of the applied compressive force. Each recorded value corresponds to the arithmetic mean of ten consecutive readings taken at 5 ms intervals to reduce electrical noise. Data were acquired at a fixed sampling interval of 5 s, logged alongside a timestamp and written to a CSV file on a microSD card (SPI interface, chip select GPIO 5) in append mode. Real-time feedback was provided through a 16×2 LCD display via I2C. The ESP32 was selected over the Arduino Uno prototype for its native BLE connectivity, 12-bit ADC resolution, compact form factor, and dual-core 240 MHz architecture. A BLE mode was implemented for real-time wireless streaming and adopted as the definitive communication protocol. The firmware was developed in C++ within the Arduino framework.

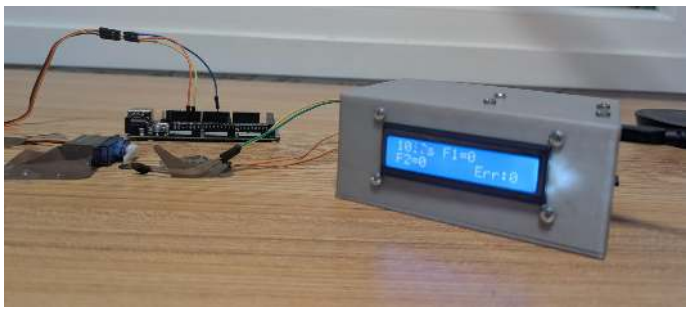
A calibration procedure was performed independently for each sensor using a micro servo motor (SG90, Tower Pro) as a controlled loading device, driven by an Arduino Mega 2560. The servo arm ($r = 15$ mm) was rigidly positioned above the sensor surface on a flat rigid base (Figure 7.7). The theoretical vertical force at each angular position was calculated from the nominal servo torque ($\tau = 1.8$ kg·cm $= 0.0177$ N·m) as

$$F = \frac{\tau}{r} \cdot \cos(\theta_{real}), \qquad \theta_{real} = 90 - \theta_{code} \tag{7.11}$$

The servo was swept in 5° steps while the ADC output was recorded simultaneously; only positions producing a non-zero ADC reading were retained.



(a) Top view of the calibration set-up



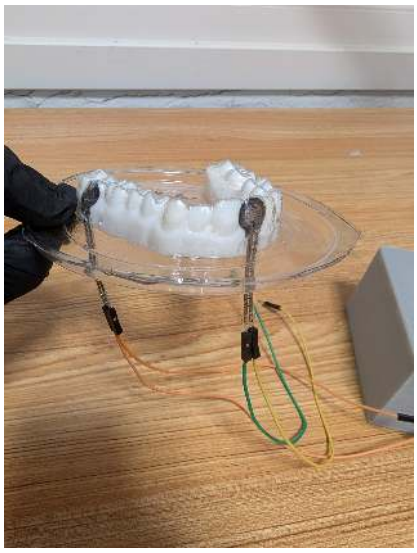
(b) lateral view of the complete calibration set-up

Figure 7.7: Calibration setup. The SG90 micro servo motor (blue, centre) is fixed above the FSR sensors using adhesive tape. Servo is controlled through an Arduino Mega 2560, while the ESP32-based acquisition unit (grey enclosure) simultaneously records the ADC output from each sensor. The LCD display shows elapsed time and real-time sensor readings.

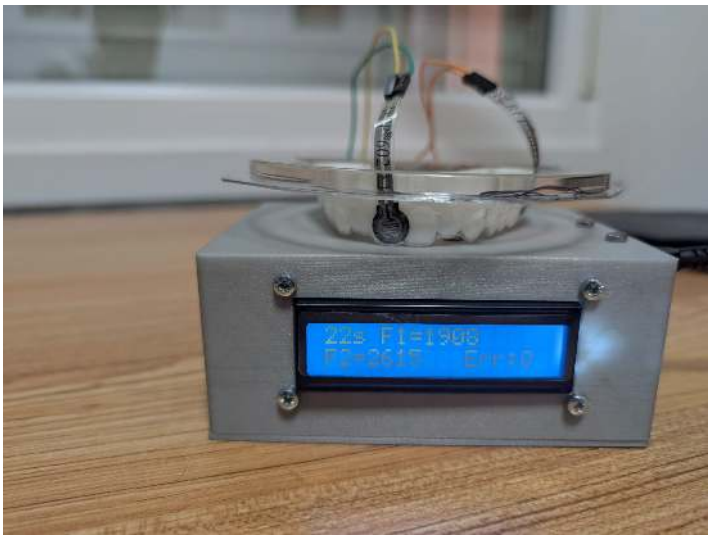
For the experimental measurements, a dental model provided by Secret Aligner was used, over which PETG aligners (Duran) were thermoformed. One sensor was placed on the buccal surface of the upper left central incisor and the second on the upper left second molar, enabling simultaneous anterior and posterior force monitoring. The complete circuit diagram and firmware source code are provided in Appendix B and openly available in the thesis repository (Available at https://github.com/Ozyman1998/Thesis/tree/main/force_sensor.

### 7.4.3. Results

Force measurements were conducted over a period of approximately 14.7 hours using a PETG Duran clear aligner thermoformed directly onto a 3D-printed model derived from an anonymized patient cast provided by Secret Aligner. No composite attachments were present on the model. Two FSR sensors were placed beneath the aligner at two tooth locations: F1 at the upper left central incisor and F2 at the upper left second molar. Measurements were taken at rest, with no masticatory activity simulated. The sensor placement and experimental set-up are illustrated in Figure 7.8.



(a) Two FSR sensors placed between the 3D-printed model and the clear aligner



(b) Complete acquisition setup during force measurement of a Duran (PETG) clear aligner

Figure 7.8: Experimental measurement setup.

A total of 10,586 valid data points were recorded at 5-second intervals, with no data corruption observed, demonstrating the reliability of the acquisition system over an extended period. Raw ADC readings were converted to force estimates in Newtons using the sensor-specific calibration functions described in Section 2. A Savitzky-Golay filter (window length = 15, polynomial order = 3) was applied to reduce high-frequency noise while preserving the underlying signal trend.

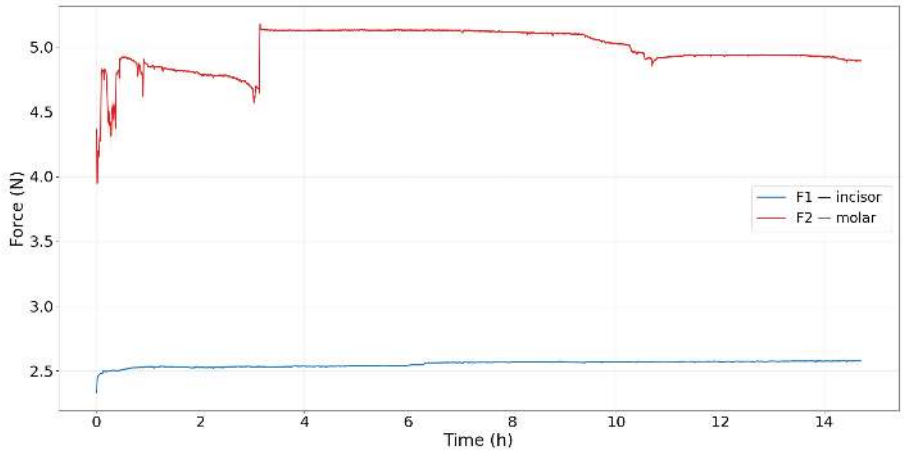



Figure 7.9: Force estimates from sensors F1 (incisor) and F2 (molar) over 14.7 hours. Duran (PETG) aligner at room temperature

As shown in the Figure 7.9, both sensors exhibited stable readings throughout the measurement period. F1 (central incisor) recorded a mean estimated force of approximately 2.55 N, while F2 (second molar) recorded a mean estimated force of approximately 4.98

N, consistently higher than F1 throughout the entire measurement period. These values are consistent with the clinically reported range of 1-10 N for thermoplastic clear aligners under passive loading conditions [65], [400], and are of the same order of magnitude as the contact forces predicted by the finite element model described in the previous chapter for a PETG aligner under identical activation conditions ($F_{\text{total}}$ = 2.094 N). The systematic difference between F1 and F2 suggests that the aligner exerts greater force at the molar region under resting conditions with this particular PETG material and no attachments present. A gradual drift was observed in both signals over time, which may reflect viscoelastic relaxation of the thermoplastic material as the aligner progressively adapts to the model geometry.

### 7.4.4. Discussion

This study presents a proof-of-concept for a low-cost, FSR-based force measurement system integrated into a clear aligner, demonstrating its feasibility for continuous data acquisition over extended periods. The system successfully recorded over 10,500 valid data points across approximately 14.7 hours without data corruption, validating both the hardware design and the data logging firmware developed for this purpose.

The results confirm that commercially available FSR sensors can be embedded beneath a PETG clear aligner and used to monitor relative force distribution between different tooth locations. The observed difference in normalized ADC values between the anterior (F1) and posterior (F2) sensors is consistent with the expected biomechanical behavior of thermoplastic aligners, which tend to exert greater forces in molar regions under passive loading conditions.

The spatial force distribution recorded by the FSR-based system is qualitatively consistent with the stress distributions predicted by the finite element model described in the previous chapter. In that model, von Mises stress concentrations in the full arch case were predominantly localized in the posterior sector, with the molar and premolar regions exhibiting systematically higher contact stresses than the anterior region under identical geometric activation. The experimental measurements reported here show an analogous pattern: the molar sensor (F2) recorded consistently higher force estimates than the incisor sensor (F1) throughout the entire 14.7-hour measurement period. Furthermore, the mean force estimated for F2 (4.98 N) is of the same order of magnitude as the total contact force predicted by the FEM for a PETG aligner ($F_{\text{total}}$ = 2.094 N), noting that the FEM value represents the total distributed contact force over the full arch while the FSR measures the local force at a single tooth location.

Although the calibration uncertainty of ±30-50% precludes a rigorous quantitative comparison, this order-of-magnitude agreement supports the physical plausibility of both the simulation results and the experimental measurements, and motivates the use of a more rigorously calibrated FSR system as a validation tool for the FEM framework in future work. Commercial FSRs are qualitative devices not originally designed for accurate

force measurement [429], and their force-resistance characteristic is known to be nonlinear, particularly at low force values and near saturation [429], [430]. The assumption of linearity adopted in the present calibration is therefore a simplification that contributes to the estimated uncertainty of ±30–50%.

Furthermore, the gradual signal drift observed in both sensors over time — indicative of viscoelastic stress relaxation in the PETG shell — corresponds directly to a known limitation of the FEM model, which captures only the instantaneous peak-force state upon aligner insertion and does not account for time-dependent material behavior.

### Comparison with existing force measurement approaches for clear aligners

Several approaches have been reported in the literature for measuring forces exerted by clear aligners, each presenting distinct advantages and limitations that are worth considering in the context of the system proposed here.

Orthodontic Measurement and Simulation Systems (OMSS) represent the most established methodology, offering high accuracy and the ability to resolve forces and moments in three dimensions. However, these systems are bulky, expensive, and require specialized operation, making them unsuitable for routine clinical use or prolonged continuous monitoring. A recent comparative study demonstrated the superior accuracy of OMSS over force-sensitive films [421], reinforcing their role as a laboratory reference standard rather than a practical measurement tool.

Force-sensitive films offer a simpler and more accessible alternative, providing spatial force distribution through colorimetric analysis when placed between the aligner and a 3D-printed model [99], [422]. Nevertheless, this approach is inherently limited to discrete, static measurements and cannot provide continuous temporal data, which is a significant constraint when studying force relaxation over the aligner wear cycle.

Electromagnetic-based sensors bonded directly to the aligner surface have also been reported, enabling force measurements on 3D-printed models [423]. While this approach holds potential for in-vivo use, no studies documenting its application in real patients have been identified in the literature to date.

FSR-based systems represent the approach most closely related to the present work. The majority of published studies employing FSRs in clear aligners rely on laboratory-fabricated silver ink sensors, which tend to be thick and introduce considerable measurement errors [76], [424], [426]. The system proposed by Xiang et al. [75] represents a notable exception, and the present work builds upon their approach by employing commercially available thin-film FSRs, which offer a significantly reduced profile and improved integration characteristics. Its primary current limitation, the absence of calibration to convert ADC readings into force units, will need to be addressed in future work before quantitative comparisons with published force values can be established.

### 7.4.5. Conclusions of in vitro measurements of forces

The following conclusions can be drawn from the work presented in this section:

A low-cost, FSR-based force measurement system for clear aligners was successfully designed, built, and validated as a proof of concept. The system integrates commercially available thin-film force sensing resistors with an ESP32 microcontroller and an SD card logging module, achieving continuous data acquisition at 5-second intervals over periods exceeding 14 hours without data corruption. The use of commercial thin-film FSRs represents a practical advancement over the laboratory-fabricated silver ink sensors reported in the literature, offering a reduced sensor profile and straightforward integration beneath a clear aligner.

Continuous force monitoring was demonstrated on a 3D-printed model using a PETG Duran aligner, showing a consistent difference between the anterior and posterior measurement locations, with the molar sensor recording higher values than the incisor sensor throughout, consistent with the expected biomechanical behaviour of thermoplastic aligners under passive loading and in qualitative agreement with the stress distributions predicted by the FEM framework. A gradual signal drift observed in both sensors is consistent with the viscoelastic relaxation of the thermoplastic as the aligner adapts to the model geometry.

The system remains subject to several limitations — a first-order servo-based calibration with an estimated uncertainty of ±30–50%, measurement at room temperature rather than under intraoral conditions, and validation on a single aligner and model ($n = 1$); which are addressed, together with the proposed calibration, environmental, and validation protocols, in the future work of Chapter 8. Nevertheless, the results confirm that the proposed approach is technically viable and provides a solid foundation for a calibrated, clinically applicable force monitoring system.

## 7.5. Life cycle assessment of clear aligners

### 7.5.1. Introduction to life cycle assessment of clear aligners

Clear aligner therapy generates a substantial volume of single-use plastic waste over the course of a complete treatment. Unlike many medical devices, clear aligners are classified Class IIa custom-made medical devices under Annex VIII, Rule 5 of Regulation (EU) 2017/745 [431]. As single-use devices, they are subject to Article 17 of the same regulation, which permits reprocessing only where expressly authorized by national law and only when demonstrably safe according to current scientific evidence [432], [433]. In practice, recycling through conventional channels is further precluded by the classification of used aligners as potentially contaminated medical waste under national healthcare waste legislation [434], which requires their disposal by incineration with energy recovery. However, most patients discard used aligners together with general household waste,

bypassing any controlled disposal pathway and substantially increasing the environmental impact of the treatment [125], [131], [435], [436].

The environmental burden of orthodontic practice has been recognized at the highest institutional level, with the World Federation of Orthodontists recently identifying material selection, recycling limitations, and waste management as priority areas for the profession [437]. The environmental burden of clear aligner therapy extends well beyond the plastic material of the devices themselves. The fabrication workflow involves several process steps with significant environmental impact that are frequently overlooked in life cycle assessments of this technique [438], [439]. The 3D-printed dental models used as thermoforming substrates are typically produced by digital light processing (DLP), which yields a hard, brittle thermoset that cannot be reprocessed or recycled [39], [435]. Although fused filament deposition (FFF) using recyclable thermoplastic filaments represents a more sustainable alternative for model production, its spatial resolution and throughput are insufficient for industrial clear aligner manufacturing. A further major source of environmental impact is the use of organic solvents - primarily isopropanol (IPA) - both for post-printing cleaning of the DLP molds and for disinfection of the thermoformed devices prior to delivery to the patient [39], [125].

Despite the clinical prevalence of clear aligner therapy, the published literature on its environmental footprint remains scarce. Existing life cycle assessment (LCA) studies have focused primarily on orthodontic retainers [439] or on directly 3D-printed aligners [438], and most have adopted simplified system boundaries that exclude key contributors such as DLP resin production, mold cleaning solvents, and packaging and logistics. Peluso et al. [440] conducted the most recent comparative review of the environmental footprint of traditional multibracket appliances and clear aligners, explicitly acknowledging that no prior study had quantified the full carbon footprint of either treatment modality under equivalent system boundaries. The present chapter addresses this gap by providing the first quantitative comparison of the carbon footprint of thermoformed clear aligner therapy versus conventional fixed appliance treatment using metallic brackets, both modeled under equivalent system boundaries including clinical visits, material fabrication, sterilization, and end-of-life waste management.

A related and equally underexplored question is the potential for reuse or repurposing of clear aligner thermoplastic waste. Used aligners are partially degraded and swollen by biological fluids during intraoral wear, rendering them unsuitable for reprocessing in a closed material loop. Multilayer thermoplastic constructs present additional challenges, as separating the constituent layers for individual recycling is neither economically nor environmentally feasible without prior chemical separation. Production scraps, by contrast, are not classified as medical materials and are therefore not subject to the same regulatory restrictions, representing a more tractable waste stream. Initiatives aimed at valorizing aligner waste have begun to emerge, including a programme at the Universidad Complutense de Madrid focused on converting used aligners into 3D printing filament [129], and partnerships between aligner manufacturers and specialist recycling companies

[125], [441].

The present chapter reports a life cycle assessment of a complete clear aligner treatment and, for the first time, a comparative LCA of conventional fixed appliance treatment with metallic brackets, both performed using OpenLCA software and the ELCD 3.2 database. The objective is to quantify the carbon footprint of each treatment modality across the full manufacturing and clinical workflow and to identify the process steps with the greatest environmental impact. Two disposal scenarios are evaluated for clear aligner therapy: an optimistic scenario in which all waste streams are incinerated with energy recovery, and a realistic scenario reflecting current disposal practices in southern Europe [442], [443]. The results provide a basis for identifying the most effective interventions to reduce the environmental burden of orthodontic treatment and for informing future sustainability strategies in the profession.

### 7.5.2. materials and methods

The life cycle assessment (LCA) of a complete clear aligner treatment was performed using OpenLCA (version 1.10, GreenDelta GmbH, Berlin, Germany), an open-source LCA software selected for its broad database compatibility, transparent modeling architecture, and high reproducibility of results [444]. The background database used was the European Life Cycle Database (ELCD) version 3.2, developed by the Joint Research Centre of the European Commission (JRC) [445]. ELCD 3.2 was selected because it was the only freely available database containing sufficient coverage of the materials and processes relevant to clear aligner manufacturing. Although more recent databases exist, most require a commercial license or lack the specific process data required for this analysis. ELCD 3.2 dates from October 2015 and represents the final stable release of the database, which was discontinued on 29 June 2018 [446].

#### System Boundary and Functional Unit

The functional unit for the clear aligner LCA was defined as one complete thermoformed clear aligner treatment comprising 60 devices. This number was selected as a conservative upper estimate of typical clinical practice: most treatments involve between 20 and 40 aligners, but complex cases may require 50 or more, and retainers and refinement aligners must also be accounted for. The system boundary includes material production and transport, device fabrication, 16 clinical visits [447], [448], and end-of-life waste management.

The functional unit for the fixed appliance LCA was defined as one complete orthodontic treatment with metallic brackets comprising 10 stainless steel archwires, 24 brackets, and the associated adhesive system, delivered over 24 clinical visits. The number of archwires is consistent with typical three-phase clinical protocols reported in the literature [8]. The system boundary includes component fabrication, sterilization and

packaging, transport from manufacturer to clinic, 24 clinical visits, and end-of-life waste management. Ligatures, bands, and auxiliary components were excluded from the inventory due to their negligible mass relative to the total system; this exclusion is declared as a limitation.

Each clinical visit was modeled with an electricity consumption of 0.3 kWh and generating 0.4 kg of mixed clinical waste (plastics, paper, packaging, gloves, and other medical disposables), modeled as incinerated in a dedicated clinical waste incineration process, following the approach used for dental procedure LCAs in the literature [449], [450].

Both LCAs share equivalent system boundaries with respect to clinical visits, material fabrication, and waste management, enabling a direct comparison of the two treatment modalities.

### Clear aligners - Material and Process Inventory

**Thermoplastic sheet** PETG and TPU were not available in the ELCD 3.2 database; PET was therefore used as a proxy material for the aligner sheets. The PET sheets were assumed to be produced and processed in China and shipped to Barcelona in 18 kg rolls, with a 10 % scrap rate in the extrusion process [435], [438]. Transport was modeled as ocean freight from Shanghai to Barcelona, followed by road transport to Madrid.

**3D printing resin** No suitable entry for DLP photopolymer resin was available in the database. Polymethyl methacrylate (PMMA) was selected as the closest available proxy for two reasons. First, both PMMA and commercial DLP dental resins belong to the methacrylate polymer family: DLP resins are typically formulated from multifunctional acrylate and methacrylate monomers and oligomers that undergo radical chain-growth photopolymerization [451], sharing the same backbone chemistry and similar carbon, hydrogen, and oxygen elemental composition as PMMA. Second, PMMA is the only acrylic polymer with a published and freely accessible LCA dataset and Environmental Product Declaration (EPD) in European databases [452], making it the most defensible proxy available within the ELCD 3.2 framework. It should be noted that this substitution introduces uncertainty in the absolute carbon footprint values associated with the resin synthesis step, as the energy intensity of photopolymerization differs from that of thermoplastic PMMA production; the resin contribution should therefore be interpreted as an order-of-magnitude estimate rather than a precise value.

The resin was assumed to be produced in China and transported to Madrid by the same route as the thermoplastic sheet, with a total quantity of 5 kg per treatment. Secret Aligner uses an industrial SLA printer capable of producing all models for a complete treatment in a single 45-minute print run at 2 kW, corresponding to an energy consumption of 1.5 kWh per treatment. A typical hollow dental model weighs approximately 15 g; accounting for approximately 30% material loss in printing supports and residual resin [451], a total of

1.35 kg of SLA resin is required to produce 60 models. Following printing, all models are cleaned with isopropanol (IPA) to remove uncured resin.

**Solvents** IPA was used both for post-printing cleaning of the DLP models and for disinfection of the thermoformed devices prior to patient delivery. A total of 1.33 kg of IPA (1.7 L) was included in the inventory, comprising 0.55 kg for post-printing cleaning and 0.78 kg for device disinfection, based on primary data from the manufacturing facility. This consumption reflects current industrial practice including solvent decantation and filtration between uses [453], which substantially reduces IPA consumption relative to the theoretical rate of approximately 10 models per litre recommended by resin manufacturers for fresh solvent. These values were provided by technical experts at Secret Aligner S.L. and reflect the company's current operational workflow.

The solvent was assumed to be manufactured in Barcelona and transported to Madrid by road. The analysis also included the PET container for the solvent, modeled from its constituent raw materials (propylene and water).

**Thermoforming and cutting.** A desktop thermoformer with a rated power of 1.5 kW and a cycle time of 2 minutes per device was modeled. After thermoforming, devices are laser cut to shape and disinfected with IPA. Finished aligners are individually packaged in polypropylene (PP) zip-lock boxes and collectively boxed in a cardboard outer carton.

A graphical representation of the complete manufacturing workflow of clear aligners is available in Figure 7.10

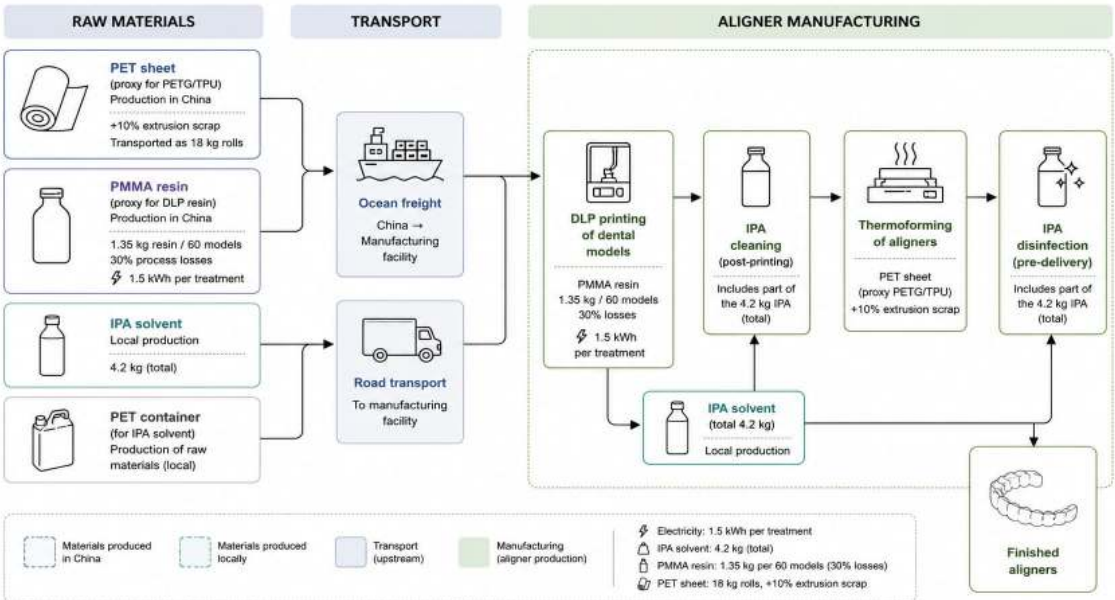



Figure 7.10: Complete clear aligner manufacturing workflow for LCA analysis. Figure generated with the assistance of ChatGPT (OpenAI, 2026).

**Clinical visits.** Sixteen clinical visits were modeled per treatment; the typical schedule for long-term clear aligner therapy according to Secret Aligner clinical experts [454], [455]. Each visit was modeled with 0.3 kWh electricity consumption and 0.4 kg of mixed clinical waste (plastics, paper, packaging, gloves, medical equipment,etc), incinerated in a dedicated clinical waste facility.

### Fixed Appliance - Material and Process Inventory

**Archwires.** Stainless steel archwires were modeled using primary process data. The fabrication of 3600 archwires requires 1 kg of hot-rolled steel coil and 2 kWh of electricity, generating 0.1 kg of metal scrap.

**Brackets.** The fabrication of 5000 stainless steel brackets requires 1 kg of hot-rolled steel coil and 1 kWh of electricity, generating 0.2 kg of metal scrap.

**Adhesive system.** The bracket bonding adhesive was modeled using an epoxy sealer as a proxy for the resin matrix and polypropylene granulate for the syringe body. Each syringe contains 3 g of adhesive; one syringe was modeled per treatment. Transport from the manufacturer to Madrid was included.

### Sterilization and packaging.

**Sterilization pouches.** Components are assembled into sterilization pouches and sterilized by ethylene oxide, with an energy consumption of 0.1 kWh per pouch. Two pouch types were modeled. The **initial pouch** (one per treatment) contains 24 stainless steel brackets, 1 adhesive syringe, and 2 archwires (upper and lower arches), and is used during the bonding appointment. The **follow-up pouch** (23 per treatment) contains 2 archwires only, one for each arch, and is used at each subsequent clinical visit. All pouches are transported from the Chinese manufacturer to Madrid by ocean freight, followed by road transport.

The complete manufacturing workflow of metallic fixed appliances is available in a resumed graphical manner in Figure 7.11

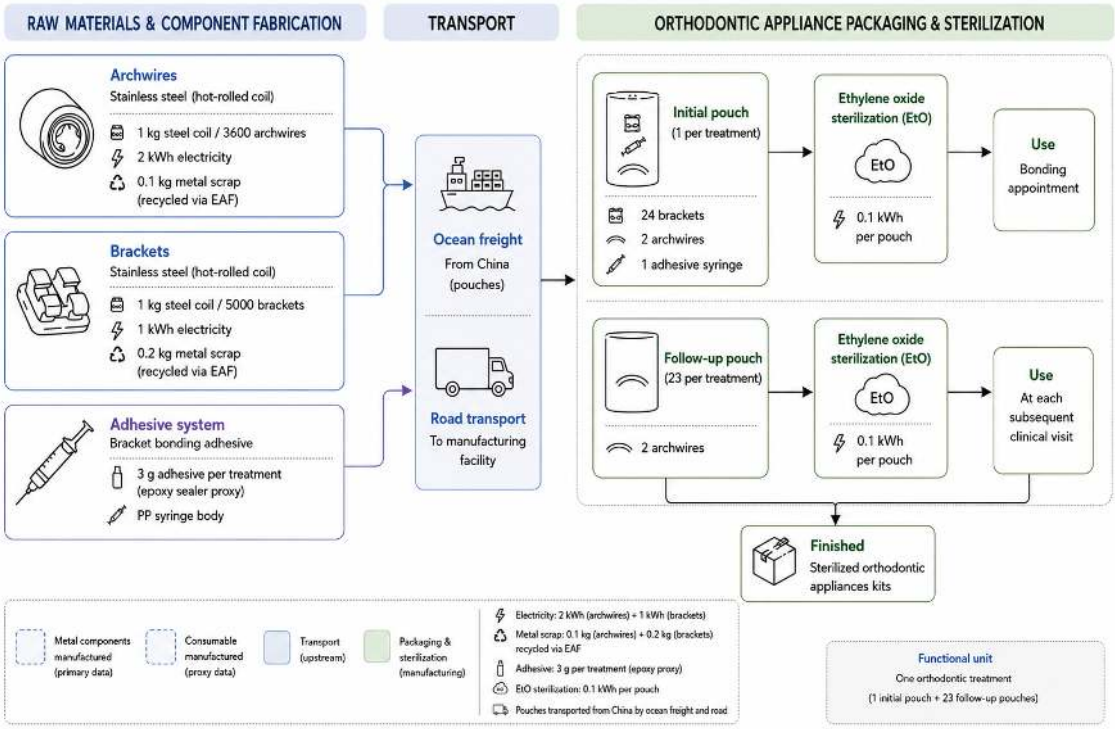



Figure 7.11: Complete fixed metallic appliances manufacturing workflow. Figure generated with the assistance of ChatGPT (OpenAI, 2026).

**Clinical visits.** Twenty-four clinical visits were modeled per treatment, consistent with typical three-phase fixed appliance protocols [8] (this number was also recommended by experts at Secret Aligner). Each visit consumes one sterilized pouch, 0.3 kWh of electricity, and 1 kg of water, and generates 0.4 kg of mixed clinical waste (plastics, paper, packaging materials, gloves, and other medical disposable equipment) and 0.002 kg of metal scrap from used archwires. Clinical waste was modeled as incinerated in a dedicated clinical waste incineration process without energy recovery credit, consistent with the management of hazardous healthcare waste in Spain [434]. Metal scrap from used archwires was modeled as recycled via the electric arc furnace route.

All 24 visits were assumed to include archwire replacement in both arches, yielding a total of 48 archwires per treatment. This represents a conservative upper estimate of material consumption, as in clinical practice not all visits involve archwire changes; the carbon footprint associated with metallic components should therefore be interpreted as an upper bound [8].

### Waste and End-of-Life Scenarios

Two end-of-life scenarios were evaluated to bound the range of plausible environmental outcomes.

In the **optimistic scenario**, all waste streams - production thermoplastic scraps, spent

solvents, cured resin molds, used aligners, and packaging materials - are modeled as incinerated with energy recovery, representing the legally mandated disposal route for medical waste in the EU [434].

In the **realistic scenario**, the waste streams are modeled according to current disposal practices in southern Europe [442], [443]. Used aligners are discarded as conventional household plastic waste rather than routed to medical waste incineration, consistent with the observation that patients typically discard aligners at home rather than returning them to the clinic. The incineration of aligner materials alongside general household waste releases up to 20 distinct potentially harmful compounds per material [436],providing further justification for modeling this stream as uncontrolled disposal rather than energy recovery.

Cured DLP resin molds, being thermoset materials that cannot be mechanically recycled, are modeled as landfilled. Thermoplastic production scraps, which are not classified as medical waste and are theoretically recyclable, are also modeled as landfilled in this scenario, reflecting actual industrial practice [125], [435]. The only waste stream retaining energy recovery treatment in the realistic scenario is the spent solvent, which is classified as hazardous chemical waste and processed accordingly.

For the **fixed appliance scenario**, clinical waste from dental visits is modeled as incinerated in a specialized clinical waste facility with energy recovery, consistent with the management of hazardous healthcare waste in Spain [434]. Metal scraps from fabrication and used archwires are modeled as exiting the system as a recovered steel waste elementary flow, consistent with the ELCD 3.2 system expansion approach for recyclable materials. Neither the energy consumption of the recycling process nor the avoided burden credit associated with secondary steel production was included in the inventory, as no suitable process was available in the ELCD 3.2 database. This exclusion is conservative with respect to the fixed appliance carbon footprint, as including the recycling credit would reduce the total result. However, given the small total mass of metal recycled per treatment the associated impact and credit are estimated to be negligible relative to the total system footprint, with a maximum contribution of less than 0.1% of the total result regardless of the recycling scenario assumed.

### 7.5.3. LCA Results

#### Clear Aligner Therapy

The total carbon footprint of a complete clear aligner treatment, including 16 clinical visits, amounts to 14.27 kg of $CO_2$ equivalent per treatment - approximately twice the carbon footprint of manufacturing a single cotton t-shirt (6.5 kg $CO_2$ eq) [456],and slightly lower than other common dental procedures: approximately 1.6 times lower than a single endodontic treatment (23.3 kg $CO_2$ eq), 2.7 times lower than a dental crown (35-44 kg $CO_2$ eq), and 4.5 times lower than a full dental prosthesis (60-70 kg $CO_2$ eq) [449]. Within

orthodontics specifically, it is 1.9 times lower than a complete fixed appliance treatment with metallic brackets (27.66 kg $CO_2$ eq), as quantified in the present study.

The total carbon footprint of the clear aligner manufacturing workflow alone-excluding clinical visits and energy recovery credits from waste incineration-amounts to 0.49 kg of $CO_2$ equivalent per treatment (3.4 % of total).

Given that dental visits account for 96.6 % of the total treatment carbon footprint, visualising all processes in a single diagram would obscure the relative contributions of the manufacturing steps. Furthermore, clinical visits are common to both clear aligner therapy and fixed appliance treatment - the number and duration of appointments being determined primarily by clinical protocols rather than by the device manufacturing process. For these reasons, the following Figures 7.12and 7.13 present the carbon footprint breakdown of the *manufacturing processes only* (0.49 kg $CO_2$ eq), excluding dental visits. This approach enables identification of optimisation opportunities within the production workflow itself. The dominant contribution of dental visits to the overall footprint is nonetheless reflected in the quantitative results and discussion presented above.

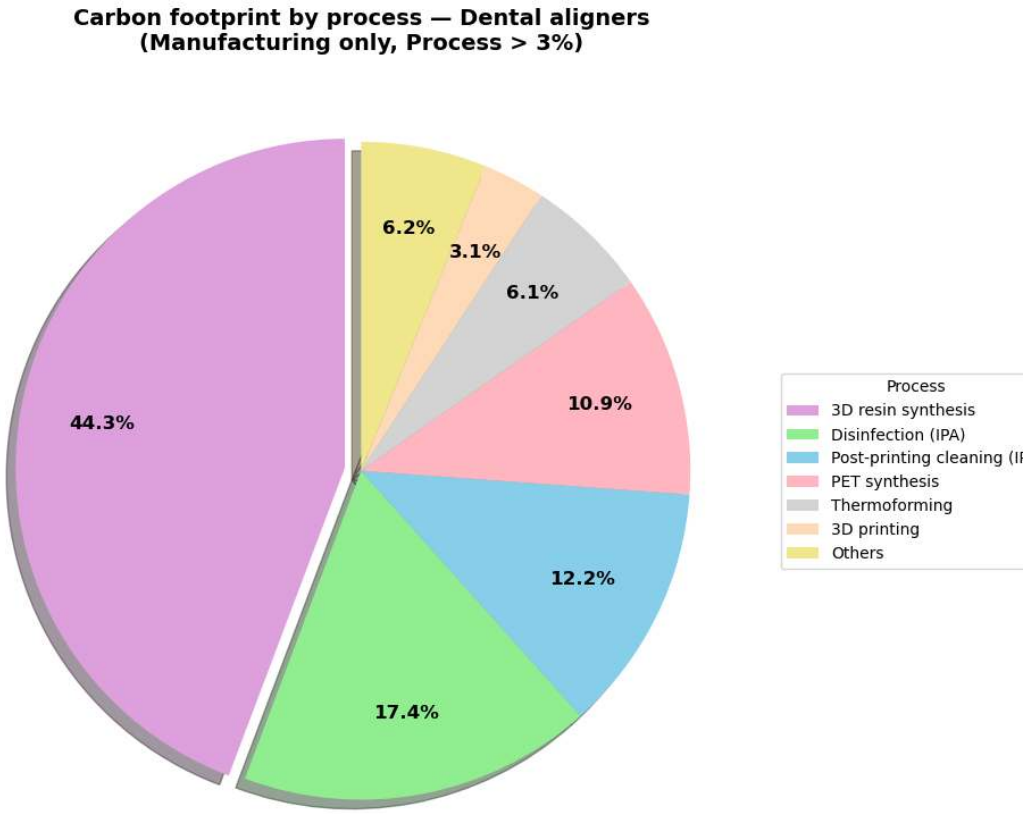



Figure 7.12: Main processes contributing to the carbon footprint of clear aligners without taking into account dental visits

Figure 7.12 shows the relative contribution of each manufacturing process. Within

the manufacturing workflow, DLP resin synthesis is the largest contributor at 44.3%, followed by device disinfection with IPA (17.4%) and post-printing mold cleaning (12.2%). Together, the two solvent-related processes account for 23.8% of the manufacturing footprint. PET synthesis contributes 10.9%, thermoforming 6.1%, and 3D printing energy consumption 3.1%. All remaining processes individually contribute less than 3% and are grouped under *Others*, with their detailed breakdown shown in Figure 7.13. Transport processes contribute less than 1% of the manufacturing footprint. A complete graphical diagram of the manufacturing workflow is available in Figure 7.10.

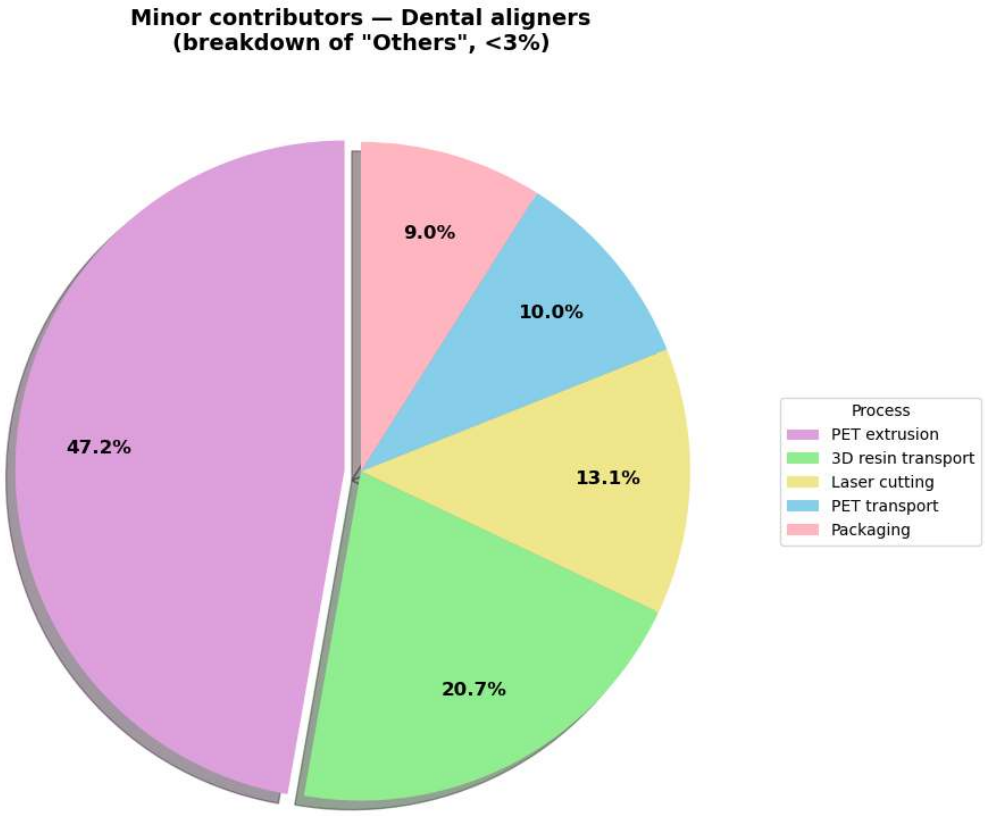



Figure 7.13: Minor processes contributing to the carbon footprint of clear aligners without taking into account dental visits

Table 7.4 quantifies the reduction in equivalent $CO_2$ emissions achievable by incinerating each waste stream with energy recovery rather than disposing of it as general waste, expressed both in absolute terms and as a percentage of the total treatment footprint (14.27 kg $CO_2$ eq.). The most impactful intervention is the proper management of clinical waste from the 16 dental visits, which alone would reduce the total treatment footprint by 14.8%, followed by 3D-printed molds (2.1%) and device disinfection solvent (1.4%). Together, controlled disposal of all waste streams would reduce the total treatment footprint by approximately 20 %. The two end of life scenarios are sketched grphically in Figure 7.14

| Waste stream | kg $CO_2$ eq. | Reduction (%) |
|---|---|---|
| Clinical waste (16 visits) | 2.11 | 14.8 |
| 3D-printed molds | 0.30 | 2.1 |
| Device disinfection solvents | 0.20 | 1.4 |
| Post-printing solvents | 0.13 | 0.9 |
| PET scraps (cutting + extrusion) | 0.08 | 0.6 |
| **Total if all optimised** | **2.82** | **19.8** |

Table 7.4: Reduction in equivalent $CO_2$ emissions achievable by incinerating each waste stream with energy recovery rather than landfilling it.

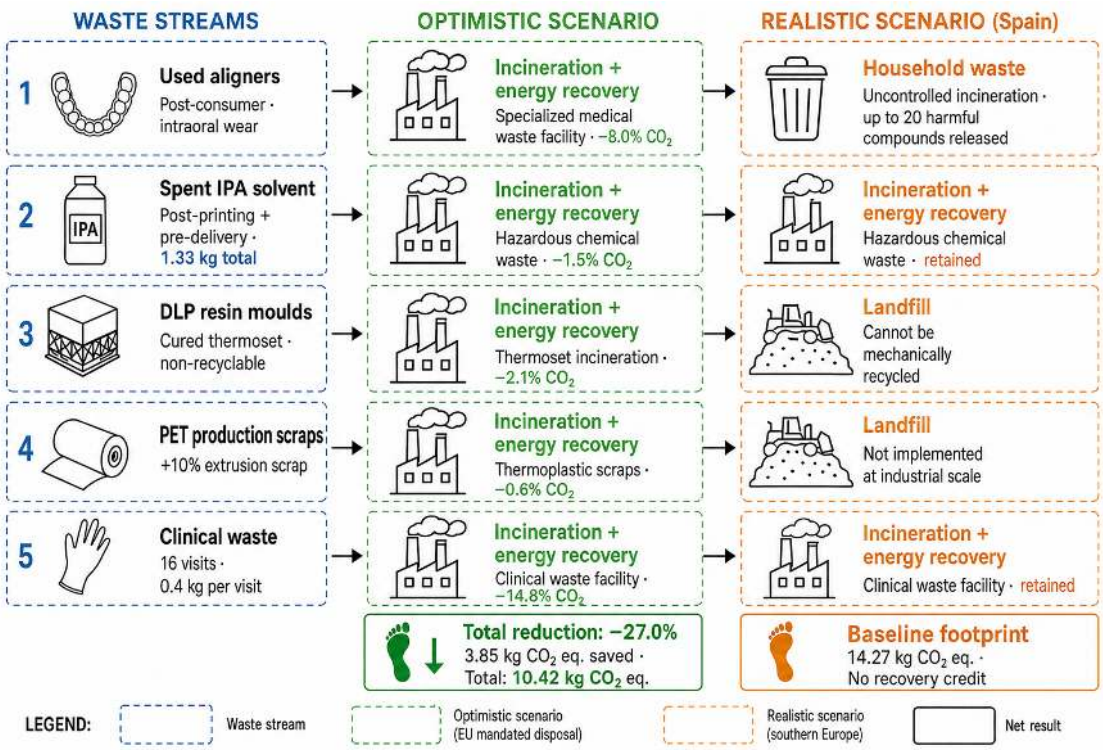



Figure 7.14: End-of-life scenarios for clear aligner therapy.Figure generated with the assistance of ChatGPT (OpenAI, 2026).

### Treatment with Metallic fixed appliances

The total carbon footprint of a complete fixed appliance treatment with metallic brackets amounts to 27.66 kg of $CO_2$ equivalent per treatment, including 24 clinical visits. The Figure 7.15 shows the relative contribution of each process.

As with clear aligner therapy, clinical visits dominate the total carbon footprint (79.1%), making direct visualisation of manufacturing process contributions difficult. Since the number and duration of appointments are determined by clinical protocols common to orthodontic practice rather than by device-specific factors, the following figures present

the carbon footprint breakdown of the *manufacturing and clinical consumables processes only* (5.78 kg $CO_2$ eq), excluding the electricity consumption of dental visits themselves. This enables direct comparison with the clear aligner manufacturing workflow presented above.

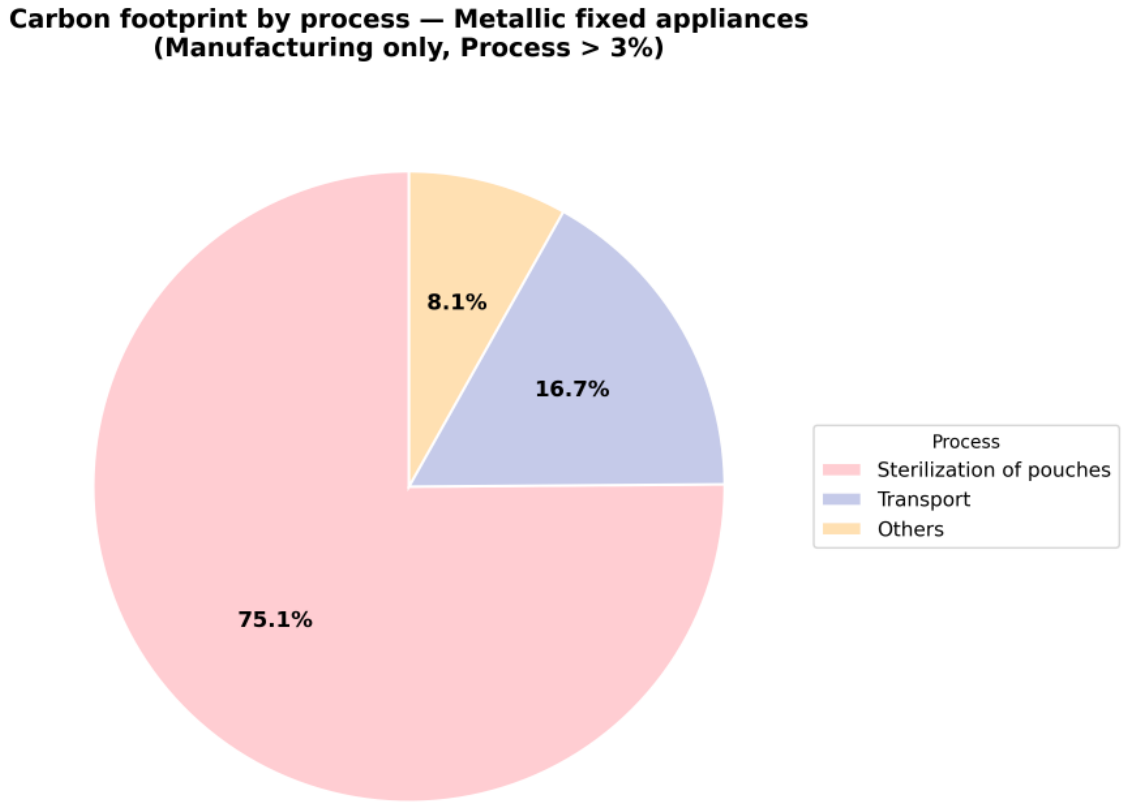



Figure 7.15: Main processes contributing to the carbon footprint of a complete metallic fixed appliances treatment without taking into account dental visits

Electricity consumed across all 24 clinical visits is by far the dominant contributor, accounting for 79.1 % of the total process footprint, reflecting the high number of appointments required for fixed appliance treatment. Sterilization of pouches constitutes the second largest contribution at 15.7 %, followed by transport at 3.5%. All remaining processes - including archwire and bracket fabrication and adhesive production - each contribute less than 2% individually and are shown in the figure 7.16. Notably, the metallic components themselves (archwires and brackets) together account for less than 0.6% of the total process footprint, confirming that the environmental burden of fixed appliance treatment is dominated by clinical management rather than by material production.

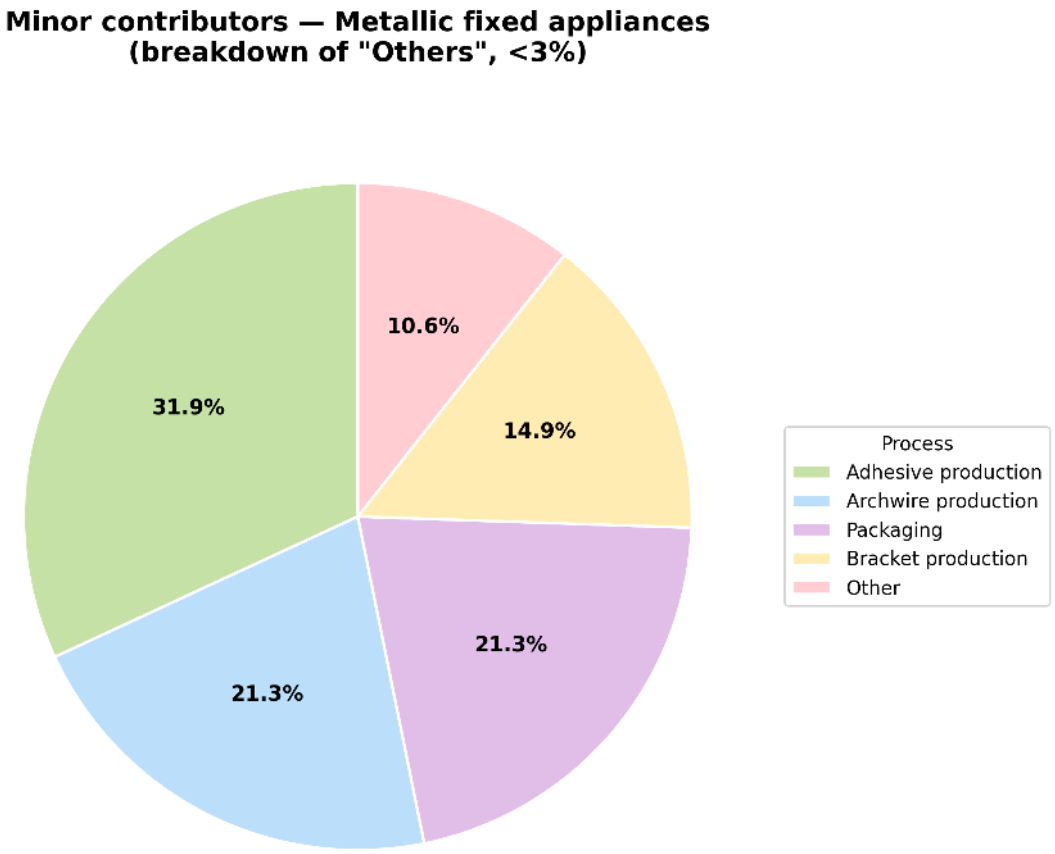



Figure 7.16: Minor processes contributing to the carbon footprint of metallic fixed appliance treatment without taking into account dental visits

Unlike clear aligner therapy, fixed appliance treatment offers very limited opportunities for energy recovery or recycling from waste streams in order to lower the environmental footprint of the process. The metallic components—archwires and brackets—exit the system as recovered steel waste and are modeled as recycled; however, given the small total mass of metal per treatment (approximately 0.016 kg), the associated environmental credit is negligible. The only waste stream with meaningful energy recovery potential is the mixed clinical waste generated at each dental visit—comprising plastics, paper, and packaging—which is modeled as incinerated with energy recovery in a specialized clinical waste facility [434]. This contrasts markedly with clear aligner therapy, where multiple waste streams—spent solvents, thermoplastic scraps, used aligners, and DLP molds—each represent independent opportunities for energy recovery or material recycling. The absence of solvent streams and the small mass of recyclable material in fixed appliance treatment mean that process-level optimization has a much more limited scope than in clear aligner therapy, and that reducing the number of clinical visits is the primary—and effectively the only—lever available to reduce the environmental footprint of fixed appliance treatment. The end-of-life treatment of waste streams from fixed appliance therapy is summarised in Figure 7.17.

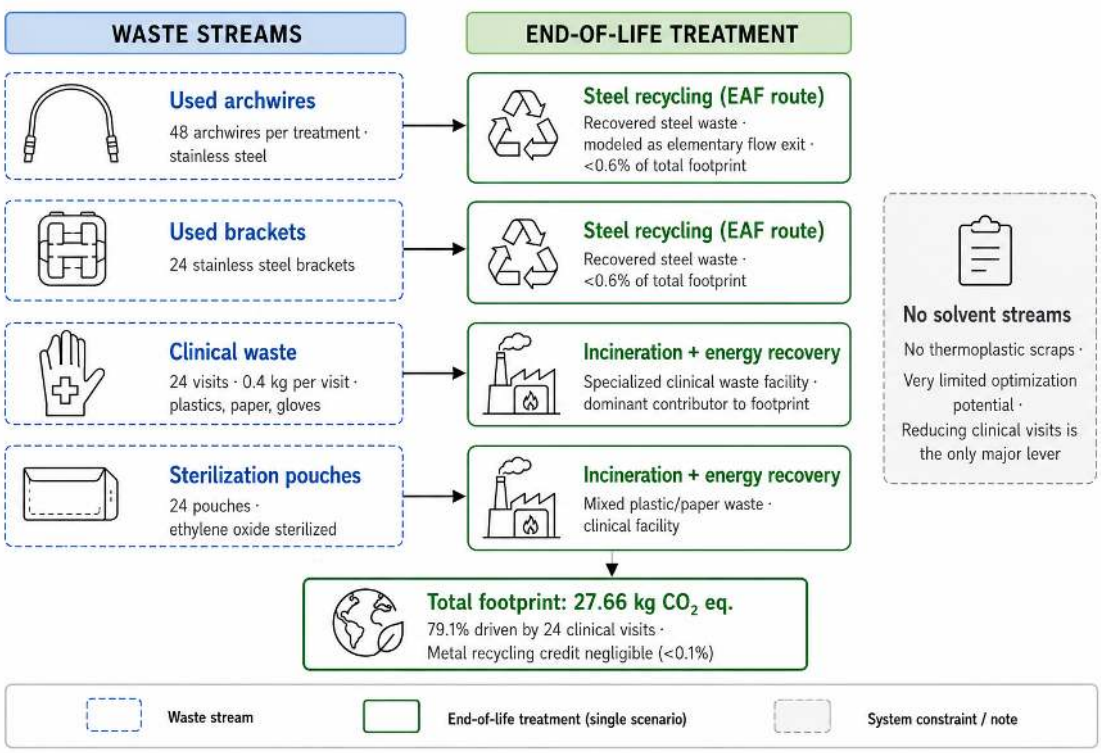



Figure 7.17: End-of-life treatment of waste streams from fixed appliance therapy with metallic brackets. Figure generated with the assistance of ChatGPT (OpenAI, 2026).

### Comparative Results

The table 7.5 summarizes the carbon footprint of both treatment modalities under equivalent system boundaries.

| Treatment modality | kg $CO_2$ eq. | Clinical visits |
|---|---|---|
| Clear aligner | 14.27 | 16 |
| Metallic fixed appliances | 27.66 | 24 |

Table 7.5: Carbon footprint of clear aligner therapy and metallic fixed appliances treatment.

Under equivalent system boundaries, including clinical visits and end-of-life waste management, fixed appliance treatment generates approximately 1.9 times more $CO_2$ equivalent than clear aligner therapy (27.66 versus 14.27 kg $CO_2$ eq). This difference is driven by the higher number of clinical visits required (24 versus 16) and the associated clinical waste streams, rather than by material production—metallic components (archwires and brackets) contribute less than 0.6 % of the total fixed appliance footprint [437], [440], [450].

### 7.5.4. LCA discussion

#### Contextualisation of results

The figure 7.18 places the carbon footprint of both treatment modalities in the context of everyday products and common dental procedures for a better understanding of the values.

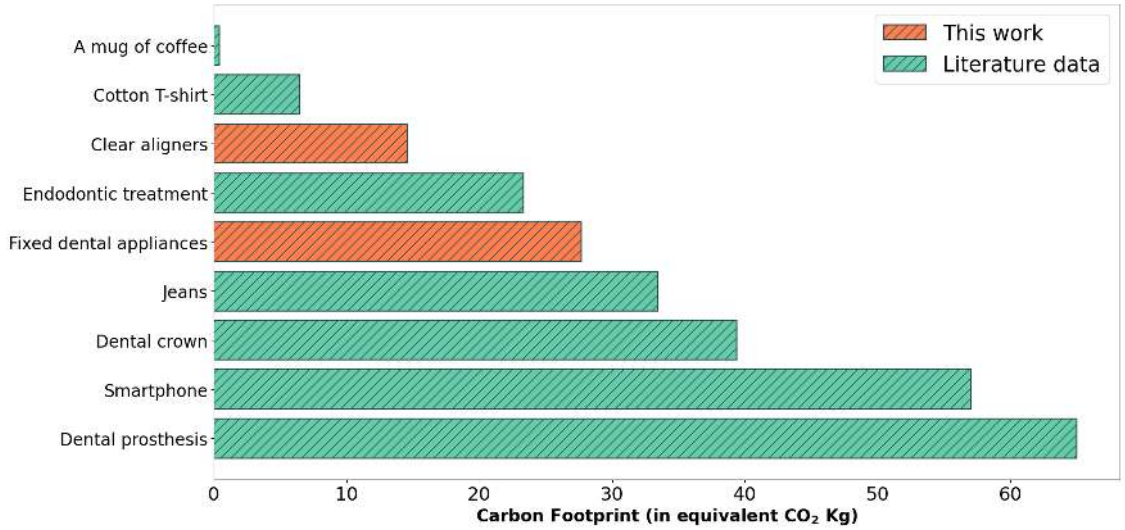



Figure 7.18: Comparative carbon footprint of a complete clear aligner treatment and a complete fixed appliance treatment with metallic brackets (both from the present study) against common everyday products[456], [457], [458], [459] and dental procedures[449].

The figure 7.18 illustrates that a complete clear aligner treatment (14.27 kg $CO_2$ eq.) has a carbon footprint comparable to that of twice times of a single cotton t-shirt and is considerably lower than other common dental procedures such as dental crowns (35-44 kg $CO_2$ eq.), or dental prostheses (60-70 kg $CO_2$ eq.) [449]. Fixed appliance treatment with metallic brackets (27.66 kg $CO_2$ eq.) falls in the same range as endodontic treatment, despite involving a fundamentally different material system, reinforcing the finding that it is clinical management rather than material selection that determines the environmental burden of orthodontic treatment in general [437], [450]. For reference, both orthodontic modalities have a lower footprint than the fabrication of a smartphone (approximately 57 kg $CO_2$ eq.) [458] or a pair of jeans (approximately 33 kg $CO_2$ eq.) [457], placing orthodontic treatment within the lower end of the spectrum of manufactured consumer goods in terms of carbon intensity.

#### Dominant process contributions and solvent management

When the complete clear aligner treatment footprint is considered - including 16 clinical visits - dental appointments emerge as the overwhelmingly dominant contributor, accounting for 96.6% of the total process footprint. This result is consistent with the broader

dental LCA literature, which has consistently identified clinical management - including energy consumption, sterilisation, and waste generation per visit - as the primary driver of the environmental burden of dental procedures [449], [450]. Reducing the number of clinical appointments through remote monitoring or optimised recall intervals therefore represents the single most impactful sustainability intervention available to orthodontic practitioners, regardless of the treatment modality chosen.

Within the manufacturing workflow (0.49 kg $CO_2$ eq., 3.4% of total), DLP resin synthesis emerges as the dominant contributor, accounting for 44.3% of the manufacturing footprint, a finding that reflects the high carbon intensity of methacrylate-based photopolymer production. Solvent-related processes remain significant but secondary: device disinfection with IPA contributes 17.4% and post-printing mold cleaning a further 12.2%, together representing 29.6% of the manufacturing burden. The present inventory is based on industrial solvent management practices that include decantation and filtration between uses, as implemented at the collaborating manufacturing facility; facilities that do not employ such recovery measures would exhibit substantially higher solvent contributions.

PET synthesis accounts for 10.9% of the manufacturing footprint, thermoforming 6.1%, and 3D printing energy consumption 3.1%. All remaining processes (including transport, laser cutting, and packaging) individually contribute less than 3% and together account for approximately 6% of the manufacturing burden.

Although clear aligner therapy generates a lower carbon footprint than fixed appliance treatment (14.27 vs 27.66 kg $CO_2$ eq.), the potential for further footprint reduction is considerably greater for clear aligners. The fixed appliance footprint is dominated by visit-related energy consumption; inherent to the treatment protocol and difficult to reduce without compromising clinical outcomes. The aligner footprint, by contrast, is driven by clinical visits (96.6%) and, within manufacturing, by resin synthesis and solvent use - both amenable to process-level improvements through the adoption of bio-based resins [451] and solvent recovery systems; and its end-of-life waste streams offer tractable recycling opportunities absent in fixed appliance treatment.

### Transport and supply chain

All transport processes contribute less than 1% of the total footprint, indicating that relocating material sourcing or manufacturing within Spain or Europe would yield negligible environmental benefit. This result is consistent with the general finding in LCA studies that transport is rarely a dominant contributor to the footprint of manufactured goods relative to material production and processing [438]. However, in-house production of thermoplastic sheet by manufacturers such as Secret Aligner would reduce supply chain dependency and lead times, and improve material traceability and quality, even if the carbon footprint reduction under the present model would be modest.

### End-of-life scenarios and disposal infrastructure

The disposal scenario has a decisive influence on the overall carbon footprint. Under the optimistic scenario, in which all waste streams are incinerated with energy recovery in specialized facilities, the manufacturing process would be rendered net carbon-negative. However, incineration must be carried out in specialized facilities to generate a positive energetic output [460] - facilities that remain scarce in southern European countries such as Spain due to economic and socio-political barriers [461]. In 2022, Spain landfilled 62% of its plastic waste, recycled 19%, and incinerated only 12% [442], [443], which is substantially below the EU average. The European Union as a whole exports 1.3 million tons of plastic waste annually to third countries for treatment under less restrictive conditions [462].

Although EU MDR Article 17 and national healthcare waste legislation require used aligners to be disposed of by incineration with energy recovery [432], [434], in practice, most patients discard used aligners with general household waste [125], [435], [436]. The uncontrolled incineration of PETG and polyurethane aligner materials alongside general household waste has been shown to release up to 20 distinct potentially harmful compounds per material [436], underscoring the environmental and public health case for controlled disposal pathways.

More controlled waste streams - cutting scraps and DLP molds - are technically amenable to proper treatment: thermoplastic scraps can be recycled or incinerated with energy recovery, and while thermoset molds cannot be mechanically recycled, incineration is technically feasible [451], [463], [464]. Neither route is currently implemented in practice at scale.

### Material reuse and scrap recovery

Production thermoplastic scraps represent the most tractable waste stream for recycling, as they are not classified as medical devices and therefore not subject to the regulatory restrictions governing post-consumer device disposal [431]. Their recovery could reduce the manufacturing carbon footprint by approximately one-third. In the case of multilayer thermoplastics, recovery of individual constituent polymers through chemical separation has been demonstrated in the preceding chapters of this thesis (Chapter 4), opening a pathway toward closed-loop material recovery. However, blending multilayer scraps without prior separation would produce PET/TPU blends with a TPU fraction below 50%- a threshold below which the blend becomes too stiff and brittle for most commercial applications [465], [466]. Separated fractions could nonetheless be repurposed for non-medical applications such as packaging.

The recycling of used aligners, while legally permissible in Spain under specific conditions [433] and technically possible under ideal conditions, as some initiatives have demonstrated [129], has significant burdens to be actually implemented. The biggest

problems are the high cost of a certified sterilization procedure and the opacity of aligner manufacturers regarding material composition. The initiatives found based their studies on a fully sorted clear aligner source, specifically aligners manufactured by the same company.

### Biodegradable and biobased alternatives

Biodegradation and composting were not modeled in the present LCA as aligners are not currently manufactured from biodegradable materials. The main candidates under investigation—PLA, PCL, and biobased PEF—do not yet meet the mechanical, optical, and durability requirements for clinical use. PLA-based materials are generally stiff, fragile, and opaque [467], [468]; PCL presents issues of high cost, low mechanical properties, rapid intraoral degradation, and a low melting point of 60 ºC [469], [470]; and PEF, while biobased with properties similar to PET, remains at an early stage of development with limited commercial availability [471].

### Comparison with published LCA studies and methodological novelty

Published quantitative life cycle assessments of orthodontic thermoplastic devices are scarce. Caelli et al. [438] performed the most methodologically comparable study, conducting a comparative LCA of thermoformed versus directly 3D-printed aligners using primary data from an Italian manufacturer. Their analysis confirmed that resin synthesis and post-printing cleaning are significant contributors regardless of manufacturing route - consistent with the present findings - but did not model solvent management in detail, end-of-life disposal scenarios, or production scrap streams. Da Tan et al. [439] assessed orthodontic retainers but excluded DLP model production, solvent use, and post-manufacturing waste streams that dominate the present analysis. Peluso et al. [440] conducted the most recent comparative review of the environmental footprint of traditional multibracket appliances and clear aligners, explicitly acknowledging that no prior study had quantified the full carbon footprint of either treatment modality under equivalent system boundaries.

Several narrative reviews have addressed the environmental burden of CAT qualitatively, reinforcing the need for better control of aligner disposal [107], [125], [436]. The World Federation of Orthodontists (WFO) recently published a white paper on sustainable orthodontic practices [437] identifying material selection, recycling limitations, and waste management as priorities.

The present study differs from all prior work in four respects. First, it models the complete thermoformed aligner workflow, including DLP resin production, IPA solvent use for both mold cleaning and device disinfection, laser cutting, and packaging—process steps absent from prior analyzes. Second, it evaluates two end-of-life scenarios explicitly calibrated to current disposal infrastructure in southern Europe. Third, it identifies

production thermoplastic scraps as a regulatory and technically tractable recycling opportunity distinct from post-consumer device waste, connecting this finding to the chemical separation of multilayer materials demonstrated experimentally in the preceding chapters of this thesis. Fourth, and most importantly, it provides the first quantitative comparison of the carbon footprint of thermoformed clear aligner therapy versus fixed appliance treatment with metallic brackets under equivalent system boundaries, demonstrating that clinical visit frequency—rather than material selection—is the primary determinant of the environmental burden of orthodontic treatment.

### 7.5.5. LCA Conclusions

The present study provides the first quantitative comparison of the carbon footprint of thermoformed clear aligner therapy and fixed appliance treatment with metallic brackets under equivalent system boundaries, the latter generating approximately twice the emissions of the former — a difference driven primarily by the number of clinical appointments rather than by material production.

In both modalities, clinical visits emerge as the dominant contributor to the total footprint, consistent with the broader dental LCA literature. Reducing clinical visit frequency through remote monitoring where clinically appropriate is therefore the single most impactful sustainability intervention available to orthodontic practitioners; technologies such as Dental Monitoring have demonstrated the potential to reduce visits by up to 50% without compromising outcomes, which would correspondingly reduce the treatment footprint.

Within the clear aligner manufacturing workflow, DLP resin synthesis and solvent use (for mold cleaning and device disinfection) together dominate the process-level footprint. The development of lower-carbon photopolymer resins and the implementation of solvent recovery therefore represent the most effective manufacturing-level interventions. All transport processes contribute less than 1% of total emissions, indicating that relocating material sourcing within Europe would yield negligible environmental benefit and should not be prioritized over process-level improvements.

The disposal scenario has a decisive influence on the overall footprint: controlled incineration with energy recovery could reduce it substantially, but the underdeveloped incineration infrastructure in southern Europe makes the realistic scenario — in which used aligners are discarded as household waste and thermoset molds are landfilled; more representative of current practice. Production thermoplastic scraps represent a particularly tractable opportunity, as they are not classified as medical devices and constitute a well-sorted, uncontaminated feedstock; in the case of multilayer materials, the chemical separation of constituent polymers demonstrated in Chapter 4 opens a pathway toward closed-loop recovery. Biodegradable alternatives remain at an early stage and do not yet meet the mechanical, optical, and durability requirements of clinical use.

In summary, the most impactful measures to reduce the environmental footprint of clear aligner treatment are, in order of priority:

1. Reduction of clinical visit frequency through remote monitoring technologies, which can reduce visits by up to 50% and correspondingly halve the treatment carbon footprint.
2. Development and adoption of lower-carbon photopolymer resins.
3. Implementation of solvent recovery and regeneration systems.
4. Recovery and recycling of production thermoplastic scraps, particularly through chemical separation of multilayer materials as demonstrated in this thesis.
5. Establishment of take-back programmes for the proper disposal of used devices.
6. Development of biocompatible biodegradable or biobased polymers and in-house material processing to reduce supply chain dependency.

All improvement measures are summarized in Figure 7.19.

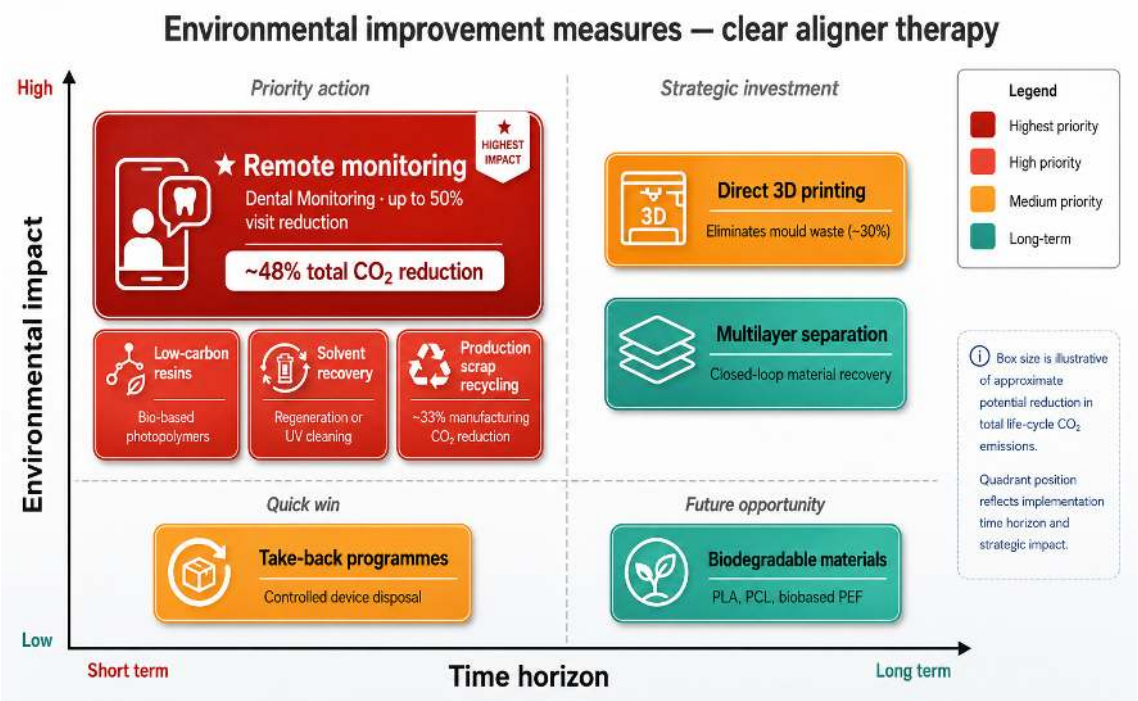



Figure 7.19: LCA improvement matrix for clear aligners. Figure generated with the assistance of ChatGPT (OpenAI, 2026)

These recommendations are consistent with the sustainability priorities identified in the World Federation of Orthodontists white paper [437] and represent actionable measures within the current regulatory and technological framework.

## 7.6. Economical feasibility of clear aligner production

This chapter assesses the economic feasibility of in-house multilayer aligner sheet production as an alternative to commercial procurement. First, the main candidate manufacturing processes are reviewed, and the rationale for the selected approach is presented. Subsequently, a cost analysis is conducted comparing in-house production costs against current market prices, and the investment recovery period is estimated based on Secret Aligner's current production volume.

### 7.6.1. Manufacturing process selection

As demonstrated, planar extrusion and thermal joining seem to produce good results; hence, this will be the focus method. Here we present a list of other methods we have studied, tried, or discarded, along with the reasons why they are not feasible. All the methods are graphically depicted in the figure 7.20.

- **SLA or DLP 3D printing**: 3D printing resins for commercially available clear aligners currently seem to have high cytotoxic behavior and to not reach the required thermophysical properties for a good clear aligner[39], [472]. In addition, tailoring the properties for each patient would be impossible, since it would mean, in some cases, reducing the polymerization time, which can lead to a great increase in cytotoxic and less controllable properties. The literature on 3D medical resins for SLA or DLP is highly controversial on the biological properties of these resins, mainly due to the effect of polymerization time on the biological and physical properties of this material. Nevertheless we tried to manufacture devices by SLA 3D printing just by mixing a TPU based resin with a PC based resin. The results show that the properties are actually a mixture of both materials but still far from the required properties for clear aligners. We envisioned other methods in order to manufacture a multilayer system, such as producing only the inner layer by 3D printing, and introducing it in a bath of outer layer, and finally polymerizing the outer layer; or printing both layers by separate and joining them by laser cutting the edges.

- **Sequential thermoforming**:Sequential thermoforming refers to either thermoform each layer individually in different and sequent thermoforming processes; or to just pile the needed layers and thermoform them being not joined. This was one of the most promising ideas, because it would be possible to actually eliminate the joining process, but also because it provides a simple manner of tailoring the properties of aligners with minimal equipment. We tried to manufacture devices by simply placing a non-joint stack of thermoplastics in the thermoformer, but also to thermoform each layer one by one. Both systems did not cast good multilayers. The plastics were easily peeled off by hand or even did not adhere. In some cases where any of the systems actually worked, many imperfections were observed in

the device, such as huge inhomogeneities, bubbling of the TPU, or puncture of the device.

- **Direct coextrusion**: Direct coextrusion refers to the conventional process of joining two or more material streams that come out of each extruder. This implies the use of at least one extruder for each different layers, but also to adapt the extrusion velocity of all the material streams to avoid inhomogeneities. For direct coextrusion, usually the extruders are connected to a common nozzle using heated hoses which increase the time the materials are at high temperature, which being particularly problematic for TPU. Direct coextrusion may be the main process in industrial production of these types of materials, but requires specialized and costly equipment that is usually produced in large quantities.[473], [474]

  A huge advantage of direct coextrusion in the case of sandwiched multilayers, is that the middle layer of the sandwich is never in contact with the external ambient and hence is fully protected from air, humidity, chemicals and microbes, being a completely aseptic layer [475]. This may be beneficial in the case of TPU which is easily degraded.

  However not all the plastics or not all the combinations of different plastics can be joined by coextrusion due to adhesion probelms. In fact it is considered that plastics with large processing temperature differences are not processable by coextrusion due to the degradation of the plastic of lower processing temperature. [473], [474]

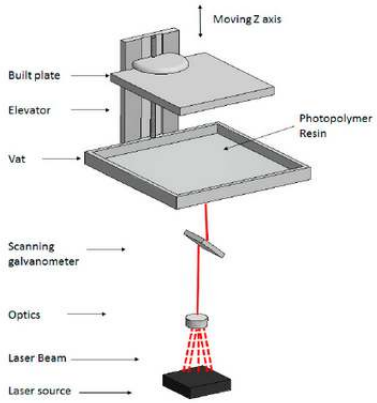
- **Indirect coextrusion/lamination**: Indirect coextrusion refers to extruding all the needed layers individually and joining them together by thermal methods which may include using a heated press as it has been done during all this thesis; or using a heated calender specially designed for this purposes.

  Indirect coextrusion has one main huge advantage over direct coextrusion which that all the extrusions may be performed with the same extruder. This process is more economically viable due to the relatively low cost of a single extruder and the joining process against the costs of the direct coextrusion process which needs at least two different extruders working at the same time. Indirect extrusion in general provides an easier, less costly and flexible processing option than direct coextrusion, although it may not reach to the production capabilities of direct methods. [476], [477], [478], [479]

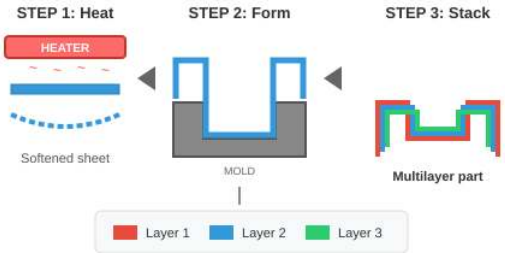
  The main drawbacks of indirect coextrusion is that the adhesion is weaker as compared with direct methods, and hence glues or adhesive layers or surface treatments are needed[479]. Also the processing times are larger and it implies more thermal cycling for the material, being possible to degrade them in the joining process. In the case of sandwiched structures, the middle layer is not aseptic and may be problematic in sensible materials as the TPU.[476], [478]

- **Extrusion lamination**: It is an indirect coextrusion process of highly importance

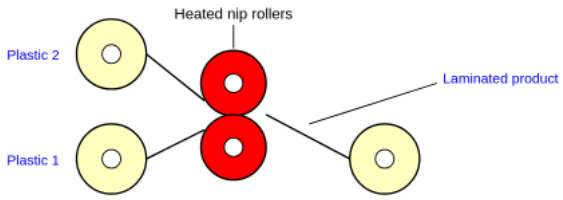
because it may reduce the manufacturing times and provide the best adhesion results, but may be the most complex indirect coextrusion method[478]. Extrusion lamination consists on extruding first the two outer layers of the material and spooling them. Then the rolls are unrolled at the same time by a specialized machine while the middle layer is extruded in the middle of both layers.[478].



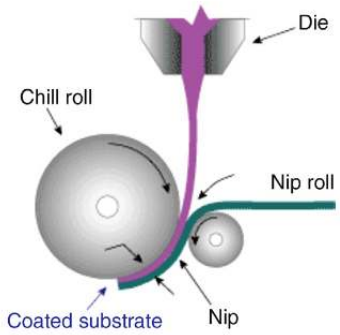

(a) SLA/DLP 3D printing[480]



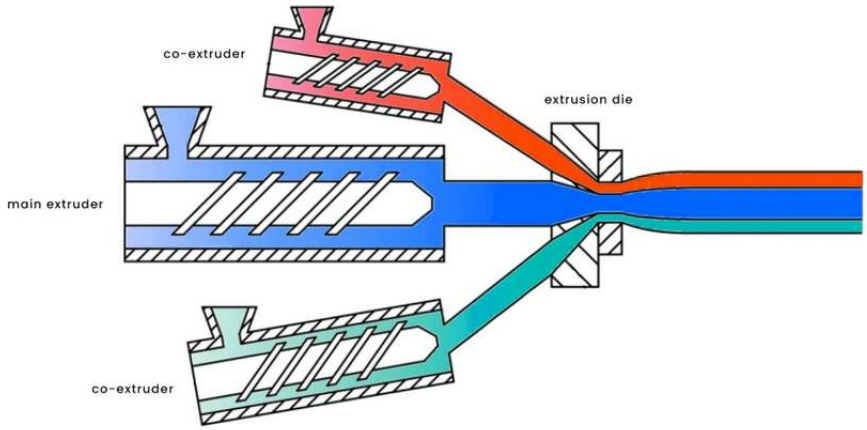

(b) Sequential thermoforming




(c) Indirect extrussion/lamination. Image adapted from [481]




(d) Extrusion-lamination.[482]




(e) coextrusion[474]

Figure 7.20: Graphical representation of all the tested or studied multilayer production methods

In light of the above, indirect extrusion combined with thermal lamination by heated rollers was selected as the most suitable process for small-to-medium scale production, offering the best balance between equipment cost, process flexibility, and achievable adhesion quality.

### 7.6.2. Raw material cost analysis

Listed below, we can see a list of the 10 most common clear aligner sheets:

| Name | Brand | Price (€/sheet) | chemistry | Country |
|---|---|---|---|---|
| Duran | Scheu Dental | 2 | PETG | Germany |
| Biolon | Dreve Dentamid | 1.30 | PETG | Germany |
| Taglus PU Flex | Taglus | 3.5 | TPU | India |
| Zendura | Bay materials | 4 | TPU | USA |
| Zendura FLX | Bay materials | 4 | Multilayer | USA |
| Maflex Premium | Maflex | 3 | Multilayer | China |
| Taglus Ultra | Taglus | 4 | Multilayer | India |
| CA Pro | Scheu Dental | 4.10 | Multilayer | Germany |
| Pro S | Prismlab | 3.50 | Multilayer | China |
| Pro S | Molekur | 3.50 | Multilayer | China |
| ClearQuartz | Straussmann | 3.5 | Multilayer | USA |

Table 7.6: Approximate prices of Clear Aligner commercial materials (January 2026). Prices shown are catalogue/retail prices. Bulk purchasing agreements may result in significantly lower unit costs (e.g., <1 €/sheet for large volumes)

As we can see in Table 7.6 the majority of the materials available on the market originate from non-European sources, which are subject to significant economic fluctuations, particularly those sourced from China and the United States. The prices reported here do not account for shipping costs or other potential economic variabilities.

Therefore, we believe that the establishment of local production, with a projected cost of 3 to 6 euros per sheet, would be economically advantageous for Secret Aligner or another European company. Although this may not immediately reduce the retail price of aligners, it would allow complete control over the production process—from raw materials to the finished device—and mitigate many of the supply chain challenges that Secret Aligner has faced in recent years. These challenges have previously compelled the company to explore alternative material sources and ultimately to consider manufacturing its own materials. We have considered various scenarios for the production that will be presented here.

The most logical manner to start would be to manufacture a generic multilayer and test its results in a real treatment.

- weight of a sheet (0.75 mm thick, 120 mm diameter): 0.11 Kg
- Number of sheets per Kg of material: 9 sheets
- Cost of 1 Kg of trilayer material ≈ 4.4 € (0.5 €/sheet)
  - Tritan copolyester 4-6 €/kg ≈ 5 €/kg
  - TPU: 2.5 - 4 €/kg ≈ 3.25 €/kg

### 7.6.3. Economic feasibility and investment recovery

A planar extrusion has an energetic cost between 0.4 and 0.6 kWh/kg, and the lamination process by heated rolls bears a similar energetic cost to single-layer extrusion; the energetic cost of multilayer production would therefore be approximately 1 kWh/kg [483]. Taking into account an average industrial electricity price in Spain during 2025 of 0.14 €/kWh and a sheet mass of 0.11 kg, the energy cost for producing a multilayer sheet amounts to 0.015 €. It must also be borne in mind that not all plastic entering the manufacturing system exits as a commercial product: approximately 10% of the input mass is converted into scraps [484], [485]. Assuming that Tritan copolyester scrap accounts for two-thirds of the total and TPU for one-third, scrap losses incur an added cost of approximately 0.05 €/sheet.

For labor costs, it has been assumed that the system would be operated by two workers at a cost of 16 €/h [486], working 8 hours per day, and that a multilayer sheet requires approximately 10 seconds to manufacture. At this rate, the system would produce 2,880 sheets per day, yielding a labor cost of 0.09 €/sheet. The breakdown of production costs is summarized in Table 7.7.

| **Cost component** | **€/sheet** |
|---|---|
| Raw materials (trilayer) | 0.50 |
| Energy (extrusion + lamination) | 0.015 |
| Scrap losses (10%) | 0.05 |
| Labor (2 workers, 16 €/h) | 0.09 |
| **Subtotal (production)** | **0.66** |
| Packaging and storage | 0.07 |
| **Total cost per sheet** | **0.73** |

Table 7.7: Estimated production costs for in-house multilayer sheet manufacturing

The estimated production cost of 0.73 €/sheet is comparable to the current bulk purchase price of approximately 0.65 €/sheet (0.70 USD) negotiated by Secret Aligner. At present market conditions, in-house production would therefore operate near break-even rather than yielding immediate cost savings. However, the catalogue prices for multilayer

materials remain in the range of 3.50-4.00 €/sheet (Table 7.6), and the current low purchase price likely reflects a supplier agreement that may not be sustainable long-term. Should bulk prices return to catalogue levels, in-house production would yield a margin of approximately 2.77-3.27 €/sheet.

Beyond direct cost considerations, in-house production provides strategic value that justifies the investment: supply chain independence from non-European sources, the ability to customize material formulations, protection against currency fluctuations and price volatility, and full traceability from raw materials to finished device.

To compute the investment recovery time, the following equipment was considered as an example:

- Equipment cost: ≈ 65,000 € (not including shipment)
  - Hartek Lab Cast Film Extrusion Line: €27,000-31,000 €
  - Hartek Film analyzer: €15,000-25,000 €
  - Hartek Three-roll calender machine: €10,000-20,000 €

Considering that there are approximately 23,000 dental clinics in Spain [487], and assuming an average of 2.92 new clear aligner treatments per clinic each month [488], the market generates around 450,000 to 500,000 cases annually. Given that a standard treatment requires an average of 35 to 40 aligners, this results in a total manufacturing demand of approximately 17 to 20 million clear aligners per year in Spain alone.

Secret Aligner reported printing around 2,000 3D molds daily in 2023 [489], [490], implying the same number of aligners thermoformed daily. Assuming that manufacturing capacity has scaled with the company's sales growth of 18.4% between 2023 and 2025 [491], and considering 230 working days per year, Secret Aligner currently manufactures approximately 550,000 aligners annually (roughly 3% of the Spanish market), with an expected growth of 100,000 aligners per year.

To obtain a realistic investment estimate, the initial capital was set at €500,000, accounting for equipment acquisition, raw material stock, and facility adaptation. Under current market conditions, in which the production cost of 0.73 €/sheet operates near break-even against the negotiated bulk price, the investment is not recovered through direct production margin but is justified instead by the strategic value of supply chain resilience, formulation control, and protection against price volatility. The economic case changes markedly if supplier prices return to catalogue levels: at a catalogue reference of 3.65 €/sheet, the resulting margin of approximately 2.92 €/sheet on 550,000 annual units would generate savings of over €1.6 million per year, yielding a payback period of under four months on the €500,000 initial investment.

### 7.6.4. Economic feasibility conclusions

The analysis presented in this chapter demonstrates that in-house production of multilayer thermoplastic sheets for clear aligners is technically feasible at the scale of a mid-sized European manufacturer such as Secret Aligner, with an estimated production cost of approximately 0.73 €/sheet. This cost is comparable to the current bulk purchase price of approximately 0.65 €/sheet (0.70 USD) negotiated by the company, indicating that in-house production would operate near break-even under present market conditions. However, catalogue prices for equivalent commercial multilayer materials remain in the range of 3.50-4.00 €/sheet; should bulk purchasing agreements become unavailable or supplier prices return to catalogue levels, in-house production would yield a margin of approximately 2.90 € per unit and a payback period of less than four months on an initial capital investment of 500,000 €.

Among the manufacturing processes reviewed, indirect extrusion combined with thermal lamination by heated rollers was identified as the most suitable approach for small-to-medium scale production. Direct coextrusion, while potentially superior in terms of interlayer adhesion and aseptic conditions for the inner layer, requires specialized and costly multi-extruder equipment that is not economically justified at the production volumes currently handled by Secret Aligner. Sequential thermoforming and direct 3D printing were discarded on the basis of insufficient interlayer adhesion and unresolved biocompatibility and mechanical property concerns, respectively.

From a strategic standpoint, the value of local production extends beyond direct cost savings. In-house manufacturing would provide Secret Aligner with full control over the supply chain—from raw material sourcing to the finished aligner sheet—thereby reducing dependence on non-European suppliers subject to significant price volatility and logistical uncertainties. This strategic independence is particularly relevant given the current geopolitical and economic context, in which supply chain disruptions have already compelled the company to seek alternative material sources. Additional benefits include the ability to customize material formulations, protection against currency fluctuations, and full traceability from raw materials to finished device.

Nevertheless, several limitations of this analysis must be acknowledged. The cost estimates rely on a number of assumptions—including labor costs (16 €/h), energy consumption, scrap rates (10%), and equipment throughput—that would require experimental validation at a pilot scale before any investment decision is made. Furthermore, the payback estimates do not account for the time and cost associated with regulatory approval and market certification, which may be substantial given the medical device classification of clear aligners. In this regard, however, the confirmed biocompatibility of both the source materials and the multilayer constructs represents a significant advantage, as it substantially mitigates one of the most demanding regulatory hurdles typically associated with the introduction of novel intraoral materials. A sensitivity analysis of the key economic parameters, as well as a pilot production trial, are strongly recommended as the

next steps prior to full-scale implementation.

# 8. GENERAL DISCUSSION AND CONCLUSIONS

The work presented in this thesis addressed a central challenge facing mid-sized clear aligner manufacturers such as Secret Aligner: the technical, biological, and economic feasibility of developing proprietary multilayer thermoplastic materials as an alternative to commercially sourced sheets, using semi-industrial means and widely available raw materials. The results obtained across the experimental chapters collectively demonstrate that this objective is achievable, and that the homemade multilayer materials developed here not only replicate the performance of commercial products but surpass them in several clinically relevant parameters.

## 8.1. Commercial material characterization

The systematic characterization of commercial aligner thermoplastics conducted in Chapter 4 addressed the first primary objective and established the reference framework against which all subsequently developed materials were evaluated. The multi-method protocol enabled the quantitative identification of the mechanical, thermal, and chemical properties that define clinical performance, and revealed several findings of direct practical relevance to Secret Aligner. The cross-sectional nanoindentation of commercial multilayer aligners — performed here for the first time — unambiguously identified the architecture of SmartTrack (Invisalign) as sTPU/PCTG/sTPU, an inverted configuration relative to the PCTG/sTPU/PCTG arrangement of Zendura FLX and Maflex Premium, and established that the outer copolyester chemistry, rather than the soft core composition, is the primary determinant of the viscoelastic force delivery profile. The master curve analysis identified Maflex Premium, the material currently used by Secret Aligner in clinical practice, as the best-performing commercial material in terms of sustained force delivery, thereby setting the performance benchmark for the homemade materials developed in the subsequent chapters.

## 8.2. Material selection, fabrication, and design framework

The second, third, and fifth primary objectives were addressed jointly in Chapter 5. Screening of candidate materials identified the Tritan copolyester grades and the polyether TPU grades as the most suitable hard and soft constituents, with EVA emerging as a promising soft layer not previously explored in commercial multilayer aligner design. The semi-industrial route of planar extrusion followed by hot-plate consolidation proved technically viable for all selected combinations, producing laminates whose composition and mechanical properties are comparable to those of commercial multilayer aligners.

The rational design framework established in this chapter is one of the most practi-

cally significant contributions of the thesis. Combining the iso-strain rule of mixtures with the exponential Shore hardness–modulus relationship, it enables the storage modulus of a laminate at oral temperature to be predicted from the Shore hardness of the soft layer alone, without iterative fabrication, and was validated across twelve configurations with a mean absolute error below 8%. Its extension to the fractional Double Zener parameters provides a further predictive layer for the therapeutic force window and residual force, enabling rational material selection for a given clinical target. The most significant mechanical finding is that the MP200-based series substantially outperforms all commercial materials in viscoelastic force delivery, confirming that the synergistic combination of a high-$T_g$ Tritan copolyester with a compliant polyether TPU interlayer produces a performance envelope beyond the reach of any currently available commercial material, and that the Tritan family represents an unexplored but highly promising hard layer chemistry.

### 8.3. Biocompatibility and intraoral stability

The sixth primary objective was addressed in Chapter 6. All tested materials — the homemade monolayers and multilayer configurations and the commercial reference Maflex Premium — satisfied the cytocompatibility criteria of ISO 10993-5 in both indirect and direct assays, and none induced a pro-inflammatory response in gingival fibroblasts under direct contact. The strong positive correlation between substrate storage modulus and fibroblast proliferation rate is a novel finding with potential clinical implications, suggesting that the mechanical properties of the aligner may influence the biological response of the gingival tissues beyond their role in force delivery. The artificial saliva absorption data confirmed the intraoral stability of all materials, with no detectable bulk hydrolysis; the unexpected stiffening of Maflex Premium after immersion, not previously reported, underscores the value of characterising commercial reference materials under simulated intraoral conditions. Taken together with the mechanical results, these findings establish the MP200-based configurations as mechanically and biologically viable candidates for clinical translation.

### 8.4. Economical feasibility of multilayer production

The ninth primary objective was addressed in Chapter 7, Section 7.6. The cost analysis demonstrated that in-house production by indirect extrusion and thermal lamination is economically viable at Secret Aligner's current production scale, with an estimated cost near the current negotiated bulk price. Under present market conditions the investment is therefore justified primarily by its strategic value — supply chain resilience, formulation control, and protection against price volatility — rather than by direct margin; should supplier prices return to catalogue levels, the same route would yield a payback period of under four months. The semi-industrial approach developed in this thesis is thus not merely a laboratory proof of concept but a commercially actionable strategy.

## 8.5. Environmental impact of clear aligners

The life cycle assessment presented in Chapter 7, Section 7.5 provided the first quantitative comparison of the carbon footprint of thermoformed clear aligner therapy and fixed appliance treatment under equivalent system boundaries, with the latter generating roughly twice the emissions of the former. The key finding is that clinical visit frequency — not material selection — is the primary determinant of the environmental burden of orthodontic treatment, with important implications for sustainability strategy: reducing the number of appointments is more impactful than any change in material chemistry or manufacturing route. Within the manufacturing workflow, DLP resin synthesis and solvent use together dominate the process-level footprint, a result not previously reported for clear aligners that identifies lower-carbon photopolymer resins and solvent recovery as the concrete, actionable targets within the production process itself.

## 8.6. Force characterization and simulation

The open-source finite element framework developed in Chapter 7, Section 7.1 constitutes the first complete open-source FEM pipeline for clear aligner biomechanics, validated across three levels of geometric complexity. Its use of elastic moduli determined experimentally at 37 ºC as simulation inputs — rather than room-temperature values from generic databases, as is common in the literature — directly connects the mechanical characterization of the preceding chapters to the computational predictions. The FSR-based force measurement system, still at proof-of-concept stage, demonstrated continuous force monitoring over a full working day and confirmed that the posterior sector experiences higher contact forces than the anterior sector under passive loading, in agreement with the stress distributions predicted by the FEM. Together, these tools establish a foundation for the experimental validation of the computational framework and for the rational optimisation of activation distance and material selection.

## 8.7. Limitations

The findings of this thesis must be read in light of a number of limitations, grouped here by domain.

### 8.7.1. Materials and fabrication

The semi-industrial fabrication route — planar extrusion followed by hot-plate consolidation — does not match the surface quality and dimensional precision of commercial co-extrusion. Air entrapment at the layer interfaces, surface irregularities, and thickness variations are consequently more prevalent in the homemade laminates than in commercial products, which precluded reliable quantitative optical transparency measurement and

introduced additional scatter into the tensile data, most notably in the yield strain. The $\varepsilon_y$ values reported for the homemade multilayers should therefore be regarded as conservative lower-bound estimates of the intrinsic elastic deformability of these material combinations. In the interest of experimental efficiency, stress relaxation of the homemade series was characterised at a single strain level (0.5%) rather than the three levels used for the commercial materials, so the strain dependence of the relaxation response was not investigated; bending modulus characterisation at 37 ºC and interfacial nanoindentation of the homemade laminates were likewise outside the scope of the present work. Finally, the biological characterisation was restricted to a representative subset of the full experimental matrix, and extrapolation of its conclusions to the remaining configurations should be made with caution.

### 8.7.2. Biological characterisation

The biological assessment employed a single cell type (human gingival fibroblasts) and did not include immune cells such as macrophages or monocytes, the primary producers of $IL\text{-}1\beta$ and $TNF\text{-}\alpha$, which would be required for a complete characterisation of the inflammatory potential. The in vitro extraction conditions (72 h at 37 ºC) and the static immersion assays do not reproduce the salivary flow, pH fluctuations, thermal cycling, mechanical loading, and microbial colonisation of the intraoral environment, and therefore represent a conservative lower bound on the degradation experienced clinically. The experimental design was not optimised for the detection of microplastic release, which consequently cannot be excluded on the basis of the present results. The correlation between substrate stiffness and fibroblast proliferation, though statistically significant, was established across a chemically heterogeneous set of materials and requires validation on substrates of defined stiffness with equivalent surface chemistry.

### 8.7.3. Computational modelling and force measurement

The finite element framework is restricted to linear elastic, isotropic, and homogeneous behaviour: it neglects the viscoelastic stress relaxation that governs time-dependent force decay, omits the periodontal ligament, models the multilayer as a single homogeneous layer, and adopts Poisson's ratios from the literature rather than from direct measurement. The reported forces and displacements therefore correspond to the initial insertion state and should not be extrapolated to the full wear period, and the reference contact force remains an empirically adjusted parameter rather than one derived from contact mechanics. The FSR-based force measurement system is at the proof-of-concept stage: measurements were conducted on a single model and aligner ($n = 1$) without composite attachments, at room temperature rather than under simulated intraoral conditions, and with a first-order servo-based calibration carrying an estimated uncertainty of ±30–50%. The rigid 3D-printed model does not replicate the compliance of the periodontal ligament, so the measured values represent the aligner-to-rigid-surface interaction rather than

the physiological force transmitted to the tooth root.

### 8.7.4. Life cycle assessment

The principal methodological limitation of the LCA is the use of proxy materials for those absent from the ELCD 3.2 database (PET for PETG and TPU, and PMMA for the DLP photopolymer resin) so that the results strictly represent a fully PET-based treatment and the resin contribution should be interpreted as an order-of-magnitude estimate. Both assessments rely on primary data from a single manufacturer and on literature-derived process data, and the realistic end-of-life scenario is based on national plastic-waste statistics rather than on primary data on patient disposal behaviour. Patient-side processes, including travel to the clinic, identified elsewhere as a dominant contributor to the footprint of dental services, were excluded from the system boundary, meaning the total footprint is likely underestimated, particularly for fixed appliance treatment with its higher visit count. The ELCD 3.2 database was discontinued in 2018 and may not reflect current manufacturing efficiencies, although the relative ranking of contributors is expected to be robust to this limitation. Finally, the functional units represent typical complete treatments but do not account for differences in treatment duration, clinical complexity, or outcome quality between the two modalities.

## 8.8. Competitiveness and future directions

The body of work presented in this thesis identifies several strategic opportunities for Secret Aligner to strengthen its competitive position in a sector dominated by large multinational suppliers. The Tritan copolyester family ( MP200 in particular) represents an unexplored high-performance hard layer chemistry whose viscoelastic force delivery substantially exceeds that of the PCTG grades used in current commercial multilayer aligners. EVA, TPU-free and more readily processed than polyether TPU, opens a category of multilayer aligner not currently available on the market and potentially advantageous for patients with TPU sensitivity. The rational design framework established here enables material development to proceed through a design-first, fabricate-second workflow, reducing the time and cost of iterative optimisation. Finally, the economic analysis confirms that the transition from commercial sheet procurement to in-house production is technically viable and strategically justified at current volumes: although in-house cost (0.73 €/sheet) is comparable to the current negotiated bulk price, the investment protects against supplier price volatility and supply chain disruption, and yields a payback period of under four months should market prices return to catalogue levels.

Realizing this potential requires further work along five interdependent lines, which together define the research agenda arising from this thesis.

### 8.8.1. Materials and process development

The most immediate priority is the optimisation of the consolidation process. Replacing hot-plate pressing with continuous hot-roll lamination of individually extruded layers, the route identified in Chapter 7 as offering the best balance of capital cost, flexibility, interlayer adhesion, and scalability, is expected to eliminate the air entrapment, surface irregularities, and thickness variations that currently limit the optical and mechanical performance of the homemade laminates, and to enable the quantitative optical transparency assessment that could not be performed on the present specimens. The MP200/52D and MP200/85A configurations are identified as the highest-priority candidates for scale-up, given their exceptional viscoelastic performance relative to commercial materials. The incorporation of Tritan copolyesters as hard layers in a co-extruded or roll-laminated architecture has not been reported in the literature and, per the patent analysis in Chapter 4, may fall outside the scope of existing claims for soft layer moduli below 50 MPa, offering a potential route to intellectual property protection. Complementary characterisation should include bending modulus at 37 ºC under controlled force ramps, multi-strain stress relaxation, and high-resolution nanoindentation across the interfacial region of the homemade laminates, whose mechanically active interphase was identified as a primary source of deviation from the iso-strain mixing rule. The design space for the soft interlayer should also be extended beyond the grades studied here, including EVA formulations with higher vinyl acetate content and alternative polyolefin elastomers.

### 8.8.2. Biological characterisation

The biocompatibility assessment should be extended from the representative subset studied here to the full multilayer matrix, and broadened beyond the single cell type (hGFs) employed. Incorporating immune cells such as macrophages and monocytes; the primary producers of $IL\text{-}1\beta$ and $TNF\text{-}\alpha$; would provide a more complete characterisation of the inflammatory potential in a physiologically representative context. In vivo assessment is a further priority: establishing validated protocols for salivary LDH-based periodontal stress monitoring in clear aligner patients would close a notable gap, as no such measurements have been reported for aligner therapy despite their established use with fixed appliances. Two dimensions of biological safety remain entirely uncharacterised and warrant dedicated study: the endocrine-disrupting potential of the constituent materials, which BPA-free status does not preclude, and the release of microplastics under conditions representative of prolonged intraoral use, combining mechanical loading, thermal cycling, salivary enzymatic activity, and microbial colonisation, with detection by micro-FTIR or Raman spectroscopy.

### 8.8.3. Computational modelling and force measurement

The open-source FEM framework should be extended from its present proof-of-concept status through five developments: incorporation of a viscoelastic constitutive law to capture force decay during clinical wear; inclusion of the periodontal ligament to enable simulation of the biological tooth movement response; a gap-dependent contact formulation to remove the empirical force calibration parameter; explicit representation of the individual layers of multilayer architectures rather than an effective homogeneous modulus; and experimental determination of the Poisson's ratio of each material by digital image correlation. A key enabler of clinically realistic simulation is the digitisation of the as-manufactured aligner geometry, for which a structured-light scanning protocol based on opaque coating of the transparent shell was partially established in this work and should be completed and integrated into the pipeline. In parallel, the FSR-based force measurement system requires a rigorous multi-point calibration against a certified reference load cell to reduce its present ±30–50% uncertainty below ±5%, together with measurement under simulated intraoral temperature, humidity, and immersion conditions, and extension beyond PETG to PCTG, hTPU, and multilayer materials. A particularly promising protocol would combine the force sensor with the aligner scanning methodology on paired specimens; one scanned as reference, one seated on a model with a known geometric discrepancy and then scanned; to quantify force output and aligner deformation simultaneously and to provide an experimental validation benchmark for the FEM predictions.

### 8.8.4. Environmental sustainability

The life cycle assessment identifies clinical visit frequency as the dominant determinant of the environmental burden of orthodontic treatment; the widespread adoption of remote monitoring, which can reduce visits by up to 50%, is therefore the single most impactful intervention and should be prioritised over changes in material or manufacturing route. Within the manufacturing workflow, the development of lower-carbon photopolymer resins for the DLP molds and the implementation of solvent recovery and regeneration systems address the two dominant process-level contributors. Production thermoplastic scraps, which are not classified as medical waste, represent the most tractable recycling opportunity; scaling the chemical separation of multilayer materials demonstrated in Chapter 4 toward closed-loop recovery, and identifying a more ecologically acceptable solvent than THF, are well-defined objectives for future work. The establishment of take-back programmes for used devices, and continued research into biocompatible biodegradable or biobased polymers that meet the mechanical and optical requirements of clinical use, would further reduce the end-of-life impact of clear aligner therapy. Finally, future LCA work should incorporate primary energy and material data from Secret Aligner's automated production workflow and, where possible, an updated open-access background database, to provide a more accurate baseline for this rapidly evolving manufacturing context.

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# A. FRACTIONAL CALCULUS IN POLYMER VISCOELASTICITY AND ANOMALOUS DIFFUSION: THEORETICAL FOUNDATIONS

## A.1. Viscoelastic modelling of thermoplastic aligners and periodontal ligament

### Physical requirements of stress relaxation functions

Any constitutive model intended to describe the time-dependent mechanical response of a polymer or soft tissue must satisfy the following physical and mathematical requirements [140], [141], [149]:

1. **Causality**: No material response can precede the applied stimulus; the relaxation function must vanish for all t<0 [140], [492].
2. **Energy dissipation**: The relaxed quantity must decrease monotonically over time, reflecting molecular reorganization from a higher to a lower entropy state [493], [494].
3. **Asymptotic equilibrium**: As $t \to \infty$ , the relaxed quantity must approach a non-negative equilibrium value [140], [141].
4. **Thermal dependence**: The relaxation function must exhibit temperature dependence consistent with the time-temperature superposition (TTS) principle, which allows measurements at different temperatures to be combined into a single master curve spanning many decades of time [142], [256].
5. **Complete monotonicity**: The successive derivatives of the relaxation function must alternate in sign, ensuring a monotonic decrease and convexity. By Bernstein's theorem [495], this is equivalent to requiring that the relaxation function can be expressed as a superposition of decaying exponentials, in other words, as a continuous or discrete relaxation spectrum.
6. **Time-frequency interconversion**: The relaxation function should admit mapping between the time domain and the frequency domain via integral transforms (Laplace, Fourier, or Mellin), enabling the combined use of stress relaxation and dynamic mechanical analysis data [142], [146].

### A.1.1. Non-Markovian dissipation and its physical origin

A dissipative process is Markovian if the stress at time t depends only on the instantaneous state of the system. The classical Newtonian dashpot ($\sigma = \eta\dot{\varepsilon}$) is the prototypical Marko-

vian element. In contrast, a non-Markovian process is one in which the stress depends on the entire deformation history:

$$\sigma(t) = \int_0^t K(t-s)\,\dot{\varepsilon}(s)\,\mathrm{d}s \tag{A.1}$$

where $K(t-s)$ is a memory kernel weighting past strain contributions. This is the structure of the generalized linear viscoelastic constitutive equation, in which the relaxation modulus $E(t)$ plays the role of K [142], [496].

Non-Markovian dissipation is the rule rather than the exception in polymeric and biological soft matter. In a polymer chain, covalent connectivity couples the motion of each monomer to its neighbors, generating a broad, continuous spectrum of internal relaxation modes [497], [498]. In the Rouse model of an unentangled chain, this yields a power-law memory kernel $K(t) \sim t^{-1/2}$ ; in the Zimm model, $K(t) \sim t^{-2/3}$ [499], [500]. When the memory kernel decays as a power law $K(t) \sim t^{-\alpha}$ , the resulting constitutive equation is precisely a fractional derivative of order $\alpha$[140], [154], providing the physical justification for fractional viscoelastic models. The fractional order $\alpha \in (0, 1)$ is a direct measure of the non-Markovian character of the material: $\alpha = 1$ recovers Newtonian dissipation, while $\alpha \to 0$ approaches purely elastic behavior [140], [501].

### Classical viscoelastic models

### Generalized Maxwell model

The generalized Maxwell model represents the material as a parallel assembly of spring-dashpot elements (see figure A.1), yielding a Prony series in stress relaxation (see figure A.2) [142], [502]:

$$E(t) = E_\infty + \sum_{n=0}^{\infty} E_n, e^{-t/\tau_n} \tag{A.2}$$

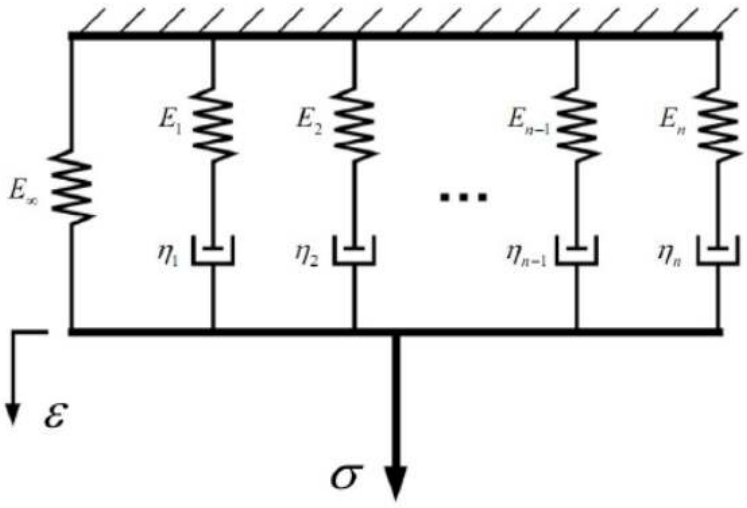



Figure A.1: Generalised Maxwell viscoelastic model [503]

The relaxation spectrum is discrete, consisting of individual characteristic times $\tau_i$. While widely implemented in commercial FEM codes due to its computational simplicity, the model requires a large number of parameters and does not yield physically interpretable quantities such as the continuous relaxation spectrum or the molecular weight distribution [147], [152].

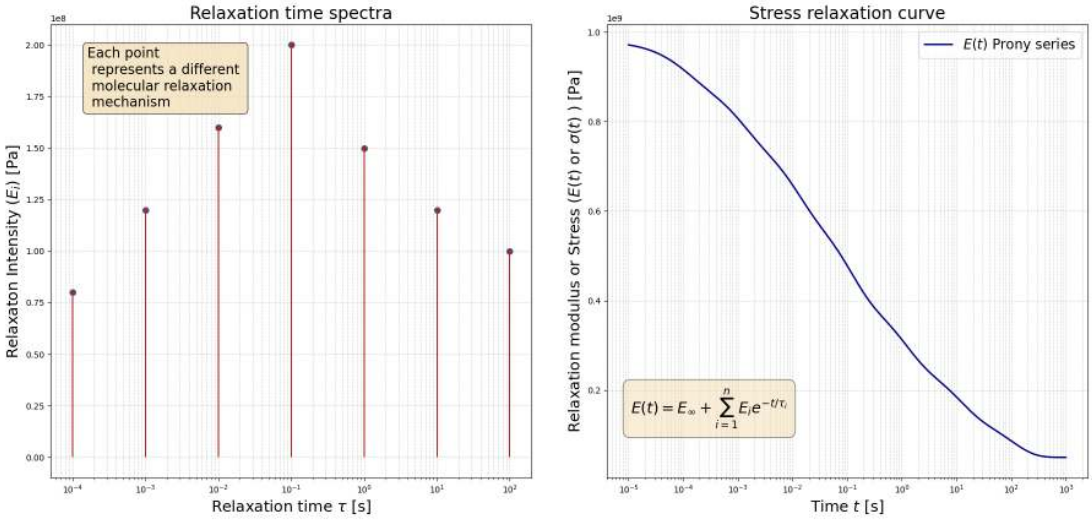



Figure A.2: Prony series relaxation times and derived stress relaxation curve

The continuous relaxation $H(\tau)$, which describes how material stiffness is distributed across all timescales, is related to the relaxation modulus and to the complex modulus through the inverse Laplace transform and the Kramers-Kronig integral, respectively [142], [146]:

$$E(t) = \int_0^\infty H(\tau)\, e^{-t/\tau}\, \mathrm{d}\ln\tau \qquad E^*(\omega) = i\omega \int_0^\infty H(\tau), \frac{\tau}{1+i\omega\tau}, \mathrm{d}\ln\tau \tag{A.3}$$

**Kohlrausch-Williams-Watts (KWW) model**

The KWW stretched exponential provides a compact empirical description of stress relaxation [504]:

$$E(t) = E_\infty + (E_0 - E_\infty), e^{-(t/\tau)^\beta} \tag{A.4}$$

The stretching exponent $\beta \in (0, 1)$ captures the deviation from a simple exponential decay. However, a closed-form relaxation spectrum does not exist in general for the KWW function [148], and the model cannot be extended to arbitrary loading histories via the Boltzmann superposition principle, precluding its use as a general constitutive equation in FEM simulations [142], [146].

### Guiu-Pratt (GP) model

The GP model, derived for thermally activated dislocation glide in crystalline metals [505], takes the form:

$$E(t) = E_0 - \frac{RT}{V_c \varepsilon_0} \ln\left(1 + \frac{t}{c}\right) \tag{A.5}$$

The logarithmic decay predicts a stress that decreases without bound as $t \to \infty$, violating the requirement of a non-negative equilibrium modulus. The model is formally nonlinear, so the Boltzmann superposition principle does not apply and time–frequency interconversion is unavailable [142], [149].

## A.1.2. Fractional viscoelastic models

The family of fractional viscoelastic models discussed in the present work is illustrated in Figure A.3, which shows the hierarchical derivation relationships between the principal models and their historical development. All fractional models share a common elementary building block — the springpot — from which progressively more complex constitutive descriptions are constructed by combining with springs and additional springpots in series or parallel configurations. The following subsections describe each model in order of increasing complexity, following the derivation tree shown in Figure A.3.

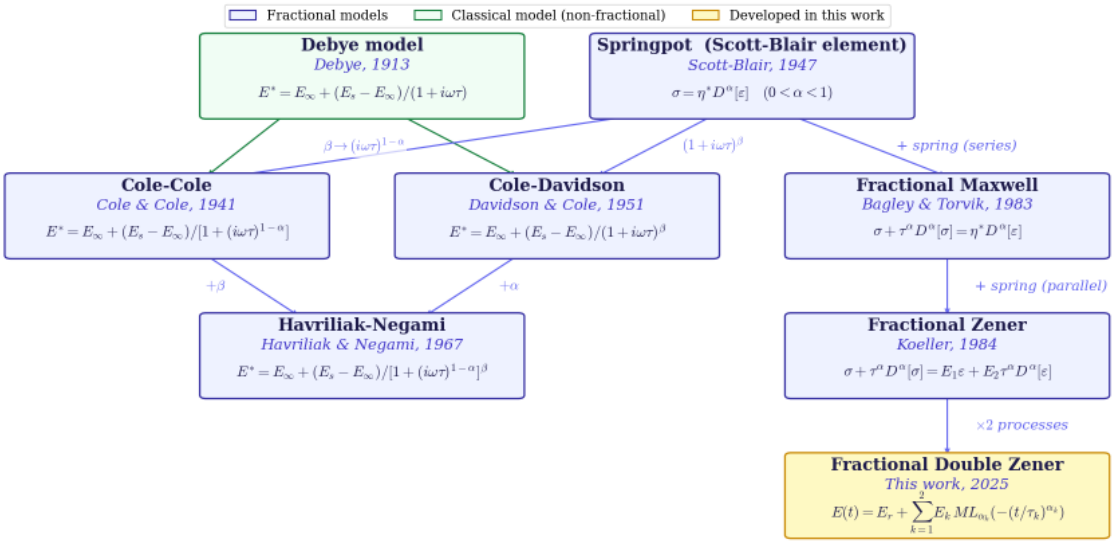



Figure A.3: Family tree of the main fractional viscoelastic models

### The Debye model and its limitations

The Debye model, introduced in 1913 [506], constitutes the canonical single-relaxation-time description of viscoelastic and dielectric relaxation. In the frequency domain, the complex modulus takes the form:

$$E^*(\omega) = E_\infty + \frac{E_s - E_\infty}{1 + i\omega\tau} \tag{A.6}$$

where $E_s$ and $E_\infty$ are the static and infinite-frequency moduli respectively and $\tau$ is a single characteristic relaxation time. In the time domain, the corresponding relaxation modulus decays as a pure exponential:

$$E(t) = E_\infty + (E_s - E_\infty)e^{-t/\tau} \tag{A.7}$$

The Debye model is the simplest constitutive description satisfying all physical requirements listed in Section A.1: it is causal, energy dissipative, admits a well-defined equilibrium modulus, and is fully interconvertible between time and frequency domains. Its relaxation time spectrum $H(\tau)$ is a delta function centred at $\tau$ corresponding to a single molecular relaxation mechanism operating at a unique timescale.

However, virtually all real polymeric and biological materials deviate systematically from Debye behaviour. Loss peaks are consistently broader and often asymmetric relative to the Lorentzian predicted by the Debye model, and the time-domain decay is persistently non-exponential, a pattern observed equally in dielectric spectroscopy and in mechanical stress relaxation [153], [154], [507]. This deviation reflects the fundamental physical reality discussed in Section A.1.1: in polymeric systems, covalent connectivity along the backbone generates a broad, continuous spectrum of internal relaxation modes rather than a single characteristic time, and the cooperative nature of segmental motion near the glass transition further widens and skews this distribution [142], [153]. The Debye model is therefore not a realistic constitutive description for thermoplastic aligner materials or for the periodontal ligament, and its failure motivated the successive development of more general models (Cole-Cole, Cole-Davidson, Havriliak-Negami, and ultimately the fractional models based on the springpot) that are capable of capturing the broad, non-exponential relaxation characteristic of these materials.

**Springpot (Scott-Blair element)**

The springpot, introduced by Scott-Blair in 1947 [150] and later formalised within the framework of fractional calculus by Koeller [151], is the fundamental rheological element of fractional viscoelasticity. Its constitutive equation interpolates continuously between a purely elastic spring ($\alpha = 0$) and a purely viscous dashpot ($\alpha = 1$):

$$\sigma(t) = \eta^* D^\alpha[\varepsilon(t)], \quad 0 < \alpha < 1 \tag{A.8}$$

where $\eta^*$ is a generalised modulus with units of $Pas^\alpha$ and $D^\alpha$ denotes the Caputo fractional derivative of order $\alpha$ defined in Equation A.24. The constitutive equations of the three elementary rheological elements (spring, dashpot, and springpot) are collected together for reference:

$$\sigma(t) = ED^0[\varepsilon(t)] \qquad \text{(spring)} \tag{A.9}$$

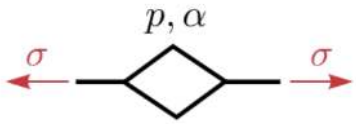
$$\sigma(t) = \eta D^1[\varepsilon(t)] \qquad \text{(dashpot)} \tag{A.10}$$

$$\sigma(t) = \eta^* D^\alpha[\varepsilon(t)], \quad 0 < \alpha < 1 \qquad \text{(springpot)} \tag{A.11}$$

The springpot satisfies all physical requirements listed in Section A: it is causal, energy dissipative, admits a well-defined equilibrium modulus ($E_\infty = 0$ for $\alpha > 0$), exhibits thermal dependence consistent with the time-temperature superposition principle, and its relaxation function is completely monotone [140], [501]. Its relaxation modulus takes the power-law form:

$$E(t) = \frac{\eta^*}{\Gamma(1-\alpha)} t^{-\alpha} \tag{A.12}$$

which decays as a power law for all t > 0, reflecting the non-Markovian memory kernel $K(t) \sim t^{-\alpha}$ discussed in Section A.1.1. The fractional order $\alpha$ carries direct physical meaning: it encodes the degree of molecular cooperativity and the breadth of the relaxation time distribution, interpolating between the delta-function spectrum of a pure spring ($\alpha = 0$)and the uniform spectrum of a pure dashpot ($\alpha = 1$).[153], [154].




(a) Springpot

Figure A.4: Schematic representation of the springpot (Scott-Blair element).

### Fractional Maxwell model

The fractional Maxwell model is constructed by placing a springpot in series with a spring [508], in direct analogy with the classical Maxwell model but replacing the dashpot with a springpot. Its constitutive equation is:

$$\sigma + \tau^\alpha D^\alpha[\sigma] = \eta^* D^\alpha[\varepsilon] \tag{A.13}$$

where $\tau = (\eta^*/E)^{1/\alpha}$ is the characteristic relaxation time. The relaxation modulus of the fractional Maxwell model takes the Mittag-Leffler form:

$$E(t) = E\ \mathrm{ML}_\alpha\left(-\left(\frac{t}{\tau}\right)^\alpha\right) \tag{A.14}$$

which transitions from a constant glassy modulus E at short times to a power-law decay $E(t) \sim t^{-\alpha}$ at long times, with no well-defined equilibrium modulus ($E_\infty = 0$). This absence of a rubbery plateau makes the fractional Maxwell model unsuitable as a standalone constitutive description for solid-like viscoelastic materials such as thermoplastic aligners, which retain a finite residual modulus after the glass transition. It serves instead as the structural precursor to the fractional Zener model, which corrects this limitation by adding a spring in parallel.

### Cole-Cole model

The Cole-Cole model was introduced by Cole and Cole in 1941 [509] in the context of dielectric spectroscopy, where it was found that the complex permittivity of many polar liquids and amorphous solids deviates systematically from the single-relaxation-time Debye model, exhibiting a symmetric broadening of the loss peak in the frequency domain. Its frequency-domain expression, adapted to mechanical viscoelasticity, is:

$$E^*(\omega) = E_\infty + \frac{E_s - E_\infty}{1 + (i\omega\tau)^{1-\alpha}} \tag{A.15}$$

where $\alpha \in (0, 1)$ controls the symmetric broadening of the relaxation process: $\alpha = 0$ recovers the Debye (single-relaxation-time) model, while increasing $\alpha$ produces progressively broader and flatter loss peaks, reflecting a wider distribution of relaxation times. In the Cole-Cole representation, where the imaginary part of the complex modulus is plotted against the real part, the model produces a perfect semicircular arc whose depression below the real axis increases with $\alpha$, a feature that provides a direct graphical measure of the breadth of the relaxation distribution [142], [509]. The Cole-Cole model was evaluated as a candidate for fitting the master curves of the commercial and homemade multilayer materials in the present work but was discarded due to its inability to capture the asymmetric relaxation spectra characteristic of these materials, which exhibit a longer tail towards slow relaxation timescales than towards fast ones.

### Cole-Davidson model

The Cole-Davidson model was proposed by Davidson and Cole in 1951 [510] to describe the asymmetric broadening of dielectric loss peaks observed in supercooled liquids and amorphous polymers, a deviation from Debye behaviour that the symmetric Cole-Cole model cannot reproduce. Its frequency-domain expression is:

$$E^*(\omega) = E_\infty + \frac{E_s - E_\infty}{(1 + i\omega\tau)^\beta} \tag{A.16}$$

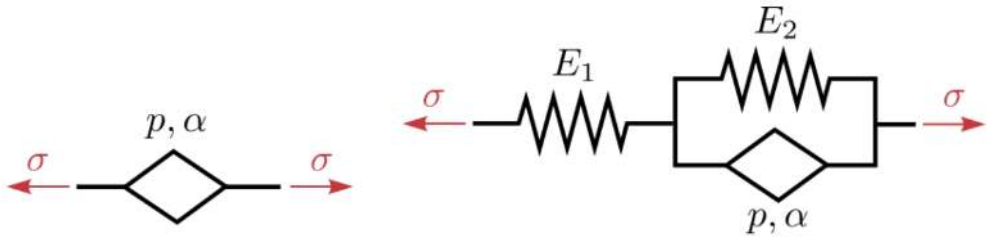
where $\beta \in (0, 1)$ controls the asymmetric skewing of the relaxation process: $\beta = 1$ recovers the Debye model, while decreasing $\beta$ produces progressively more asymmetric loss peaks with a steeper high-frequency flank and a more gradual low-frequency tail — consistent with cooperative relaxation processes in which the fastest modes relax nearly exponentially while the slowest modes are strongly retarded by intermolecular constraints [142], [153]. In the Cole-Cole representation, the Cole-Davidson model produces a skewed arc rather than a semicircle, with the asymmetry directly reflecting the value of $\beta$. The Havriliak-Negami model discussed below generalises both the Cole-Cole and Cole-Davidson models by incorporating both $\alpha$ and $\beta$ simultaneously.

### Fractional Zener model

The fractional Zener model, also known as the fractional standard linear solid, was formalised within the framework of fractional calculus by Koeller in 1984 [151], building on earlier work by Bagley and Torvik [508]. It is constructed by placing a spring in parallel with the fractional Maxwell element, correcting the absence of a rubbery equilibrium modulus in the latter. Its constitutive equation is:

$$\sigma + \tau^{\alpha} D^{\alpha}[\sigma] = E_1 \varepsilon + E_2 \tau^{\alpha} D^{\alpha}[\varepsilon] \tag{A.17}$$

where $E_1$ is the rubbery modulus, $E_1 + E_2$ is the glassy modulus, $\tau$ is the characteristic relaxation time, and $\alpha \in (0, 1)$ controls the breadth and shape of the relaxation process. The fractional Zener model is graphically represented in Figure A.5.




(a) Springpot (b) Fractional Zener model

Figure A.5: Schematic representation of the springpot and the fractional Zener model.[511]

Applying a step-strain excitation $\varepsilon(t) = \varepsilon_0 H(t)$ to Equation A.17 yields the relaxation modulus expressed in terms of the one-parameter Mittag-Leffler function [140], [145]:

$$E(t) = E_1 + E_2 \, \mathrm{ML}_{\alpha}\left(-\left(\frac{t}{\tau}\right)^{\alpha}\right) \tag{A.18}$$

The Mittag-Leffler function, illustrated in Figure A.6, generalizes the exponential: it transitions from a stretched-exponential decay at short times to a power-law tail at long times [140], [512], naturally encoding the non-Markovian dissipation present in polymeric systems. At t = 0, $E(0) = E_1 + E_2$ (glassy modulus); as $t \to \infty$, $E(\infty) \to E_1$ (rubbery modulus). The continuous relaxation time spectrum $H(\tau)$ can be extracted analytically from the Mittag-Leffler function, enabling direct comparison with molecular weight distributions [146], [513].

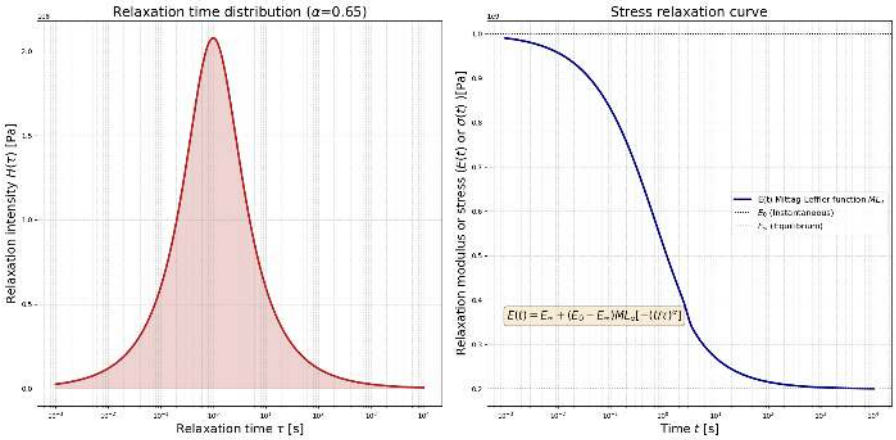



Figure A.6: Mittag-Leffler relaxation time distribution and corresponding stress relaxation curve for the fractional Zener model.

The fractional Zener model satisfies all physical requirements listed in Section A.1 and admits a well-defined Boltzmann hereditary integral formulation, making it suitable for finite-element implementation under arbitrary loading histories [140], [514]. Its four parameters $(E_1, E_2, \tau, \alpha)$ are physically interpretable: $E_1$ and $E_2$ quantify the residual and relaxing elastic contributions, respectively; $\tau$ governs the timescale of force decay, and $\alpha$ encodes the degree of molecular cooperativity. The fractional Zener model was evaluated as a candidate for fitting the master curves of all materials in the present work but was found to provide insufficient flexibility to simultaneously capture both the fast local segmental relaxation and the slow cooperative glass transition relaxation process, motivating the development of the fractional double Zener model.

### Havriliak-Negami model

The Havriliak-Negami (HN) model [515] is the most general of the empirical fractional models discussed here, incorporating both the symmetric broadening parameter $\alpha$ of the Cole-Cole model and the asymmetric skewing parameter $\beta$ of the Cole-Davidson model simultaneously:

$$E^*(\omega) = E_\infty + \frac{E_s - E_\infty}{\left[1 + (i\omega\tau)^{1-\alpha}\right]^\beta} \tag{A.19}$$

Setting $\beta = 1$ recovers the Cole-Cole model; setting $\alpha = 0$ recovers the Cole-Davidson model; and setting both $\alpha = 0$ and $\beta = 0$ recovers the Debye model. The time-domain solution of the HN model cannot be expressed in closed form via elementary functions; it is given by a series expansion corresponding to the three-parameter Mittag-Leffler function [516], [517]:

$$\mathrm{ML}^\gamma \alpha, \beta(z) = \frac{1}{\Gamma(\gamma)} \sum n = 0^\infty \frac{\Gamma(n+\gamma), z^n}{n!, \Gamma(n\alpha + \beta)} \tag{A.20}$$

Despite its generality, the HN model presents significant practical limitations: parameter identification from experimental data is highly ambiguous and ill-conditioned [146],

[518], and establishing a clear relationship between its parameters and molecular properties is difficult [142], [153]. The HN model was evaluated as a candidate for fitting the master curves of all materials in the present work and achieved good fits across all configurations; however, it was ultimately discarded in favor of the fractional double Zener model on the grounds of superior physical interpretability and parameter uniqueness of the latter.

### Fractional Double Zener model

The fractional double Zener (DZ) model was developed in the present work (2025) as an extension of the fractional Zener model to simultaneously capture two distinct relaxation processes operating at different timescales, a slow cooperative process associated with the glass transition of the hard layer and a fast local segmental process governing insertion mechanics. The model consists of a single elastic element connected in series with two fractional Zener elements, as illustrated in Figure A.7.

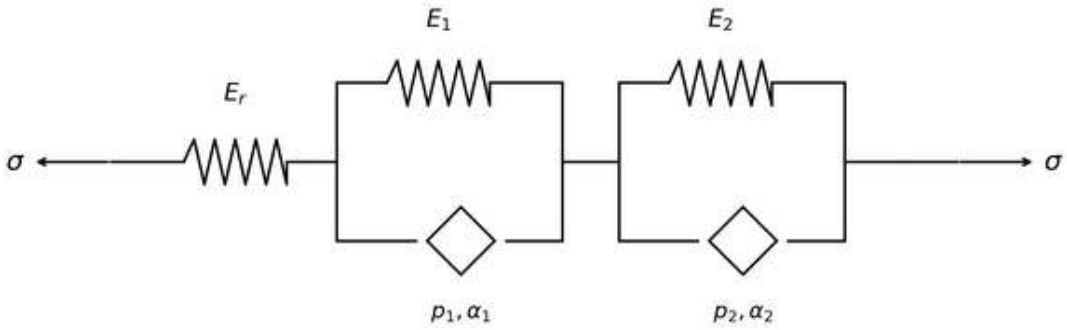



Figure A.7: Schematic representation of the fractional double Zener model.

The relaxation modulus of the DZ model is obtained by superposition of two Mittag-Leffler contributions:

$$E(t) = E_r + E_1 \, \mathrm{ML}\alpha_1 \left( -\left( \frac{t}{\tau_1} \right)^{\alpha_1} \right) + E_2 \, \mathrm{ML}\alpha_2 \left( -\left( \frac{t}{\tau_2} \right)^{\alpha_2} \right) \tag{A.21}$$

The model contains seven physically interpretable parameters, whose clinical significance in the context of clear aligner therapy is summarised in Table A.1.

| Parameter | Physical meaning | Clinical relevance | Units |
|---|---|---|---|
| $E_r$ | Rubbery equilibrium modulus | Residual force after prolonged wear | MPa |
| $E_1$ | Magnitude of slow relaxation | Total force lost at glass transition | MPa |
| $\alpha_1$ | Breadth of slow relaxation | Shape of force decay over wear cycle | – |
| $\tau_1$ | Timescale of slow relaxation | Therapeutic force window | h |
| $E_2$ | Magnitude of fast relaxation | Force at insertion | MPa |
| $\alpha_2$ | Breadth of fast relaxation | Comfort profile upon seating | – |
| $\tau_2$ | Timescale of fast relaxation | Duration of insertion discomfort | s |

Table A.1: Physical meaning and clinical relevance of the seven parameters of the fractional double Zener model in the context of clear aligner therapy.

The slow relaxation process (process 1) represents the cooperative motion of large polymer chain segments and is associated with the glass transition of the material. The fast relaxation process (process 2) arises from localized, non-cooperative motion of short chain segments below the glass transition. From a clinical standpoint, $\tau_1$ is the parameter most directly relevant to sustained force delivery — it governs the gradual force decay over the wearing period; while $\tau_2$ determines the immediate mechanical response upon aligner insertion and hence the patient's seating comfort. The breadth parameters $\alpha_1$ and $\alpha_2$ control how abruptly or gradually each process dissipates its associated force: lower values indicate a more broadly distributed, gradual decay, while values approaching 1 indicate a near-exponential, abrupt decay. Among the four candidate models evaluated in the present work; Cole-Cole, fractional Zener, DZ, and Havriliak-Negami, the DZ model was selected as the preferred constitutive description on the grounds of its superior balance between fitting accuracy, parameter parsimony, and physical interpretability, as discussed in Chapters 4 and 5.

The fractional double Zener model represents the most complete constitutive description employed in the present work for the mechanical characterization of thermoplastic aligner materials. Remarkably, this same mathematical framework extends beyond mechanical viscoelasticity to describe anomalous transport phenomena in polymeric media, including the water sorption behavior of thermoplastic aligner materials under simulated intraoral conditions.

### A.1.3. Anomalous diffusion and the classification of transport regimes

When the timescales of molecular diffusion and polymer chain relaxation are comparable, the transport behavior deviates from the Fickian ideal and is classified as anomalous. The classical framework for classifying diffusion regimes in polymers, introduced by Alfrey, Gurnee and Lloyd [519], distinguishes between three limiting cases based on the relative rates of diffusion and relaxation:

1. **Case I (Fickian diffusion)**: The diffusion rate is much slower than chain relax-

ation, so the polymer matrix equilibrates instantaneously with the local concentration field. Mass uptake scales as $t^{1/2}$ at short times.

2. **Case II diffusion**: The diffusion rate is much faster than chain relaxation, so the transport front advances at a constant velocity into the dry polymer. Mass uptake scales linearly with time ($t$). A sharp concentration front separates a fully swollen outer layer from a dry inner core.

3. **Anomalous diffusion**: The two timescales are comparable, producing intermediate behavior with a time exponent between 0.5 and 1. Both concentration-gradient-driven diffusion and relaxation-driven swelling contribute to mass transport.

A further regime, commonly referred to as pseudo-Fickian or surface-dominated diffusion, is observed when mass transport is governed primarily by adsorption at the polymer surface rather than by bulk diffusion [520]. In this case the time exponent is substantially below 0.5, reflecting the limited penetration of the diffusing species into the interior of the material. This regime is characteristic of low-permeability, glassy thermoplastics, including the copolyesters and TPU grades studied in the present work, in which the high chain packing density and low free volume restrict bulk water transport to the near surface region.

### A.1.4. The Ritger-Peppas fractional diffusion model

A unified mathematical framework capable of describing all of the above regimes was proposed by Ritger and Peppas [156], [157], based on the power-law equation:

$$\frac{M(t)}{M_\infty} = kt^n \tag{A.22}$$

where k is a kinetic constant that depends on the geometry and diffusivity of the system, and n is a dimensionless transport exponent that encodes the diffusion mechanism: $n = 0.5$ for Fickian diffusion, $0.5 < n < 1$ for anomalous diffusion,$n = 1$ for Case II diffusion, and $n < 0.5$ for surface-dominated transport [156]. While this empirical model provides a convenient classification tool, it does not account for the saturation behavior observed at long immersion times and is therefore only applicable to the early stages of absorption ($M(t)/M_\infty \leq 0.6$).

A more complete description, applicable across the full absorption curve including the saturation plateau, is obtained by formulating the transport problem in the framework of fractional calculus [140], [521]. The fractional differential equation governing anomalous diffusion in a viscoelastic medium takes the form [156], [522]:

$$D_t^\alpha [M(t)] = \Delta M_\infty \tau^{-\alpha} \tag{A.23}$$

where $D_t^\alpha$ denotes the Caputo fractional derivative of order $\alpha$ with respect to time, $\Delta M_\infty$ is the saturation weight gain, and $\tau$ is a characteristic absorption time. The Caputo fractional

derivative of order $\alpha \in (0, 1)$ is defined as [140], [358]:

$$D_t^{\alpha}[f(t)] = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{f'(s)}{(t-s)^{\alpha}}, \mathrm{d}s \tag{A.24}$$

Unlike the classical integer-order derivative, the Caputo operator is non-local in time: the value of $D_t^{\alpha}[f(t)]$ depends on the entire history of $f$ up to time t, weighted by the power-law kernel $(t - s)^{-\alpha}$. This hereditary character is physically appropriate for transport in viscoelastic media, where the memory of past swelling states influences the current transport rate, a direct analog of the non-Markovian dissipation discussed in Section A in the context of mechanical viscoelasticity [140], [521].

Integrating Equation A.23 and applying the initial condition $M(0) = M_0$ yields the closed-form solution [156], [522]:

$$M(t) = M_0 + \frac{\Delta M_{\infty}}{\Gamma(1+\alpha)} \left(\frac{t}{\tau}\right)^{\alpha} \tag{A.25}$$

where $\Gamma$ is the Euler gamma function, defined for positive real arguments as $\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$, which generalizes the factorial to non-integer values of $\alpha$ and ensures dimensional consistency across all diffusion regimes. The factor $1/\Gamma(1 + \alpha)$ arises naturally from the integration of the Caputo derivative and modulates the amplitude of the absorption curve as a function of the transport exponent.

### A.1.5. Physical interpretation of the model parameters

The three parameters of Equation A.25 each carry distinct physical meanings:

- $\Delta M_{\infty}$ is the saturation weight gain, representing the total mass of liquid absorbed by the material at thermodynamic equilibrium. It is governed primarily by the affinity between the polymer and the solvent, the free volume of the glassy matrix, and — in the case of semicrystalline materials — the fraction of the amorphous phase accessible to the diffusing species.

- $\tau$ is the characteristic absorption time, representing the timescale over which the material absorbs the bulk of its equilibrium liquid uptake. Materials with $\tau \geq 1$ *month* are generally considered low-permeability plastics with respect to water [302], as the characteristic diffusion time substantially exceeds the clinically relevant wear cycle of 10 to 14 days.

- $\alpha$ is the fractional transport exponent, which encodes the nature of the diffusion mechanism. Its physical interpretation follows the classification of Alfrey et al. [519] and Ritger and Peppas [156]: values near 0 indicate predominantly surface-dominated sorption with negligible bulk penetration; values near 0.5 correspond to classical Fickian diffusion driven by a concentration gradient; values between

0.5 and 1 indicate anomalous diffusion in which chain relaxation and concentration gradient-driven transport contribute comparably; and values approaching or exceeding 1 correspond to Case II or super-Case II diffusion, characteristic of glassy polymers immersed in strong plasticizing solvents [523], [524]. The $\alpha$ values of 0.08-0.13 measured for all materials in the present work confirm that water transport in these thermoplastics is predominantly surface-dominated, with negligible bulk penetration over the clinically relevant timescale, a finding consistent with their classification as low-permeability plastics.

### A.1.6. Relationship to the viscoelastic framework

It is instructive to note the formal analogy between the fractional diffusion model presented here and the fractional viscoelastic models discussed in Section A. In both cases, a Caputo fractional derivative of order $\alpha$ replaces the integer-order derivative of the classical model, introducing a power-law memory kernel that reflects the non-Markovian character of the underlying physical process. In the viscoelastic case, this memory arises from the broad spectrum of polymer chain relaxation timescales; in the diffusion case, it arises from the coupling between molecular transport and the slow relaxation of the polymer matrix. The fractional order $\alpha$ plays an analogous role in both contexts: it measures the degree to which the current response is influenced by the history of the system, and its value between 0 and 1 interpolates continuously between the purely elastic (infinite memory) and purely dissipative (memoryless) limits. This mathematical unity between viscoelastic relaxation and anomalous diffusion in polymers is not coincidental — both phenomena arise from the same underlying physical origin, namely the broad, hierarchical spectrum of molecular relaxation timescales that characterizes amorphous and semicrystalline thermoplastics [140], [154], [521].

# B. COMPLETE LIST OF COMMERCIAL CLEAR ALIGNER MATERIALS EVALUATED

Table B.1 lists all commercial clear aligner materials supplied by Secret Aligner S.L. for evaluation in the present work, obtained from 14 different sources worldwide. The table includes materials that were fully characterized as part of the screening study described in Chapter 4, as well as materials received for reference or partial characterization only. The subset of materials selected for the systematic mechanical and thermal screening is reported separately in Table 4.1 of Chapter 4.

| Name | Brand | Thickness (mm) | Chemistry | recommended use | Origin |
|---|---|---|---|---|---|
| Duran | Scheu-Dental | 0.76 | PETG | aligner/retainer | Germany |
| Biolon | Dreve | 0.76 | PETG | aligner/retainer | Germany |
| Leone | Leone | 1 | PETG | aligner/retainer | Italy |
| Attachment | Tristar | 0.31 | PETG | Template | Malaysia |
| Smilekeeper | Tristar | 0.76 | TPU | Aligner/retainer | Malaysia |
| PU flex | Taglus | 0.76 | TPU | Aligner/retainer | India |
| Premium | Taglus | 0.76 | PETG | Aligner/Retainer | India |
| Tuff | Taglus | 0.8 | PETG | Retainers | India |
| Ultra | Taglus | 0.76 | ML | Aligners | India |
| Ultra2 | Taglus | 0.76 | ML | aligners | India |
| Essix C+ | Dentsply Sirona | 1 | PP/PE | Guards/retainers | USA |
| Zendura FLX | Bay materials | 0.76 | ML | Aligners | USA |
| Zendura Viva | Bay materials | 0.76 | ML | Aligners | USA |
| Zendura Trak | Bay materials | 0.76 | PETG | Retainers | USA |
| Zendura AT | Bay materials | 0.31 | TPU | Template | USA |
| Premium | Maflex | 0.76 | ML | Aligners | China |
| Premium Plus | Maflex | 0.76 | ML | Aligners | China |
| Attachment | Maflex | 0.3 | TPU | Aligners | China |
| Retainer | Maflex | 0.76 | PETG | Retainer | China |
| TO40 | Maflex | 0.8 | PCTG | Retainer | China |
| C+ | Prismlab | 0.76 | PCTG | Aligner/retainer | China |
| Pro S | Prismlab | 0.76 | ML | Aligners | China |
| Pro S+ | Prismlab | 0.76 | ML | Aligners | China |
| Ounora | Prismlab | 0.76 | ML | Aligners | China |
| Multilayer | Filtzwiesser | 0.76 | ML | Aligners | Germany |
| Pro S | Molekur | 0.76 | ML | Aligners | China |
| Pro S+ | Molekur | 0.76 | ML | Alignesr | China |
| Express | Molekur | 1 | TPU | Retainers | China |
| Splint | SmileFind | 0.8 | PETG | Guard/retainer | China |
| Vivak MED | Exolon | 0.8 | PETG | Aligner/retainer | Belgium |
| Inspiria MED | Exolon | 0.8 | PCTG | Aligner/retainer | Belgium |

Table B.1: List of all commercial clear aligner materials supplied by Secret Aligner

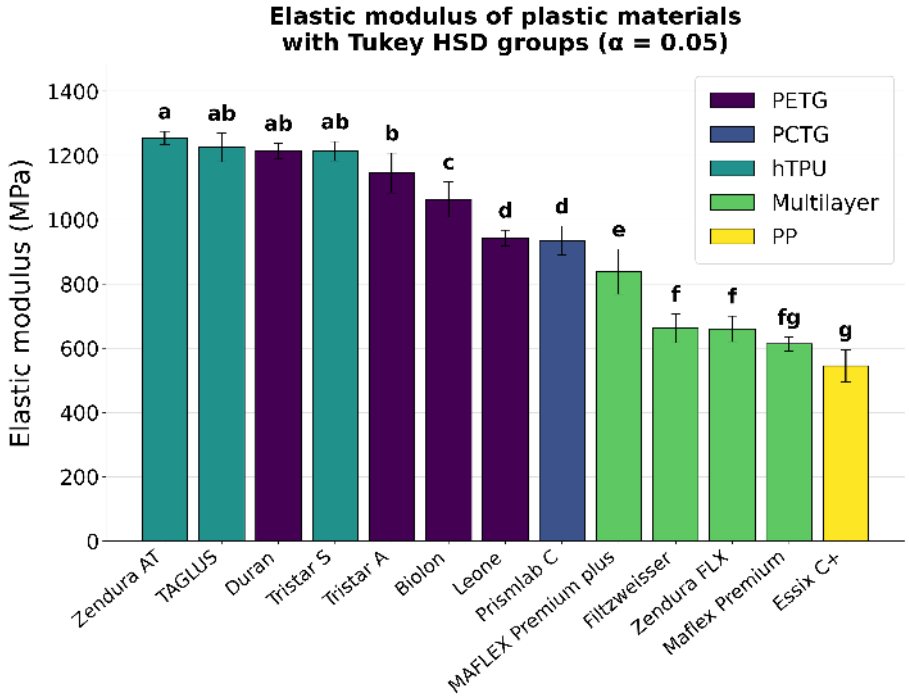
# C. EXTRUSION PROFILES

| Material | Speed (rpm) | T1 (ºC) | T2 (ºC) | T3 (ºC) | T4 (ºC) | T5 (ºC) | T6 (ºC) | T Nozzle (ºC) | Nozzle aperture (mm) | T Roller (ºC) |
|---|---|---|---|---|---|---|---|---|---|---|
| MP100 | 100 | 180 | 195 | 215 | 230 | 245 | 260 | 270 | 0.3 | 60 |
| MP150 | 100 | 180 | 195 | 215 | 230 | 245 | 260 | 270 | 0.3 | 70 |
| MP200 | 100 | 180 | 195 | 215 | 230 | 245 | 260 | 270 | 0.3 | 70 |
| TPU85A | 50 | 180 | 185 | 190 | - | - | - | 185 | 0.25 | 50 |
| TPU85A | 50 | 180 | 185 | 190 | - | - | - | 185 | 0.5 | 50 |
| TPU 52D | 50 | 190 | 195 | 205 | - | - | - | 200 | 0.3 | 40 |
| TPU52D | 50 | 190 | 195 | 200 | - | - | - | 195 | 0.3 | 40 |
| TPU52D | 50 | 190 | 195 | 200 | - | - | - | 195 | 0.25 | 40 |
| TPU70D | 40 | 200 | 210 | 215 | - | - | - | 205 | 0.2 | 50 |

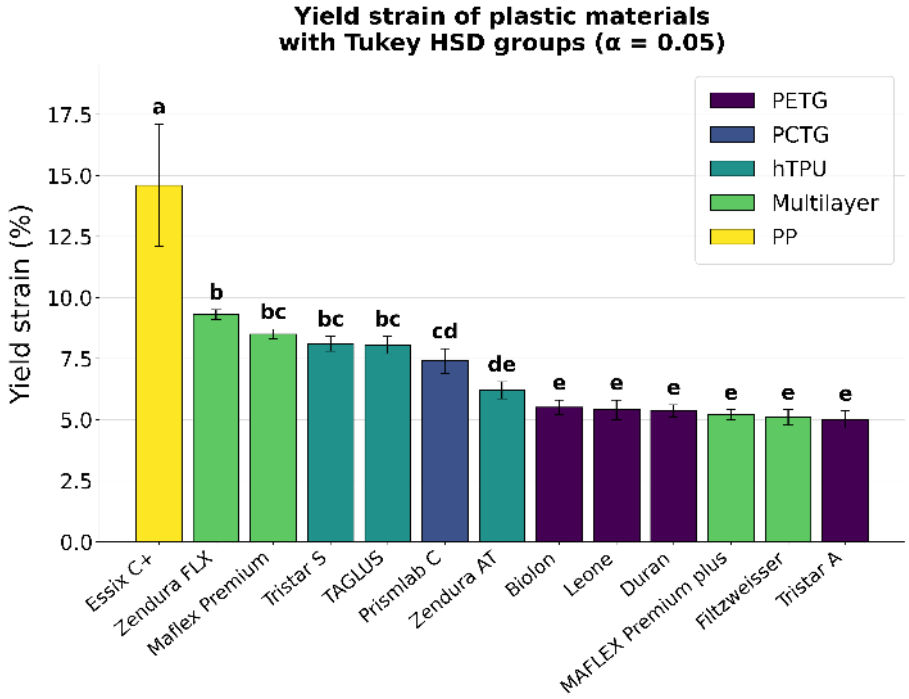
Table C.1: Extrusion thermal profiles

# D. STATISTICAL ANALYSIS OF UNIAXIAL TENSILE PROPERTIES

The following figures support the statistical analysis of the uniaxial tensile tests discussed in Figure D.1, Bars are arranged in descending order and different letters indicate statistically significant differences (Tukey's HSD test, $\alpha = 0.05$).




(a) Elastic Modulus




(b) Yield strain

Figure D.1: Tensile parameters of as-received commercial materials. Materials are arranged in descending order for both parameters. Different letters indicate statistically significant differences as determined by Tukey's HSD test (p = 0.05)

# E. DETERMINATION OF THE ACTIVATION ENERGY BY THE COATS-REDFERN METHOD

The activation energy ($E_a$) of the main thermal degradation process reported for all as-received and thermoformed materials (Table 4.17) was obtained from the TGA curves using the integral method of Coats and Redfern [525], [526]. This appendix details the kinetic model on which the method rests, the integration leading to the linearized expression used for the regression, and the criteria applied when selecting the region of each thermogram to be analyzed.

## E.1. Kinetic model

The thermal degradation of a polymer can be described by a single-step, Arrhenius-type kinetic model in which the rate of conversion is the product of a temperature-dependent kinetic constant $K(T)$ and a function of the degree of conversion $f(\alpha)$:

$$\frac{d\alpha}{dt} = K(T)\, f(\alpha) = \frac{A(1-\alpha)}{\beta} \operatorname{Exp}\left(-\frac{E_a}{RT}\right) \tag{E.1}$$

Here $\alpha$ denotes the degree of conversion during degradation, that is, the fraction of material degraded throughout the experiment; $A$ is the pre-exponential factor, also referred to as the collision frequency factor; $E_a$ is the activation energy; $R$ is the universal gas constant (KJ/mol K); $\beta = dT/dt$ is the constant heating rate applied during the experiment; and $f(\alpha)$ is the reaction model, determined by the reaction order of degradation. Because the experiments were carried out at a constant heating rate, the time derivative in Equation E.1 can be expressed as a function of temperature through $d\alpha/dt = \beta\,(d\alpha/dT)$.

## E.2. Reaction order

The reaction order characterizes the dependence of the degradation rate on the remaining polymer fraction during the thermal process, and defines the reaction model as $f(\alpha) = (1-\alpha)^n$. A zero-order reaction implies a constant degradation rate, independent of the polymer concentration; a first-order reaction indicates a rate directly proportional to the remaining polymer fraction; a second-order reaction implies that the rate is proportional to the square of the remaining polymer amount; and so forth [527], [528].

For polymeric materials, first-order kinetics is most commonly assumed [529], [530], as it has been shown that a first-order approximation to an $n^{th}$-order thermogravimetric curve remains a good approximation over a wide range of heating rates, with the error involved in assuming first-order kinetics being small for most practical purposes [529]. Accordingly, throughout this work $f(\alpha)$ was taken as $f(\alpha) = 1-\alpha$.

## E.3. Integration: the Coats-Redfern equation

Introducing $f(\alpha) = (1 - \alpha)$ into Equation E.1 and separating variables gives

$$\frac{d\alpha}{1-\alpha} = \frac{A}{\beta}\,\mathrm{Exp}\!\left(-\frac{E_a}{RT}\right)dT. \tag{E.2}$$

Integrating the left-hand side from 0 to $\alpha$ and the right-hand side from 0 to $T$,

$$-\mathrm{Ln}(1-\alpha) = \frac{A}{\beta}\int_0^T \mathrm{Exp}\!\left(-\frac{E_a}{RT}\right)dT. \tag{E.3}$$

The temperature integral in Equation E.3 has no closed-form analytical solution. Coats and Redfern evaluated it through its asymptotic expansion and retained the leading term,

$$\int_0^T \mathrm{Exp}\!\left(-\frac{E_a}{RT}\right)dT \;\approx\; \frac{RT^2}{E_a}\left(1-\frac{2RT}{E_a}\right)\mathrm{Exp}\!\left(-\frac{E_a}{RT}\right) \;\approx\; \frac{RT^2}{E_a}\,\mathrm{Exp}\!\left(-\frac{E_a}{RT}\right), \tag{E.4}$$

where the second approximation is justified because $2RT/E_a \ll 1$ over the temperature range of the main degradation process for the materials studied. Substituting Equation E.4 into Equation E.3, dividing by $T^2$, and taking natural logarithms yields the linearized Coats-Redfern equation:

$$\mathrm{Ln}\!\left[-\frac{\mathrm{Ln}(1-\alpha)}{T^2}\right] = \mathrm{Ln}\!\left(\frac{AR}{E_a\beta}\right) - \frac{E_a}{RT}. \tag{E.5}$$

A plot of $\mathrm{Ln}\left[-\mathrm{Ln}(1-\alpha)/T^2\right]$ against $1/T$ is therefore a straight line whose slope, $-E_a/R$, provides the activation energy, while the intercept, $\mathrm{Ln}(AR/E_a\beta)$, provides the pre-exponential factor.

For completeness, the general form for an arbitrary reaction order $n \neq 1$ follows the same procedure with $\int d\alpha/(1-\alpha)^n = \left[1-(1-\alpha)^{1-n}\right]/(1-n)$, leading to

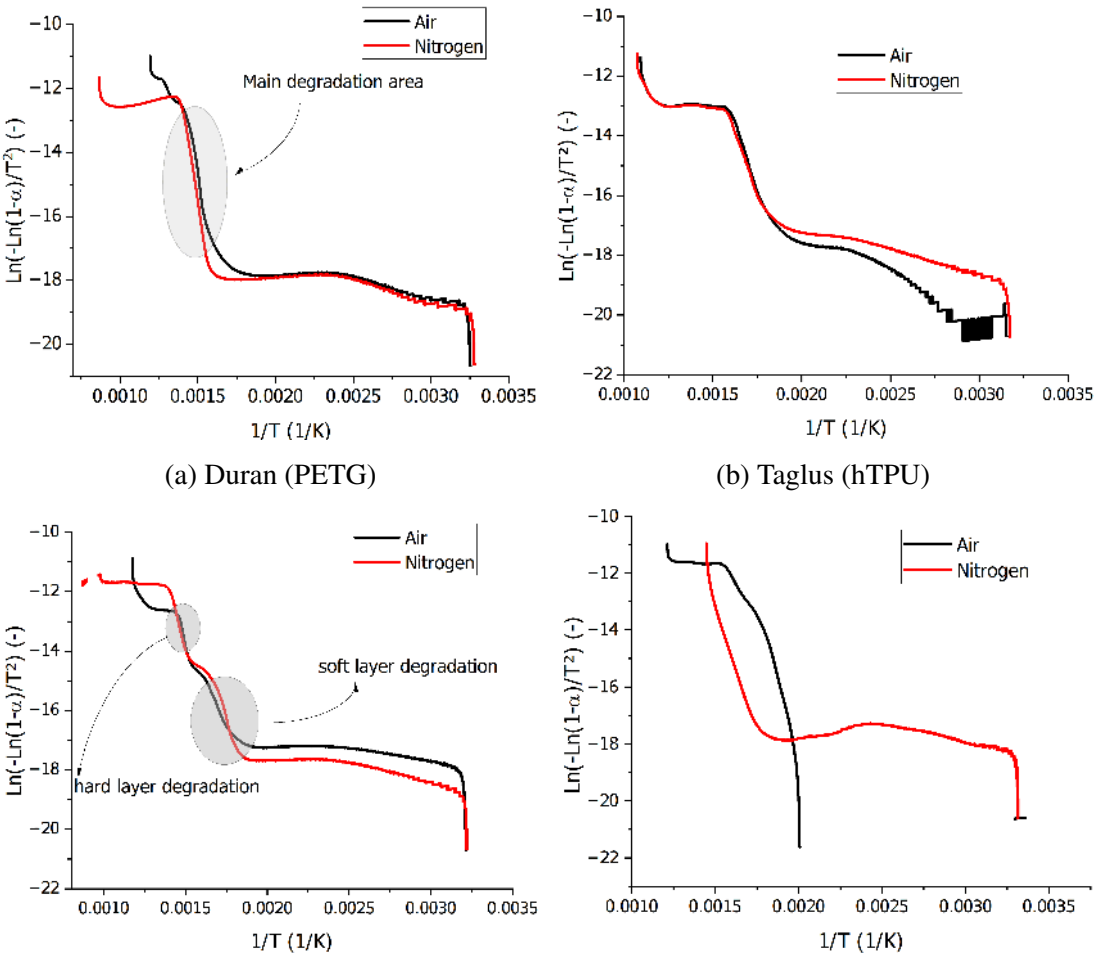
$$\mathrm{Ln}\!\left[\frac{1-(1-\alpha)^{1-n}}{T^2(1-n)}\right] = \mathrm{Ln}\!\left(\frac{AR}{E_a\beta}\right) - \frac{E_a}{RT}, \tag{E.6}$$

which reduces to Equation E.5 in the first-order limit.

## E.4. Selection of the analysis region

Figure E.1 shows representative Coats-Redfern plots for the different thermoplastics. Only the main degradation region was fitted. The processes visible on the left-hand side of each plot correspond to the charring degradation of PETG discussed previously and, in the case of the hTPU materials, to the minor co-monomers; these were not analyzed owing to the lack of accuracy in that region. The part of the plot to the right of the main degradation region corresponds to a pre-degradation stage in which the conversion $\alpha$ is very low and noisy, and was likewise excluded from the fit. For Zendura FLX (ML) it

was possible to separate the contributions of the different layers and obtain the activation energy of each one.




(a) Duran (PETG)

(b) Taglus (hTPU)

(c) Zendura FLX (ML)

(d) Essix C+ (PP)

Figure E.1: Coats–Redfern plots of the main degradation region in air

# F. THREE-POINT BENDING STRESS RELAXATION AT 4% AND 6% STRAIN

The three-point bending stress relaxation results at 2% strain, representative of the elastic regime, are reported and discussed in the main text (Section 4.3). This appendix reports the full results at the 4% (transitional) and 6% (post-yield) strain levels, including the complete relaxation and recovery parameters for all materials and the detailed material-by-material analysis.

## F.1. 4% strain

Increasing the applied strain to 4% reveals substantially greater differences between materials than were observed at 2%, both in stress relaxation and in strain recovery (Figure F.1, Table F.1). At this level the materials operate in or near the transitional regime between elastic and anelastic deformation, and the divergence of the relaxation curves reflects the increasing influence of viscoelastic and plastic mechanisms specific to each material class. Minor experimental artefacts attributable to specimen slippage are present in some curves, but all experiments yielded data of sufficient quality for analysis.

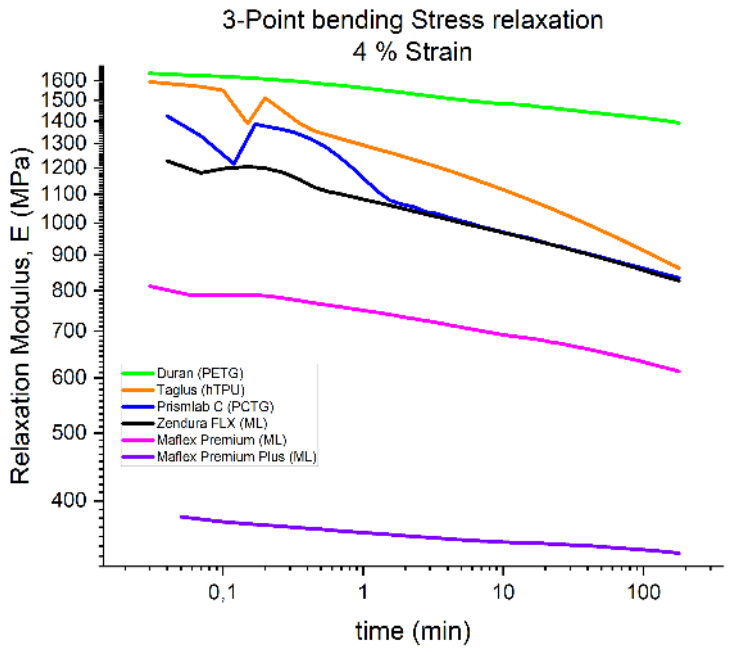



Figure F.1: Stress relaxation curves at 4% strain in 3PB

The most notable feature is the markedly different decay behaviour of Taglus (hTPU), which shows the highest decay exponent ($a = 0.1$) and the lowest retained modulus after

3 h (53%). These values must be interpreted with caution: the 18 N force limit of the equipment was reached before the target strain was fully attained, so the reported $E_0$ (1425 MPa), paradoxically the highest in the experiment, and inconsistent with the lowest $E_0$ shown by Taglus at 2% strain, is an artefact of the truncated loading ramp rather than an intrinsic stiffening. The Taglus results at 4% are therefore not fully reliable.

| Material | E0 (MPa) | $\varepsilon_0$ (%) | a | $E(3h)/E(0)$ (%) | Strain Recovery (%) |
|---|---|---|---|---|---|
| Duran (PETG) | 1392 | 3.98 | 0.02 | 57 | 55 |
| Taglus (hTPU) | 1425 | 3.88 | 0.1 | 53 | 65 |
| Prismlab C (PCTG) | 1227 | 3.91 | 0.05 | 63 | 72 |
| Zendura FLX (ML) | 1342 | 4.04 | 0.05 | 61 | 75 |
| Filtzwiesser (ML) | 1320 | 3.98 | 0.04 | 60 | 65 |
| Maflex Premium (ML) | 812 | 4.05 | 0.04 | 70 | 82 |
| Maflex Premium Plus (ML) | 390 | 3.98 | 0.01 | 86 | 56 |

Table F.1: Stress relaxation and strain recovery parameters for the 4% strain experiment in 3PB

Within the monolithic group, Duran (PETG) and Prismlab C (PCTG) retain 57% and 63% of their initial moduli, with strain recovery values of 55% and 72% respectively. The substantially lower recovery of Duran indicates that PETG has entered its plastic deformation regime at this strain, whereas PCTG, by virtue of its higher yield strain, remains predominantly within the recoverable anelastic region.

Within the multilayer group the response depends strongly on outer-layer chemistry. Filzwiesser (PETG/sTPU/PETG) retains 60% of its modulus and recovers 65% of the strain; its retained modulus is close to that of monolithic Duran, consistent with the dominant role of its PETG outer layers in governing the bending response. Zendura FLX (PCTG/sTPU/PCTG) shows intermediate behaviour, retaining 61% and recovering 75%, while Maflex Premium (PCTG/sTPU/PCTG) presents the most favourable profile at this strain level, retaining 70% and achieving the highest strain recovery of all materials (82%). The superior recovery of the PCTG/sTPU/PCTG laminates relative to both the PETG/sTPU/PETG architecture and monolithic PETG shows that the choice of outer-layer copolyester, and not merely the presence of a soft interlayer, is a determining factor in recovery performance at clinically relevant strains. This consistently higher recovery of the multilayers relative to the monolithic sheets is attributed to the elastomeric sTPU interlayer, whose elastic recoil drives recovery of the laminate beyond what monolithic copolyester or hTPU sheets achieve alone.

Maflex Premium Plus (sTPU/hTPU/sTPU) again stands apart from all other materials, showing the lowest $E_0$ of the dataset (390 MPa) yet the highest retained modulus ratio (86%). The low $E_0$ is consistent with the neutral axis of the laminate passing through the centre of its hTPU core, which, operating beyond its elastic limit at 4% strain, presents a substantially reduced apparent bending modulus, while the compliant outer sTPU lay-

ers contribute negligibly to the measured stiffness. Its strain recovery, however, drops markedly to 56%, indicating that a significant fraction of the deformation imposed on the hTPU core becomes irrecoverable within the 5-minute recovery window.

### F.2. 6% strain

At 6% strain the differences between materials become most pronounced, and several limitations of the three-point bending configuration become apparent at this deformation level (Figure F.2, Table F.2).

Taglus (hTPU) could not be characterised at this strain, as the 18 N force limit was again reached before the target deformation was achieved, precluding the acquisition of reliable relaxation or recovery data. This confirms that monolithic hTPU is unsuitable for bending stress relaxation experiments at strains of 4% and above in 3PB using the present setup, and its entry is therefore absent from Table F.2.

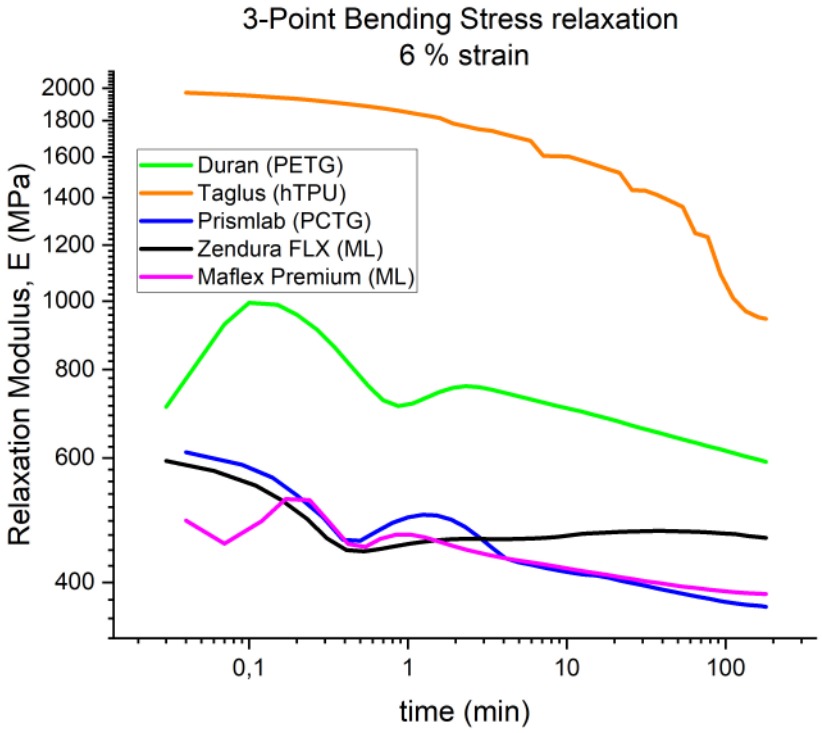



Figure F.2: Stress relaxation curves at 6% strain in 3PB

Among the remaining materials, $E_0$ now spans almost an order of magnitude, from 260 MPa (Maflex Premium Plus) to 1220 MPa (Duran), reflecting the increasing sensitivity of the apparent initial modulus to strain as materials progressively enter their post-yield regimes. Duran (PETG) retains the highest $E_0$ but shows the most pronounced relaxation, retaining only 49% of its initial modulus after 3 h and recovering just 35% of the applied strain; the lowest recovery value observed across all bending experiments; consistent with

6% strain placing PETG well within the plastic deformation plateau identifiable in the tensile stress-strain curves (Figure 4.13). Prismlab C (PCTG) shows a markedly lower $E_0$ (610 MPa) yet retains 60% of its modulus and recovers 44% of the strain, values superior to those of Duran, consistent with PCTG possessing a higher yield strain than PETG.

| Material | E0 (MPa) | $\varepsilon_0$ (%) | a | $E(3h)/E(0)$ (%) | Strain Recovery (%) |
|---|---|---|---|---|---|
| Duran (PETG) | 1220 | 6.45 | 0.06 | 49 | 35 |
| Taglus (hTPU) | 1935 | 3.39 | - | - | - |
| Prismlab C (PCTG) | 610 | 6.00 | 0.04 | 60 | 44 |
| Zendura FLX (ML) | 606 | 5.95 | 0.04 | 77 | 40 |
| Filzwiesser (ML) | 492 | 5.96 | 0.06 | 51 | 39 |
| Maflex Premium (ML) | 525 | 6.04 | 0.03 | 72 | 41 |
| Maflex Premium Plus (ML) | 260 | 5.79 | 0.01 | 91 | 55 |

Table F.2: Stress relaxation and strain recovery parameters for the 6% strain experiment in 3PB

Within the multilayer group, the architectural differentiation established at 4% strain becomes sharper. Filzwiesser (PETG/sTPU/PETG) retains only 51% of its initial modulus and recovers 39% of the strain, values closely matching those of monolithic Duran (49% and 35% respectively), confirming that its PETG outer layers are operating deep within their plastic regime at 6% strain and that the soft sTPU core provides negligible additional recovery in bending at this deformation level. In marked contrast, Zendura FLX and Maflex Premium, both employing PCTG outer layers, retain 77% and 72% of their initial moduli respectively and recover approximately 40% of the strain. The convergence of $E_0$ between Filzwiesser (492 MPa), Zendura FLX (606 MPa), and Maflex Premium (525 MPa), combined with the divergence in retained modulus ratio (51% versus 77% and 72%), provides the clearest evidence in the dataset that the choice of outer-layer copolyester (PETG versus PCTG) is the dominant determinant of long-term force retention and recovery capacity at high bending strains, independently of the soft-core composition.

Maflex Premium Plus (sTPU/hTPU/sTPU) again exhibits the most distinctive profile: the lowest $E_0$ (260 MPa), the highest retained modulus ratio (91%), and the highest strain recovery (55%). The extremely low $E_0$ reflects the dominant contribution of the hTPU core operating beyond its elastic limit; the high retained modulus ratio reflects its superior capacity to sustain residual stress over time; and the 55% recovery, though the highest among all materials at 6% strain, remains limited by the plastic deformation accumulated in the hTPU core at this deformation level.

### F.3. Summary of the bending stress relaxation behaviour

Across the three strain levels investigated, the 3PB experiments show a consistent picture. At 2% strain (main text) all materials behave similarly, retaining 82-87% of their modulus after 3 h and recovering 87-92% of the strain, and the soft interlayer of the laminates plays no discernible role. As strain increases to 4% and 6%, material-dependent differences emerge: monolithic PETG loses both retained modulus and recovery with increasing strain as irreversible plastic deformation develops beyond its yield point, whereas PCTG, with its higher yield strain, maintains superior recovery at equivalent strains. Among the multilayers, outer-layer chemistry is the dominant factor: Filzwiesser (PETG outer layers) mirrors the response of monolithic Duran at both 4% and 6% strain, whereas the PCTG/sTPU/PCTG laminates (Zendura FLX and Maflex Premium) retain substantially more modulus and recover better, an effect attributable to the higher yield strain of the PCTG outer layers rather than to the sTPU interlayer per se. Maflex Premium Plus (sTPU/hTPU/sTPU) behaves throughout as a thin hTPU monolayer, with consistently low absolute stiffness, the highest retained modulus ratios, and intermediate recovery. Finally, because the stiff outer layers carry most of the bending stress while the soft core is loaded predominantly in shear, the three-point bending configuration is intrinsically limited for characterising these multilayer materials; this limitation, compounded by specimen instability at higher strains, motivated the adoption of tensile stress relaxation as the primary characterisation mode (main text).

# G. TENSILE STRESS RELAXATION AT 1.5% STRAIN AND FORCE-CEILING LIMITATIONS

The tensile stress relaxation results at 0.5% strain (elastic baseline) and at 2% strain (the decisive comparison between the two PCTG/sTPU/PCTG laminates) are reported and discussed in the main text (Section 4.4). This appendix reports the intermediate 1.5% (transitional) strain level in full, and details the 18 N force-ceiling artifact that affects the stiffer materials at the 1.5% and 2% strain levels.

## G.1. Relaxation and recovery at 1.5% strain

At 1.5% strain the differences between materials become more pronounced (Figure G.1, Table G.1). The ordering of $E_0$ is broadly preserved, with Taglus highest (1442 MPa) and Prismlab C next (1001 MPa); Duran shows a reduced $E_0$ (884 MPa, versus 937 at 0.5%), consistent with the onset of viscoelastic softening, while Maflex Premium Plus reaches 850 MPa and the remaining multilayers 544-616 MPa. Taglus and Prismlab C could not reach the target strain, as the 18 N force ceiling was reached first (achieving only 0.92% and 1.35% respectively), and the Taglus curve is further affected by an experimental discontinuity visible in Figure G.1; the origin and bounds of this force-ceiling artifact are detailed in Section G.2.

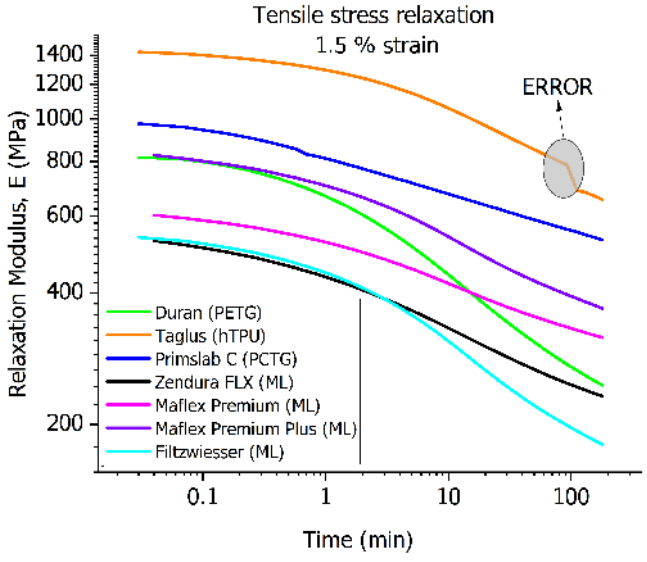



Figure G.1: Stress relaxation curves at 1.5% strain in tensile configuration

Regarding relaxation, the retained modulus of Duran drops sharply relative to 0.5% (from 50 to 27%), with the highest decay exponent at this strain ($a = 0.21$) and a cor-

respondingly low strain recovery (22%), indicating that PETG has entered a regime of accelerated viscoelastic relaxation. Prismlab C, by contrast, shows the most favourable behaviour ($a = 0.08$; 52% retained, 51% recovery), subject to the force-ceiling caveat noted above. Among the multilayers, Zendura FLX shows the lowest retained modulus (42%) and recovery (36%), while Maflex Premium and Maflex Premium Plus are intermediate (approximately 51% and 43%); Maflex Premium Plus again tracks the monolithic materials more closely than the other laminates, consistent with its hTPU-dominated response.

| Material | E0 (MPa) | $\varepsilon_0$ (%) | a | $E(3h)/E(0)$ (%) | Strain Recovery (%) |
|---|---|---|---|---|---|
| Duran (PETG) | 884 | 1.48 | 0.21 | 27 | 22 |
| Taglus (hTPU) | 1442 | 0.92 | - | - | - |
| Prismlab C (PCTG) | 1001 | 1.35 | 0.08 | 52 | 51 |
| Zendura FLX (ML) | 544 | 1.51 | 0.13 | 42 | 36 |
| Filzwiesser (ML) | 552 | 1.51 | 0.18 | 33 | 27 |
| Maflex Premium (ML) | 616 | 1.51 | 0.11 | 51 | 50 |
| Maflex Premium Plus (ML) | 850 | 1.49 | 0.13 | 43 | 43 |

Table G.1: Relaxation and strain recovery parameters for the 1.5% strain experiment in tensile configuration

## G.2. Force-ceiling limitations

In several of the tensile stress relaxation experiments the 18 N force limit of the DMTA was reached before the target strain could be attained, which affects the validity of the extracted parameters for the stiffer materials at the 1.5% and 2% strain levels. This section describes the nature of the resulting artefact and the extent to which the affected data can be interpreted.

### G.2.1. Nature of the artifact

When the force ceiling is reached, the instrument maintains a constant load while continuing to displace the specimen, effectively transitioning from the prescribed constant-strain relaxation protocol to a creep experiment. The resulting curves therefore do not represent true stress relaxation at a fixed strain, and their $E(3h)/E(0)$ and strain recovery values are not directly comparable with those of the materials that reached the target strain. Because a lower applied strain inherently produces less relaxation and greater recoverability, the parameters extracted from affected experiments are systematically optimistic. It is important to note that this behaviour is not attributable to grip slippage or to statistical outliers, but reflects an intrinsic mechanical characteristic of the materials themselves: they generate forces exceeding 18 N at moderate strain levels under the prescribed displacement ramp.

### G.2.2. Quantitative bounds

At 1.5% strain, Prismlab C (PCTG) reached a maximum strain of 1.35% rather than the 1.5% target, a 10% shortfall. Its reported retained modulus ratio (52%) and strain recovery (51%) are consequently expected to be artificially favourable, since a lower applied deformation produces less relaxation and greater recoverability. Given the non-linear dependence of these parameters on strain, a simple proportional correction is not warranted; however, the 10% shortfall provides a lower bound on the degree of overestimation, and the true values at 1.5% are likely to be meaningfully lower. For Taglus (hTPU) the discrepancy is considerably larger: the achieved strain (0.92%) represents only 61% of the target, so the reported $E_0$ (1442 MPa) cannot be considered representative of the material response at 1.5% strain, and the remainder of the relaxation curve is unreliable.

At 2% strain the force ceiling was reached before the target in four of the six materials, which attained actual strains of 1.04% (Duran), 1.12% (Taglus), 1.42% (Prismlab C), and 1.59% (Maflex Premium Plus). Valid relaxation parameters could therefore be obtained only for Zendura FLX and Maflex Premium, both of which reached the full 2% target. The relaxation data for Maflex Premium Plus are reported with caution given the 20% strain shortfall, while those for Duran, Taglus, and Prismlab C are considered unreliable and are excluded from the main-text discussion.

### G.2.3. Clinical corollary

Although these experiments do not yield valid relaxation parameters for the affected materials, the underlying behaviour is itself mechanically informative. Materials that exceed the 18 N instrument ceiling at moderate strain ( Taglus (hTPU) and Prismlab C (PCTG)) in particular, are capable of generating forces that may surpass the physiological limits recommended for orthodontic tooth movement, especially if the prescribed activation distance is not carefully controlled. Stiffer materials thus leave a narrower margin for dimensional inaccuracy before clinically excessive forces are produced, underscoring the importance of material selection and activation magnitude in clear aligner treatment planning.

# H. DECLARATION OF USE OF GENERATIVE ARTIFICIAL INTELLIGENCE TOOLS

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| YES | NO |
|---|---|

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Please be aware that the use of AI carries some risks and may lead to a series of serious academic consequences: the thesis will not be evaluated by the University if the use of AI involves the use of confidential data, copyrighted materials, or personal data, used without complying with the conditions required in each case (authorization of the interested parties, authorization of the owners, following the instructions of the University).

| **Question** | |
|---|---|
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| YES, I have used this data with permission | NO, I have not used confidential data |
| 2. In my interaction with Generative AI tools, I have submitted **copyrighted materials** with the permission of those concerned. | |
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| | |
|---|---|
| 3. In my interaction with Generative AI tools, I have submitted **personal data** with the consent of the data subjects. | |
| YES, I have used this data with the authorization of the interested parties and in accordance with the instructions contained in the guide approved by the University. | NO, I have not used personal data |
| 4. My use of the Generative AI tool has **respected its terms of use**, as well as the essential ethical principles, not being maliciously oriented to obtain an inappropriate result for the work presented, that is to say, one that produces an impression or knowledge contrary to the reality of the results obtained, that supplants my own work or that could harm people**.** | |
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- *Supporting reflection in relation to the development of the work: iterative process of analysing alternatives and approaches using AI*

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| *No analysis was developed by AI during this thesis. Documentation was not sourced by any AI tool.* |

- *Revision or rewriting of previously drafted paragraphs*

| |
|---|
| *Claude (Anthropic, versions Sonnet and Opus, 2025-2026) was used to assist with proofreading, grammar correction, and language refinement throughout the manuscript.* |

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***Develop specific content***

*Generative AI has been used as a support tool for the development of the specific content of the dissertation, including:*

- *Assistance in the development of lines of code (programming)*

| |
|---|
| *Claude (Anthropic, versions Sonnet and Opus, 2025-2026) was used as assistant in python code development, specifically to improve image clarity and aesthetics* |

- *Generation of diagrams, images, audios or videos*

| |
|---|
| *Claude (Anthropic, versions Sonnet and Opus, 2025-2026) was employed to generate explanatory workflow diagrams.*<br>*ChatGPT (OpenAI, GPT-4o, 2026) was used to generate additional illustrative figures.*<br>*All the prompts for figures and workflow diagram generation were elaborated by the autor.* |

- *Optimisation processes*

| |
|---|
| *No optimization process was done by an AI tool during this thesis* |

- *Data processing: collection, analysis, cross-checking of data...*

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- *Inspiration of ideas in the creative process*

| |
|---|
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Please provide a personal assessment (free format) of the strengths and weaknesses you have identified in the use of Generative AI tools in the development of your work. Mention if it has helped you in the learning process, or in the development or drawing conclusions from your work.

During the thesis it was discovered that although AI tools may be helpful in coding checking and code troubleshooting, the whole code must be developed first by an expert with a clear focus.
Image generation resulted to be relatively easy and accurate provided a highly specific and elaborated prompt given to the AI.
Regarding the spelling correction and grammar checking, I found it really helpful on finding spelling errors, synonyms, and shortening phrases. However it required human supervision to avoid AI to change the meaning and focus of the texts in some cases.